



CD200 Series Digital Pulse Servo Drive User Manual

Preface

Thank you for choosing the CD200 digital pulse servo drive!

The CD200 digital pulse servo drive is a general-purpose, high-performance AC servo drive developed by Simphoenix. It supports communication protocols such as Modbus and CANopen, and by adopting the corresponding communication interface, multiple servos can be networked and operated in conjunction with an upper computer.

This series features the latest model-based tracking control and gain adjustment-free functionality, enabling active identification and correction of deviations in input commands for more responsive equipment performance and greater motion precision. Through real-time command reconstruction and dynamic compensation, it effectively eliminates jitter and jerk during movement, ensuring smooth operation across the entire speed range from low to high, and significantly improving machining and control accuracy of the equipment. In addition, functions such as single-parameter auto-tuning, 1-second inertia identification, online load measurement, online resonance suppression, end vibration suppression, and pulse/bus communication make servo control convenient and easy to use. Paired with the CM10 series high-response servo motors, the system delivers reliable and stable performance, making it suitable for automation equipment in typical industries including packaging, food production, CNC cutting, textiles, machine tools, and wood carving. With a high-performance solution, it enables fast and precise position control, speed control, and torque control.

This manual is the product user manual for the CD200 digital pulse servo drive, providing content and precautions covering product selection, installation, wiring, basic debugging, function descriptions, parameter descriptions, and troubleshooting. For first-time users, please read this manual carefully. For further information regarding product usage, please contact our technical staff for more detailed and professional support! Thanks for your use!

The content of this manual is subject to change without prior notice as Simphoenix Electric continuously optimizes product functionality and performance. For the latest updates, please refer to our official website: www.simphoenix.com.cn.

If you identify any discrepancies in this manual or encounter issues with the product, please contact our 24/7 customer service team:

- **Hotline:** 400-8819-800
- **Email:** Marketing@sunfars.com

We highly value your feedback and appreciate your trust in Simphoenix Electric. Thank you for your support!

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Chapter 1 Safety Precautions

1.1 Safety Statement

- This chapter explains the safety precautions that must be observed for the correct use of this product. **Read and understand all safety instructions thoroughly before use.** Failure to comply with the matters stipulated in the safety precautions may result in serious injury and even death, or equipment damage.
- The "DANGER", "WARNING", and "CAUTION" notices in this manual do not represent all safety matters that should be observed; they serve only as a supplement to all safety precautions.
- This product should be used in an environment that meets the design specifications. Otherwise, it may cause malfunctions. Functional abnormalities or component damage caused by non-compliance with relevant regulations are not covered by the product quality warranty.
- Our company will not assume any legal responsibility for personal safety accidents, property damage, etc., caused by non-compliance with the contents of this manual or by operating the product in violation of regulations.

Safety Level Definitions



Danger

Indicate that failure to follow instructions will result in serious physical injury or even death.



Warning

Indicate that failure to follow instructions may result in serious physical injury or even death.



CAUTION

Indicate that failure to follow instructions may result in minor physical injury or equipment damage.

Safety Precautions

- In this manual, product illustrations may sometimes show the product with its cover or safety guard removed to display detailed parts.
- When using this product, always ensure that the cover or guard is properly installed and operate according to the instructions in the manual.
- Product diagrams in this manual are for illustration purposes only and may differ slightly from the product you ordered. Please refer to the actual product ordered.
- Operators must take mechanical protective measures to ensure personal safety. Please wear and use necessary protective equipment, such as safety shoes, safety clothing, safety glasses, protective gloves, and arm sleeves.

Unboxing Inspection



Warning

- Unpacking and acceptance: If the product or its accessories are found to be damaged, rusted, or show signs of prior use during unpacking, do not install!
- If water ingress, missing components, or damaged components are found inside the product during unpacking, do not install!
- Please carefully check the packing list. If the packing list does not match the product name, do not install!



CAUTION

- Before unpacking, please check if the external packaging of the equipment is intact, free from damage, wetting, dampness, or deformation.
- Please open the packaging in sequential order, and strictly prohibit violent striking!
- When unpacking, please check the surface of the equipment and accessories for any residual damage, rust, or impact Marks.
- After unpacking, please carefully compare with the packing list to verify the quantity of equipment and accessories, and whether all documents are complete.

Storage and Transportation



Warning

- Always use professional lifting equipment and have large or heavy products moved by qualified professionals with operating licenses. Otherwise, there is a risk of injury or product damage!
- Before vertically lifting the product, confirm that the front cover, terminal block, and other components of the product are securely fastened with screws. Otherwise, there is a risk of components falling off, leading to personal injury or product damage!
- No personnel are allowed to stand or remain under the product when it is being lifted by lifting equipment.
- When lifting the product with steel wire ropes, lift it smoothly and at a constant speed. Do not subject the product to vibration or impact, do not turn the product over, and do not keep the product suspended for extended periods. Otherwise, there is a risk of personal injury or product damage!



CAUTION

- Always lift and place the product gently when moving it, and always be aware of objects underfoot to prevent tripping or falling. Otherwise, there is a risk of injury or product damage!
- When carrying the product by hand, always grasp the product casing firmly to prevent components from falling off. Otherwise, there is a risk of injury!
- Please strictly follow the product's required storage and transportation conditions for storage and transportation. Otherwise, there is a risk of product damage.
- Avoid storing and transporting in locations exposed to water splashes or rain, direct sunlight, strong electric fields, strong magnetic fields, or severe vibrations.
- Avoid storing the product for more than 3 months. If stored for an extended period, more stringent protection and necessary inspections should be performed.
- Please package the product strictly before vehicle transportation. For long-distance transportation, enclosed containers must be used.
- It is strictly prohibited to transport this product mixed with equipment or items that may affect or damage this product.

During installation



Danger

- Only professional personnel who have received training related to electrical equipment and possess electrical knowledge are allowed to operate. Non-professional personnel are strictly prohibited from operating!



Warning

- Before installation, be sure to carefully read the product user manual and safety precautions!
- Do not install this product in locations with strong electric fields or strong electromagnetic wave interference!
- Before performing installation work, ensure that the mechanical strength of the installation location is sufficient to support the weight of the equipment, otherwise, it may lead to mechanical hazards.
- When performing installation work, do not wear loose clothing or jewelry, otherwise, there may be a risk of electric shock!
- When installing the product in an enclosed environment (such as inside a cabinet or chassis), ensure adequate cooling with cooling devices (such as cooling fans or air conditioners) to meet installation environment requirements. Otherwise, it may lead to product overheating or fire.
- It is strictly prohibited to modify this product!
- It is strictly prohibited to tighten or loosen the fixing bolts and red-Marked bolts of product parts and components!
- When this product is installed in a cabinet or terminal equipment, the cabinet or terminal equipment needs to provide corresponding protective devices such as fire-resistant enclosures, electrical protective enclosures, and mechanical protective enclosures. The protection level should comply with relevant IEC standards and local legal and regulatory requirements.
- When installing equipment that generates strong electromagnetic interference, such as transformers, please install shielding protection devices to prevent this product from malfunctioning!
- Please install the product on flame-retardant materials such as metal. Do not allow flammable materials to come into contact with the product or attach flammable materials to the product, otherwise, there is a risk of fire.

 CAUTION

- When performing installation work, cover the top of the product with cloth or paper to prevent metal chips, oil stains, water, and other foreign matter from entering the product during drilling, which could lead to product failure. After the work is completed, remove the cover to prevent it from blocking the ventilation holes, which could affect the drive's heat dissipation and lead to abnormal product heating.
- Resonance may occur when a machine operating at a constant speed is switched to variable speed operation. In this case, installing anti-vibration rubber under the motor frame or using the vibration suppression function can Validationonly reduce resonance.

Wiring

 Danger

- Non-professional personnel are strictly prohibited from performing equipment installation, wiring, maintenance, inspection, or component replacement!
- Before wiring, disconnect the power to all equipment. After power is disconnected, residual voltage may remain in the internal capacitors of the equipment. Please wait at least the time specified on the product's warning label before performing wiring or other operations. Measure the DC voltage of the main circuit and confirm it is below a safe voltage, otherwise there is a risk of electric shock.
- Perform un-wiring, removal of product covers, or touching of circuit boards only when power is disconnected, otherwise there is a risk of electric shock.
- Always ensure proper grounding of the equipment and product, otherwise there is a risk of electric shock.

Safety Precautions

- It is strictly prohibited to connect the input power to the output terminals of the equipment or product, otherwise it will cause equipment damage and even fire.
- When connecting the drive equipment to the motor, always ensure that the phase sequence of the product and motor terminals are accurate and consistent to avoid reverse rotation of the motor.
- When connecting the drive equipment to the motor, always ensure that the phase sequence of the product and motor terminals are accurate and consistent to avoid reverse rotation of the motor.
- Please tighten terminal screws according to the specified tightening torque in the manual. Insufficient or excessive tightening torque may lead to overheating, damage to the connections, and a risk of fire.
- After wiring is complete, ensure that all cables are correctly wired and that there are no loose screws, washers, or exposed cables inside the product, otherwise there may be a risk of electric shock or product damage.

 CAUTION

- Please follow the steps specified by electrostatic discharge (ESD) prevention measures and wear an anti-static wrist strap when performing wiring or other operations to avoid damaging the equipment or internal circuitry of the product.
- When wiring the control circuit, use a shielded twisted pair cable and connect the shield to the product's grounding terminal for grounding, otherwise the product may malfunction.

Powering up

 Warning

- After wiring and parameter settings are complete, perform a test run to confirm that the machine can operate safely, otherwise it may result in personal injury or equipment damage.
- Before powering on, ensure that the rated voltage of the product matches the power supply voltage. Incorrect power supply voltage carries a risk of fire.
- Before powering on, ensure that no personnel are around the product, motor, or machinery, otherwise it may result in personal injury or death.

During running



Danger

- Non-professional personnel are strictly prohibited from operating the product, otherwise there is a risk of personal injury or death!
- It is strictly prohibited to touch any wiring terminals, or dismantle any devices or components of the equipment and product while they are operating, otherwise there is a risk of electric shock!



Warning

- It is strictly prohibited to touch the equipment casing, fan, or resistors to test temperature, otherwise it may cause burns!
- During operation, prevent other items or metal objects from falling into the equipment, otherwise it may cause fire or product damage!

Maintenance



Danger

- Non-professional personnel are strictly prohibited from performing equipment installation, wiring, maintenance, inspection, or component replacement!
- It is strictly prohibited to perform equipment maintenance while power is on, otherwise there is a risk of electric shock!
- After disconnecting power to all equipment, please wait at least the time specified on the product's warning label before performing equipment maintenance or other operations.
- When using a PM motor, even if the product's power is off, induced voltage will be generated at the motor terminals during motor rotation. Do not touch the motor terminals, otherwise there may be a risk of electric shock.



Warning

- Please perform daily and periodic inspections and maintenance on the equipment and product according to the equipment maintenance and care requirements, and keep maintenance records.

Repair



Danger

- Non-professional personnel are strictly prohibited from performing equipment installation, wiring, maintenance, inspection, or component replacement!
- It is strictly prohibited to perform equipment repair while power is on, otherwise there is a risk of electric shock!
- After disconnecting power to all equipment, please wait at least the time specified on the product's warning label before performing equipment inspection, repair, or other operations.



Warning

- Please report equipment repairs according to the product warranty agreement.
- When a fuse blows, circuit breaker trips, or earth leakage circuit breaker (ELCB) trips, please wait at least the time specified on the product's warning label before restoring power or operating the machine, otherwise it may result in personal injury or death and equipment damage.
- If the equipment malfunctions or is damaged, it is imperative to have professional personnel troubleshoot and repair the equipment and product according to the repair guidelines, and keep repair records.
- Do not continue to use a damaged machine, otherwise it may cause personal injury or further damage to the product.
- After replacing equipment, be sure to re-check the equipment wiring and reset parameters.

Scrapping



Warning

- Please scrap equipment and products according to relevant national regulations and standards to avoid property damage or casualties!
- Please dispose of and recycle scrapped equipment and products in accordance with industrial waste treatment standards to prevent environmental pollution.

1.2 Other Precautions

Precautions for Drives Equipped with Dynamic Braking Function

- Dynamic braking is only for emergency stops in case of malfunction or sudden power failure. Do not frequently trigger malfunctions or power failures.
- At high speeds, ensure an operation interval of at least 5 minutes for the dynamic braking function, otherwise it may damage the internal dynamic braking circuit.
- Commonly seen in rotating mechanical structures, when dynamic braking stops the motor, the motor has already stopped rotating, but it continues to rotate due to the load on the shaft. In this state, the motor is driven by an external load and is in a power generation state, and short-circuit current flows through the dynamic brake. If continuously driven externally, the drive may smoke or catch fire, and the motor itself may also burn out.

Safety Signs

To ensure safe operation, always comply with the safety labels affixed to the equipment. Do not damage or remove safety labels. Safety label descriptions are as follows:

Safety Signs	Description of Safety Label Content
 危险 DANGER  高压注意 Hazardous Voltage  高温注意 High Temperature	• To prevent electric shock, always connect the grounding terminal properly. Please operate strictly according to the instructions in the user manual. • Never fail to connect Protective Earth(PE) terminal. Read the manual and follow the safety Instructions Before use. • Do not touch terminal parts within 15 minutes after power is disconnected, otherwise it may cause electric shock. • Do not touch terminals within 15 minutes after Disconnect the power, Risk of electric shock. • Do not touch the heat sink after powering on, otherwise it may cause burns. • Do not touch heatsink when power is ON, Risk of burn.

Chapter 2 Product Information

2.1 Servo Drive and Motor Model Description

2.1.1 Drive Model

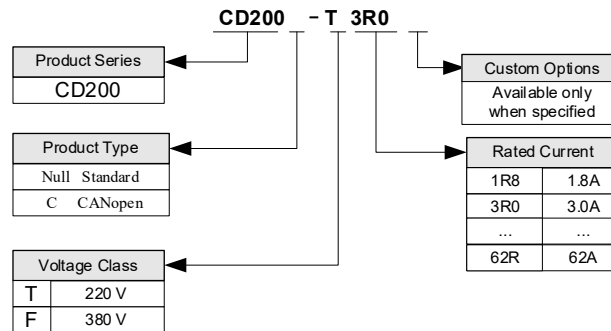


Figure 2-1 Servo Drive Model Naming Rule

Rated Voltage	Drive Model	Rated Current (A)	Maximum Compatible Motor Power (kW)
Single-phase AC220V	CD200□-T1R8	1.8	0.2
	CD200□-T3R0	3.0	0.75
	CD200□-T4R0	4.0	1.0
Single-phase/three-phase AC220V	CD200□-T4R5	4.5	1.0
	CD200□-T5R5	5.5	1.3
	CD200□-T7R5	7.5	2.0
	CD200□-T10R	10	2.1
	CD200□-T13R	13	2.3
Three-phase AC380V	CD200□-F4R0	4.0	1.5
	CD200□-F6R5	6.5	2.3
	CD200□-F8R5	8.5	3.0
	CD200□-F12R	12	4.5
	CD200□-F17R	17	4.4
	CD200□-F22R	22	5.5
	CD200□-F27R	27	7.5
	CD200□-F38R	38	15
	CD200□-F52R	52	22
	CD200□-F62R	62	30

2.1.2 Motor Model

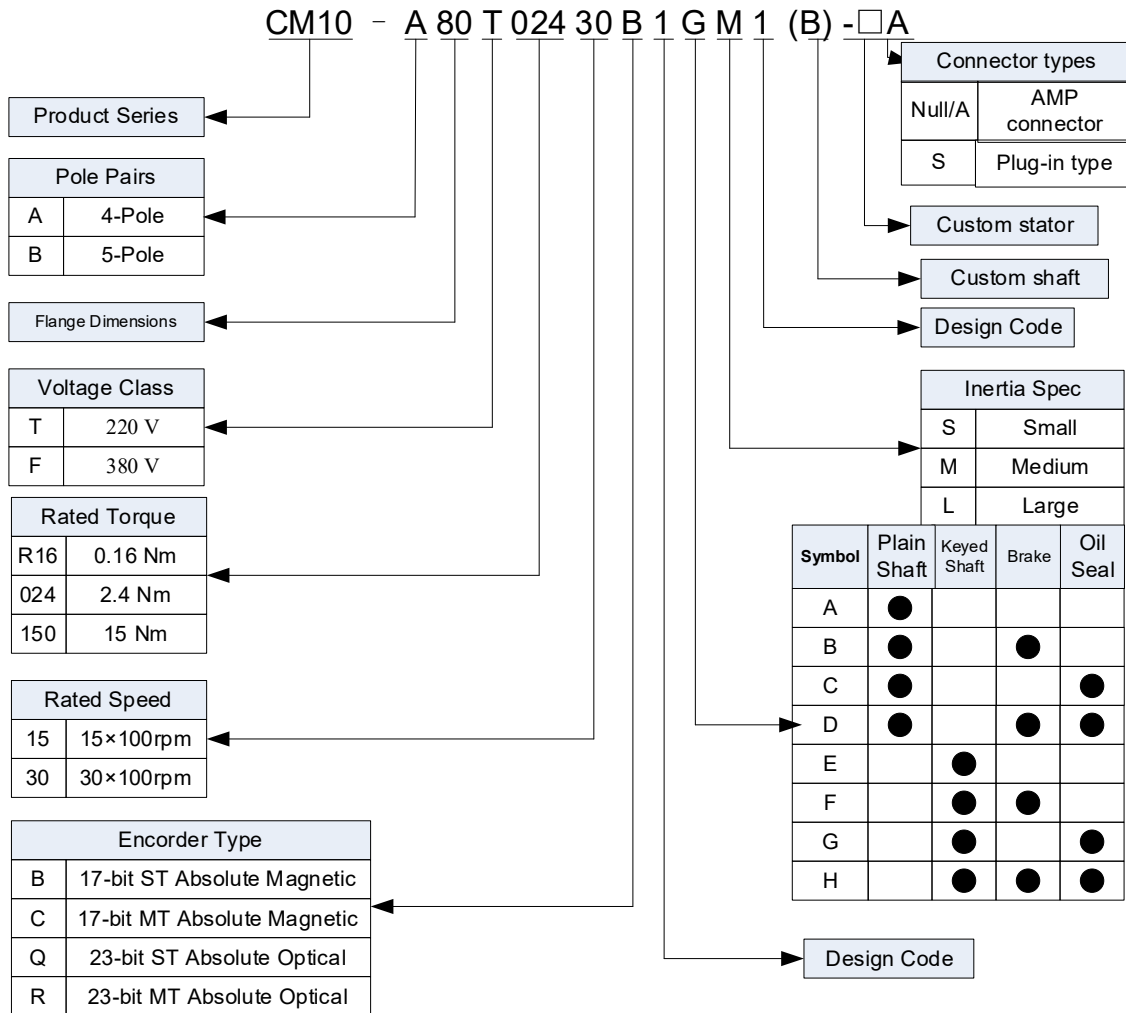


Figure 2-2 Servo Motor Model Naming Rule

Motor Model	CM10-			
	B60TR6430B	B60T01330B	B80T02430B-3A	B80T03230B-4A
Voltage (V)	AC220			
Frame Size	60	60	80	80
Rated Output Power (kW)	0.2	0.4	0.75	1.0
Rated Current (A)	1.4	2.6	3.0	4.5
Rated Torque (N·m)	0.64	1.27	2.4	3.2
Maximum Torque (N·m)	1.92	3.81	7.2	9.6
Rated Speed (r/min)	3000	3000	3000	3000
Maximum Speed (r/min)	6000	6000	3500	3500
Rotor Inertia (Kg·m ² ×10 ⁻⁴)	0.29	0.52	1.48	1.93
Torque Constant (N·m/A)	0.46	0.49	0.8	0.71
Back EMF (V/1000r/min)	31.7	31	53	50
Line Resistance (Ω)	8	4.3	3.4	1.85
Line Inductance (mH)	15	6.7	11.2	19.9
Number of Pole Pairs	5	5	5	5
Insulation Level	F	F	F	F
Protection Level	IP65	IP65	IP65	IP65
Body Length L (mm)	78 (110)	94 (126)	105 (142)	114 (152)

Note: For body length, the value in parentheses is the length with the holding brake function.

Motor Model	CM10-			
	A130T05025	A130T060025	A130T07725	CM10-A130T10015
Voltage (V)	AC220			
Frame Size	130	130	130	130
Rated Output Power (kW)	1.3	1.5	2.0	1.5
Rated Current (A)	5.0	6.0	7.5	6
Rated Torque (N·m)	5.0	6.0	7.7	10
Maximum Torque (N·m)	12.5	15.0	18.7	25
Rated Speed (r/min)	2500	2500	2500	1500
Maximum Speed (r/min)	3000	3000	3000	2000
Rotor Inertia ($\text{Kg}\cdot\text{m}^2\times 10^{-4}$)	10.7	12.9	14.1	18.8
Torque Constant (N·m/A)	1.0	1.0	1.03	1.67
Back EMF (V/1000r/min)	65	65	68	108
Line Resistance (Ω)	1.6	1.3	1.2	1.85
Line Inductance (mH)	8.0	6.2	5.8	9.9
Number of Pole Pairs	4	4	4	4
Insulation Level	F	F	F	F
Protection Level	IP65	IP65	IP65	IP65
Body Length L (mm)	171 (224)	179 (224)	192 (229)	209 (265)

Note: For body length, the value in parentheses is the length with the holding brake function.

Motor Model	CM10-			
	A130F06025	A130F15015	A180F19015	A180F21520
Voltage (V)	380			
Frame Size	130	130	180	180
Rated Output Power (kW)	1.5	2.3	3	4.5
Rated Current (A)	4	5	7.5	9.5
Rated Torque (N·m)	6	15	19	21.5
Maximum Torque (N·m)	15	30	57	64.5
Rated Speed (r/min)	2500	1500	1500	2000
Maximum Speed (r/min)	3000	2000	1800	2150
Rotor Inertia ($\text{Kg}\cdot\text{m}^2\times 10^{-4}$)	12.9	25.5	63.5	72.7
Torque Constant (N·m/A)	1.5	3	2.53	2.26
Back EMF (V/1000r/min)	108	180	166	140
Line Resistance (Ω)	3.1	3.2	1.33	0.84
Line Inductance (mH)	17.1	19	14.2	8.4
Number of Pole Pairs	4	4	4	4
Insulation Level	F	F	F	F
Protection Level	IP65	IP65	IP65	IP65
Body Length L (mm)	179(224)	231(282)	205(252)	215(262)

Motor Model	CM10-			
	B180F28415	A180F35015	A180F48015	A200F70015
Voltage (V)	380			
Frame Size	180	180	180	200
Rated Output Power (kW)	4.4	5.5	7.5	11
Rated Current (A)	16.5	12	20	21
Rated Torque (N·m)	28.4	35	48	70
Maximum Torque (N·m)	85.2	105	120	175
Rated Speed (r/min)	1500	1500	1500	1500
Maximum Speed (r/min)	3000	1750	1750	1800
Rotor Inertia (Kg·m ² ×10 ⁻⁴)	88.5	114	137.3	97.7
Torque Constant (N·m/A)	1.72	2.92	2.4	3.33
Back EMF (V/1000r/min)	104	181	171	220
Line Resistance (Ω)	0.31	0.78	0.56	0.95
Line Inductance (mH)	3.7	8	6.1	10.3
Number of Pole Pairs	5	4	4	4
Insulation Level	F	F	F	F
Protection Level	IP65	IP65	IP65	IP65
Body Length L (mm)	232(279)	260(307)	305(352)	438(538)

Note: For body length, the value in parentheses is the length with the holding brake function.

2.2 Technical Specifications of Drive

Item		Description	
Basic Specifications	Main Power Supply	Single-phase 220-240V AC, -15%~+10% (50/60Hz) Three-phase 380-415V AC, -15%~+10% (50/60Hz)	
	Control Method	FOC+SVPWM	
	Encoder Feedback	Serial Communication Encoder: 17bit-26bit optional	
	Protection	Overcurrent, abnormal voltage, overload, input/output phase loss, motor stall, overspeed, abnormal pulse command, Brake resistor overload, drive overheating, encoder abnormality, etc.	
	Operating Conditions	Temperature	Operating temperature: 0°C~+45°C; Ambient temperature: +45°C~50°C; Please derate usage, current derates 2% for every 1°C increase; Storage temperature: -20°C~+60°C
		Humidity	Relative humidity: below 90%RH (non-condensing)
		Vibration	0.5g (4.9m/S ²)
		Protection	IP20
		Altitude	Below 1000m (>1000m, please derate usage)
		Others	1: No electrostatic interference, strong electric fields, strong magnetic fields, radiation, etc. 2: No corrosive gases, flammable gases, no water, oil, or chemical splashes. 3: In an environment with little dust, dirt, or salt, and no metal powder.
Control Signal	Digital Input	8 DI (DC24V optocoupler isolation, supporting NPN and PNP)	
	Digital Output	4 DO (Open collector output, load capacity 50mA, voltage range 5V-28V)	

Speed/ Torque Control Mode	Load Variation Rate	0.0%-100% load: ≤0.5% (at rated speed)	
	Voltage Variation Rate	Rated voltage ±10%: 0.5% (at rated speed)	
	Speed Control Range	1:5000 (The lower limit of the speed control range is the value under non-stop conditions at rated torque load)	
	Torque Control Accuracy	±2%	
	Multi-segment Speed Command	Achieve 0-31 segment speed selection using DI signal combinations	
Position Control Mode	Pulse Command	Input Pulse Mode	"Pulse + Direction", "A, B phase quadrature pulse", "CW/CCW pulse" "Digital Pulse Bus"
		Input Type	Differential input, open collector, pulse bus
		Input Pulse Frequency	Differential input: Quadrature 500Kpps, pulse width cannot be less than 1μs Open collector: Maximum pulse frequency for single channel 200Kpps, pulse width cannot be less than 2.5μs
	Position Output	Output Type	A phase/B phase: Differential output Z phase: Differential output or open collector output
		Frequency Division Ratio	Arbitrary frequency division
	Multi-segment Position Command	Achieve 0-31 segment position selection using DI signal combinations	
	Full-Closed Loop	Support external ABZ type and communication encoders(only CD200C)	
	Communication Function	RS485	Link Layer Protocol
Application Layer Protocol			Modbus-RTU, TLink protocol (used for digital pulse)
Baud Rate			4800 bps, 9600 bps, 19200 bps, 38400 bps, 57600 bps, 115200 bps
Duplex Mode			Half-duplex
Multi-station Communication Axis Count			Maximum 32 stations
CANopen		CD200C Supported	
Type-C		For communication with PC (X Servo Configurator)	
Internal Function	Model-based tracking control	Actively identifies and corrects deviations in input commands, enabling more responsive equipment performance and greater motion precision. Through real-time command reconstruction and dynamic compensation, it effectively eliminates jitter and jerk during movement, ensuring smooth operation across the entire speed range from low to high, and significantly improving the machining and control accuracy of the equipment.	
	Adjustment-free function	Automatically identifies load characteristics, optimizes parameters with a single click, and eliminates the need for complex manual tuning.	
	Vibration Inhibiting	Two vibration suppression notch filters: Settable vibration suppression frequency and intensity Two notch filters: Settable notch frequency, width, and depth	
	Overtravel Prevention	Positive limit, negative limit, and software limit	
	Virtual Braking	In some applications, the motor can be used for virtual regenerative braking, replacing the braking resistor	
	LED Display	Main power CHARGE, 6-digit LED display	
	Others	Digital pulse bus control, gain adjustment, inertia identification, mechanical frequency analysis, alarm record, and JOG operation, etc.	

2.3 Names of Servo Drive Components

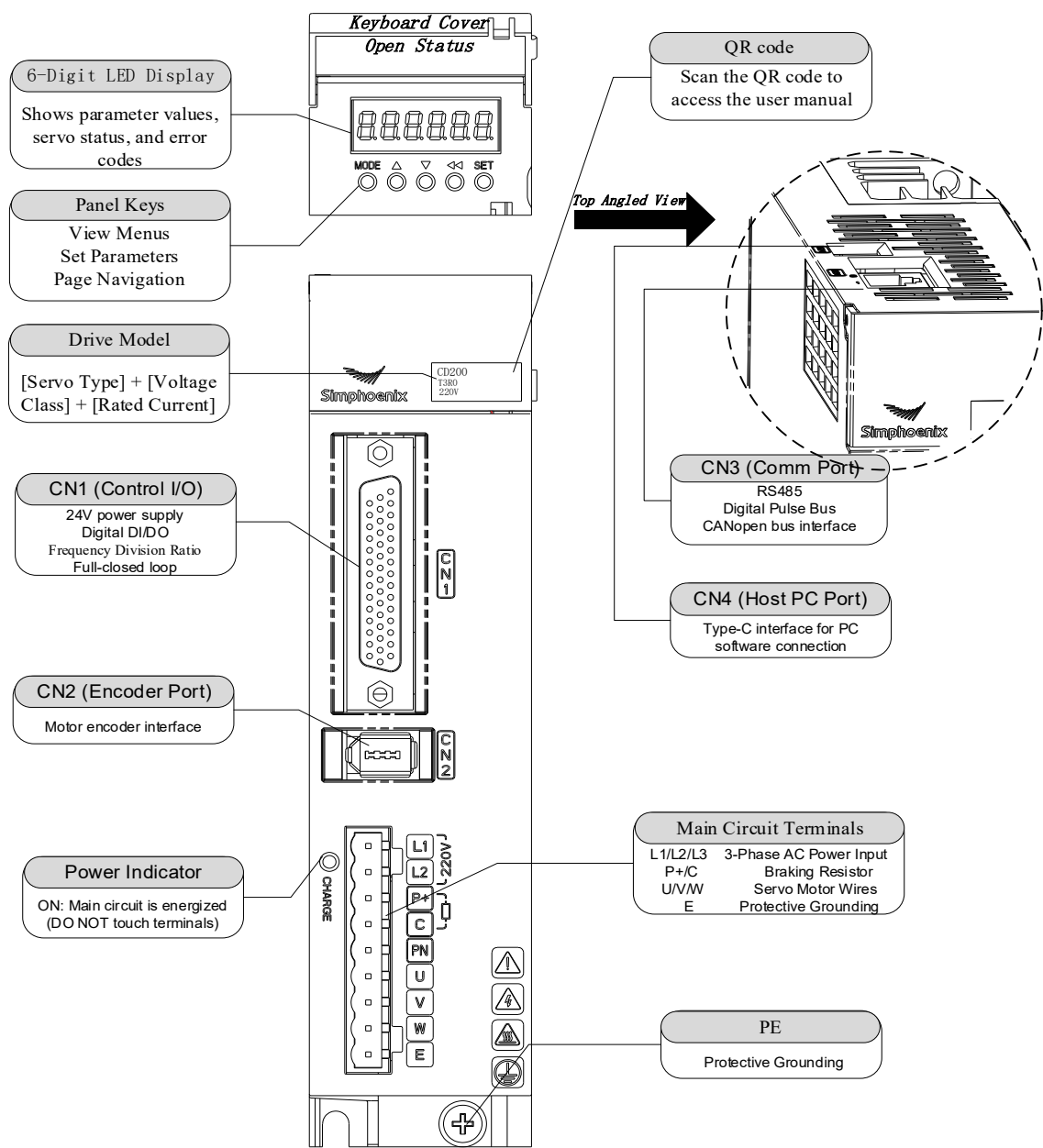
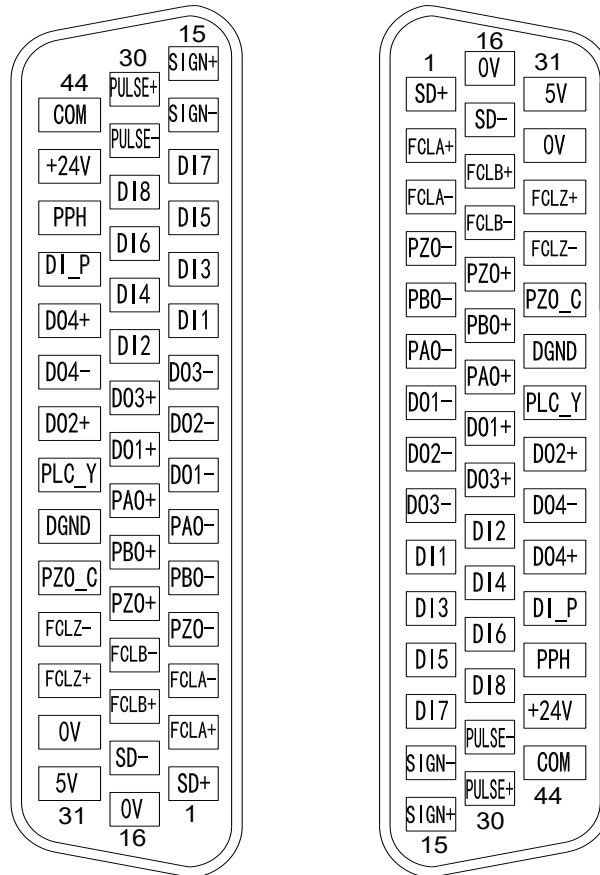


Figure 2-3 Names of CD200 Drive Components

2.4 Servo Drive Terminal Layout

2.4.1 CN1 Control Signal Interface

The control signal terminals provide the necessary signals for connection with the upper computer controller. A DB44 connector is used, and the pin distribution and wiring diagram are as follows:



For T3R0 and lower models For T4R5 and higher models:

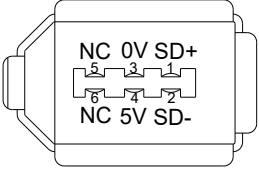
Figure 2-4 CD200 Servo Drive CN1 Terminal Pin Distribution Diagram

● CN1 Control Signal Terminal Plug Pin Distribution and Function Description

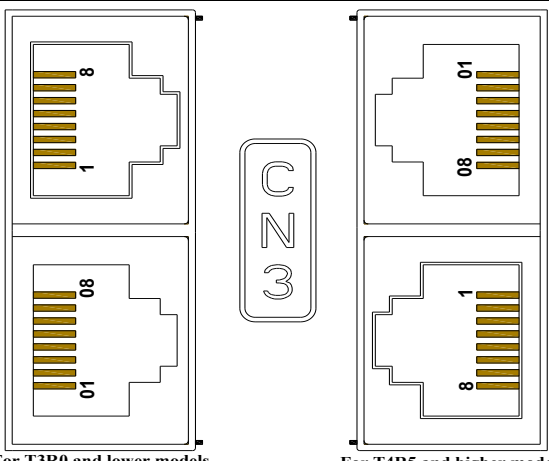
Signal Name		Pin No.	Function Description	
Power Supply	+24V	43	Internal 24V power supply, voltage range +20V-28V, maximum output current 100mA	
	COM	44	Internal 24V power ground	
Position Command	PULSE+	30	Pulse Command Input Method: · Differential drive input · Open collector	Input Pulse Form: · Pulse + direction · A, B phase quadrature pulse · CW/CCW pulse
	PULSE-	29		
	SIGN+	15		
	SIGN-	14		
	PPH	42	Command pulse power input interface (2KΩ resistor already in series inside the drive)	
	PLC_Y	37	Open collector pulse input pull-up/pull-down interface	
Digital Input	DI1	10	Digital input, default function: Servo enable	
	DI2	25	Digital input, default function: Fault reset	
	DI3	11	Digital input, default function: Positive limit	
	DI4	26	Digital input, default function: Negative limit	
	DI5	12	Digital input, default function: Origin signal	
	DI6	27	Digital input, default function: Start origin search	
	DI7	13	Digital input, default function: Emergency stop	
	DI8	28	Digital input, default function: Interrupt position table	
	DI P	41	Digital Input Common Terminal	

Signal Name		Pin No.	Function Description		
Digital Output	DO1+	23	Digital output, default function: Ready		
	DO1-	7			
	DO2+	38	Digital output, default function: Fault alarm output		
	DO2-	8			
	DO3+	24	Digital output, default function: Target position (speed) reached		
	DO3-	9			
	DO4+	40	Digital output, default function: Brake control		
	DO4-	39			
Secondary Encoder	5V	31	Secondary encoder 5V power supply, maximum output current 150mA		
	0V	16/32	Secondary encoder 5V power ground		
	SD+	1	Secondary encoder communication input	Expandable communication type full closed-loop function (Only CD200C supported)	
	SD-	17			
Frequency Division Output	PAO+	22	A Phase Frequency Division Output Signal	A, B phase quadrature frequency-divided pulse output signals	
	PAO-	6			
	PBO+	21	B Phase Frequency Division Output Signal		
	PBO-	5			
	PZO+	20	Z Phase Frequency Division Output Signal	Origin Pulse Output Signal	
	PZO-	4			
	PZO_C	35	Z pulse open collector output	Origin Pulse Open Collector Output Signal	
	D_GND	36	Signal Ground		
Full Closed-Loop	FCLA+	2	Full closed-loop encoder A-phase differential input interface		CD200C Supported
	FCLA-	3			
	FCLB+	18	Full closed-loop encoder B-phase differential input interface		
	FCLB-	19			
	FCLZ+	33	Full closed-loop encoder Z-phase differential input interface		
	FCLZ-	34			

2.4.2 CN2 Encoder Signal Interface

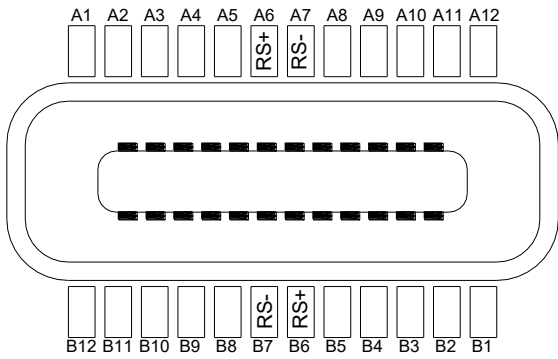
		Pin	Signal	Description
		1	SD+	Data transmit +
		2	SD-	Data transmit -
		3	0V	Encoder 5V power
		4	5V	
		5	NC	-
		6	NC	-
PE	Casing			

2.4.3 CN3 Communication (RS485, CANopen) Interface

Terminal Position Number CN3	Pin	Signal	Description
 For T3R0 and lower models For T4R5 and higher models	1/01	NC	-
	2/02	NC	-
	3/03	NC	-
	4/04	RS+	RS485 Differential Input Signal Port
	5/05	RS-	
	6/06	GND	Signal Ground
	7/07	CAN+	CAN Differential Input Signal Port
	8/08	CAN-	
	PE	Casing	-

- Please use twisted-pair cables for RS485 and CAN differential signal lines.
- Always ensure reliable grounding of the shielded mesh layer, otherwise it may cause false alarms from the drive.

2.4.4 CN4 Communication (Upper Computer) Interface

Terminal Position Number CN4	Pin	Signal	Description
	A1-A5	-	-
	B1-B5	-	-
	A6/B6	RS+	RS485+
	A7/B7	RS-	RS485-
	A8-A12	-	-
	B8-B12	-	-

- One end of the connecting cable is Type-C, the other end is USB. Shielded cables with ferrite cores are recommended.
- Cable length $\leq 3\text{m}$.

2.5 Servo Cables

Servo Cable Naming Rule

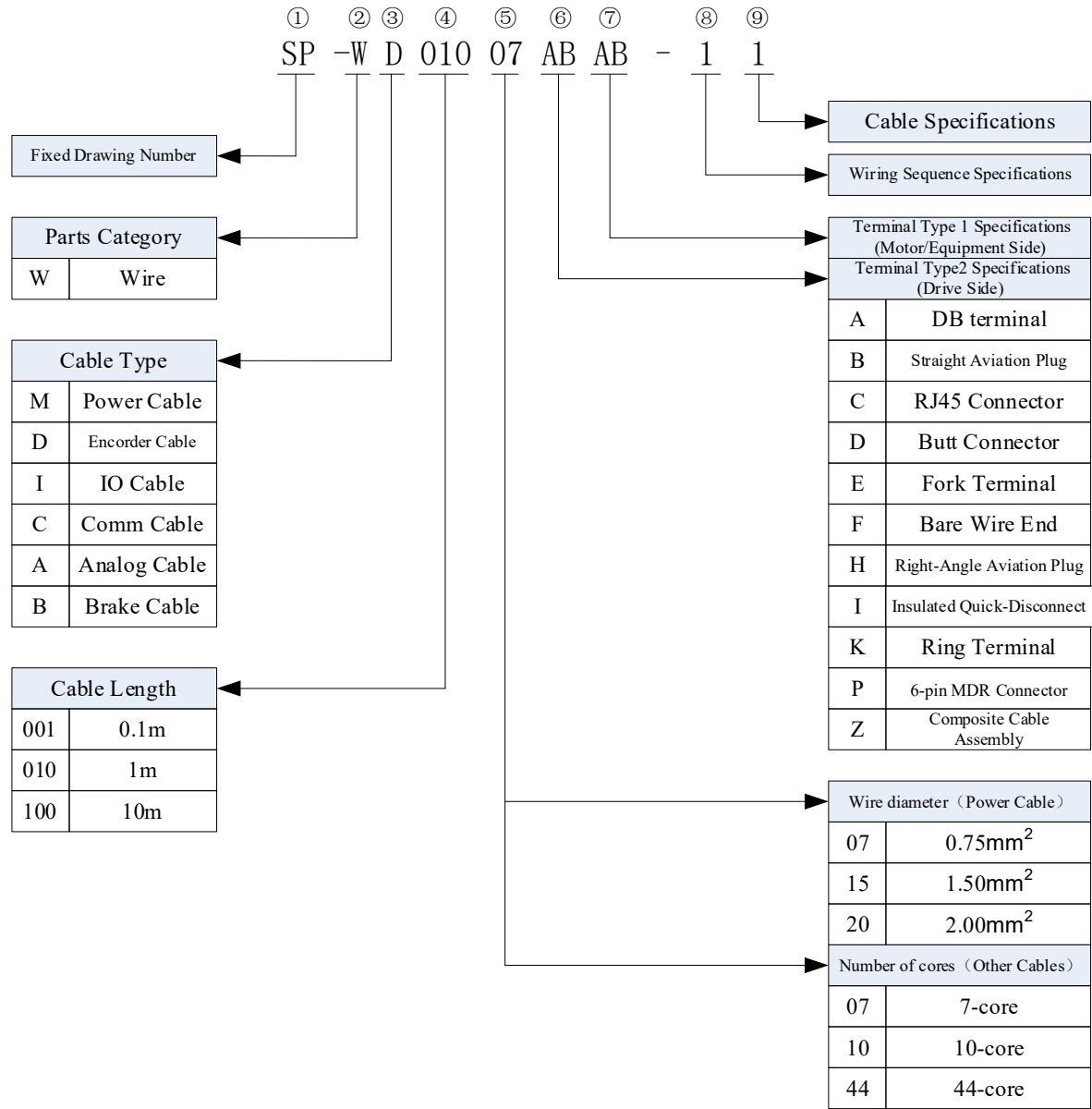


Figure 2-5 Servo Motor Cable Naming Rule

Note:

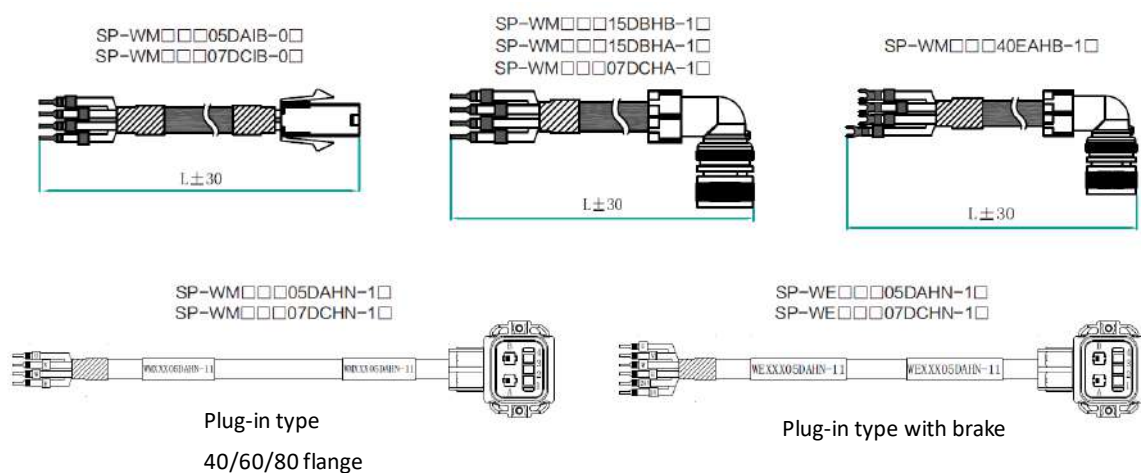
- ⑧ Wire sequence parameter: To distinguish the wire sequence of motor power cables, for the CD200 series compatible with a 110 flange (using an aviation plug), the power cable fixed selection is 1
- ⑨ Cable material parameter: 1 = Ordinary cable, 2 = Highly flexible cable

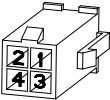
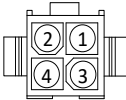
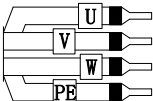
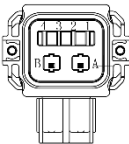
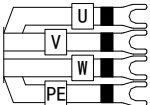
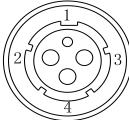
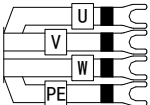
※ Note:

- "XXX" in power cables and encoder cables refers to the cable length. Users should order according to their needs; 3m/5m/10m are commonly stocked.
- For multi-turn absolute encoders, please select 7-core cables (with battery box). For single-turn absolute encoders, please select 5-core cables (without battery box).
- The motors in the table above are standard motors for each drive. Other compatible motors can be found in Table 2.1.2 Servo Motor Technical Parameters.

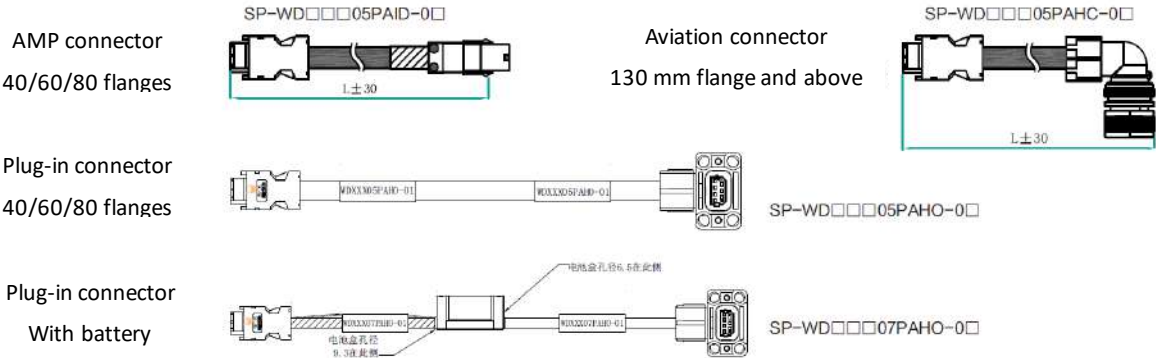
Power	Motor Model	Power Cables	Encoder Cable
T1R8	CM10-B40TR3230	SP-WMXXX05DAIB-01 (AMP connector) SP-WMXXX05DAHN-01 (Plug-in type)	SP-WDXXX05PAID-01 (AMP connector) SP-WDXXX05PAHO-01 (Plug-in type)
	CM10-B60TR6430		
T3R0	CM10-B60T01330	SP-WMXXX07DCIB-01(AMP connector) SP-WMXXX07DCHN-01(Plug-in type)	SP-WDXXX05PAID-01(AMP connector) SP-WDXXX05PAHO-01(Plug-in type)
	CM10-B80T02430		
T4R0	CM10-B80T03230	SP-WMXXX07DCHA-11	SP-WDXXX05PAHC-01
T4R5	CM10-B80T03230		
T5R5	CM10-B130T05430	SP-WMXXX15DBHA-11	SP-WDXXX05PAHC-01
T7R5	CM10-A130T07725		
T10R	CM10-B130T08415	SP-WMXXX15DBHA-11	SP-WDXXX05PAHC-01
T13R	CM10-B130T11515		
F4R0	CM10-A130F10015	SP-WMXXX15DBHA-11	SP-WDXXX05PAHC-01
F6R5	CM10-B130F08415		
F8R5	CM10-B130F11515	SP-WMXXX15DBHB-11	SP-WDXXX05PAHC-01
F12R	CM10-B180F18615		
	CM10-A180F21520	SP-WMXXX40EAHB-11	SP-WDXXX05PAHC-01
F17R	CM10-B180F28415		
F22R	CM10-A180F35015	SP-WMXXX15DBHB-11	SP-WDXXX05PAHC-01
F27R	CM10-A180F48015	SP-WMXXX40EAHB-11	SP-WDXXX05PAHC-01

2.5.1 Power Cables

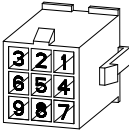
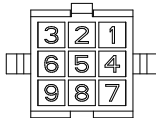
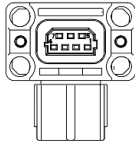
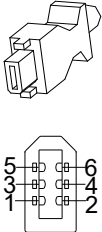
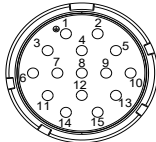
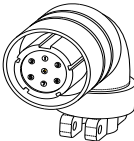
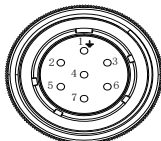


Motor Terminal Wire Sequence					Drive Side
Motor Flange Type	Terminal View	Terminal Number	Pin	Definition	Terminal Number
40/60/80 Using AMP plug			1	PE	
			2	U	
			3	V	
			4	W	
The 80-flange using a small aviation plug to replace the AMP plug is mainly for motor applications involving reciprocating motion and harsh conditions like high temperature and high humidity.					
40/60/80 Using Plug-in connector			1	U	
			2	V	
			3	W	
			4	PE	
			A	24V	
			B	0V	
80 (using small aviation plug) 130/180 Using aviation plug			1	PE	
			2	U	
			3	V	
			4	W	

2.5.2 Encoder Cables



(1) Encoder Cable Terminal Definition

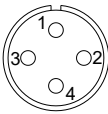
Motor Side					Drive Side	
Motor Flange Type	Terminal Shape	Terminal Number	Pin	Definition	Pin	Terminal Number
40/60/80 Using AMP plug			1	PE	PE	
			2	5V	4	
			3	0V	3	
			4	SD+	1	
			5	SD-	2	
			6	E+	NC	
			7	E-	NC	
			8	NC	NC	
			9	NC	NC	
The 80-flange using a small aviation plug to replace the AMP plug is mainly for motor applications involving reciprocating motion and harsh conditions like high temperature and high humidity.						
40/60/80 Using Plug-in connector			1	PE	PE	
			2	E-	NC	
			3	E+	NC	
			4	SD-	2	
			5	0V	3	
			6	SD+	1	
			7	5V	4	
			8	NC	NC	
			9	NC	NC	
80 (using small aviation plug)			1	PE	NC	
			2	E-	NC	
			3	E+	NC	
			4	SD-	2	
			5	0V	3	
			6	SD+	1	
			7	5V	4	
130/180 Using aviation plug			1	PE	NC	
			2	E-	NC	
			3	E+	NC	
			4	SD-	2	
			5	0V	3	
			6	SD+	1	
			7	5V	4	

Encoder Wiring Precautions: Please reliably ground the encoder shielding mesh on both the drive and motor sides, otherwise it may cause false alarms from the drive.

(2) Encoder Cable Battery Selection

Battery Model	Battery Parameters	Rated Value	ReMarks
EVE Energy EVE ER14505 AA (Recommended)	Battery Output Voltage (V)	3.6	-
	Battery Capacity (mAh)	2700	-
	Battery Low Voltage Warning Value (V)	3.1	Alarm Code AL.0001
	Battery Operating Ambient Temperature (°C)	-20~+85	-
	Battery Storage Ambient Temperature (°C)	≤30	-

2.5.3 Brake Terminal Definition

Motor Type	Brake Terminal Model	Motor Side Terminal	Pin	Definition
40/60	172233-1		1	24V
			2	0V
80/130	XS12K3P		1	24V
			2	0V
			3	NC
180	XS16K4TM		1	24V
			2	0V
			3	NC
			4	NC

2.6 Regenerative Braking Resistor Selection

When the motor's output torque is opposite to its operating speed direction, the motor will be in a power generation state. This regenerative energy will cause the bus voltage to rise, and its magnitude depends on the inertia of the motor rotor and the load. If the system inertia is small, the internal bus capacitors of the drive can absorb the regenerative energy. However, if the system inertia is large, the bus capacitors are insufficient to absorb the regenerative energy, and it must be dissipated through a braking resistor. Otherwise, excessive bus voltage rise will cause the drive to report an overvoltage shutdown or even damage.

Formula for calculating regenerative energy from rotor inertia:

$$E_r = J \times V^2 / 182, \text{ unit J}$$

Where J is rotor inertia, unit $\text{kg} \cdot \text{m}^2 \times 10^{-4}$ and V is motor rated speed, unit rpm.

Formula for calculating braking resistor capacity:

$$P_r = 2 \times (nE_r - E_c) / T, \text{ unit W}$$

Where nE_r is the total load inertia, E_c is the maximum braking energy that the bus capacitor can absorb, and T is the action cycle.

Drive Model		Built-in Regenerative Braking Resistor Specifications		Minimum allowable external resistance (Ω)	Maximum Braking Energy E_c that Capacitor can Absorb (J)
		Resistance value (Ω)	Capacity (W)		
Single-phase AC220V	CD200□-T1R8	N/A	N/A	50	11
	CD200□-T3R0	N/A	N/A	50	16
	CD200□-T4R0	50	50	50	16
Single-phase/ Three-phase AC220V	CD200□-T4R5	50 (optional)	40	50	19
	CD200□-T5R5	50 (optional)	40	25	29
	CD200□-T7R5	25 (optional)	100	25	34
	CD200□-T10R	25 (optional)	100	15	48
	CD200□-T13R	25 (optional)	100	15	60
Three-phase AC380V	CD200□-F4R0	100 (optional)	100	80	33
	CD200□-F6R5	100 (optional)	100	60	33
	CD200□-F8R5	50 (optional)	100	40	33
	CD200□-F12R	50 (optional)	100	40	48
	CD200□-F17R	40 (optional)	150	40	60
	CD200□-F22R	30 (optional)	150	20	80
	CD200□-F27R	30 (optional)	150	20	96
	CD200□-F38R	N/A	N/A	10	144
	CD200□-F52R	N/A	N/A	10	192
	CD200□-F62R	N/A	N/A	10	240

When using an external braking resistor, connect the resistor to the P+ and C terminals, and ensure that P+ and D are open-circuited; The external braking resistor must be greater than the resistance values listed in the table, otherwise it may cause damage to the drive;

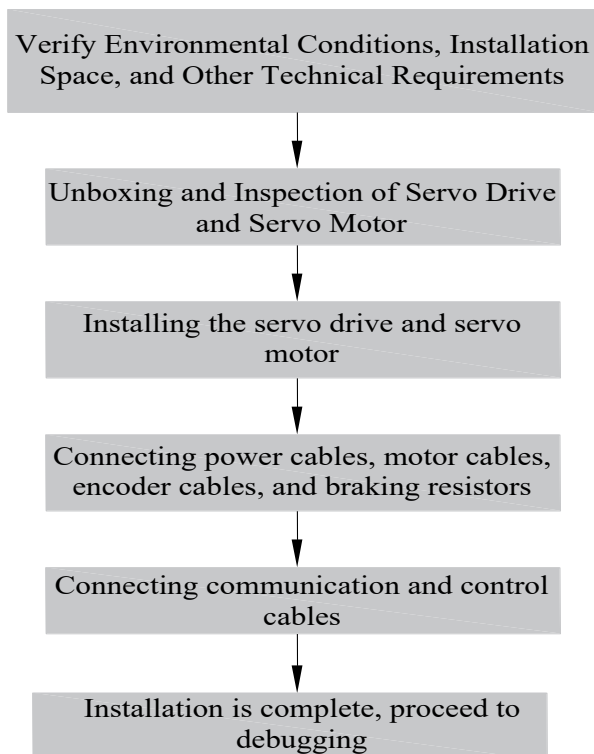
Chapter 3 Installation

3.1 Installation Notes

Users should pay special attention to the following:

- Always comply with the installation direction requirements in this manual, otherwise it may lead to product malfunction or damage.
- It is strictly prohibited to install or operate equipment that is damaged or has missing components, otherwise it will cause personal injury.
- It is strictly prohibited to install this product in locations where water may splash or in corrosive environments, otherwise it will lead to product malfunction.
- It is strictly prohibited to install this product near flammable gases or combustible materials, otherwise it will lead to fire or electric shock.
- Please install this product in an installation cabinet that provides fire and electrical protection, otherwise it may lead to fire.
- Please ensure the specified clearance between the servo drive and the inner surface of the control cabinet, as well as other machinery, otherwise it may lead to fire or product malfunction.
- It is strictly prohibited to place heavy objects on the product, otherwise it may cause personal injury or product damage.
- It is strictly prohibited to apply excessive impact force to the equipment, otherwise it may cause product damage.
- It is strictly prohibited to block the intake and exhaust ports of the drive, and do not allow foreign objects to enter the products interior, otherwise it may lead to fire or product malfunction.
- Servo motors can be installed horizontally and vertically. When connecting to machinery, use an elastic coupling if possible, and ensure the axis of the servo motor is aligned with the axis of the mechanical load. If the concentricity deviation is too large, it will cause mechanical vibration and damage to the servo motor bearings.
- Axial striking is strictly prohibited during installation, otherwise it is highly likely to damage the servo motor's encoder.
- If the connection between the drive and motor exceeds 20 meters, please use thicker U/V/W connection cables and encoder connection cables.

3.2 Installation Flowchart



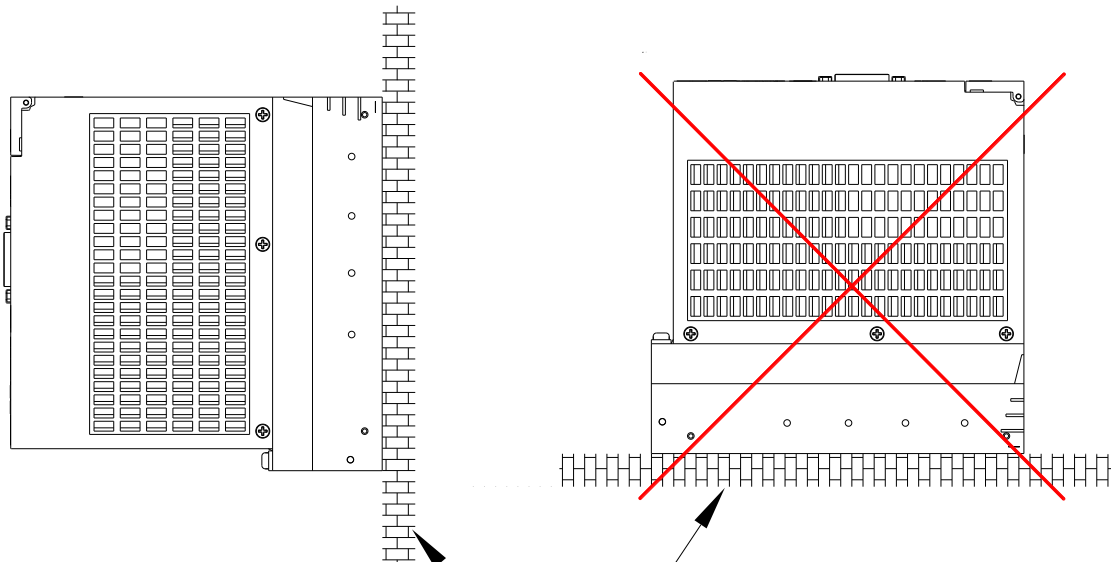
3.3 Installation Requirements

3.3.1 Installation Environment Requirements

Item	Requirements
Mounting Place	Indoor
Grid Overvoltage	Overvoltage Category III (OVCIII)
Altitude	Maximum altitude up to 1000m - Above 1000m, derate by 10% for every 1000m increase - For altitudes exceeding 2000m, please contact the manufacturer
Temperature	- Installation/Operation temperature: 0°C-+50°C. No derating needed for 0°C-+45°C. For temperatures above 45°C, derate usage by 2% for every 1°C increase - Storage/Transportation temperature: -40°C-+70°C - To improve the machine reliability, please use this product in locations where temperature does not change drastically - When used in enclosed spaces such as control cabinets, use cooling fans or air conditioning to keep the equipment's intake air temperature below 45°C. Otherwise, it may lead to overheating or fire - Install the product on the surface of flame-retardant material, with sufficient space around it for heat dissipation. - Please prevent the product from freezing
Ambient Humidity	Below 90% RH, non-condensing
Storage Humidity	Below 90% RH, non-condensing
Vibration Resistance Strength	During transportation: Conform to IEC60721-3-2:1997 Class 2M3 During operation: Conform to IEC60721-3-3:2019 Class 3M2
Protection Level	IP20
Environment	Pollution Degree 2 or below Please install the product in the following locations: - Locations not exposed to direct sunlight, free from dust, corrosive gases, flammable and explosive gases, oil mist, water vapor, dripping water, or salt - Please install in a place where vibration is unlikely (especially away from equipment like punching machines) - No metal powder, oil, water, or other foreign matter should enter the interior of the product - Locations free from radioactive substances, flammable materials, harmful gases and liquids, and with low salt corrosion - Do not install the product on flammable materials such as wood - Do not use in a vacuum environment

3.3.2 Installation Space Requirements

(1) Installation Direction

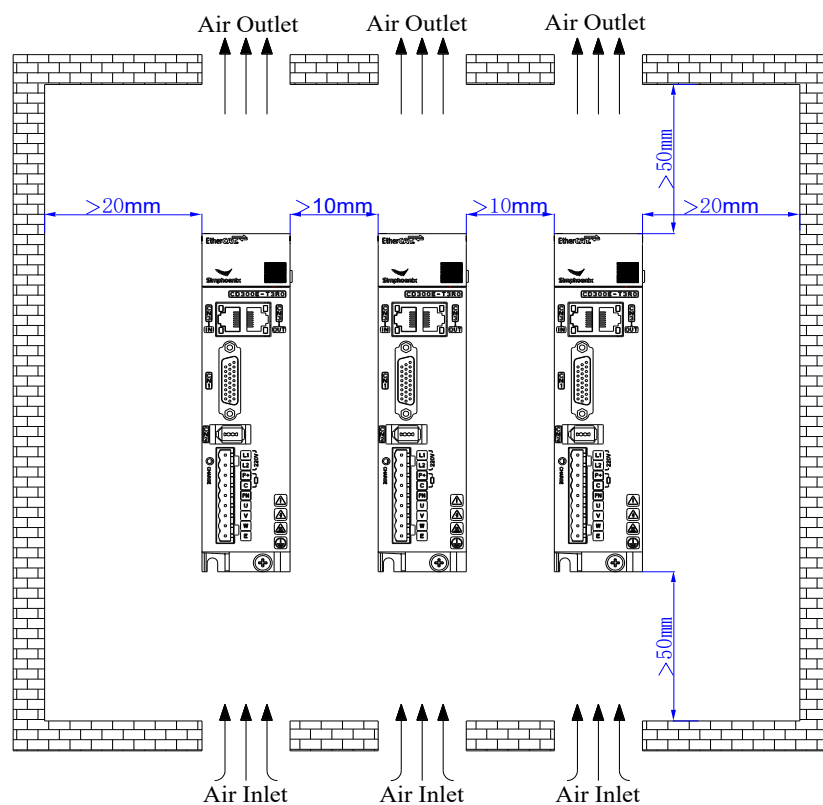


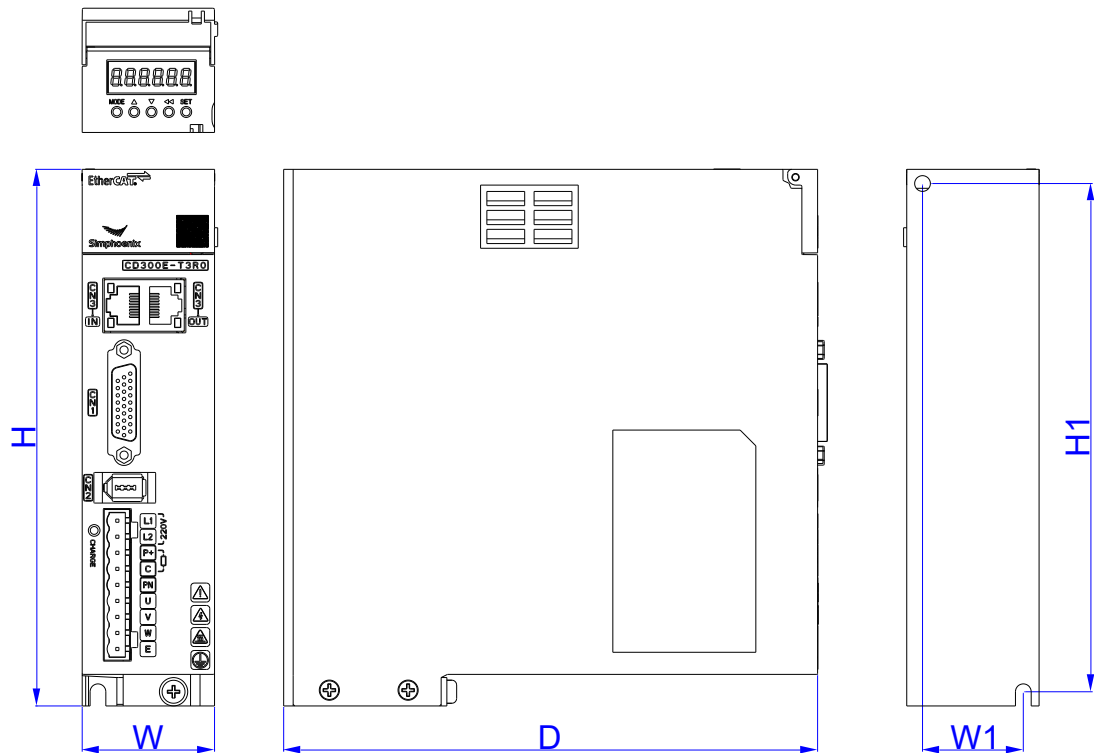
Installation Plane

The drive only supports vertical installation. Incorrect installation direction may lead to overheating or damage to the drive.

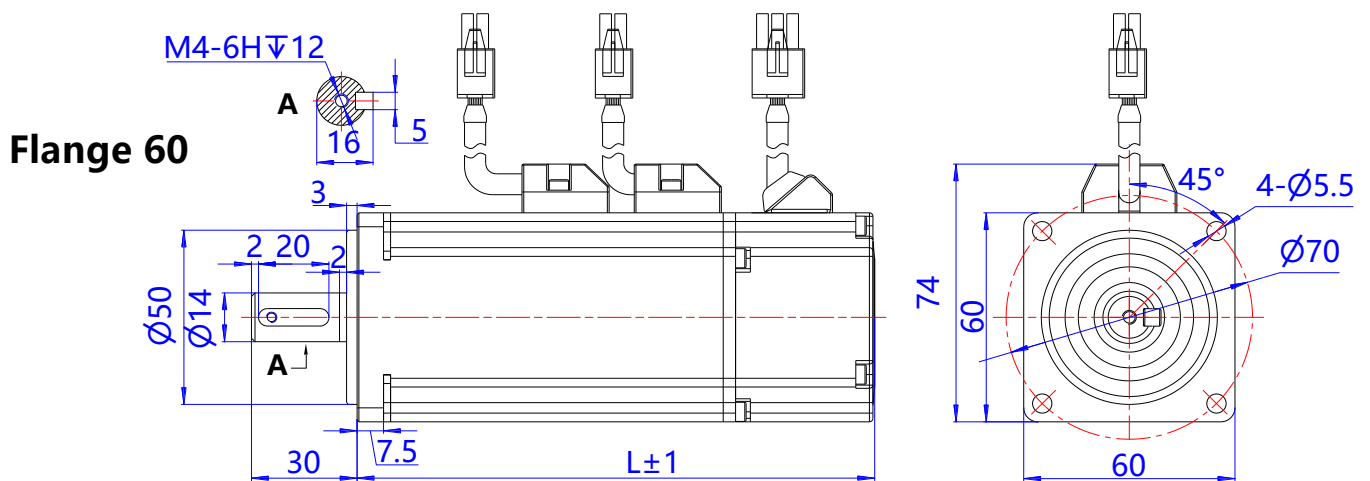
(2) Installation Clearance

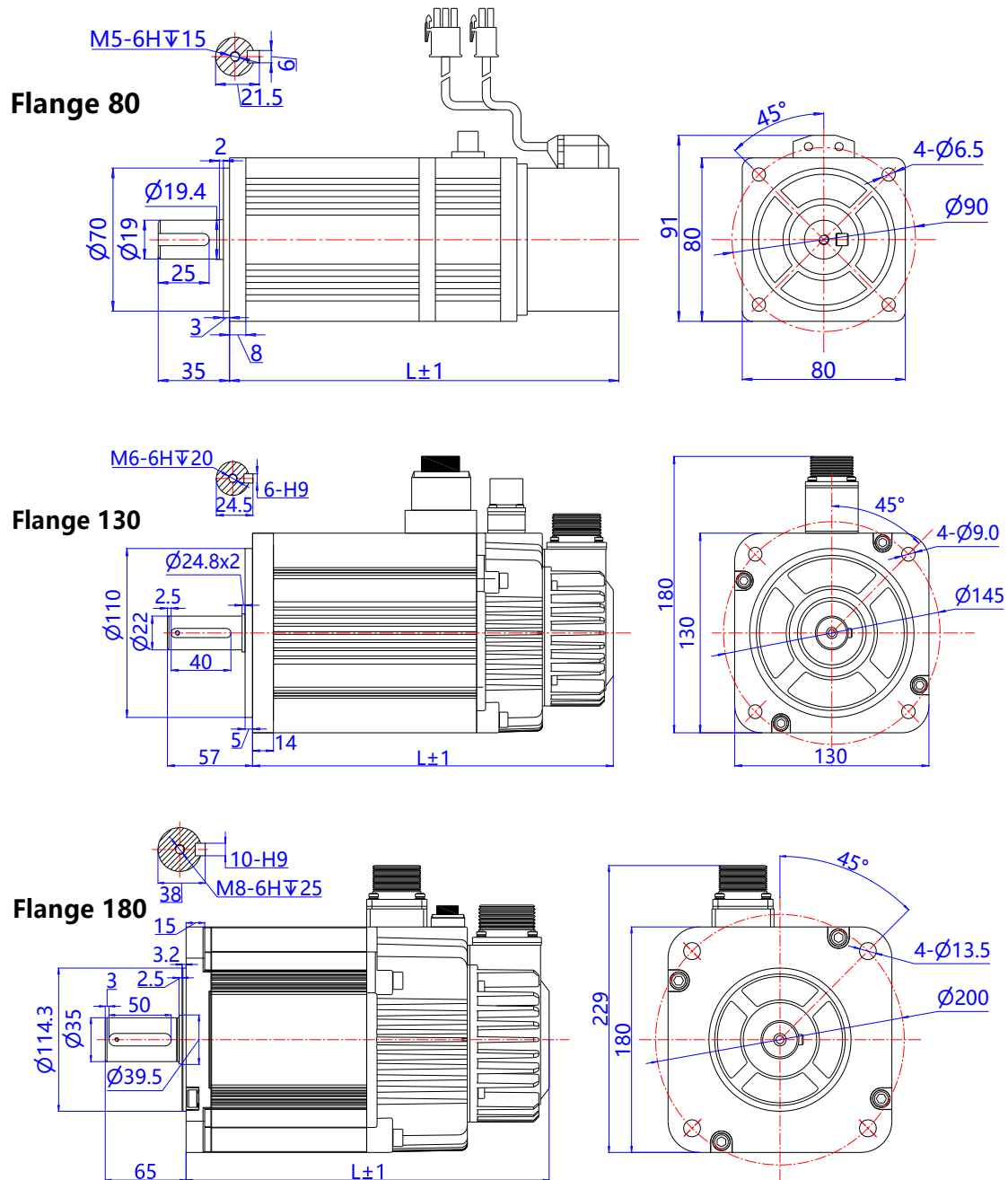
To ensure good cooling circulation, when installing the AC servo drive, there must be sufficient space above, below, left, and right from adjacent items and baffles (walls), otherwise it may cause malfunction. Its intake and exhaust holes must not be blocked, nor should it be placed on its side, otherwise it may cause malfunction. When multiple servo drives are installed side by side in a control cabinet, users should observe the installation clearances shown in the following diagram:



(3) Drive Dimensions

Drive Model	W1 (mm)	W (mm)	H1 (mm)	H (mm)	D (mm)	Screw Specifications	Weight (KG)
T1R8/T3R0/T4R0	32	42	161	170	170	M4	1
T4R5/T5R5/T7R5	40	50	161	170	170	M4	1.3
T10R/T13R	64	80	186	195	182	M4	2.1
F4R0/ F6R5/ F8R5/ F12R	64	80	186	195	182	M4	2.1
F17R/F22R/F27R	70	95	263	276	227	M4	4.9
F38R/F52R/F62R	100	150	410	426	250	M6	12.7

(4) Motor Dimensions



3.4 Inspection after Installation

S/N	Content	Confirm
1	Check if power input terminals (L1, L2, L3) are connected correctly	☐
2	Check if P+, D, C, and PN are connected correctly	☐
3	Check if the phase sequence of the servo drive output terminals (U, V, W) is consistent with that of servo motor cable (U, V, W)	☐
4	Confirm that servo drive power input terminals (L1, L2, L3) and main circuit output terminals (U, V, W) are not short-circuited	☐
5	Ensure that the encoder is correctly wired and operating	☐
6	Ensure no oil, water, or foreign contaminants inside the motor socket or at the motor installation location, otherwise it may cause short circuits	☐
7	Ensure that motor casing is grounded, and the ground wire is always connected to earth	☐

Chapter 4 Electrical Design Guidelines

4.1 Wiring Precautions

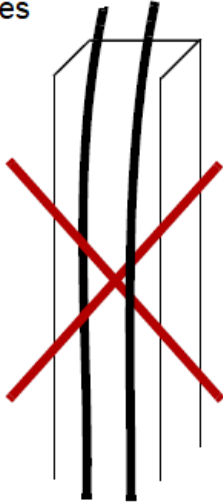
Danger

- After the power is turned off, high voltage may still remain inside the servo drive. Do not touch the power terminals. Inspection operations can only be performed after confirming that the CHARGE indicator light has extinguished.
- Confirm that the phase sequence wiring of the U, V, W terminals of the servo motor is correct. Incorrect wiring may cause the motor to not rotate or to run away.
- Do not connect the input power cable to the output terminals U, V, W, otherwise it will damage the servo drive.
- Ensure reliable grounding of the servo drive and motor, otherwise there is a risk of electric shock.

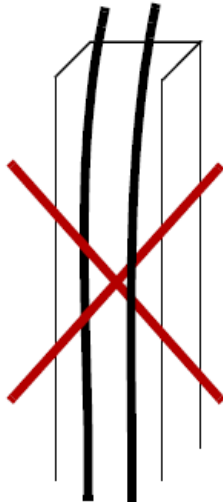
Attention

- When an external braking resistor is used, it should be connected to the P+ and C terminals (at this time, P+ and D terminals must be open-circuited). If the internal braking resistor is used, P+ and D terminals should be short-circuited (at this time, P+ and C terminals must be open-circuited).
- Neither run power cables and signal lines through the same conduit, nor bundle them together. When wiring, power cables and signal lines should be separated by at least 30cm, otherwise it may lead to malfunctions.
- Do not frequently turn the power ON/OFF. If repeated continuous ON/OFF of power is necessary, limit it to less than once per minute. Because the power section of the servo drive contains large capacitors, a large charging current will flow when power is turned ON (the charging time is several hundred milliseconds). Therefore, frequently turning ON/OFF of power will cause performance degradation of the main circuit components inside the servo drive.
- The maximum wiring length for input/output signal cables is 3m, and the maximum length for main circuit cables and encoder cables is 30m. Please use twisted-pair and shielded signal lines with isolated grounding.
- Please use thick wire for grounding cables (2.0mm² or larger) whenever possible.
- If the servo motor and mechanical system are insulated from each other, please ground the servo motor directly.

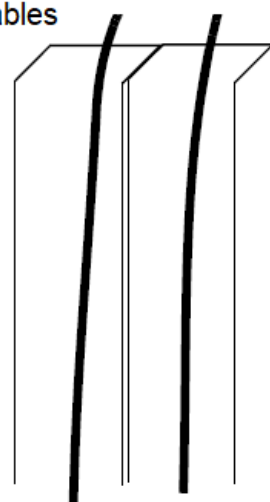
Control & Signal Power Cable
Cables



400V AC 24V DC



Control & Signal Power Cable
Cables



4.2 Main Circuit Terminals and Wiring

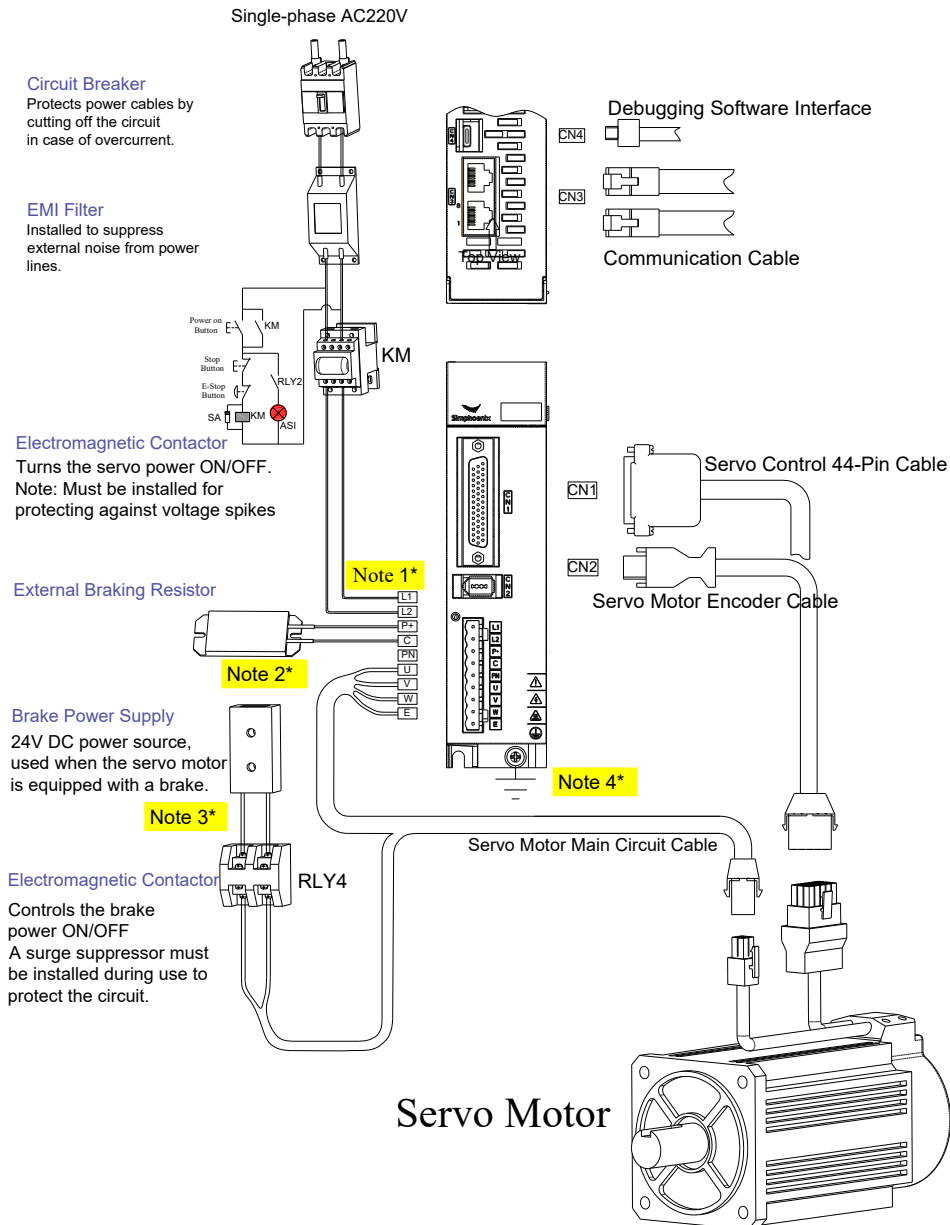
4.2.1 Main Circuit Terminals

Symbol	Name	Drive Model	Function and Connection
L1 L2 (L3)	Main Circuit Power Input Terminal	T1R8/T3R0/T4R0	Single-phase AC220V power input (no L3 terminal)
		T5R5~T13R	Single-phase/three-phase AC220V power input
		F4R0-F62R	Three-phase AC 380V Power Input
P+ D C	External Braking Resistor Connection Terminals	T1R8/T3R0	Internal braking: None External braking: Braking resistor between P+ and C
		T4R0	Internal braking: 50Ω,50W External braking: Braking resistor between P+ and C
		T5R5~T13R F4R0-F27R	Internal braking: Optional Short-circuit between P+ and D External braking: Connect braking resistor between P+ and C, and open circuit between P+ and D
		F38R-F62R	Internal braking: None External braking: Connect braking resistor between P+ and C, and open circuit between P+ and D
U V W	Motor Connection Terminals	Power cable connection terminals of the servo drive, connected to the U/V/W phases of the motor, respectively	
P+ PN	Common DC Bus Terminals	Common DC bus terminals of the servo drive, used for common bus when multiple drives are connected in parallel	
N1 N2	External Reactor Terminals	By default, N1 and N2 are short-circuited. When high-order harmonics of the power supply need to be suppressed, remove the short-circuit between N1 and N2 and connect an external DC reactor	
PE	Ground Terminal	Connect to the ground terminal of the power supply and the ground terminal of the motor	

4.2.2 Main Circuit Cable Diameter Specifications

Drive Model	Power Input Cable	Motor Connection Cable	Braking Resistor Connection Cable	Protective Earth Wire
	L1/L2/L3	U/V/W	P+/D/C	PE
CD200□-T1R8	0.50 mm ²	0.50 mm ²	0.50 mm ²	0.50 mm ²
CD200□-T3R0	0.50 mm ²	0.50 mm ²	0.50 mm ²	0.50 mm ²
CD200□-T4R0/T4R5	0.75 mm ²	0.75 mm ²	0.75 mm ²	0.75 mm ²
CD200□-T5R5/T7R5	0.75 mm ²	0.75 mm ²	0.75 mm ²	0.75 mm ²
CD200□-T10R/T13R	1.5 mm ²	1.5 mm ²	1.5 mm ²	1.5 mm ²
CD200□-F4R0	1.5 mm ²	1.5 mm ²	1.5 mm ²	1.5 mm ²
CD200□-F6R5	1.5 mm ²	1.5 mm ²	1.5 mm ²	1.5 mm ²
CD200□-F8R5	1.5 mm ²	1.5 mm ²	1.5 mm ²	1.5 mm ²
CD200□-F12R	1.5 mm ²	1.5 mm ²	1.5 mm ²	1.5 mm ²
CD200□-F17R	4.0mm ²	4.0mm ²	4.0mm ²	4.0mm ²
CD200□-F22R	4.0mm ²	4.0mm ²	4.0mm ²	4.0mm ²
CD200□-F27R	4.0mm ²	4.0mm ²	4.0mm ²	4.0mm ²
CD200□-F38R	6.0mm ²	6.0mm ²	6.0mm ²	6.0mm ²
CD200□-F52R	10.0mm ²	10.0mm ²	10.0mm ²	10.0mm ²
CD200□-F62R	16.0mm ²	16.0mm ²	16.0mm ²	16.0mm ²

4.2.3 Single-Phase 220V Wiring (T1R8/T3R0/T4R0)



- **KM:** Electromagnetic Contactor **D:** Freewheeling Diode **SA:** Surge Suppressor
RLY2/RLY4: Relays **ASI:** Alarm Status Indicator

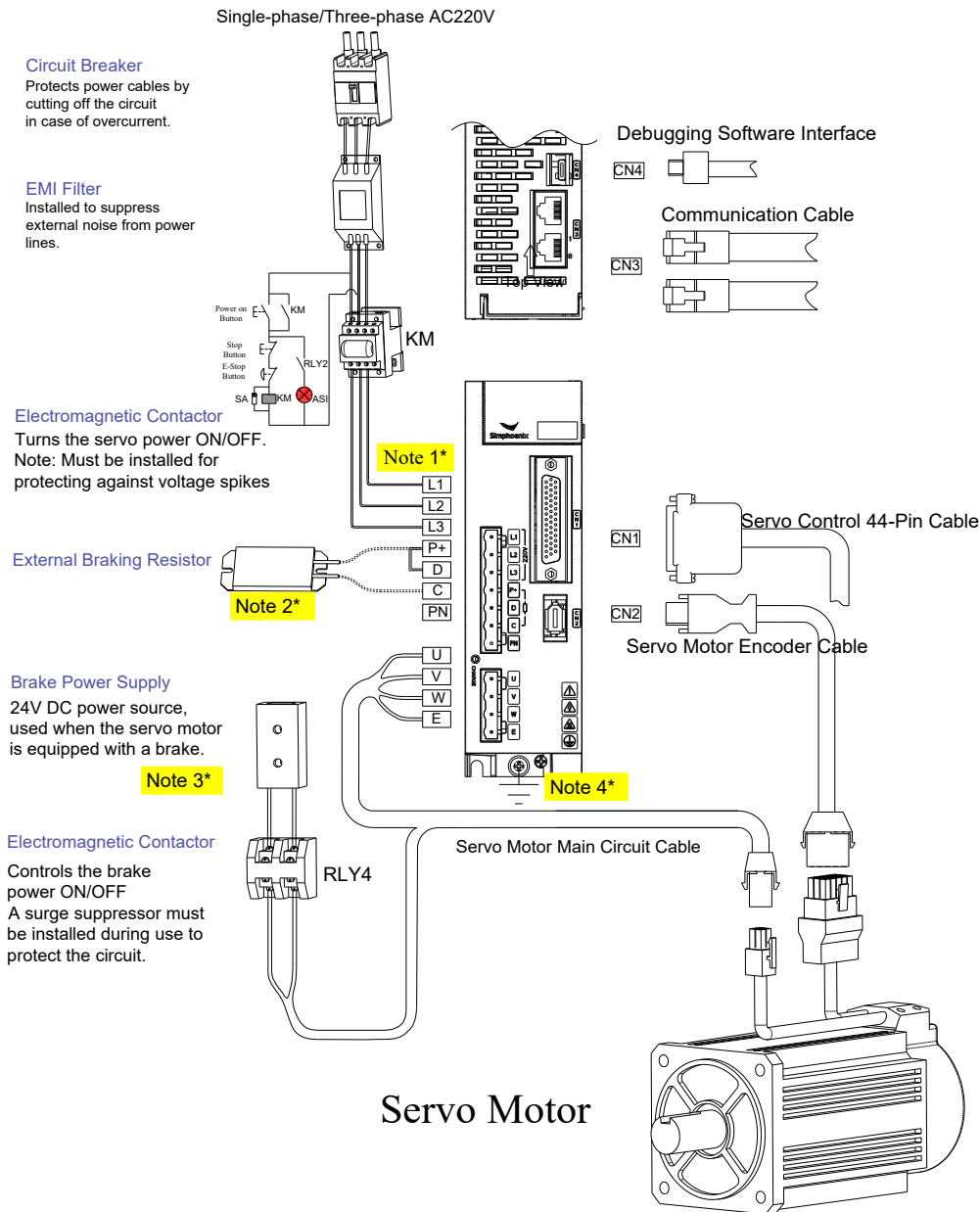
Note 1*: T1R8/T3R0/T4R0 accepts single-phase AC 220V power input.

Note 2*: T1R8/T3R0/T4R0 has no internal braking resistor. Connect an external braking resistor between **P+** and **C** terminals.

Note 3*: The **24V DC brake power supply** must be user-provided.
Galvanically isolated from the control signal power (12–24V).

Note 4*: Motor cable shield must terminate at the product's **PE output terminal**.
 Main circuit **PE input terminal** must connect to the control cabinet's **grounding busbar** via a protective earth conductor.

4.2.4 Single-Phase/Three-Phase 220V Wiring (T4R5/T5R5/T7R5)



- **KM:** Electromagnetic Contactor **D:** Freewheeling Diode **SA:** Surge Suppressor
RLY2/RLY4: Relays **ASI:** Alarm Status Indicator

Note 1*: T4R5/T5R5/T7R5 accepts either three-phase AC 220V or single-phase AC 220V power supply

Note 2*: When using external braking resistor: Connect between P+ and C terminals, Meanwhile keep P+ and D terminals open-circuit;

When no external braking resistor is used: Must short-circuit P+ and D terminals

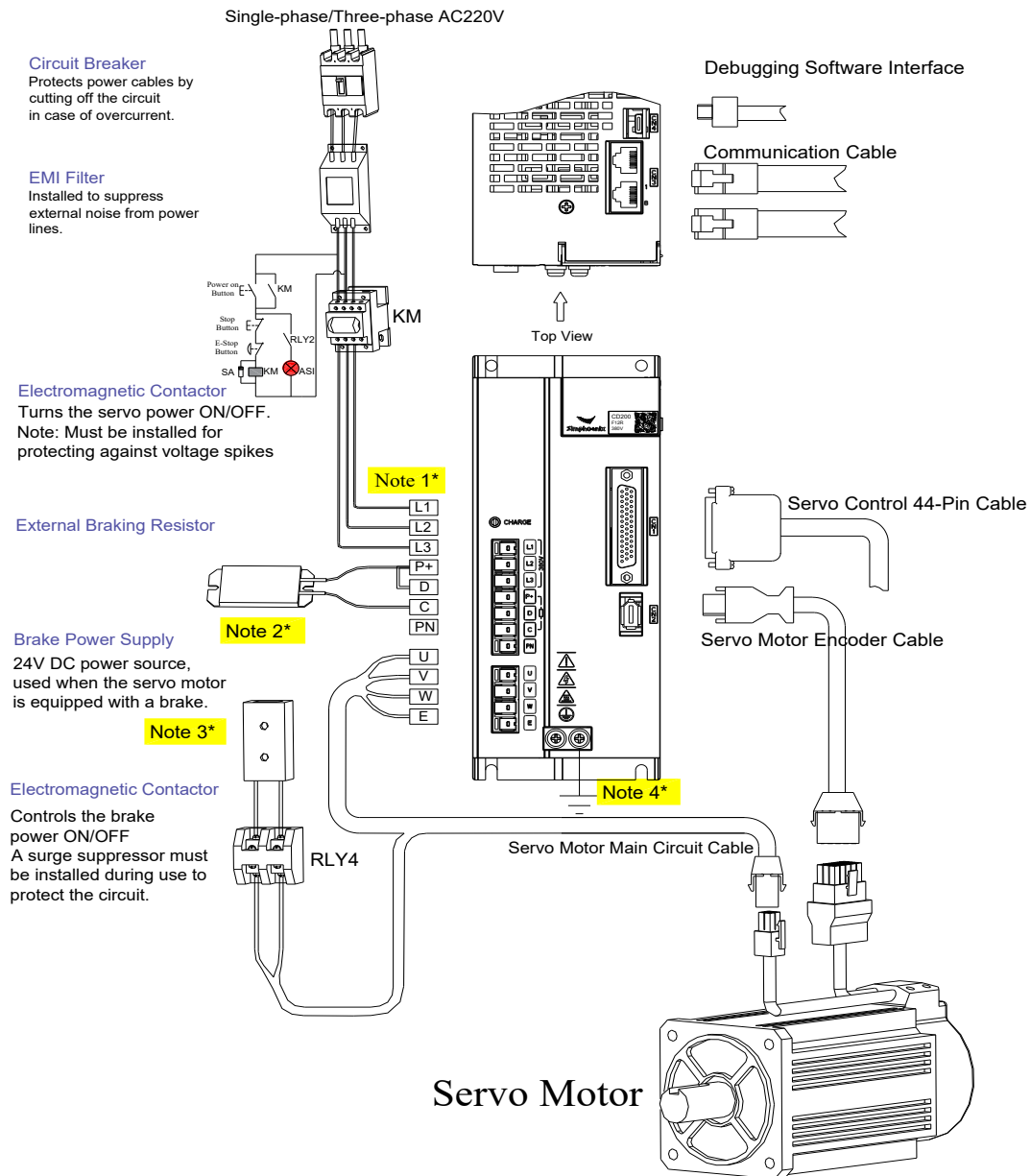
Note 3*: The **24V DC brake power supply** must be user-provided.

Galvanically isolated from the control signal power (12–24V).

Note 4*: Motor cable shield must terminate at the product' s **PE output terminal**.

Main circuit **PE input terminal** must connect to the control cabinet' s **grounding busbar** via a protective earth conductor.

4.2.5 Single-Phase/Three-Phase 220V Wiring (T10R ~ T13R)



• **KM:** Electromagnetic Contactor **D:** Freewheeling Diode **SA:** Surge Suppressor

RLY2/RLY4: Relays **ASI:** Alarm Status Indicator

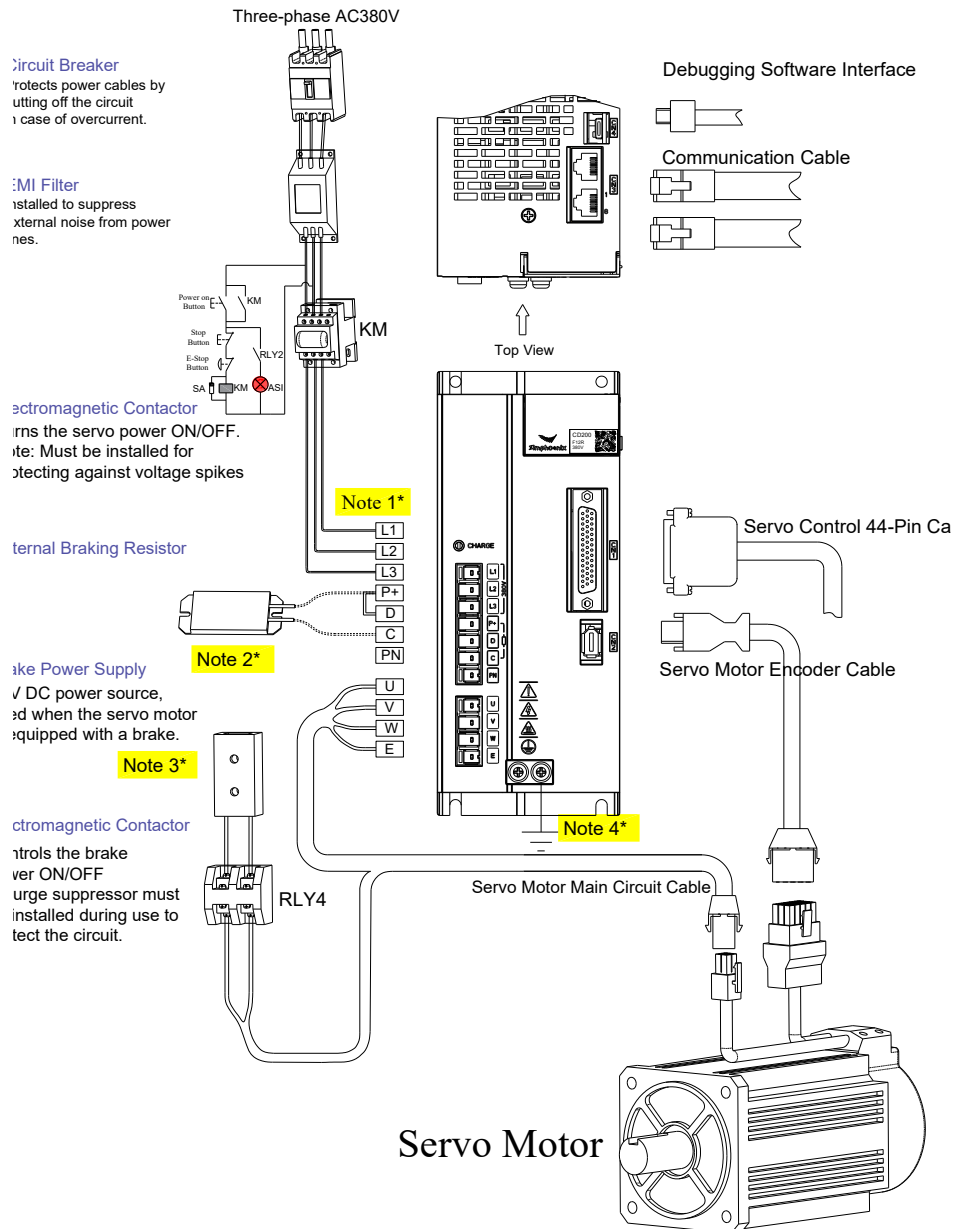
Note 1*: T10R~T13R accepts three-phase AC220V power supply

Note 2*: When using external braking resistor:
Connect between P+ and C terminals, Meanwhile keep P+ and D terminals open-circuit;
When no external braking resistor is used: Must short-circuit P+ and D terminals

Note 3*: The 24V DC brake power supply must be user-provided.
Galvanically isolated from the control signal power (12~24V).

Note 4*: Motor cable shield must terminate at the product's PE output terminal.

4.2.6 Three-Phase 380V Wiring (F4R0 ~ F12R)



● **KM:** Electromagnetic Contactor **D:** Freewheeling Diode **SA:** Surge Suppressor

RLY2/RLY4: Relays **ASI:** Alarm Status Indicator

Note 1*: **F4R0~F12R** accepts three-phase AC 380V power supply

Note 2*: When using external braking resistor:

Connect between P+ and C terminals, Meanwhile keep P+ and D terminals open-circuit;

When no external braking resistor is used: Must short-circuit P+ and D terminals

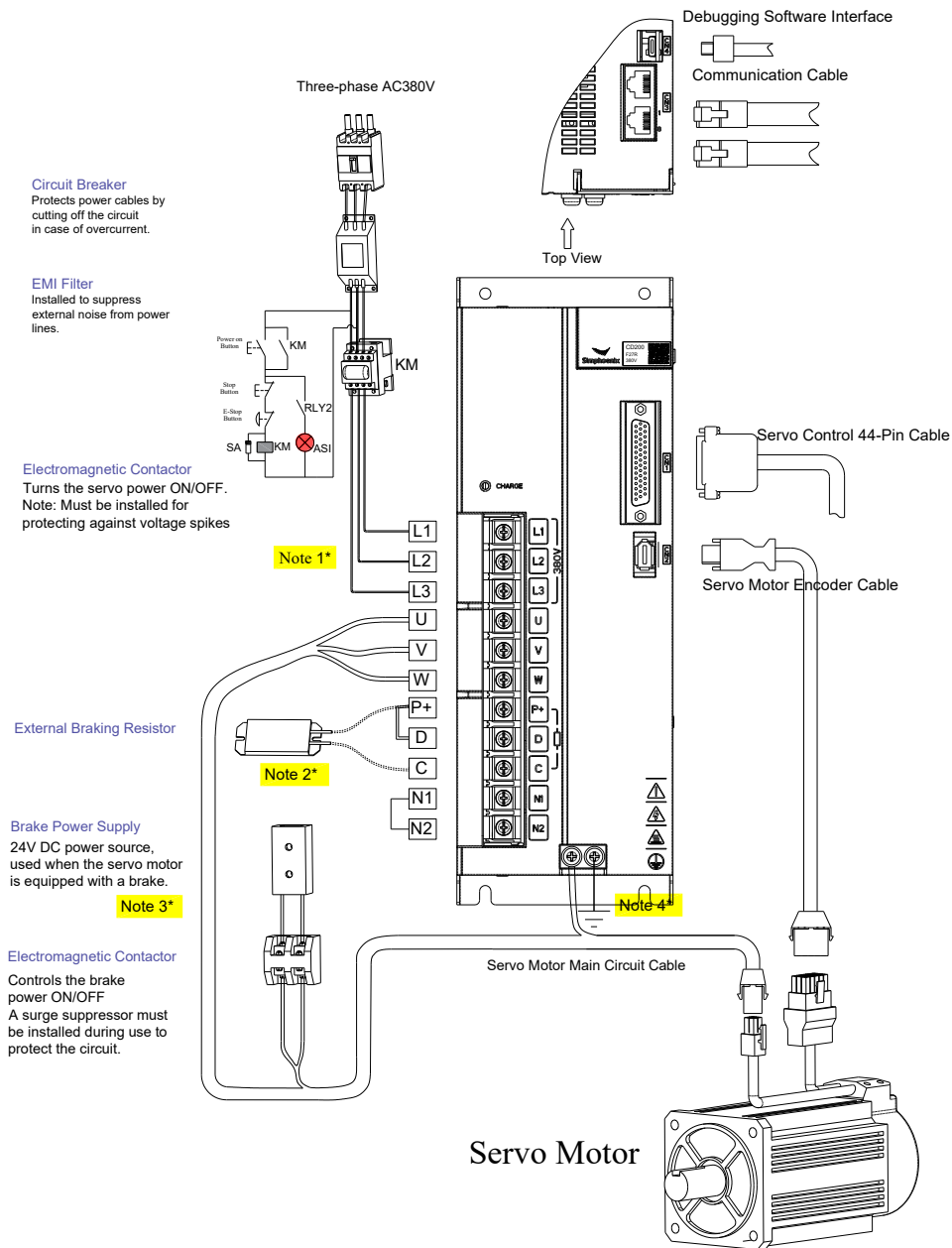
Note 3*: The **24V DC brake power supply** must be user-provided.

Galvanically isolated from the control signal power (12~24V).

Note 4*: Motor cable shield must terminate at the product's **PE output terminal**.

Main circuit **PE input terminal** must connect to the control cabinet's **grounding busbar** via a protective earth conductor.

4.2.7 Three-Phase 380V Wiring (F17R ~ F27R)



KM: Electromagnetic Contactor **D:** Freewheeling Diode **SA:** Surge Suppressor

RLY2/RLY4: Relays **ASI:** Alarm Status Indicator

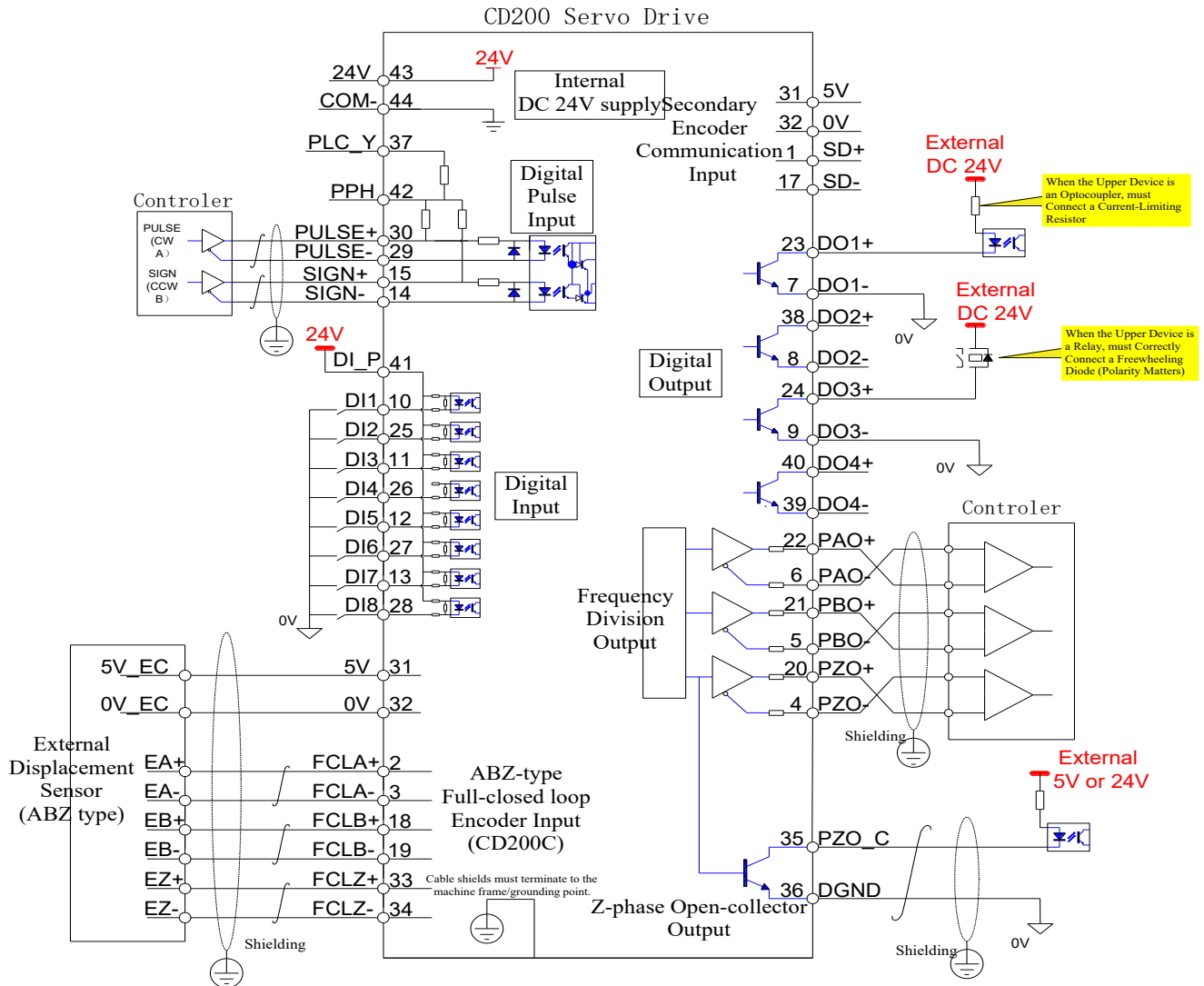
Note 1*: **F17R0~F27R** accepts three-phase AC 380V power supply

Note 2*: When using external braking resistor:
Connect between P+ and C terminals, Meanwhile keep P+ and D terminals open-circuit;
When no external braking resistor is used: Must short-circuit P+ and D terminals

Note 3*: The **24V DC brake power supply** must be user-provided.
Galvanically isolated from the control signal power (12~24V).

Note 4*: Motor cable shield must terminate at the product' s **PE output terminal**.
Main circuit **PE input terminal** must connect to the control cabinet' s **grounding busbar** via a protective earth conductor.

4.3 Control Circuit Wiring



※ Description:

- [1] **Internal 24V Power Supply Specifications:**
Voltage Range: 20V~28V DC, Maximum Current: 100mA
- [2] **Pulse Signal Wiring Requirements:**
Cable Type: Twisted-pair shielded cable
Shielding: Shield layer **must be connected to PE at both ends**
DGND must be reliably connected to the host controller's signal ground
- [3] **DO (Digital Output) Power Supply:**
User-provided power supply range: 5V~24V DC
DO Port Limits: Max Voltage: 30V DC,
Max Current: 50mA.
- [4] **Frequency Division Output Wiring Requirements:**
Cable Type: Twisted-pair shielded cable
Shielding: Shield layer **must be connected to PE at both ends**
DGND must be reliably connected to the host controller's signal ground

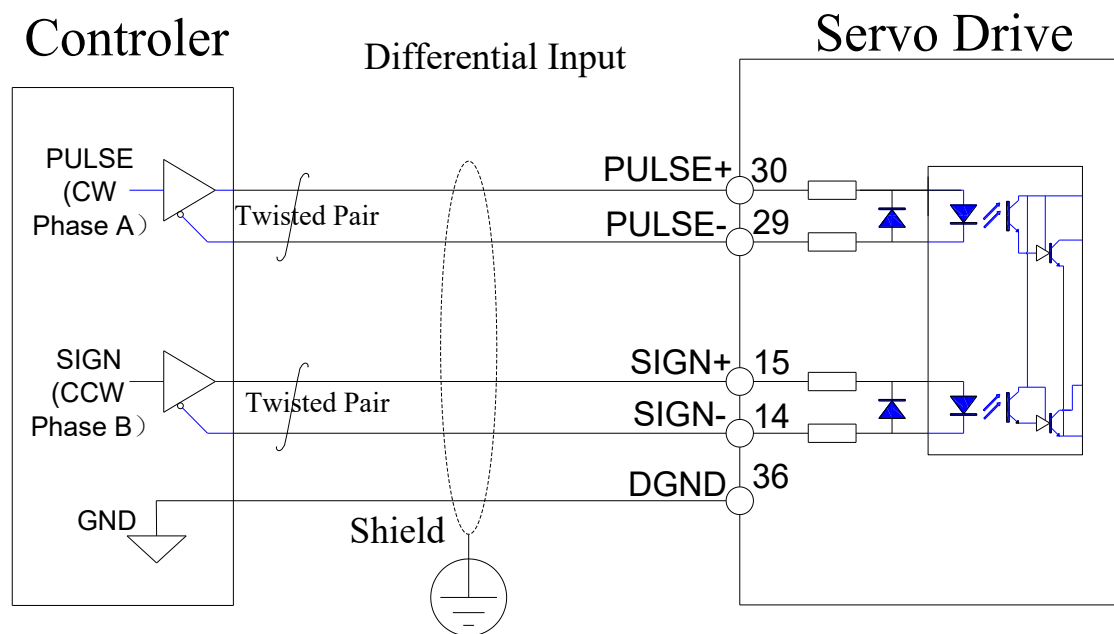
4.3.1 Pulse Input Interface

Description of Position Command Pulse Input Signal

Signal Name		Terminal Number	Function	
Internal Power Supply	+24V	43	Internal 24V power supply, voltage range +20V-28V, maximum output current 100mA	
	COM	44	Internal 24V power ground	
Position Command	PULSE+	30	Pulse Command Input Method: · Differential drive input · Open collector	Input Pulse Form: · Pulse + direction · A, B phase quadrature pulse · CW/CCW pulse · Digital Pulse Bus
	PULSE-	29		
	SIGN+	15		
	SIGN-	14		
	PPH	42	Command pulse power input interface (2KΩ resistor already in series inside the drive)	
	PLC_Y	37	Open collector pulse input pull-up/pull-down interface	

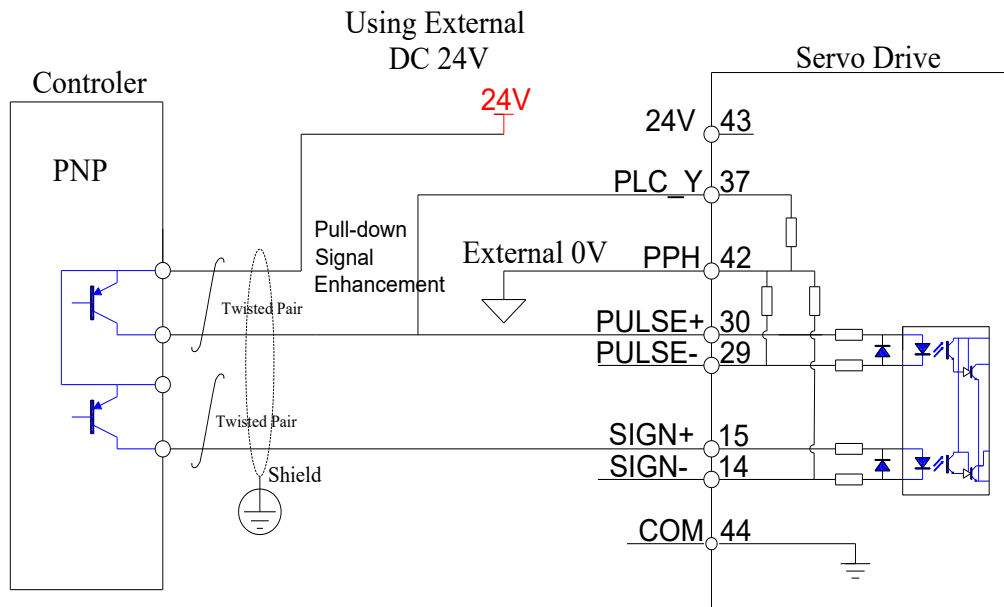
(1) Differential Input

The pulse input voltage of the differential input is 5V. This input method offers the best noise immunity and is recommended for priority use



Be sure to connect the 5V power ground of the upper device to the DGND of the drive to reduce noise interference

b. Upper device is PNP-type output: PLC_Y and PULSE+ are connected together to the PLC output port to enhance PLC pull-down current capability and reduce noise interference



III. Using external 24V power supply (using external current limiting resistors, use with caution)

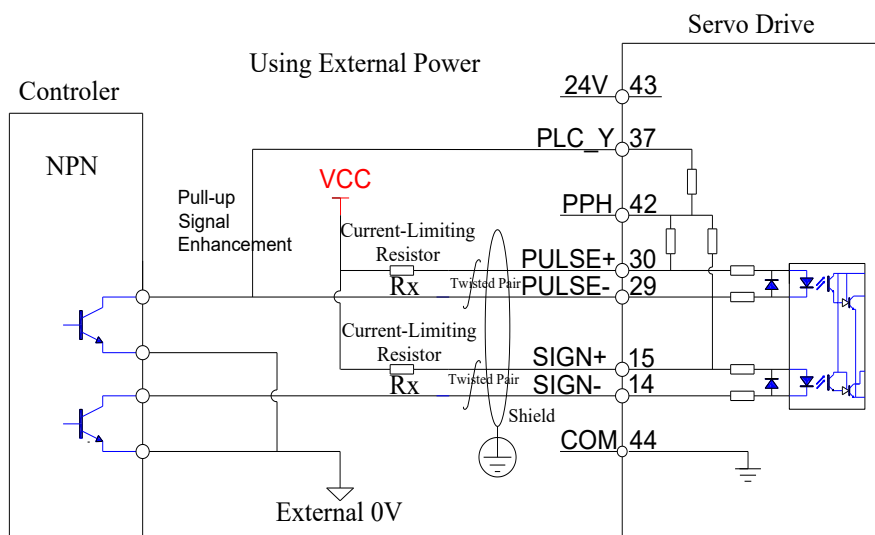
If an external resistor method is used to receive the input signal of the external open collector, please correctly connect current limiting resistors in series in the circuit. Otherwise, it may lead to abnormal pulse reception or even port damage. The selection of current limiting resistors must meet the following conditions, and each port must be connected with its own current limiting resistor; **do not share or omit connections**

$$\frac{V_{cc} - 1.4V}{R_x + 200} = 10 \text{ mA}$$

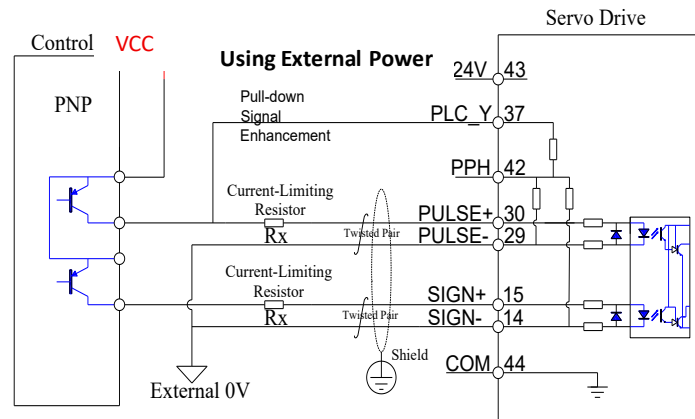
Recommended resistance values:

Vcc	Rx	Power of Rx
24V	2K	0.5W
12V	800R	0.5W
5V	0K	0W

a. Upper device is NPN-type output: PLC_Y and PULSE- are connected together to the PLC output port to enhance PLC pull-up current capability and reduce noise interference



b. Upper device is PNP-type output: PLC_Y and PULSE+ are connected together to the PLC output port to enhance PLC pull-down current capability and reduce noise interference



4.3.2 Digital Input (DI)

(1) DI Terminal Pin Definition

The digital input interface of the servo drive adopts a bidirectional optocoupler design, allowing users to connect via relays or open collector transistors. When connecting with relays, please select relays for small currents, otherwise it may result in poor contact or failure of the circuit to conduct.

Note:

1. When using an external power supply, disconnect 24V from the DI_P terminal
2. Mixing PNP and NPN inputs between multiple DI terminals is not supported
3. Do not reuse functions for different DI terminals

Signal Name		Pin Number	Default Function
Digital Input	DI1	10	Digital input, default function: Servo enable
	DI2	25	Digital input, default function: Fault reset
	DI3	11	Digital input, default function: Positive limit
	DI4	26	Digital input, default function: Negative limit
	DI5	12	Digital input, default function: Origin signal
	DI6	27	Digital input, default function: Start origin search
	DI7	13	Digital input, default function: Emergency stop
	DI8	28	Digital input, default function: Interrupt position table
	DI_P	41	Digital Input Common Terminal
Power Supply	+24V	43	Internal 24V Power Supply
	COM	44	

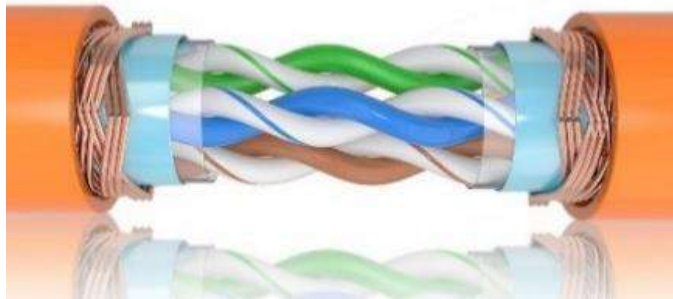
(2) Digital Input (DI) Function Definition Table

S/N	Function	Function Description	ReMarks
0	No Function		
1	Servo Enable	Control the servo to enter the running or standby status	Only when there is no fault can it be enabled
2	Fault Reset	Control the servo to switch back to the ready status from fault status	Before resetting, please ensure that the fault has been cleared
3	Position Table Index 0	In PP mode, for position table mode task index selection, the index value composed of the states of "Position Table Index 0-4" selects the starting sequence number for position table execution tasks	Position table mode requires at least one input terminal function configured as "Position Table Index". "Index 31" is a special function configuration
4	Position Table Index 1		
5	Position Table Index 2		
6	Position Table Index 3		
7	Position Table Index 4		
8	Operating Mode Index 0	The index value composed of the states of "Operating Mode Index 0/1" selects the servo operating mode	The corresponding operating mode for DI terminal input can be set via "DIN Operating Mode 0-4"
9	Operating Mode Index 1		

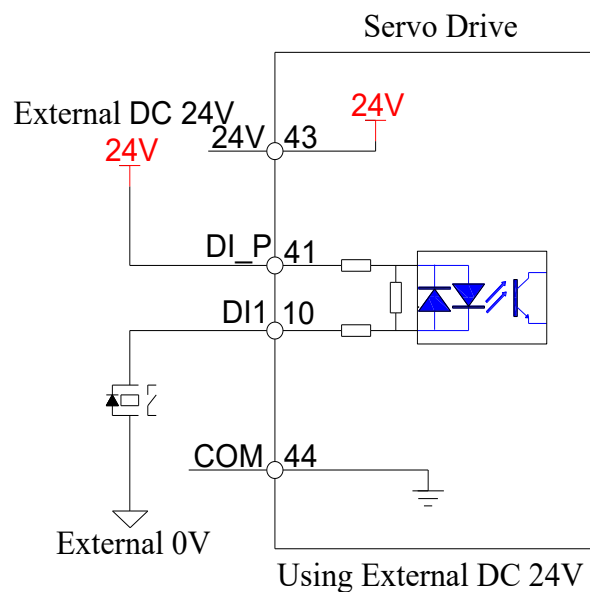
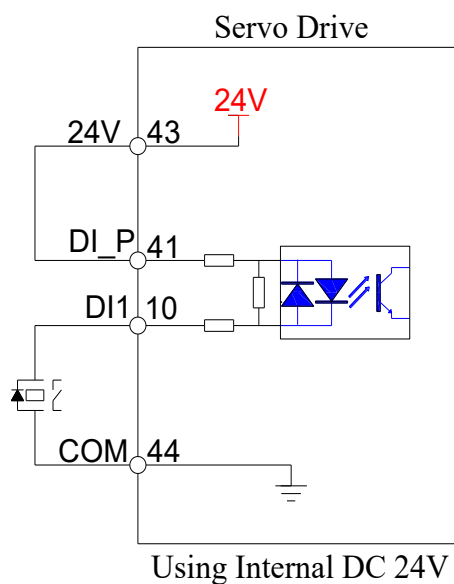
S/N	Function	Function Description	ReMarks
10	Gain Index 0	The index value composed of the states of "Gain Index 0/1" selects the internal gain group	Each gain group parameter includes: position loop bandwidth, speed loop proportional gain, speed loop integral gain, speed loop integral limit, and speed loop PDFF coefficient
11	Gain Index 1		
12	Integral Clear	Clear "Speed Loop Integral Output" to zero.	
13	Positive Limit	Positive limit switch input, "----PL" is displayed after triggering	To prevent accidents caused by abnormal signal triggering or broken wires, use "normally closed contacts" for limit switches. Positive and negative limit functions are invalid in torque mode
14	Negative Limit	Reverse limit switch input, "nL----" is displayed after triggering	
15	Origin Signal	Origin mode origin switch signal	Origin switch input
16	Start Origin Search	Origin finding action trigger signal in origin mode	
17	Command Reversal	Reverse the command target	Control loop command and feedback in reverse direction
18	Emergency Stop	Execute emergency stop action according to the set method	The emergency stop method can be set via "Emergency Stop Option" With some emergency stop options, motion will resume after the emergency stop signal disappears.
19	Trigger 0	Trigger oscilloscope data acquisition	Click "re-read data" to display data acquired at the trigger point
21	Condition 0	Judgment condition for the next index in position table mode. Setting control register bit 9 and bit 10 to 1 enables condition 0 and condition 1 inputs, respectively. When both are active, bit 11 can select the execution logic as condition 0 condition 1 (bit 12 set to 0), or condition 0 & condition 1 (bit 12 set to 1)	If both condition 0 and condition 1 are inactive, the execution condition remains 1; If condition 0 is active and condition 1 is inactive, the execution condition is condition 0; If condition 0 is inactive and condition 1 is active, the execution condition is condition 1.
22	Condition 1		
23	Pause	Pause command execution in PV and PP modes	Set "Control Word 6040" bit 8 to 1. After release, incomplete commands will continue to execute
24	Pulse gear	With the gear expansion coefficient, it enables switching of the electronic gear ratio	In pulse mode, when the terminal input is active, the gear ratio switches to (gear expansion coefficient numerator / gear expansion coefficient denominator) × electronic gear ratio.
25	Activate Position Table	Run the position table task	
26	Interrupt Position Table	Interrupt the position table running	After running the current index, the entire position table running ends. The position table will rerun from the beginning after the next activation
27	Pause Position Table	Pause the position table running	After running the current index, the next index will pause, and resume running until the pause signal is cleared
28	Probe 1	The probe function can record the position and time information of the rising and falling edges of the probe signal when triggered, and can also record the number of triggers	Probe enable and trigger methods are configured via "Probe Settings"
29	Probe 2		
30	Full/Half Closed-loop Switching	Enables full closed-loop control	
31	Pulse input disable	Ignores all received pulse commands when active	

(3) DI Terminal Wiring Usage

It is recommended to use shielded cables for control cables, and the shield layer should be reliably connected to the drive enclosure (E) using a 360° circumferential connection for better EMC anti-interference performance.

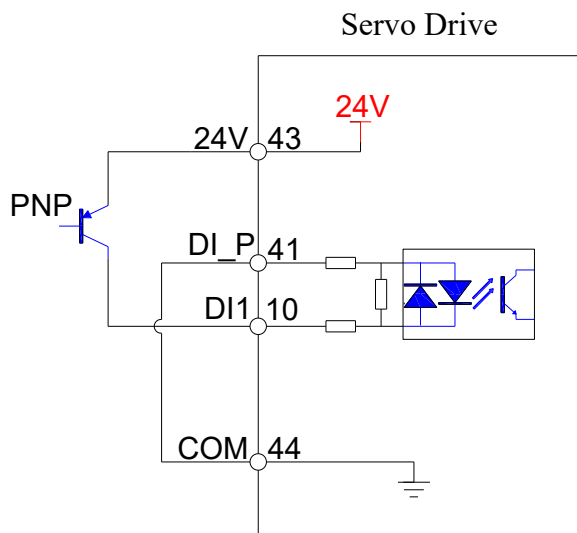
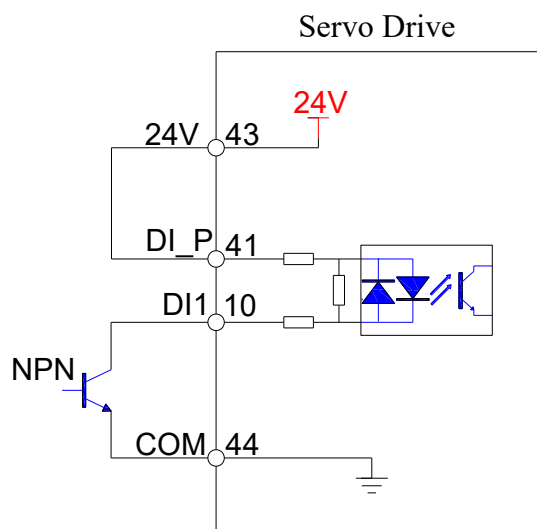
**I. If the higher-level device is a relay output:**

Taking DI1 as an example, various wiring methods are as follows. The interface circuits for DI1 to DI8 are identical.

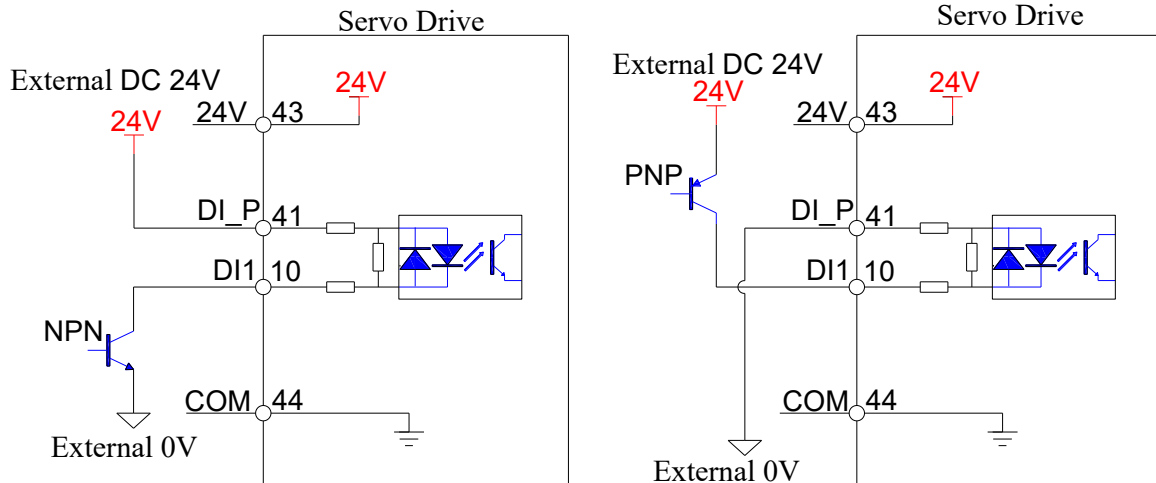
**II. If the higher-level device is an open collector output:**

Taking DI1 as an example, various wiring methods are as follows. The interface circuits for DI1 to DI8 are identical.

- a. When using the internal 24V power supply of the drive:



- b. When using the external 24V power supply:



4.3.4 Digital Output (DO)

The digital output interface of the CD200 servo drive adopts a differential output design.

Note:

1. The maximum allowed voltage for the output transistor of the servo drive is DC28V.
2. The maximum allowed current for the digital output transistor of the servo drive is DC50mA.

(1) DO Terminal Pin Definition

Signal Name	Pin Number	Function
Digital Output	DO1+	23
	DO1-	7
	DO2+	38
	DO2-	8
	DO3+	24
	DO3-	9
	DO4+	40
	DO4-	39
Power Supply	+24V	43
	COM	44

(2) Digital Output (DO) Function Definition Table

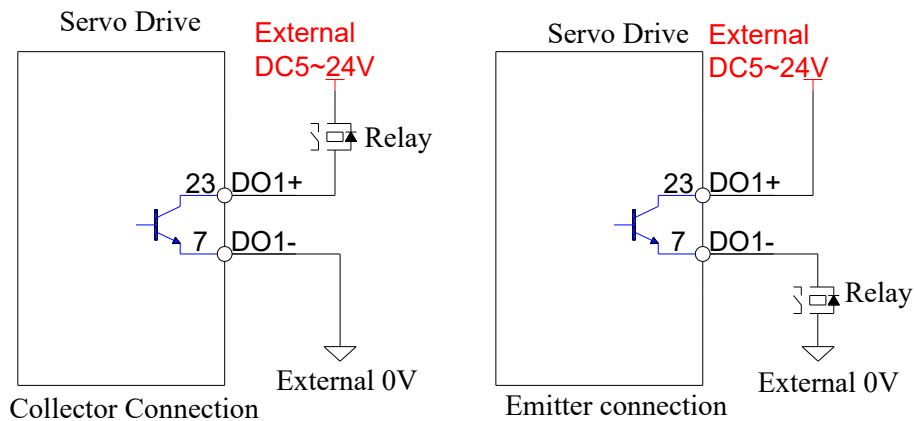
S/N	Function	Function Description	ReMarks
0	No Function	No Function	
1	Ready	Servo ready, no fault	
2	Fault	Servo fault alarm output	
3	Target Reached	Reached the set target position or target speed	Output after the servo is enabled
4	Speed Saturation	Speed reaches or exceeds "Maximum Speed Command"	The position mode is active
5	Current Saturation	Current reaches or exceeds "Current Command Limit Value"	
6	Motor Zero Speed	The motor speed is zero	Independent of servo enable status
7	Brake Control	Brake control output after servo enable	
8	Dynamic Braking	Bus voltage exceeds "Braking Point"	The braking resistor is working
9	Z Signal	Motor encoder Z signal output	
10	Motor Overspeed	The motor speed exceeds "Maximum Speed Setting" in torque mode	

S/N	Function	Function Description	ReMarks
11	Enabling	The servo is in enable status	
12	In Limit	The servo is in the positive/negative limit status of software/hardware	
13	Origin Found	Origin finding completed in origin mode	Origin mode active
14	Origin Search	Origin finding action in progress in origin mode	Origin mode active
18	Current Reached	The current is greater than "Current Comparison Value"	

(3) DO Terminal Wiring Usage

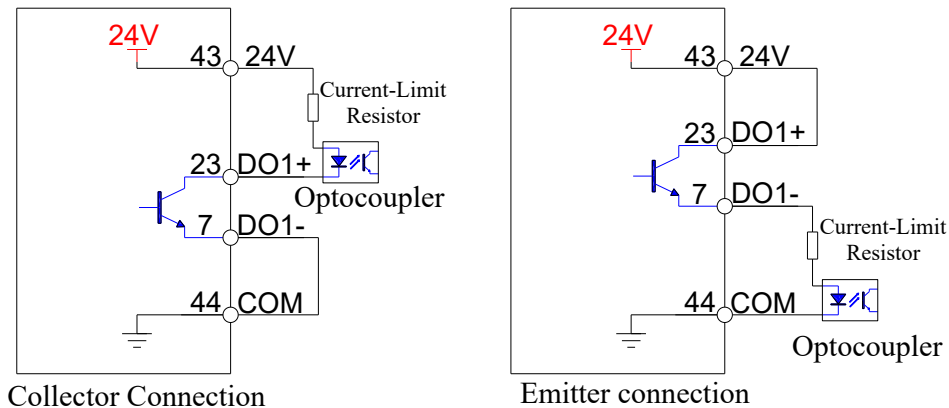
Taking DO1 as an example, various wiring methods are as follows. The interface circuits for DO1 to DO4 are identical.

I. If the higher-level device is a relay input: Please confirm that the correct polarity of the freewheeling diode

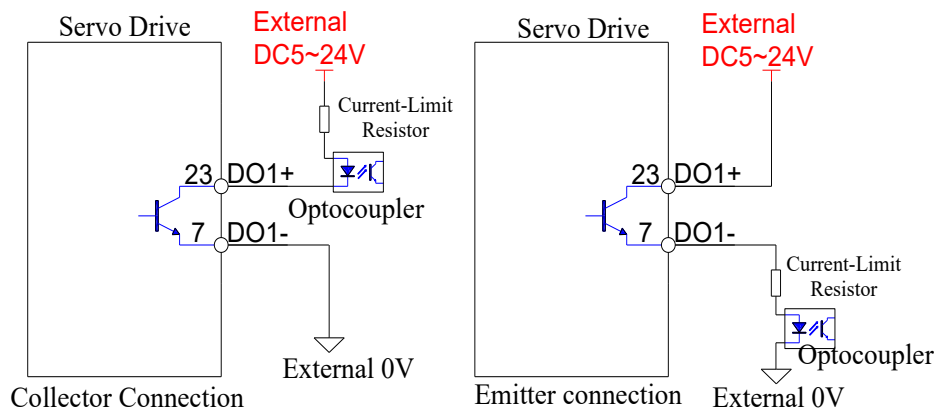


II. If the higher-level device is an optocoupler input: Please ensure that the current-limiting resistor is correctly connected

a. When using the internal 24V power supply of the drive:



b. When using the external 24V power supply:



4.3.5 Encoder Frequency Division Output

The encoder frequency division output signal of the drive is a differential signal, typically used as a position feedback signal when the higher-level device performs position control. On the higher-level device side, please use a differential or optocoupler receiving circuit.

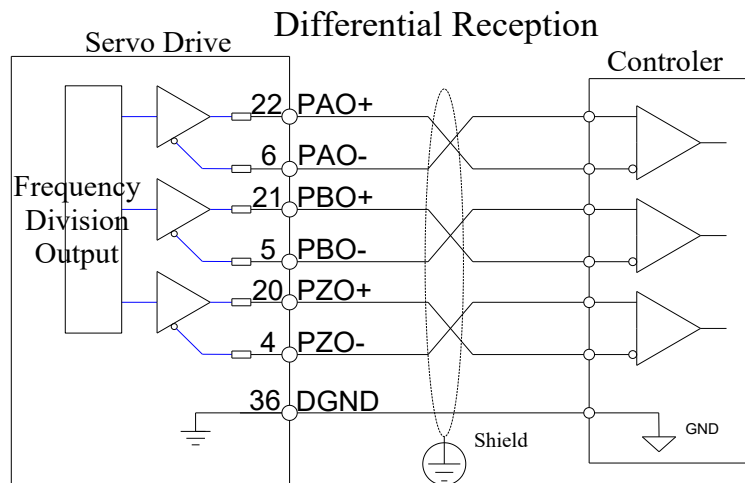
※ **Note:**

1. The maximum allowed voltage for the internal output circuit of the servo drive is DC28V.
2. The maximum allowed current for the internal output circuit of the servo drive is DC20mA.

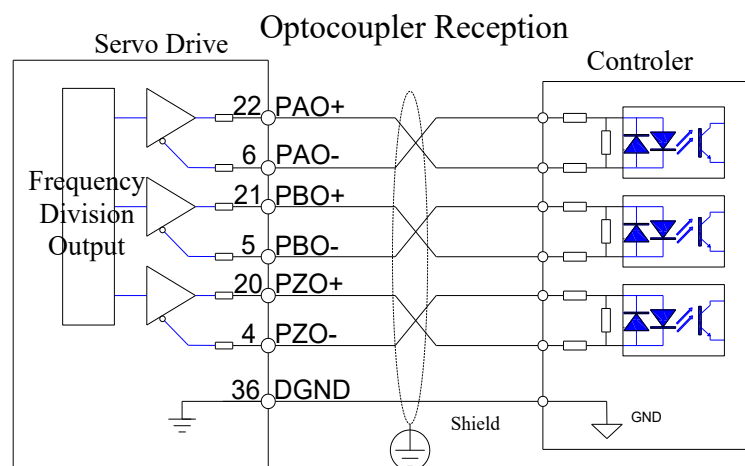
Signal Name	Pin Number	Function	
Frequency Division Output	PAO+	22	A Phase Frequency Division Output Signal
	PAO-	6	
	PBO+	21	B Phase Frequency Division Output Signal
	PBO-	5	
	PZO+	20	Z Phase Frequency Division Output Signal
	PZO-	4	
	PZO_C	35	Z pulse open collector output
	D_GND	36	
			Signal Ground
			Origin Pulse Open Collector Output Signal

(1) Encoder Frequency Division Output Differential Reception

When using differential reception on the host device side, a terminal resistor must be installed between the differential input circuits, with the resistance value set according to actual conditions. Additionally, pin 36 (GND) of the driver must be short-circuited to the GND of the differential receiver on the host device.

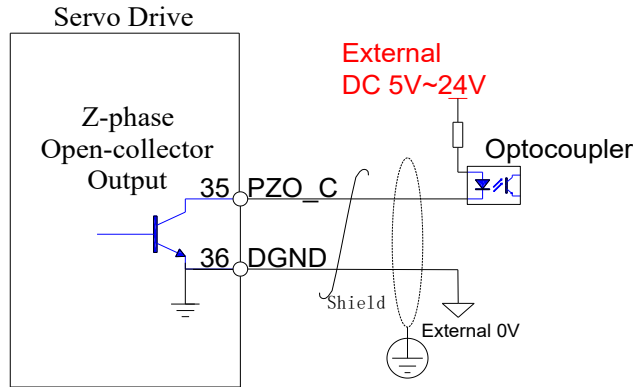


(2) Encoder Frequency Division Output Optocoupler Reception



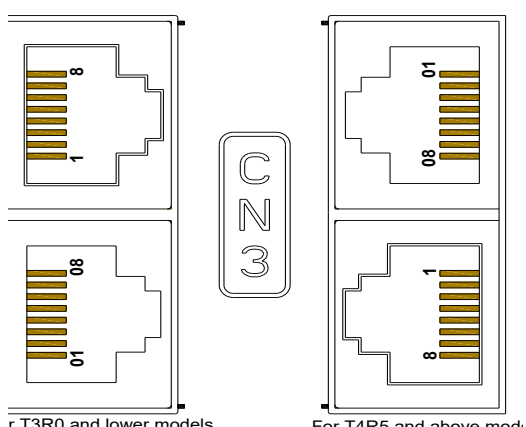
(3) Encoder Frequency Division Z-Phase Open Collector Output

Additionally, the Z-phase of the encoder frequency division output signal provides an open collector output signal, which can serve as a feedback signal when the servo drive and the higher-level device form a position control system. On the higher-level device side, please use an optocoupler circuit, relay circuit, or bus receiver circuit to receive it.



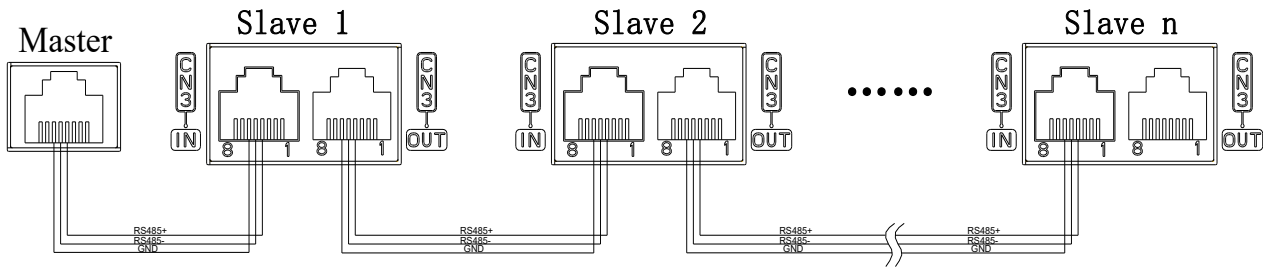
4.3.6 Communication Interface Wiring

Communication interface CN3 consists of two identical communication interfaces connected in parallel internally, enabling communication between the drive and PCs, PLCs, and other drives

Terminal Position Number CN3	Pin	Signal	Description
	1/01	NC	-
	2/02	NC	-
	3/03	NC	-
	4/04	RS+	RS485 Differential Input Signal Port
	5/05	RS-	
	6/06	GND	Signal Ground
	7/07	CANH	CAN Differential Input Signal Port (CD200C series supported)
	8/08	CANL	
	PE	Casing	-

(1) RS485 Communication

When using RS485 for network communication, CN3-IN connects to the output of the PLC/controller or the CN3-OUT of the previous drive; CN3-OUT connects to the CN3-IN of the next drive.

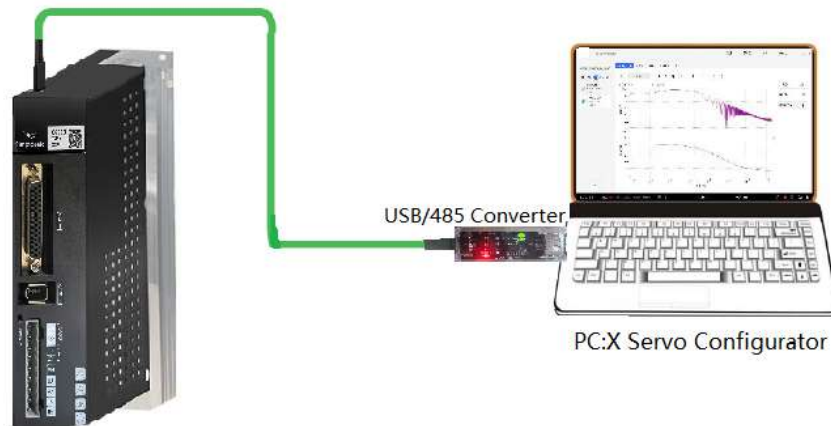


Communication Connection Descriptions:

- Please use shielded twisted-pair cables. Connectors with metal shielding are recommended to prevent signal interference.
- Connect 120Ω termination resistors at both ends of the RS485 bus to prevent RS485 signal reflection.
- When there are many nodes, please choose a daisy-chain topology for the RS485 bus. All node signal grounds should be connected together, supporting a maximum of 32 nodes.

(2) Communication Interface with Upper Computer

As shown in the figure below, the communication interface CN4 on the top of the servo drive allows connecting the drive to the dedicated debugging tool X Servo Configurator for operations such as parameter setting & monitoring, gain adjustment, troubleshooting, and firmware upgrades.



4.4 Anti-Interference Measures

As a key component of motion control systems, servo drives have high requirements for anti-interference. Variations in peripheral wiring and grounding methods for servo drives may lead to noise affecting the normal operation of the system. Therefore, correct grounding methods and wiring procedures must be adopted.

Grounding Procedures

To prevent potential electromagnetic interference problems, please ground according to the following methods:

- Always use a single-point grounding method, ensuring that the protective earth (PE) conductor has sufficient conductivity and complies with local regulatory requirements.
- When multiple units are connected in parallel, connect the PE cable from the input power supply to the grounding copper bar of the control cabinet, and connect the grounding terminal of the drive to the copper bar inside the control cabinet via a protective earth conductor. The grounding copper bar of the control cabinet should then be connected to the metal enclosure of the control cabinet.
- When the servo motor is grounded mechanically, switching interference currents may flow from the main circuit of the servo unit through the parasitic capacitance of the servo motor. To prevent this phenomenon, ensure that the E terminal of the main circuit cable connected to the servo motor is connected to the E grounding terminal of the servo unit. Additionally, the grounding terminal must be grounded.
- The shield wire of the encoder cable connected to the servo motor must be connected to the plug housing (casing).
- If input/output signal cables are susceptible to noise interference, use a 360° grounding method for the shield at the cable entry point to reduce electromagnetic radiation and interference.
- If the main circuit cable of the servo motor is encased in a metal conduit, ensure single-point grounding for both the metal conduit and the grounding box.
- When installing inside the control cabinet, the cabinet should be divided into multiple EMC areas or separated into multiple control cabinets based on the strength of interference sources:
 - When using multiple control cabinets, they should be connected with grounding wires of at least 16mm² cross-sectional area to achieve equipotential bonding between cabinets
 - Within a single control cabinet, equipment should be partitioned based on signal strength, and devices in different areas of the control cabinet should be equipotentially bonded
 - All communication and signal cables routed out of the control cabinet must be properly shielded

Chapter 5 Display and Operation

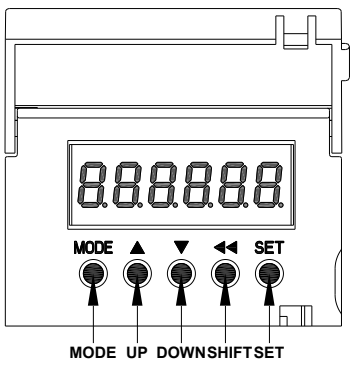
The CD200 can perform operations such as viewing and setting drive parameters, checking alarm information, monitoring operating parameters, and adjusting servo parameters through the following two methods:

- Operation panel of the drive
- Upper computer software X Servo Configurator (recommended)

5.1 Operation Panel

5.1.1 Front Panel Composition

The CD200 servo drive panel consists of a display (6-digit, 8-segment LED digital tube) and 5 buttons.

Front Panel Appearance	Symbol	Name	Function
	MODE	Mode Switch Button	Menu/Setting Toggle Exit current parameter/function operation, and return to previous menu
	▲	Up Button	Submenu toggle, increase value High/low digit display toggle
	▼	Down Button	Submenu toggle, decrease value High/low digit display toggle
	◀◀	Shift Button	Input digit (flashing) shift Enter parameter monitoring menu
	SET	OK Button	Enter next menu Confirm input Long press for fault reset

5.1.2 Panel Operation Process

The operation panel allows switching between basic modes, as well as status monitoring, parameter setting, and fault display, etc.

(1) Main Menu

When the drive is powered on, the display LEDs will light up fully for about two seconds. Then, if the drive has no abnormal alarms, it will enter the main menu; otherwise, it will display the corresponding abnormal alarm code. This menu has 4 states.

1. Normal State

The factory default display is the current speed. The content displayed in this state is defined in the F007 group. You can also enter the Jog and constant display interface using the "▲+SET" shortcut key, then scroll up to Pr7.01 to set the constant display content.

2. Drive Fault Display State. Highest priority

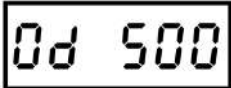
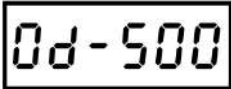
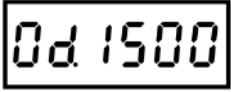
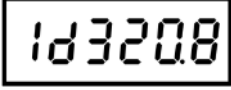
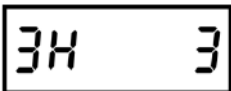
Once the drive experiences a fatal fault, when the drive transitions from a non-fault state to a fault state, it will directly jump to the fault display state, regardless of the current display state. **Faults are displayed in the format 'Er.XXXX' accompanied by 650ms flashing.** After troubleshooting the fault, long-press "SET" for about 3 seconds in the fault state to perform a control word "0x86" operation, i.e., attempt a fault reset.

3. Warning Display State

The warning state is superimposed on the normal state, displaying "AL.00XX" and flashing. It flashes once every 350ms, for a total of three flashes, then displays the normal screen for about 10 seconds. In the warning state, press "SET" to enter the warning clear interface, displaying "CLrUU". At this point, pressing "SET" will attempt to clear the warning. Warnings caused by low battery power of encoder will cause the digital tube to alternately flash 3 times with a 200ms interval. Other warnings will clear automatically.

4. Status Monitoring

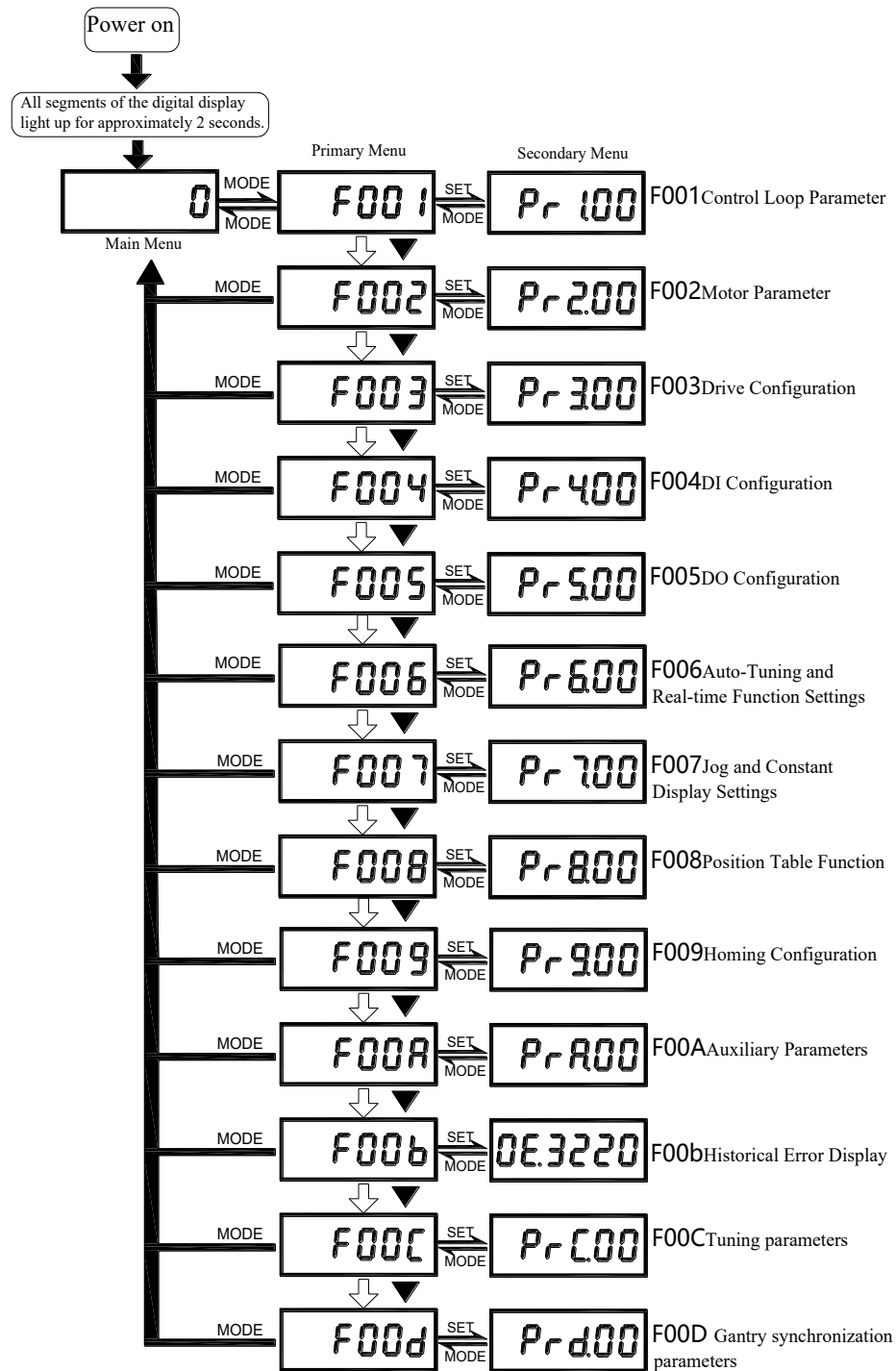
From the main menu, press the "◀◀" key to enter the status monitoring menu bar. The displayed parameters can be used to monitor the operating status of the servo drive. Specific explanations for monitoring display are as follows:

Display	Description		ReMarks
	Speed 500 r/m	The leftmost digital tube displays a number corresponding to different monitoring values, which corresponds to the constantly displayed parameter monitoring setting	Monitored parameter corresponding value 0: Speed 1: Bus voltage 2: Drive temperature 3: Input terminal status 4: Output terminal status 5: Drive load rate 6: Motor load rate Other monitoring parameters can be extended
	Speed -500 r/m		
	Speed -1500 r/m	The second digital tube from the left displays "d", indicating that the parameter is displayed in decimal; displaying "H" indicates that the parameter is displayed in hexadecimal	
	Bus voltage 320.8V		
	DI1, DI2 with input	If the displayed value is negative, a "dot" lights up directly next to the highest digit, or in the lower right corner of the second digital tube from the left if the value exceeds four digits	

(2) Primary Menu

From the main menu, press the "MODE" key to enter the primary submenu. Use "▲" or "▼" to cycle through the primary submenu. After switching to the desired primary submenu, press "SET" to enter the secondary submenu. Press "MODE" in the primary menu to return to the main menu.

- Use the "▲+SET" key to directly switch to the "Jog and Constant Display Setting Parameter Group". Note that pressing the "▲" key in the Jog interface can switch to the corresponding Jog speed settings.
- Use the "▼+SET" key to directly switch to the "Auto-tuning Parameter Group", Pr6.05 is displayed, allowing quick access to offline inertia measurement. The operation process is as follows:

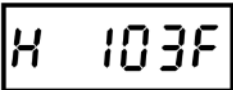
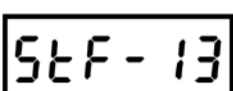
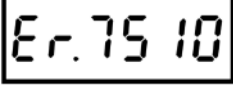
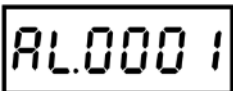


(3) Secondary Menu

The secondary submenu corresponds to the parameter list of the primary menu, displayed as "PrX.xx". Where, "X" represents the group, "xx" represents the parameter number in the current parameter group, and "." indicates the editable parameter.

When entering the secondary menu, the rightmost digital tube will flash. The flashing cursor can be moved using the "◀◀" key. If the cursor is on the units digit, pressing "▲" or "▼" keys will cycle through all members of the current group. If the cursor is on the tens digit, pressing "▲" or "▼" keys will limit the display content to the valid range of the current group and will not cycle.

5.1.3 Parameter Display Description

Display	Description	ReMarks
	The leftmost digital tube displays "H", indicating that the parameter is displayed in hexadecimal	Hexadecimal Data Display When displaying 32-bit data, by default, the lower 16 bits are displayed first, by switching with "▲" or "▼" keys
	The two leftmost digital tubes display "HH", indicating that the higher 16 bits of a 32-bit data are displayed	
	The two leftmost digital tubes display "HL", indicating that the lower 16 bits of a 32-bit data are displayed	
	32-bit data will be split into higher 5 bits and lower 5 bits The topmost segment of the leftmost digital tube for the higher 5 bits lights up	Decimal Data Display If there is a negative sign, it is next to the most significant digit. Switch between higher and lower 5 bits with "▲" or "▼" keys
	The bottommost segment of the leftmost digital tube for the lower 5 bits lights up	
	The rigidity level parameter is displayed as "5tF-xx", where "xx" is the rigidity level value. It needs to be directly modified using the "▲" or "▼" keys, and will take effect immediately.	Rigidity Level Adjustment The range is from 0-31, and the rigidity level must be adjusted step by step to prevent control issues.
	PPPP1 indicates that a position operation is about to be performed. After pressing SET to enter, PP1.01 will be displayed. The first "1" indicates that the current edited item is a position, and the "01" after it indicates that the first position is being edited, and so on	In position table editing status, PPPPx will be displayed, where "x" represents the group. For details, please refer to 9.1.8 F008 Parameter Group Settings
	Once the drive experiences a fatal fault, it will directly jump to the fault display state, regardless of the current display state. The fault display format is "Er.XXXX", where "XXXX" is the alarm code	Flash with a 650ms cycle
	The warning state is superimposed on the normal state, displaying "AL.00XX" and flashing	Flash once every 350ms, for a total of three times, then display the normal screen for about 10 seconds
Note: ● Enable state: The dot on the rightmost digital tube continuously flashes ● When not in the main menu, a timeout counter starts. If no keyboard operation occurs within 30 seconds, it automatically returns to the main menu		

5.1.4 Parameter Setting Description

When entering the parameter editing state, it always starts from the lowest digit, and the edited position will flash rapidly

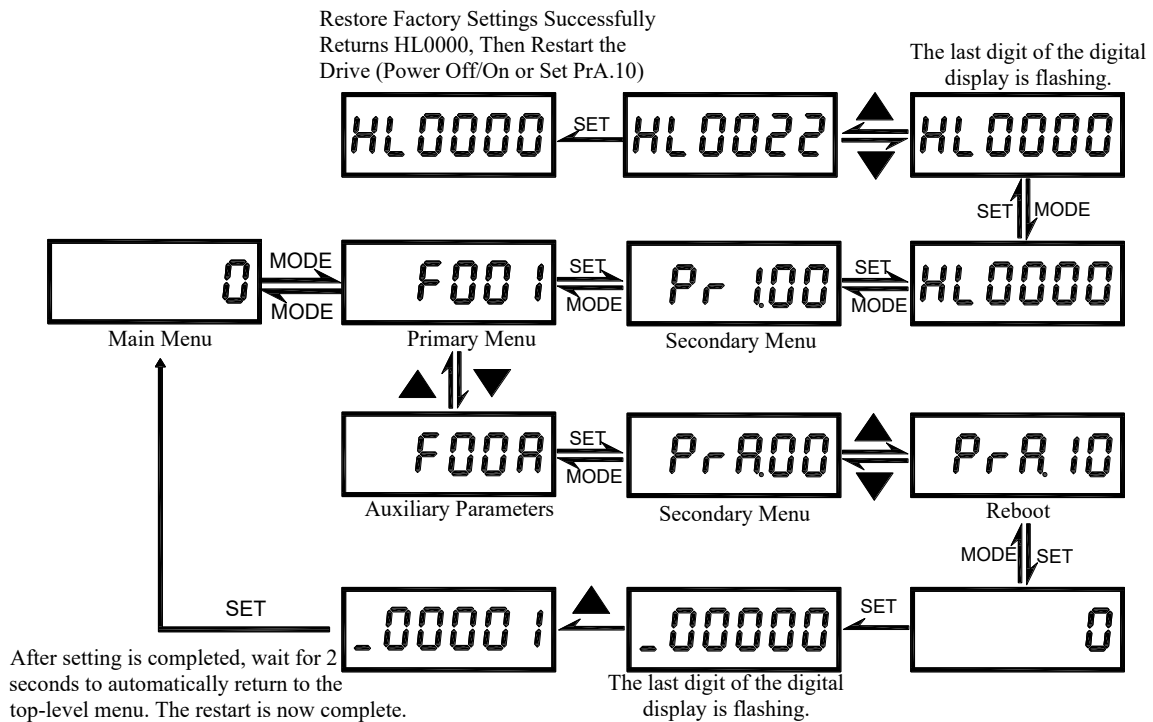
Display	Description	ReMarks	
	Signed data, occupying the right 4 digital tubes, and the leftmost one is the sign bit	8-bit Data Editing When editing the sign bit, pressing "▲" or "▼" will toggle the sign between "-" and "0"	Decimal Display
	Unsigned data, occupying the right 3 digital tubes when entering editing state		
	Signed data, fully occupying 6 digital tubes when entering editing state, and the leftmost one is the sign bit	16-bit Data Editing When editing the sign bit, pressing "▲" or "▼" will toggle the sign between "-" and "0"	
	Unsigned data, occupying 5 digital tubes when entering editing state		
	Signed data, occupying two screens when entering editing state, and splitting into lower 5 bits and higher 5 bits. The higher 5 bits will carry the sign bit on the leftmost digital tube. When editing the lower 5 bits, the leftmost digital tube will not enter the editing flashing state. When editing the higher 5 bits, the leftmost digital tube is the sign bit and also represents the higher 5 bits, and will enter the editing state.	32-bit Data Editing Long press "◀◀" key to toggle between lower five bits and higher five bits When editing the sign bit, pressing "▲" or "▼" will toggle the sign between "-" and "=" If the entire data is 0, editing the sign bit is invalid	
	Unsigned data, occupying two screens when entering editing state, and splitting into lower 5 bits and higher 5 bits, without a sign bit. Therefore, the cursor will not move to the leftmost digital tube; the leftmost digital tube only displays a horizontal line indicating higher or lower 5 bits		
	16-bit data, when entering editing state, the leftmost digital tube displays "H". Parameter settings occupy the right 4 digital tubes, and each digit is toggled between 0 and F by "▲" or "▼"	16-bit Data Editing	Hexadecimal Display
	32-bit data, occupying two screens when entering editing state, and splitting into lower 16 bits and higher 16 bits, without a sign bit. Each digit is toggled between 0 and F by "▲" or "▼"	32-bit Data Editing Long press "◀◀" key to toggle between lower 16 bits and higher 16 bits	

Note:

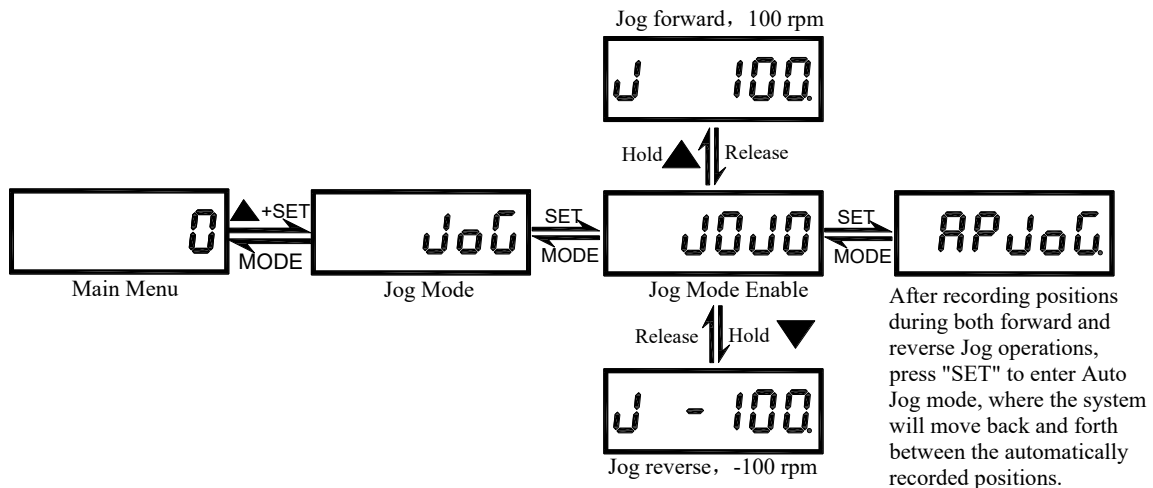
- After editing, press the "SET" key to modify the data. At this time, the data validity will be checked. If incorrect data is entered, such as data exceeding the range, it will enter the abnormal parameter operation state, and after flashing "d Err" twice, it will return to the parameter display state.
- If correct data is entered, it will directly revert to the parameter display state. If the parameter is not editable, "ro" will be displayed and flash twice, then it will revert to the parameter display state.

5.1.5 Parameter Setting Examples

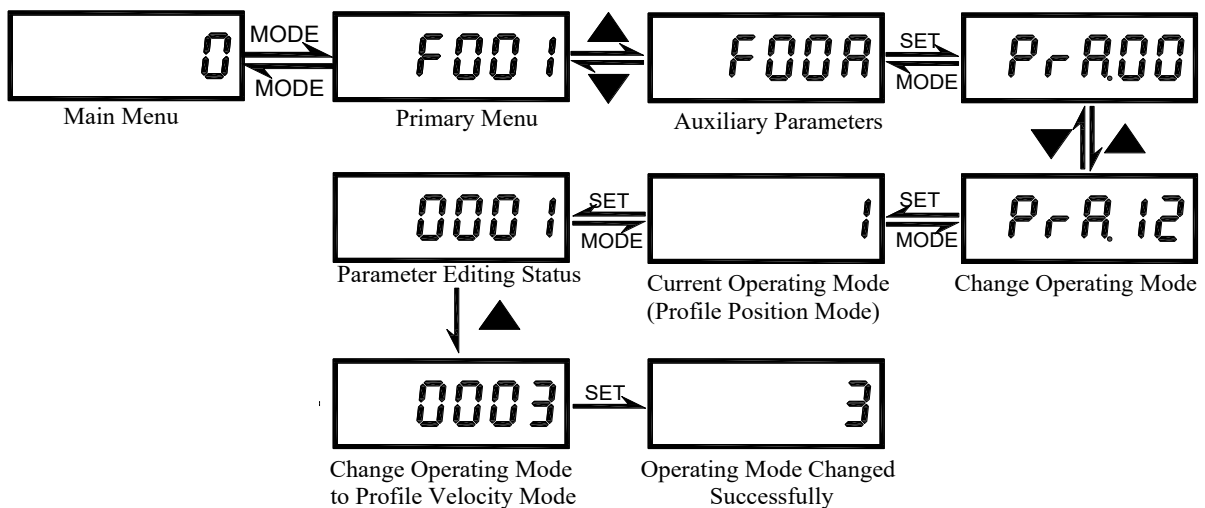
(1) Restore Factory Settings

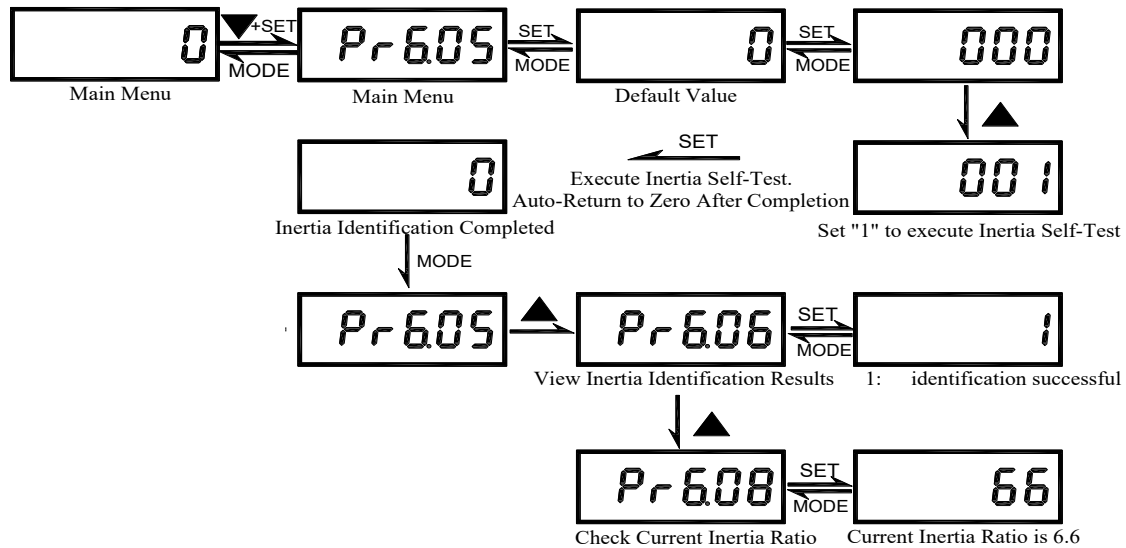


(2) Jog Mode Setting

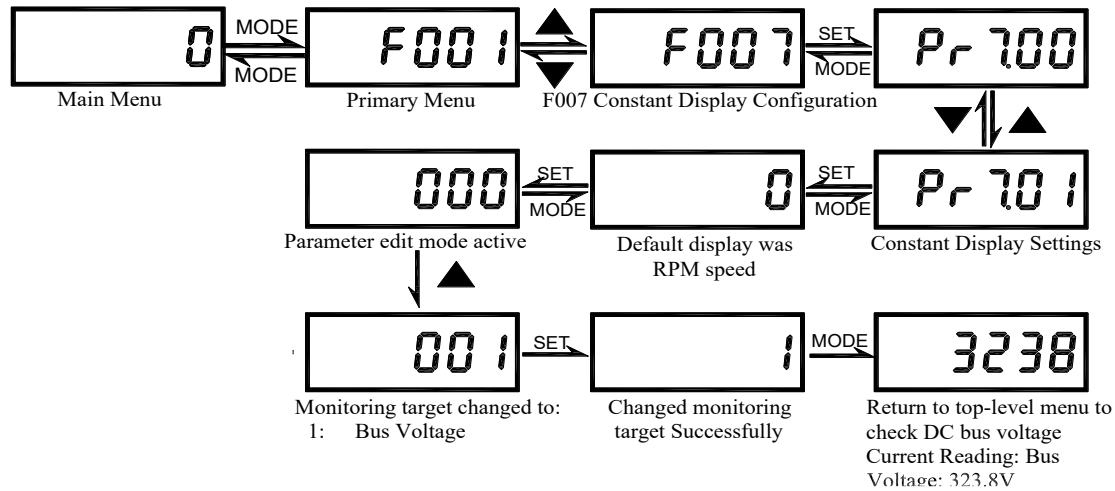


(3) Change Operating Mode



(4) Motor Parameter Auto-tuning**(5) Constant Monitoring Setting**

Constant monitoring display can be set by modifying Pr7.01. For example, to change the constant display from RPM speed to bus voltage, follow these steps:



No.	Index H	Parameter Name	Setting/Help Information	Default Value	Type	Validation
Pr7.01	310024	Constant Monitoring Object	0: Display RPM speed 1: Display bus voltage 2: Display temperature 3: DIN final input signal 4: DOUT final output status 5: Drive load rate 6: Motor load rate 7: Encoder single-turn position (H) 8: Current display Other monitoring parameters are expandable.	0	U8	Immediate

5.2 Software Overview

Communication Guide for CD200 Series Servo Drives with PC:

Please use the **factory-provided USB/RS485 isolated converter** for connection.



USB/RS485 isolated converter

The drive-side interface is **Type-C port** with **RS485 communication** support.

CN4	Pinout	Signal	Description
	A1-A5	-	-
	B1-B5	-	-
	A6/B6	RS+	RS485+
	A7/B7	RS-	RS485-
	A8-A12	-	-
	B8-B12	-	-

5.2.1 Software Installation

(1) X Servo Configurator Installation

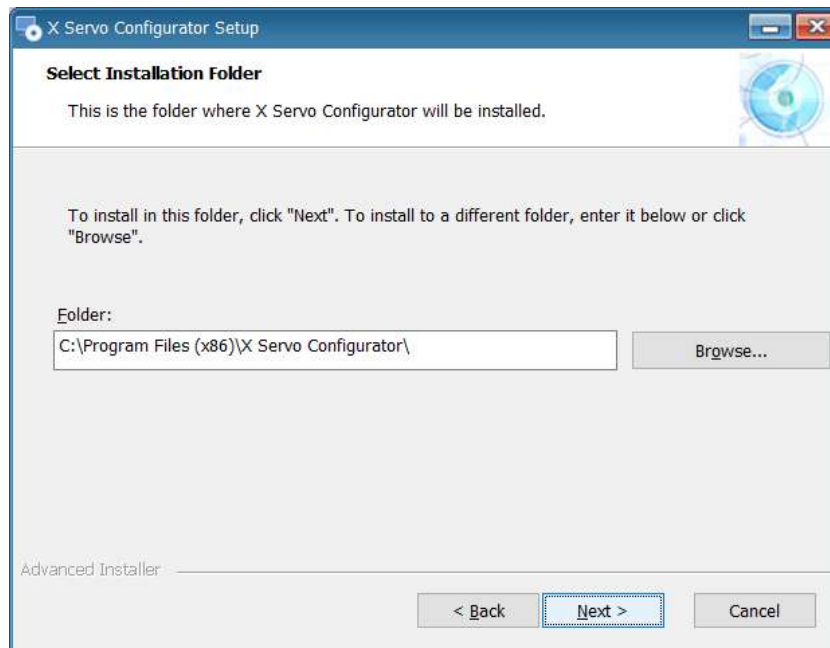
1. Run the Installer

Locate the installation file X Servo Configurator Setup.exe. Double-click the file to launch the setup wizard. Then click “Next”.

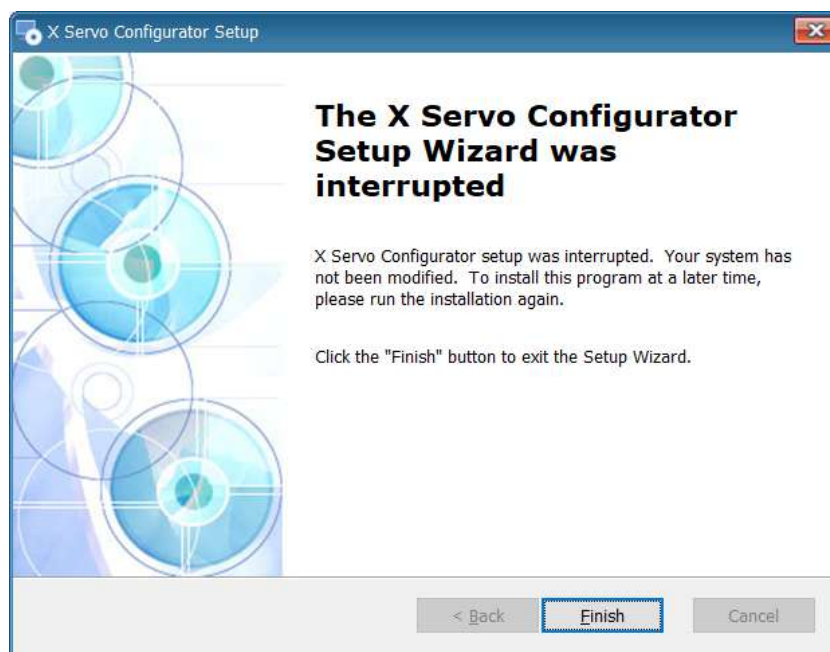


2.Installation Path Selection

Recommendation:Use the default installation path (C:\Program Files (x86)\X Servo Configurator\) for optimal compatibility and stability.



After the installation is complete, click "Finish."



A shortcut will be created on the desktop automatically.

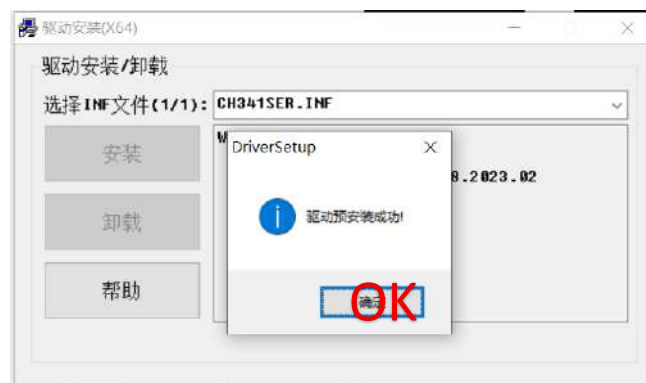


(2) Installation of USB/485 Converter Driver

Double-click the "CH341SER.exe" file to start the installation, then click "Install".



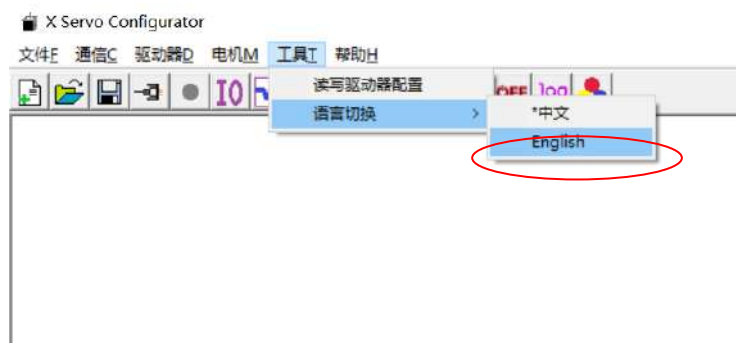
The driver installation is complete. Click "OK" to confirm.



5.2.2 Connection established.

Launch X Servo Configurator.

First of all, select the language: Navigate to the 'Tools' menu, then choose 'English' from the language dropdown options."



Locate the "Communication" option in the menu bar and click "Communication Settings" to establish the connection.

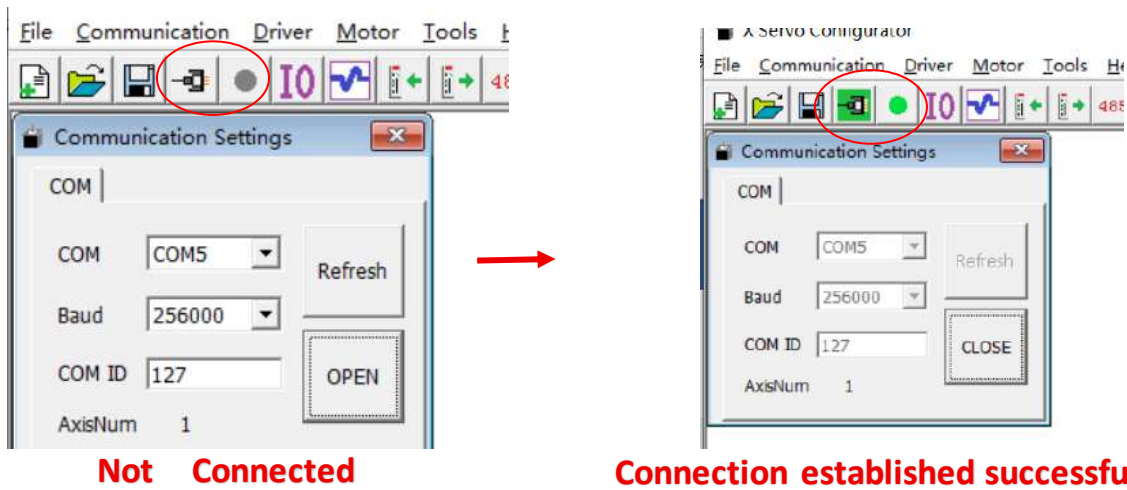
In the "COM" dropdown, select "USB Serial Port" or click "Refresh" to auto-detect available ports.

Set the appropriate **Baud Rate** (default: *256000*) and **Communication ID** (default: *127*).

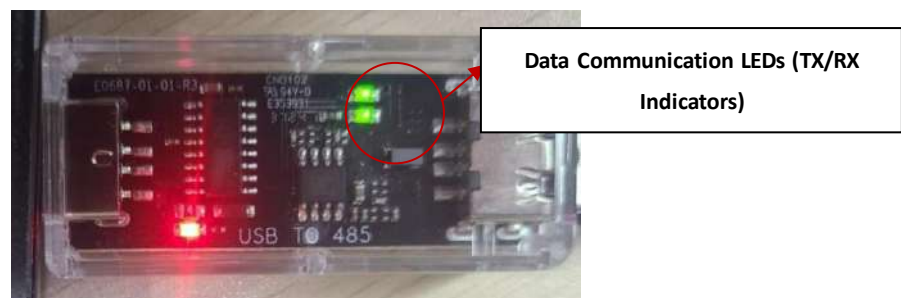
Click "Open" to initiate the connection.

When the connection is established, the 'Communication Settings' button will turn **green** (as shown below). After the servo drive

successfully communicates with the host PC, click the button  to **toggle** (disable/enable) the connection."



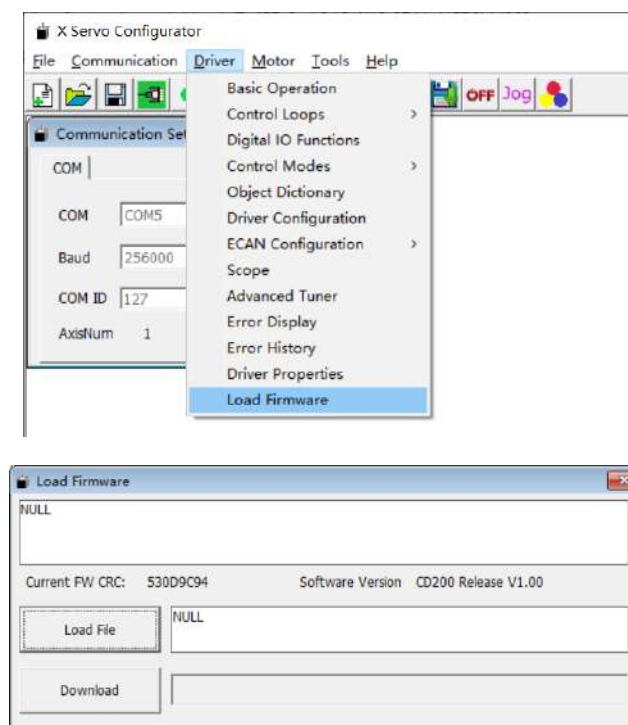
When the connection is normal, the TX/RX indicators on the USB/485 converter will illuminate.



5.2.3 Software Function Overview

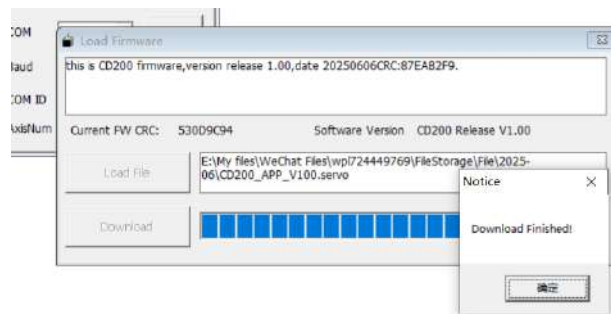
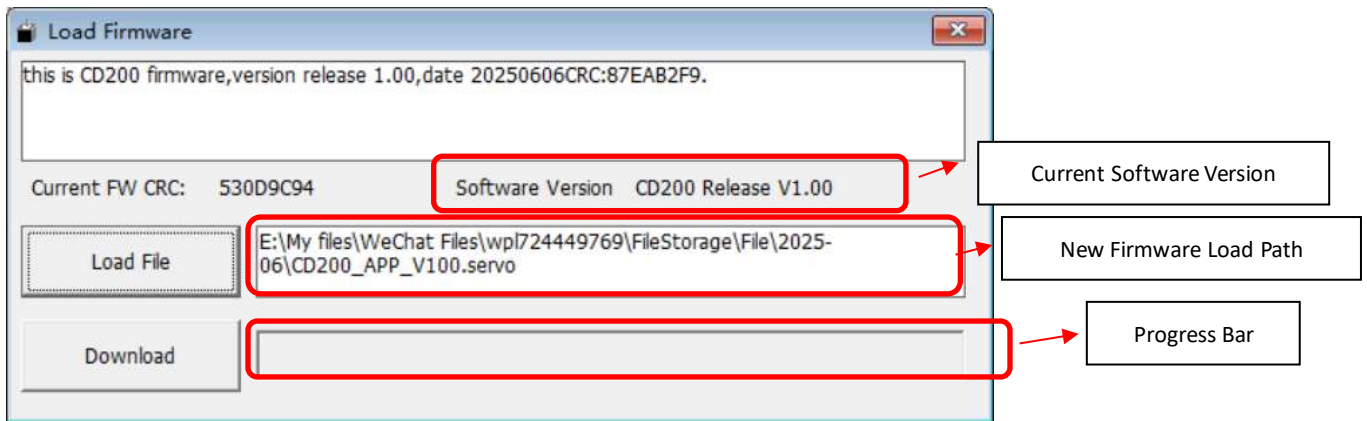
(1) Firmware Update(Drive Software Version Update)

Click **Driver** in the menu bar → **Load Firmware** to open the firmware update toolbar.

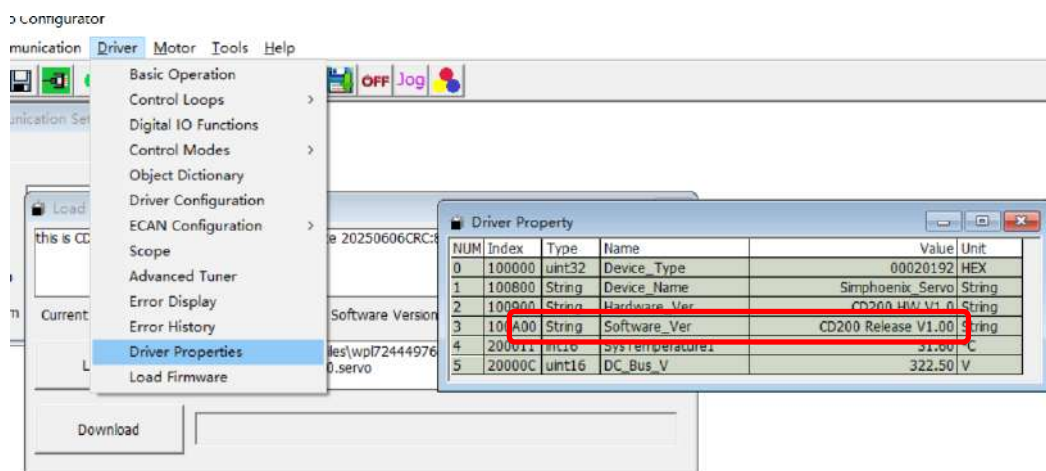


Click "**Load File**" to select the new firmware file, then click "**Download**" and wait for the process to complete.

(Before performing the firmware update, please back up the parameters. Refer to the **Batch Parameter Read/Write** operation for details.)



When the "Download Successful" window appears, go to the menu and select **Driver** → **Drive Properties** to verify whether the current version has been successfully updated.

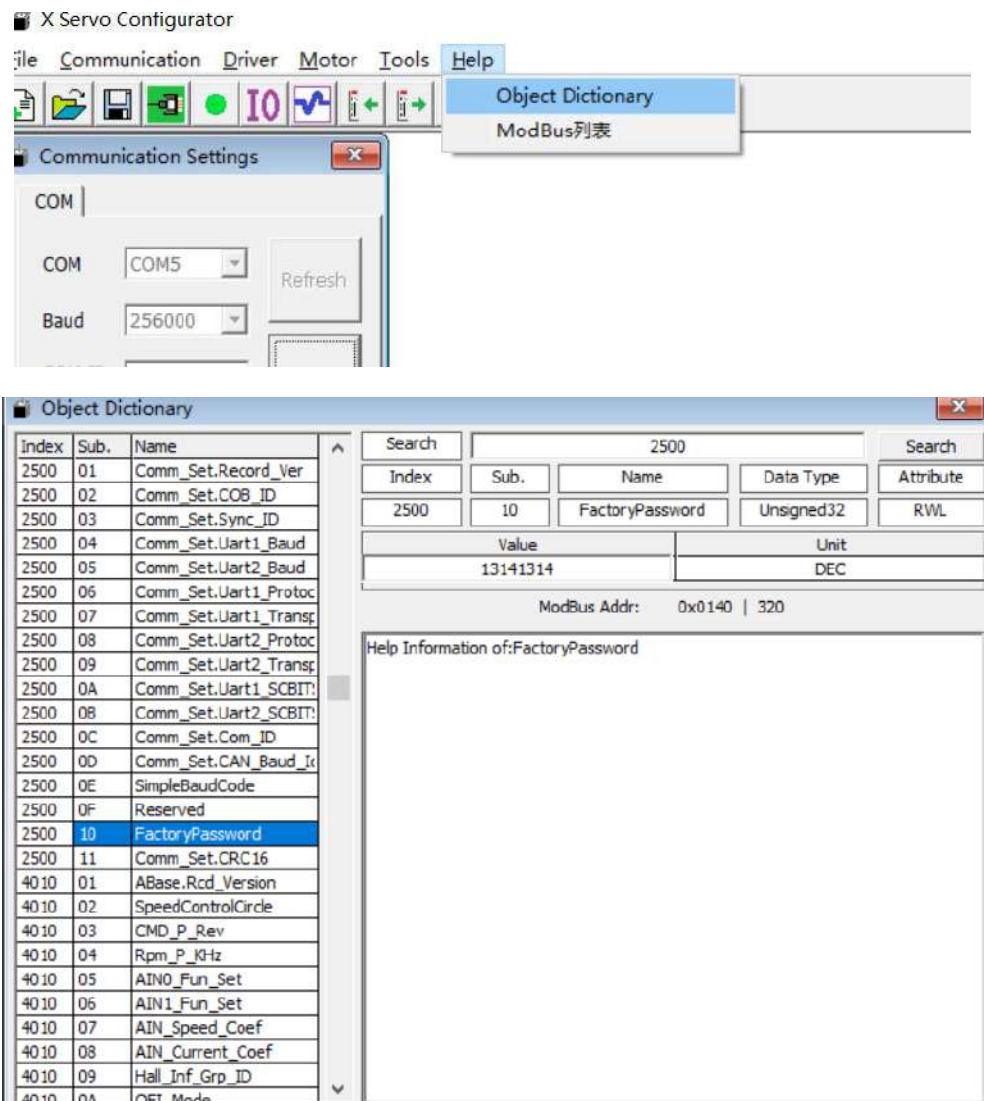


After updating the software version, an Er.6301 alarm (Memory Fault) may occur. Click 'Restart'  in the menu bar to resolve.

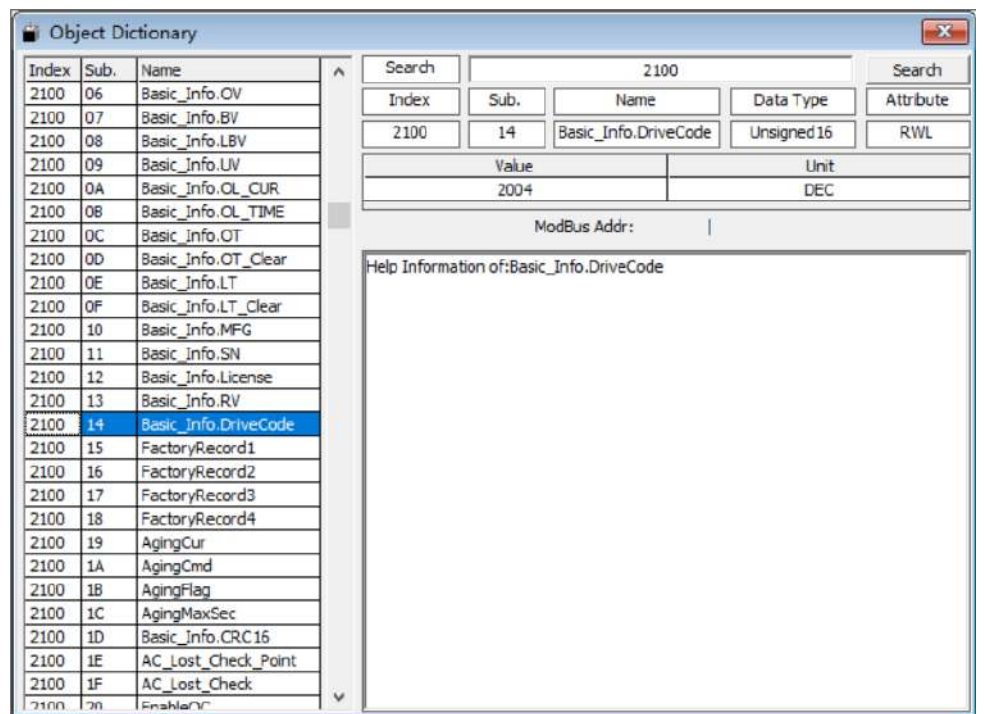
If the following status occurs after the update, a drive reset operation is required.



Search the object dictionary at address 2500h-10h for the manufacturer password, input the default code 13141314



Then, set the corresponding drive code 2100h-14h (e.g., CD200-T3R0 = 2004). Reboot the drive after configuration.



(2) Parameter configuration

1. Basic Parameter Settings

Access Parameter List:

Navigate to **Driver** → **Basic Operation** to open the list of commonly used configuration parameters.

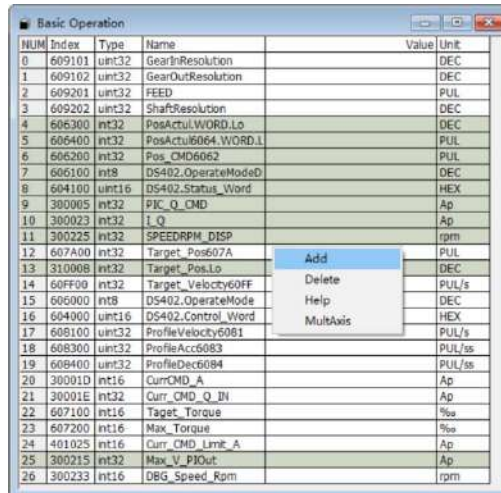
Add Parameters:

Right-click anywhere in the table → Select "Add" to open the Object Dictionary.

Choose the desired parameters and double-click to add them to the Basic Operation table.

View Parameter Help:

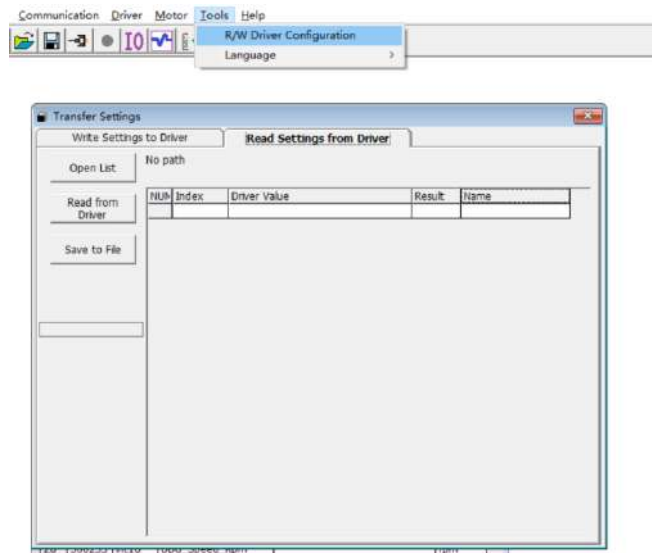
Right-click on a parameter → Select "Help" to see detailed configuration instructions.



NUM	Index	Type	Name	Value	Unit
0	609101	uint32	GearInResolution		DEC
1	609102	uint32	GearOutResolution		DEC
2	609201	uint32	FEED		PUL
3	609202	uint32	ShaftResolution		DEC
4	606300	int32	PosActuL.WORD.Lo		DEC
5	606400	int32	PosActuL.WORD.H		PUL
6	606200	int32	Pos_CMD6062		PUL
7	606100	int8	DS402_OperateMode0		DEC
8	604100	uint16	DS402_Status_Word		HEX
9	300005	int32	PIK_Q_CMD		Ap
10	300023	int32	I_Q		Ap
11	300225	int32	SPEEDRPM_DISP		rpm
12	607A00	int32	Target_Pos607A		PUL
13	310008	int32	Target_PosLo		DEC
14	60FF09	int32	Target_Velocity60FF		PUL/s
15	606000	int8	DS402_OperateMode		DEC
16	604000	uint16	DS402_Control_Word		HEX
17	608100	uint32	ProfileVelocity6081		PUL/s
18	608300	uint32	ProfileAcc6083		PUL/ss
19	608400	uint32	ProfileDec6084		PUL/ss
20	300010	int16	Curr_CMD_A		Ap
21	30001E	int32	Curr_CMD_Q_IN		Ap
22	607100	int16	Target_Torque		%a
23	607200	int16	Max_Torque		%a
24	401025	int16	Curr_CMD_Limit_A		Ap
25	300215	int32	Max_V_PIOut		Ap
26	300233	int16	DBG_Speed_Rpm		rpm

Batch Parameter Read/Write Function

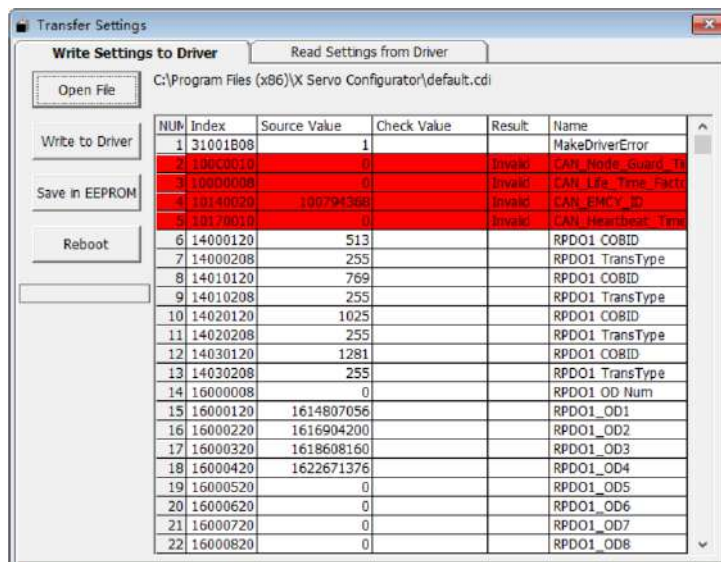
Click **Tools** → **Read/Write Drive Configuration**, or use the toolbar icon  **Read Parameters**  **Write Parameters**



Click **open list**, select the "存储列表.cdo" file (C:\Program Files (x86)\X Servo Configurator) to load the drive parameter configuration list, then click "Read Data" to obtain the drive configuration parameters.

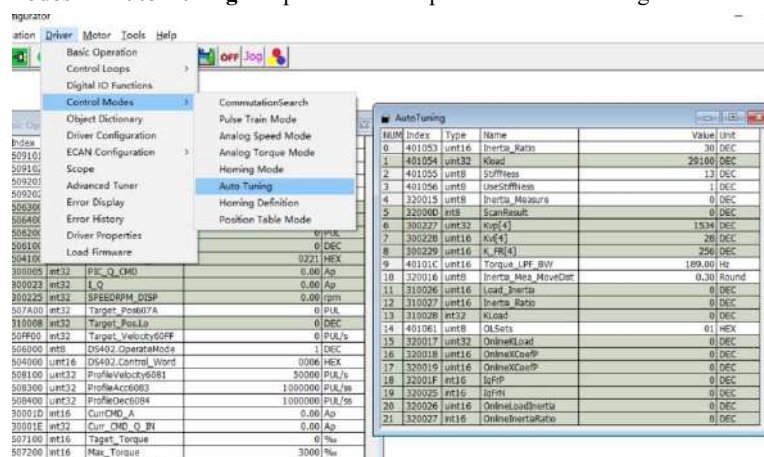
Click **Save File** to generate a ".cdi" file. When you need to configure the same parameters to other drives, select **Write Drive Configuration**, open the saved file, and click **Write to Drive**. After successful writing, click "Save Data to EEPROM", then click **Reboot** once to complete the data writing to the drive.

(Note: Parameters Marked in red are currently not supported by the drive and can be ignored.)

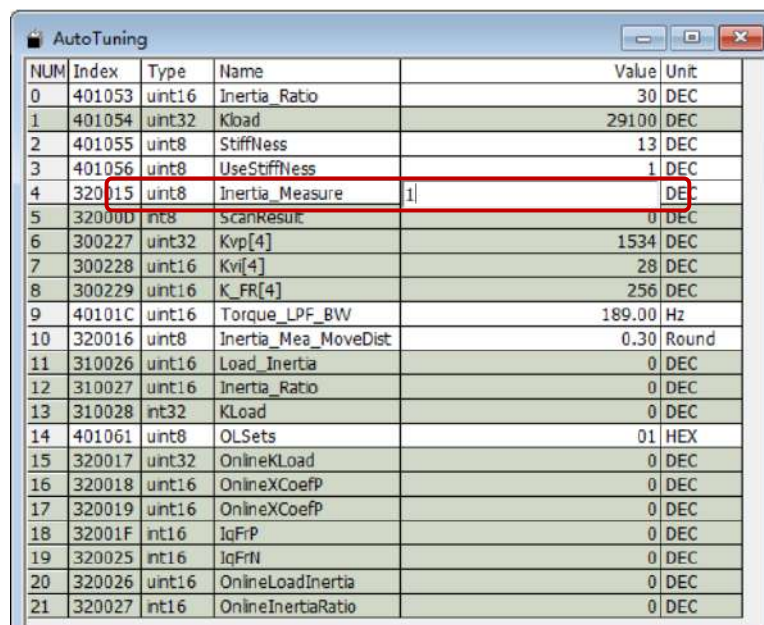


(3) Inertia Measurement

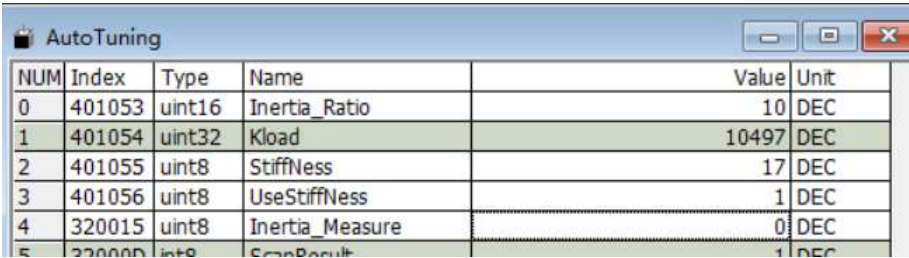
Click **Driver** → **Control Modes** → **Auto-Tuning** to open the motor parameter auto-tuning interface.



By default, the rigidity level is used. Enter '1' in the inertia_measure field and press Enter to start motor inertia identification."



After the measurement is completed, you may view parameters such as the current inertia ratio, torque inertia coefficient, and rigidity level.



NUM	Index	Type	Name	Value	Unit
0	401053	uint16	Inertia_Ratio	10	DEC
1	401054	uint32	Kload	10497	DEC
2	401055	uint8	StiffNess	17	DEC
3	401056	uint8	UseStiffNess	1	DEC
4	320015	uint8	Inertia_Measure	0	DEC
5	320000	uint8	ScanResult	1	DEC

The measurement results are displayed in **Object 3200h-0Dh**.

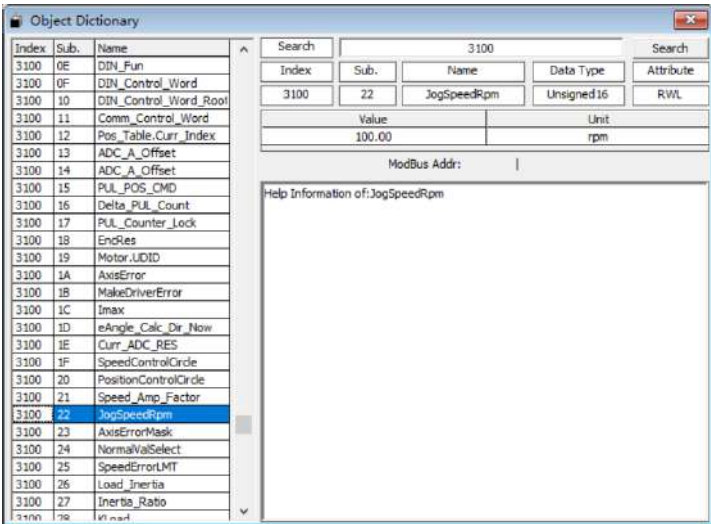
No.	Index	Parameter Name	Set/Display Value
Pr6.06	3200h-0Dh	Inertia Measurement Results	0: Test not started or in progress 1: Success (updates rigidity list) -1: Speed amplitude too small -2: Current amplitude too small -3: Inertia ratio exceeds 25:1 -4: Inertia ratio exceeds 40:1

(4) Test Run in Jog Mode

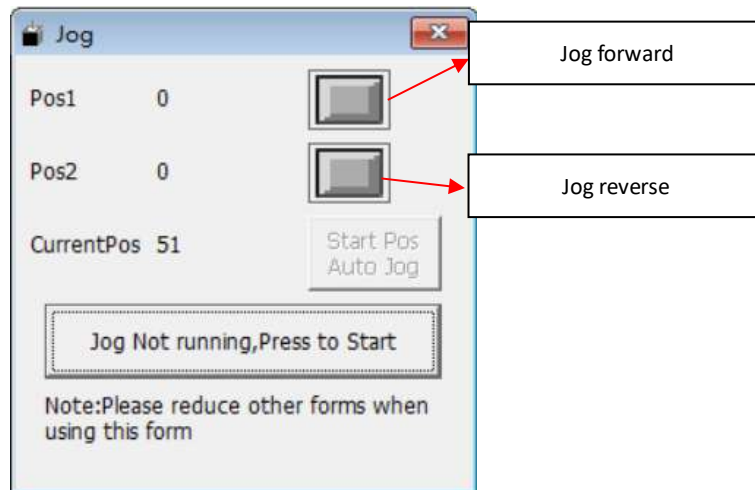
Click the menu bar button  to open the Jog mode debugging interface.



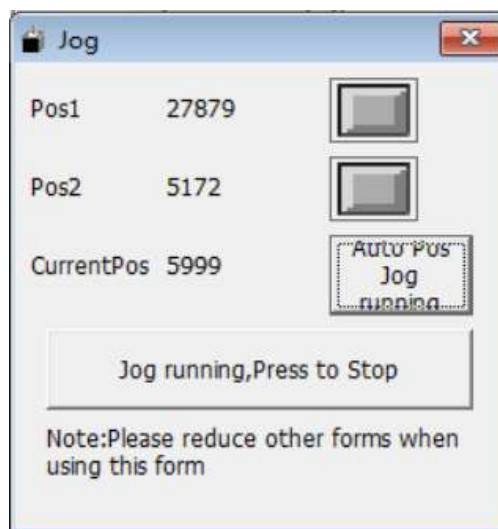
The default speed for Jog mode is set to 100 rpm. The Jog mode speed can be adjusted by modifying Object 3100h-22h.



Click "Jog not running, press to start" to enable Jog mode. Then, press the forward/reverse jog buttons to perform jog operation.



After both Position 1 and Position 2 data have been recorded, click the "Start Auto Jog" function to activate automatic reciprocating movement of the motor between the preset Position 1 and Position 2 range.



(5) Digital I/O Configuration

Click on **Driver** → **Digital IO functions** or click the shortcut toolbar button  to open the Digital IO Settings interface.

T Mode: This mode determines the form in which the input signal triggers. Options include **Low** (low-level trigger), **High** (high-level trigger), **Rise** (rising-edge trigger), and **Fall** (falling-edge trigger).

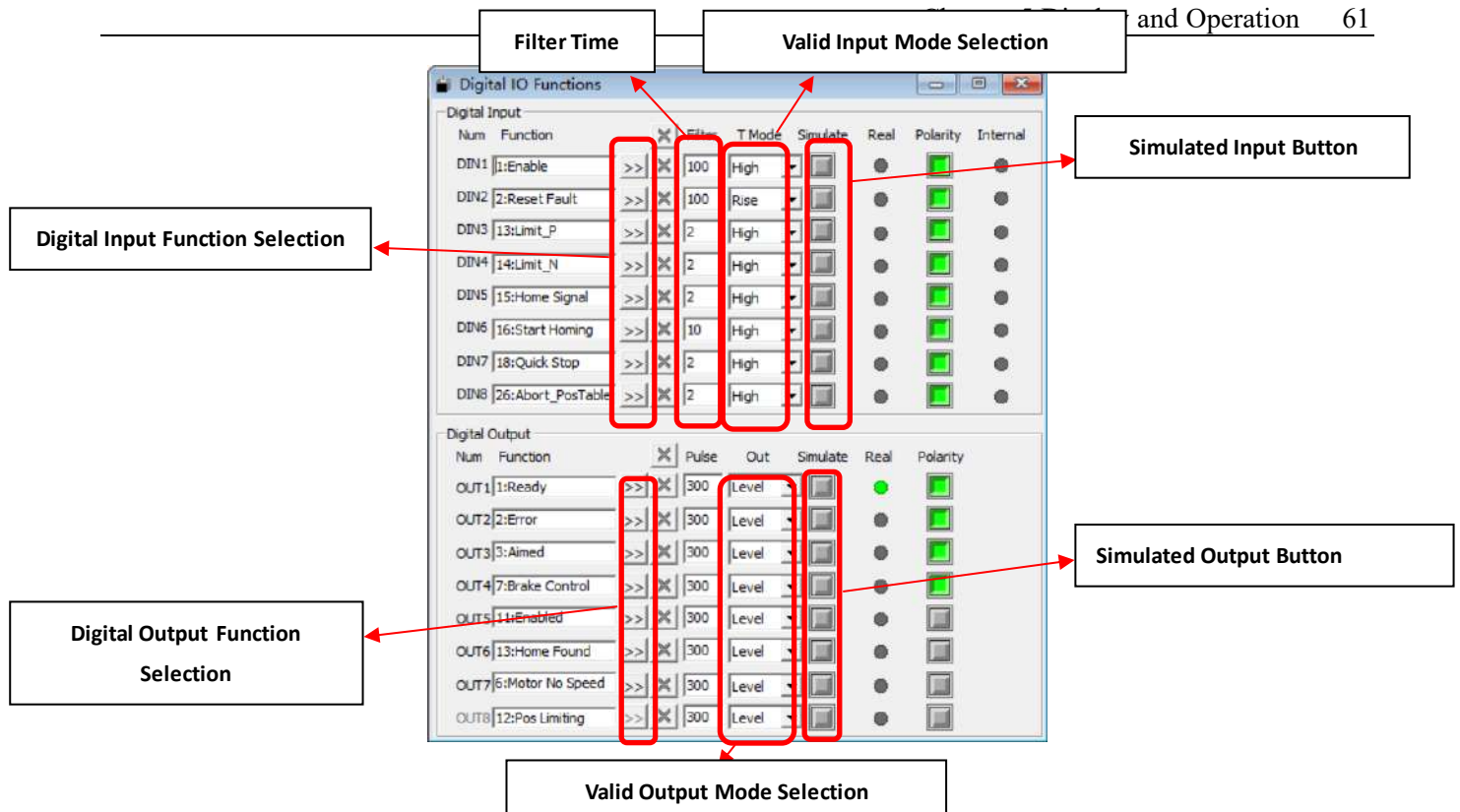
Actual Input: Displays the real-time input status.

Level: Controls the switching of active-level logic.

Valid Input: The result of the combined action of simulation, actual input, and polarity.

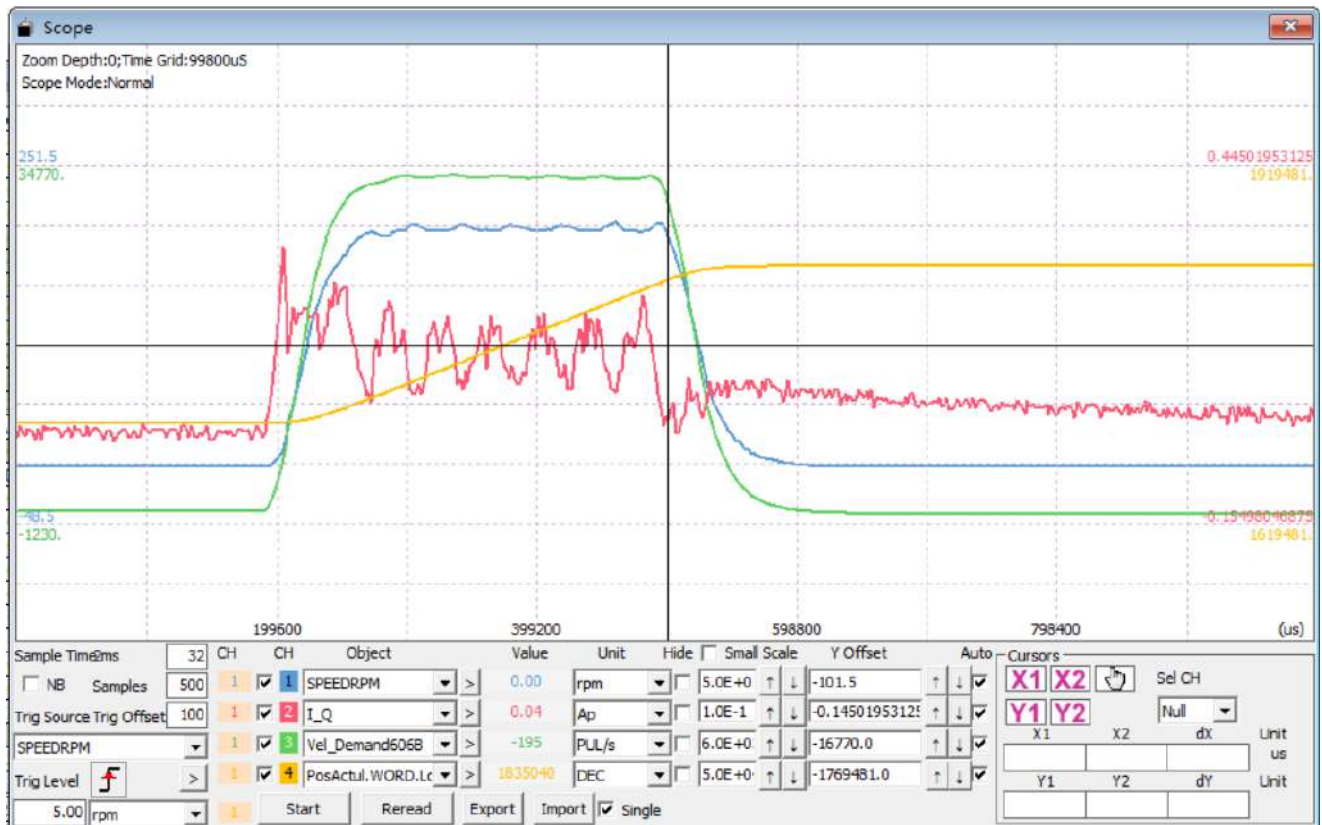
● Indicates that the current state is **inactive** (logical state 0).

● Indicates that the current state is **active** (logical state 1).



(6) Oscilloscope Function

Click Driver → Scope or click the **shortcut toolbar button**  to open the oscilloscope interface.



Sampling Configuration

Parameter	Description	Example Value
Sampling Period	Minimum sampling interval: 62.5μs (configurable in integer multiples)	62.5μs × 4 = 250μs
Sample Count	Total data points captured per trigger event	1024 points
Trigger Position	Number of pre-trigger samples collected before trigger activation	300 points

Trigger Configuration

Trigger Source: You can select common trigger sources from the dropdown menu, or click the ">" button to choose from the object dictionary. Like RPM Speed

Trigger Level: **Trigger Comparison Value Setting**, like 200 RPM

Trigger Edge:



Rising Edge (Trigger when value exceeding 200 RPM)



Falling Edge (Trigger when value **falls below** 200 RPM)

Note: Data acquisition initiates when trigger conditions are satisfied.

Multi-Channel Setup

- **Maximum Channels:** 4 simultaneous inputs
- **The total combined data width of all acquired signals must not exceed 64 bits, with a maximum of 32-bit data type per individual channel.**

You can select common trigger sources from the dropdown menu, or click the ">" button to choose from the object dictionary.

Acquisition Modes

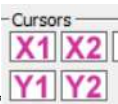
Mode	Functionality
☒ Single	One-time capture per trigger
☐ Single	Repeated capture while conditions persist

Display Controls

Zoom Functions:

Zoom Out: Right-click + drag upper-left

Zoom In: Right-click + drag lower-right



Cursor: Select the desired cursor by clicking .

Move the cursor: Move the mouse close to the cursor—when it turns red, hold down the left mouse button to drag it. The cursor measurement data is displayed as follows.

Cursors				
X1	X2		Sel CH	
Y1	Y2		CH1	
X1	X2	dX	Unit	
258000	748000	490000	us	
Y1	Y2	dY	Unit	
226.48	-23.48	249.960	rpm	

Waveform Movement

Click and drag  to horizontally scroll the zoomed-in waveform.

Export/Import

- Export:** Save sampled data as a **.scope** file.
- Import:** Load a **.scope** file to display its waveform.

Reload Data

Read the most recently collected data from the drive and regenerate the waveform for display.

Auto Mode



The oscilloscope automatically adjusts the scale and offset for optimal waveform display.

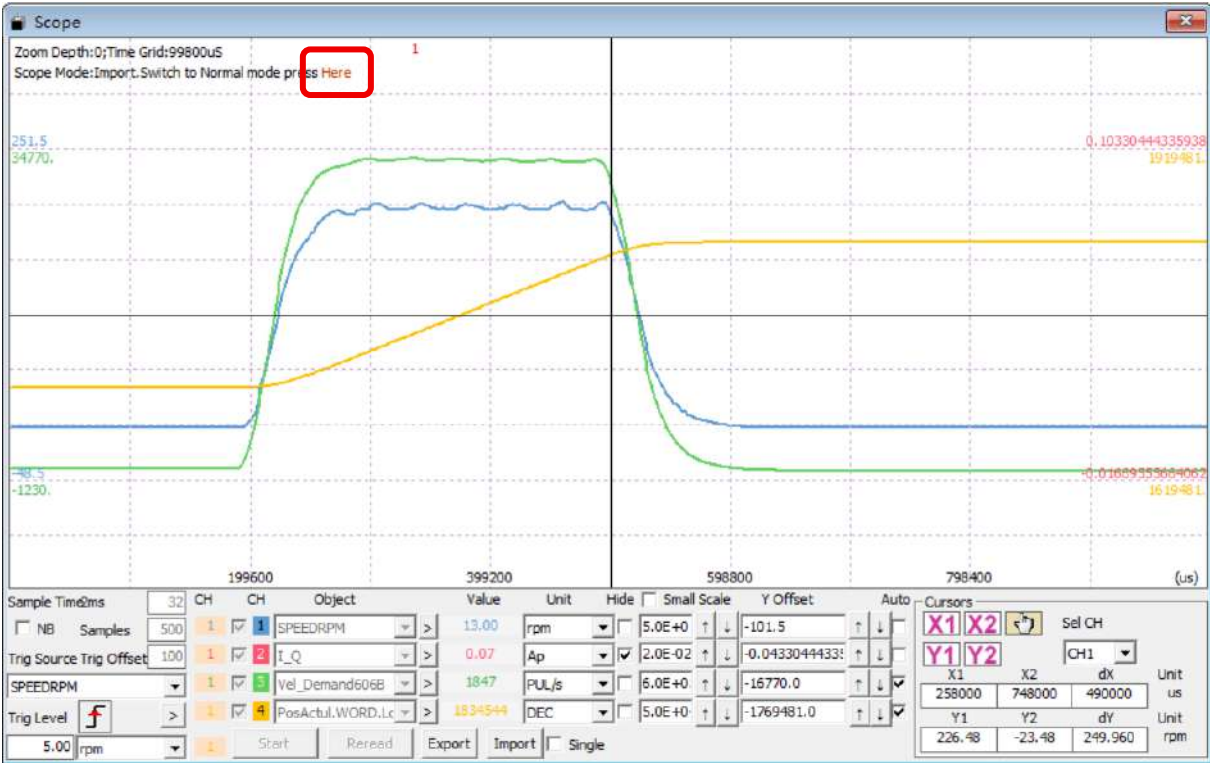


The waveform is displayed with fine scaling and follows the manually set Y-axis offset range.

Oscilloscope Mode

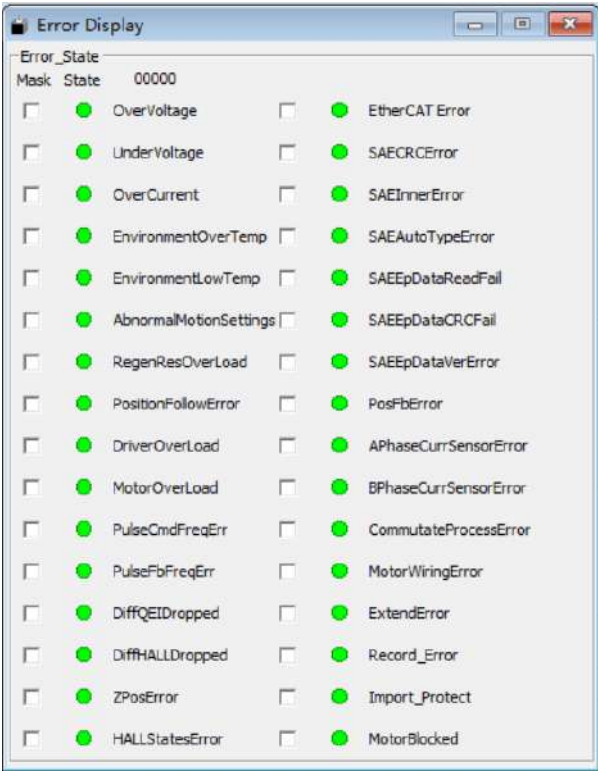
Import Mode (left diagram):

The oscilloscope is in read-only mode (displaying an imported waveform).Functional buttons are disabled.Click **"Here"** to return to **Normal Mode**.

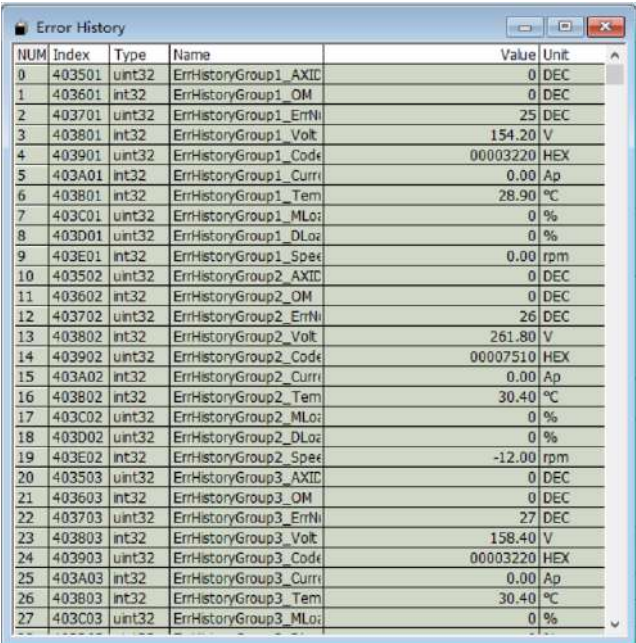


(7) Fault Display

Click **Driver** → **Error Display**, or select the Error Display button  on the Quick Access Toolbar to open the Error Display interface (as shown below). The displayed error messages can be used to diagnose and resolve faults.



Click **Driver**→ **Error history** to view the most recent 8 fault records. A higher recorded fault count value indicates a more recent occurrence time.



NUM	Index	Type	Name	Value	Unit
0	403501	uint32	ErrHistoryGroup1_AXIC	0	DEC
1	403601	int32	ErrHistoryGroup1_OM	0	DEC
2	403701	uint32	ErrHistoryGroup1_ErrNi	25	DEC
3	403801	int32	ErrHistoryGroup1_Volt	154.20	V
4	403901	uint32	ErrHistoryGroup1_Code	00003220	HEX
5	403A01	int32	ErrHistoryGroup1_Curri	0.00	Ap
6	403801	int32	ErrHistoryGroup1_Tem	28.90	°C
7	403C01	uint32	ErrHistoryGroup1_MLoi	0	%
8	403D01	uint32	ErrHistoryGroup1_DLoi	0	%
9	403E01	int32	ErrHistoryGroup1_Spee	0.00	rpm
10	403502	uint32	ErrHistoryGroup2_AXIC	0	DEC
11	403602	int32	ErrHistoryGroup2_OM	0	DEC
12	403702	int32	ErrHistoryGroup2_ErrNi	26	DEC
13	403802	int32	ErrHistoryGroup2_Volt	261.80	V
14	403902	uint32	ErrHistoryGroup2_Code	00007510	HEX
15	403A02	int32	ErrHistoryGroup2_Curri	0.00	Ap
16	403802	int32	ErrHistoryGroup2_Tem	30.40	°C
17	403C02	uint32	ErrHistoryGroup2_MLoi	0	%
18	403D02	uint32	ErrHistoryGroup2_DLoi	0	%
19	403E02	int32	ErrHistoryGroup2_Spee	-12.00	rpm
20	403503	uint32	ErrHistoryGroup3_AXIC	0	DEC
21	403603	int32	ErrHistoryGroup3_OM	0	DEC
22	403703	uint32	ErrHistoryGroup3_ErrNi	27	DEC
23	403803	int32	ErrHistoryGroup3_Volt	158.40	V
24	403903	uint32	ErrHistoryGroup3_Code	00003220	HEX
25	403A03	int32	ErrHistoryGroup3_Curri	0.00	Ap
26	403803	int32	ErrHistoryGroup3_Tem	30.40	°C
27	403C03	uint32	ErrHistoryGroup3_MLoi	0	%

(8) Homing Mode Configuration

Click **Driver** → **Control Modes** → **Homing Definition**, configure encoder index signals, home switches, and limit switches as needed for homing triggers and reference signals.

For detailed application methods, see **Section 8.3.7: Home Mode (HM)** in the manual.

Related Parameters:

No.	Index	Parameter Name	Set/Display Value	Unit	Type	Validation
Pr9.01	607C00	Homing offset	This parameter allows the alignment of the final origin position with the physical signal trigger point by applying a programmable displacement value	PUL	S32	Immediate
Pr9.03	609901	Homing Retract Speed	A high-speed motion phase during homing operations, used for coarse origin search.	PUL/s	U32	Immediate
Pr9.04	609902	Homing Approach Speed	A low-speed motion phase for precise origin positioning	PUL/s	U32	Immediate
Pr9.05	609903	Automatic Homing	0: Disabled 1: On Power-Up 2: On Origin Loss		U8	Reboot
Pr9.06	609A00	Homing Acceleration	The acceleration/deceleration rate applied during homing search motions	rpss	U32	Immediate
Pr9.07	609904	Homing Current Limit	The maximum motor current allowed during mechanical limit-based origin searches	0.1A	U16	Immediate
Pr9.08	609905	Homing Offset Mode	0: Move to the offset position, then zero the position 1: Do not move; directly set the current position as negative offset		U8	Immediate
Pr9.10	609907	Homing Offset Positioning Speed	The controlled velocity used when moving to the origin offset position during homing	rpm	U16	Immediate

(9) Position Table Mode

click **Driver** → **Control Modes** → **Position Table Mode** to open the Position Table Mode interface.

(Note: Position Stream Mode and I/O Position Mode require Profile Position Mode as the base, while I/O Velocity Mode requires Profile Velocity Mode.)

CTL Reg of index:0															
Bit0-4:Next Index		Bit5	Bit6	Bit7	Bit8:Next/Stop		Bit9:Cond 0		Bit10:Cond 1		Bit11:And/Or	Bit12-13:MODE		Bit14-15:StartCond.	
0		0	0	0	0		0		0		0	0		0	
Idx	MODE	StartCond.	Pos inc	Speed rpm	Delay ms	Acc idx	Dec idx	CTL Reg	Loops	Rest	Acc rps/s	Dec rps/s			
0	A	Ignore	0	0	0	0	0	0	0	0	0	0			
1	A	Ignore	0	0	0	0	0	0	0	0	0	0			
2	A	Ignore	0	0	0	0	0	0	0	0	0	0			
3	A	Ignore	0	0	0	0	0	0	0	0	0	0			
4	A	Ignore	0	0	0	0	0	0	0	0	0	0			
5	A	Ignore	0	0	0	0	0	0	0	0	0	0			
6	A	Ignore	0	0	0	0	0	0	0	0	0	0			
7	A	Ignore	0	0	0	0	0	0	0	0	0	0			
8	A	Ignore	0	0	0	0	0	0	0	0	0	0			
9	A	Ignore	0	0	0	0	0	0	0	0	0	0			
10	A	Ignore	0	0	0	0	0	0	0	0	0	0			
11	A	Ignore	0	0	0	0	0	0	0	0	0	0			
12	A	Ignore	0	0	0	0	0	0	0	0	0	0			
13	A	Ignore	0	0	0	0	0	0	0	0	0	0			
14	A	Ignore	0	0	0	0	0	0	0	0	0	0			
15	A	Ignore	0	0	0	0	0	0	0	0	0	0			
16	A	Ignore	0	0	0	0	0	0	0	0	0	0			
17	A	Ignore	0	0	0	0	0	0	0	0	0	0			
18	A	Ignore	0	0	0	0	0	0	0	0	0	0			
19	A	Ignore	0	0	0	0	0	0	0	0	0	0			
20	A	Ignore	0	0	0	0	0	0	0	0	0	0			
21	A	Ignore	0	0	0	0	0	0	0	0	0	0			
22	A	Ignore	0	0	0	0	0	0	0	0	0	0			
23	A	Ignore	0	0	0	0	0	0	0	0	0	0			
24	A	Ignore	0	0	0	0	0	0	0	0	0	0			
25	A	Ignore	0	0	0	0	0	0	0	0	0	0			
26	A	Ignore	0	0	0	0	0	0	0	0	0	0			
27	A	Ignore	0	0	0	0	0	0	0	0	0	0			
28	A	Ignore	0	0	0	0	0	0	0	0	0	0			
29	A	Ignore	0	0	0	0	0	0	0	0	0	0			
30	A	Ignore	0	0	0	0	0	0	0	0	0	0			
31	A	Ignore	0	0	0	0	0	0	0	0	0	0			

Current Index: 0

Run Mode: 0:位置流移

TrigMode: 0:触发信号

TrigDelay: 50

AutoRun: 加不启用

Read Table: xxx

Write Table: xxx

Import Table

Export Table

Clear Table

Position Table Mode is a function designed for position control, allowing the execution of up to 32 independent tasks in this mode.

Each task contains multiple key parameter settings, such as target position, operating speed, acceleration, and deceleration. Additionally, tasks include configurations such as whether to proceed to the next task upon completion, the index value of the next task, the trigger condition for executing the next task, and the maximum number of task cycles.

To use this function, users must first configure the "Enable Position Table" signal and the index signal at the input ports to ensure the Position Table Mode can start and execute properly. Beyond this, other position table-related functions in the system can be flexibly defined based on specific application requirements to meet different control and task demands.

This highly configurable approach ensures tasks are executed efficiently according to predefined parameters and conditions, while also providing extensive customization capabilities for complex motion control applications.

Control Register Settings

Bits 0-4: Next Task – Defines which task will execute after the current task completes.

Bits 5-7: Reserved, undefined.

Bit 8: Continue/Stop

1: Proceed to the next task (if **Bits 9-11** are all 1 and the cycle count is not reached).

0: Stop after the current task completes.

Bit 9: Condition 0 Control

1: Condition 0 is active.

0: Condition 0 is inactive.

Bit 10: Condition 1 Control

- 1: Condition 1 is active.
- 0: Condition 1 is inactive.

Bit 11: Condition 0 & 1 Logic

- 1: The next task index proceeds **only if both Condition 0 and Condition 1 are met**.
- 0: The next task index proceeds **if either Condition 0 or Condition 1 is met**.

Special Cases:

If **either Condition 0 or 1 is inactive**, Bit 11 is ignored.

If **both are inactive**, the position table runs **without condition restrictions**.

If **only Condition 0 is active**, the task loops while Condition 0 (I/O trigger) remains active.

If **only Condition 1 is active**, the task runs **only once** when Condition 1 is met.

Bits 12-13: Positioning Mode

- 0:(Mode A): **Absolute Position** – The set value is the target absolute position.
- 1:(ModeRN): **Relative to Target** – The set value is the offset from the current **target** position.
- 2:(ModeRA): **Relative to Actual** – The set value is the offset from the current **actual** position.

Bits 14-15: Task Trigger Behavior

When the "Enable Position Table" signal is triggered:

If **no task is running**: Bits 14-15 are ignored.

If **a task is running**:

- 0: **Ignore new task**, continue current task.
- 1: **Execute new task immediately after current task completes**.
- 2: **Interrupt current task and execute new task immediately**.

Position Table Parameters

Index (Idx): Task index range (0–31).

Position (inc): Target position (incremental/absolute based on mode).

Speed (rpm): Velocity to reach the target position.

Delay (ms): Time delay before jumping to the next task (controls task transition timing).

Acc Index & Dec Index: Acceleration/deceleration profile indices (supports 8 predefined profiles, configurable in the interface).

	Acc rps/s	Dec rps/s
0	10	10
1	20	20
2	30	30
3	40	40
4	50	50
5	60	60
6	70	70
7	80	80

Loop: The loop function defines cycle limits for tasks.

- 0: indicates infinite looping.
- 1: ≥ 1 means the position table will stop when either the current task's cycle count or the next index's cycle count reaches the set value.

Remaining Cycles:

When the loop is set to 0, the remaining execution count is invalid.

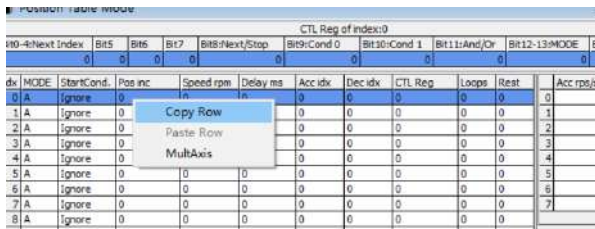
When the position table starts, it loads the preset cycle count. Each task execution decrements the remaining count by 1. Execution stops when the count reaches 0.

Copy/Paste:

Position table task information can be copied to another row.

Right-click the source task row and select "**Copy Row**".

Select the target row, then click "**Paste Row**".



The screenshot shows the 'Position Table Mode' window. At the top, there are several status bars: 'CTL Reg of Index:0', 'H0-4Next Index', 'Bits', 'Bit7', 'Bits/Next/Stop', 'Bit9:Cond 0', 'Bit10:Cond 1', 'Bit11:And/Or', and 'Bit12-13:MODE'. Below these is a table with columns: 'dx', 'MODE', 'StartCond', 'Pos inc', 'Speed rpm', 'Delay ms', 'Acc idx', 'Dec idx', 'CTL Reg', 'Loops', 'Rest', and 'Acc rps/s'. The table contains 8 rows of data. A right-click context menu is open over the first row, showing options: 'Copy Row', 'Paste Row', and 'MultiAxis'.

dx	MODE	StartCond	Pos inc	Speed rpm	Delay ms	Acc idx	Dec idx	CTL Reg	Loops	Rest	Acc rps/s
0	A	Ignore	0	0	0	0	0	0	0	0	0
1	A	Ignore	0	0	0	0	0	0	0	0	1
2	A	Ignore	0	0	0	0	0	0	0	0	2
3	A	Ignore	0	0	0	0	0	0	0	0	3
4	A	Ignore	0	0	0	0	0	0	0	0	4
5	A	Ignore	0	0	0	0	0	0	0	0	5
6	A	Ignore	0	0	0	0	0	0	0	0	6
7	A	Ignore	0	0	0	0	0	0	0	0	7
8	A	Ignore	0	0	0	0	0	0	0	0	

Current Index: Indicates the currently executing indexed task.

RUN Mode:

0: Position Stream Mode: Operates according to the conditions and parameters set in the position table mode.

1: I/O Position Mode:

Only relates to Position (inc), Speed (rpm), Delay (ms), and acceleration/deceleration.

Position (inc) determines the movement direction.

Speed (rpm) sets the operating speed.

Delay (ms) controls the duration of operation.

Note: I/O Position Mode only takes effect when the operating mode is set to 1 (PP) .

2:I/O Speed Mode:

Only relates to Speed (rpm), Delay (ms), and acceleration/deceleration.

Speed (rpm) determines both the operating speed and direction.

Delay (ms) sets the duration of operation.

Note: I/O Speed Mode only functions when the operating mode is 3.(PV)

Trigger Mode:

(Note: Trigger mode can only be modified when the operation mode is set to 1: I/O Position or 2: I/O Speed)

0:Signal Trigger: The position table operation is triggered by I/O input signals

1:Time Trigger: The operation is triggered after a specified delay

(Note: When using time trigger, set "Delay (ms)" to 0)

Trigger Delay:Only Validation when trigger mode is set to Time Trigger

Auto Run:

0: Auto run disabled

1: Automatically execute position table on power-up

2: Automatically execute position table after fault clearance

3: Automatically execute homing then run position table after power-up

Configuration Operations:

Read Table: Load position table data from drive to the configuration window

Write Table: Save position table data from window to the drive

Import Table: Import existing .pft file to the position table window

Export Table: Export current position table data as .pft file

Clear Table: Erase all data in the position table window

5.3 Quick Start for CD200 Servo Drive

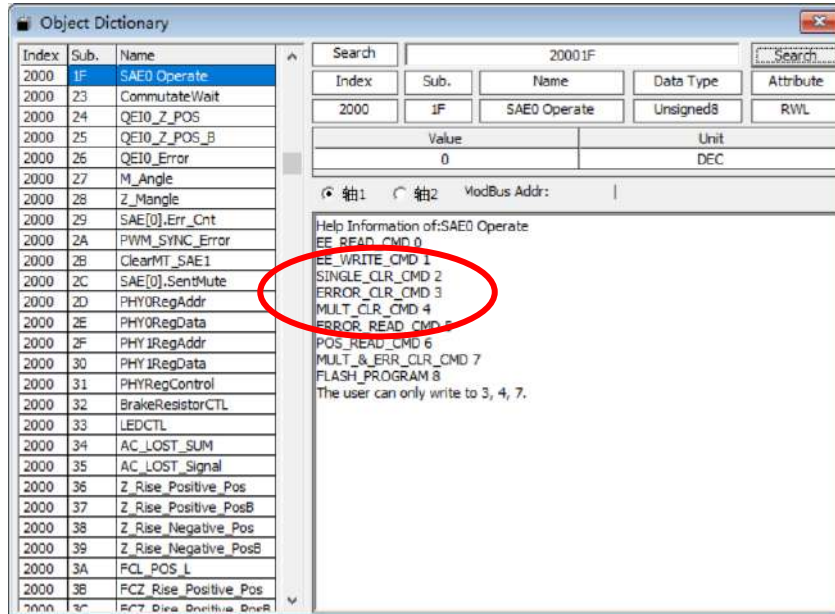
5.3.1 Drive Initialization Operation

If error "Er.FF02: multi-turn encoder data anomaly" appears after connection, you need to perform an encoder reset operation.

Reset method:

Open the object dictionary, search for SAE0 operation or 2000 1F, then enter the corresponding value in the "Value" field and press Enter to confirm.

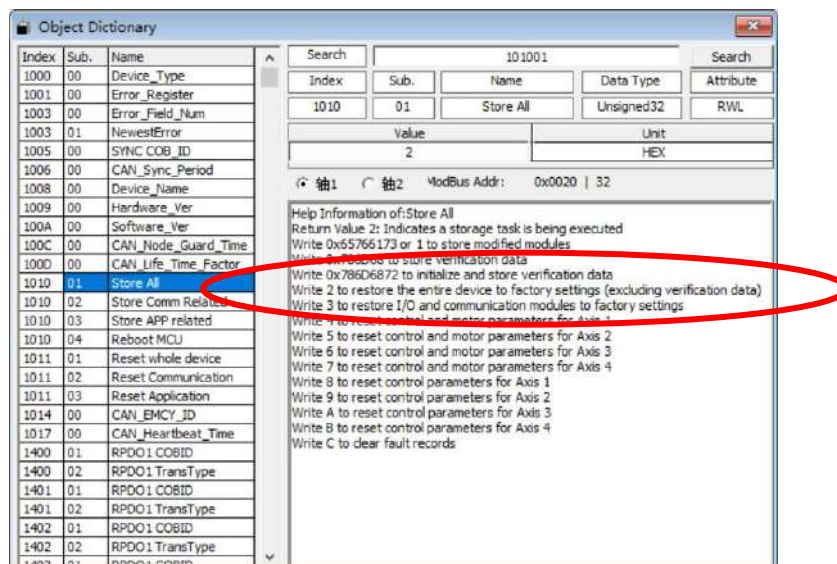
Eg. Write 3 or 7 can clear the encoder error.



Then reboot the drive.



If other alarms occur, try writing "2" to parameter 1010 01 to restore factory settings and clear the alarms.



5.3.2 Set the Basic Parameters

(1) Electronic Gear Ratio (Feed Rate) Setting

For CD200 Servo System, the feed rate (electronic gear ratio) defines the command pulses per motor revolution.

The **feed (6092-01h)** must match the **pulses/rev sent by the controller**.

Configuration Rules:

If the controller sends:

131,072 pulses/rev (17-bit: 2^{17}) → Write 131072 to 6092-01h.

8,388,608 pulses/rev (23-bit: 2^{23}) → Write 8388608 to 6092-01h.

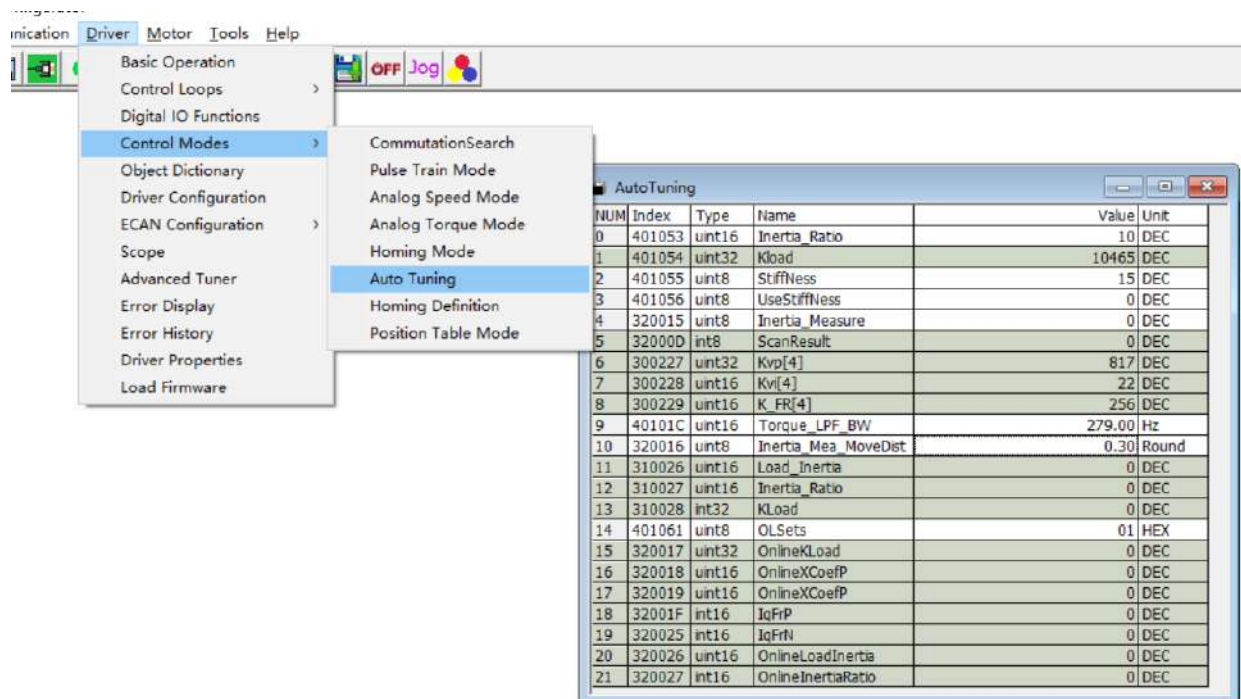
0	609101	uint32	GearInResolution		1	DEC
1	609102	uint32	GearOutResolution		1	DEC
2	609201	uint32	FEED	10000	PUL	
3	609202	uint32	ShaftResolution		1	DEC
4	606300	int32	PosActul.WORD.Lo		3	DEC

The value "10000" in the figure above means the drive receives 10,000 command units per revolution of the motor shaft.

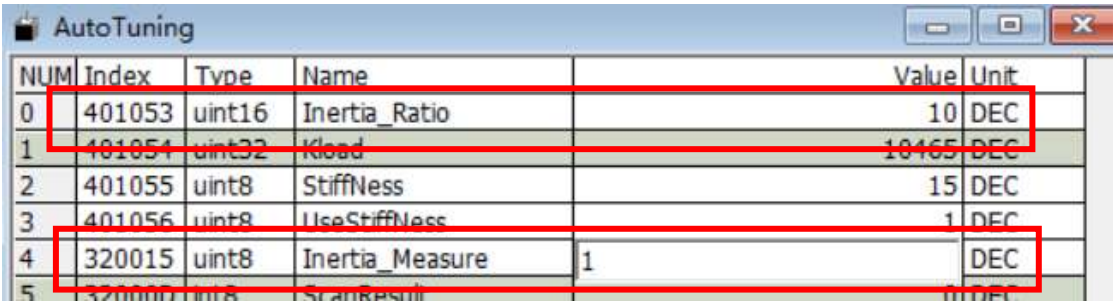
(2) Inertia Identification

After setting the feed rate (electronic gear ratio), perform auto-tuning for inertia identification:

Click **Driver**→ **Control Modes**→ **Auto-Tuning**



Auto-tuning primarily involves setting two parameters: **Stiffness Level** and **Inertia Identification**. By default, the Stiffness Level is used. When initiating Inertia Identification, the corresponding Stiffness Level will be automatically matched—for most applications, this auto-matched value is usually suitable.



NUM	Index	Type	Name	Value	Unit
0	401053	uint16	Inertia_Ratio	10	DEC
1	401054	uint32	Load	10465	DEC
2	401055	uint8	StiffNess	15	DEC
3	401056	uint8	UseStiffNess	1	DEC
4	320015	uint8	Inertia_Measure	1	DEC
5	320016	uint8	StiffNess	1	DEC

The Stiffness

Level is a set of control loop parameters based on empirical values. After the Stiffness Level is matched, you must **manually re-enter the value and press Enter** to confirm and activate it. Once the identification is complete, check that the **Inertia Ratio** value has changed accordingly.

For general applications, after the auto-matching of stiffness level is completed, you may incrementally increase the stiffness level step by step (the motor will begin to vibrate when exceeding optimal stiffness, at which point you should reduce the level). If satisfactory performance is achieved after a few increments, no further manual adjustment of control loop parameters is required.

(3) Adjustment-free function

Through real-time command reconstruction and dynamic compensation, it effectively eliminates jitter and jerk during motion, ensuring smooth operation across the entire speed range from low to high, and significantly improving the machining and control accuracy of the equipment. When the adjustment-free function is enabled, it automatically identifies load characteristics, and parameter optimization can be completed with simple settings using just two parameters – rigidity level and adjustment-free level – eliminating complex manual tuning. It also actively identifies and suppresses mechanical resonance, reduces operational vibration, and greatly enhances system stability. Even when the load inertia is 30 times greater than that of the motor, it can be easily adapted, making commissioning simple and efficient.

Object Dictionary

Index	Sub.	Name
4010	56	UseStiffNess
4010	57	CurrCompValA
4010	58	CurrCompOpt
4010	59	CurrCompTc
4010	5A	HomeInStepMode
4010	5B	PulRegenRes
4010	5C	TDTC
4010	5D	DTC_Mode
4010	5E	PwmFreqCtlMode
4010	5F	FrictionP
4010	60	FrictionN
4010	61	OL Safe

Search

UseStiffNess

Search

Index	Sub.	Name	Data Type	Attribute
4010	56	UseStiffNess	Unsigned8	RWL

Value	Unit
2	DEC

ModBus Addr: 0x0200 | 512

Help Information of:UseStiffNess
0: Do not use stiffness tables
Other: Use stiffness tables
When not using a stiffness table, the user needs to manually adjust the control loop parameters

Object Dictionary

Index	Sub.	Name
4010	AE	tunelessStiff
4010	AF	EGearNumEx
4010	B0	EGearDenEx
4010	B1	DFrikCoef
4010	B2	ABSCPMODE
4010	B3	ABSCPSpeed
4010	B4	MaxABSError
4010	B5	ABSCPSlaveID
4010	B6	GravityCurr

Search

tunelessStiff

Search

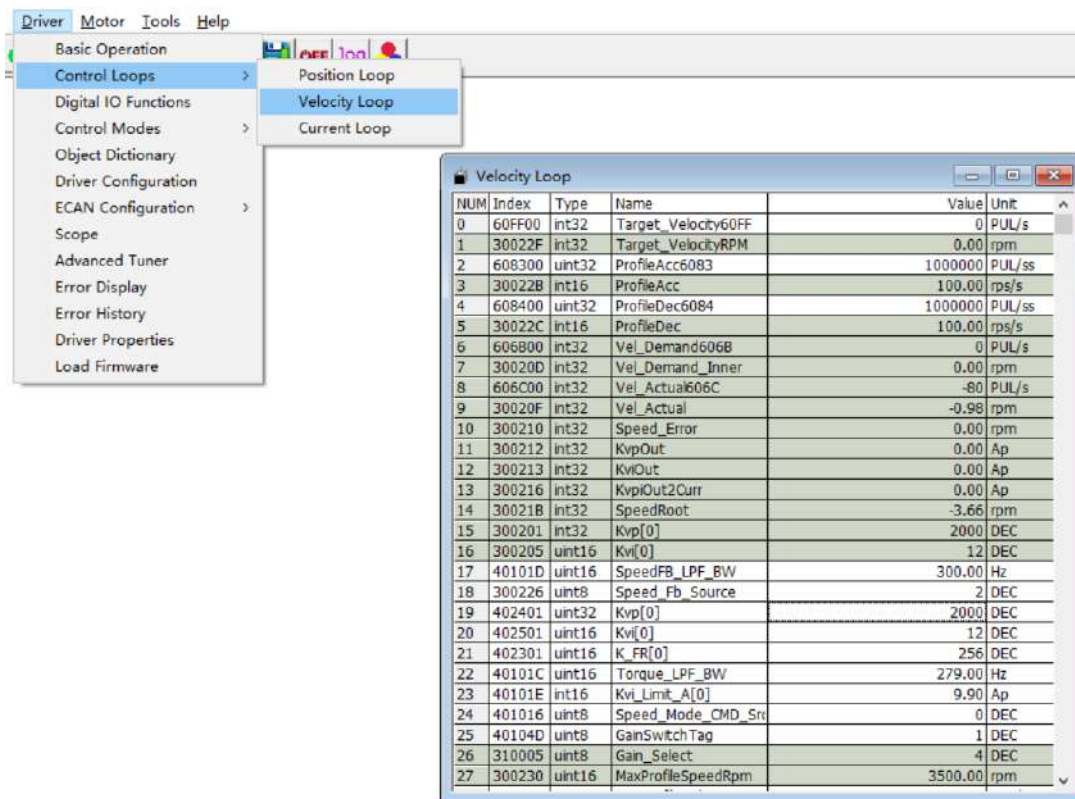
Index	Sub.	Name	Data Type	Attribute
4010	AE	tunelessStiff	Unsigned8	RWL
Value		Unit		
6		DEC		
ModBus Addr: 0x850 0				
Help Information of:tunelessStiff				

Using rigidity level settings 2, 3, or 4 to enable the adjustment-free function, the system can automatically accommodate complex mechanical systems with varying rigidity and inertia. The settings 2, 3, and 4 correspond to relatively high, high, and very high adjustment-free response levels, respectively. It is necessary to comprehensively consider factors such as inertia matching, mechanical rigidity, and external disturbances when selecting the appropriate adjustment-free mode and adjustment-free level. The adjustment-free level can be set from 1 to 12, corresponding to position loop response from low to high.

(4) Manual Gain Tuning Procedure

However, if the automatically matched stiffness level fails to meet control requirements, manual adjustment of the control loop parameters becomes necessary. When manually configuring these parameters, you must first set the UseStiffLevel parameter (401056) to 0 (to disable automatic stiffness control).

First, attempt to configure the speed loop parameters to enhance speed loop response.



Initially set the speed loop integral gain (Ki,402501) to a relatively small value, such as 2. Then gradually increase the speed loop proportional gain (Kp,402401) until either:

1. The system response meets requirements, OR
2. Mechanical vibration occurs (at which point slightly reduce the gain)"

17	40101D	uint16	SpeedFB_LPF_BW	300.00	Hz
18	300226	uint8	Speed_Fb_Source	2	DEC
19	402401	uint32	Kvp[0]	2000	DEC
20	402501	uint16	Kvi[0]	12	DEC
21	402301	uint16	K_FR[0]	256	DEC
22	40101C	uint16	Torque_LPF_BW	279.00	Hz
23	40101E	int16	Kvi_Limit_A[0]	9.90	Ap
24	401016	uint8	Speed_Mode_CMD_Src	0	DEC
25	40104D	uint8	GainSwitchTag	1	DEC

There is a useful method to find a property **Optimal Speed Loop Gains**. You can manually set the stiffness level to examine the control loop parameters under different inertia conditions at various stiffness levels, thereby identifying the appropriate speed loop gain. As shown in the figure below, **Stiffness Level 19** corresponds to a **speed loop proportional gain (Kp) of 1839**. You can begin testing by setting the speed loop gain to **1800** as an initial value to verify its suitability.

1	401054	uint32	Kload	10465	DEC
2	401055	uint8	StiffNess	19	DEC
3	401056	uint8	UseStiffNess	1	DEC
4	320015	uint8	Inertia_Measure	0	DEC
5	32000D	int8	ScanResult	0	DEC
6	300227	uint32	Kvp[4]	1839	DEC
7	300228	uint16	Kvi[4]	30	DEC
8	300229	uint16	K FR[4]	256	DEC

Once the **speed loop gains (Kp/Ki)** are properly tuned, proceed to configure the **position loop parameters** for optimal motion control performance.

The screenshot shows the 'Position Loop' configuration window in the Simphoenix software. The window contains a table of parameters with columns for NUM, Index, Type, Name, Value, and Unit. The parameters are as follows:

NUM	Index	Type	Name	Value	Unit
0	300301	uint32	Kpp[0]	50309	DEC
1	300309	int32	Pos_Error	0	DEC
2	30030A	int32	KppOut	0.00	rpm
3	60F400	int32	Pos_Error60F4	0	PUL
4	402001	uint16	KppBW[0]	7.64	Hz
5	606500	uint32	FollowErrWindow6065	50000	PUL
6	300316	uint32	FollowErrWindow	655360	DEC
7	606600	uint16	FollowErrTWindow	10.00	ms
8	606700	uint32	PositionWindow6067	10	PUL
9	30001C	int16	eAngleX	467	DEC
10	606800	uint16	PositionTWindow	15.00	ms
11	40102C	uint16	Pos_Cmd_FT_Coef_LP	1	DEC
12	40102D	uint16	Pos_Cmd_FT_Coef_MAF	1	DEC
13	40101F	int16	VelFF_Permillage	100	%
14	401020	int16	VelFF_Filter_BW	100.00	Hz
15	40104D	uint8	GainSwitchTag	1	DEC
16	310005	uint8	Gain_Select	4	DEC
17	300230	uint16	MaxProfileSpeedRpm	3500.00	rpm
18	30022A	uint16	ProfileSpeedRpm	199.00	rpm
19	300228	int16	ProfileAcc	100.00	rps/s
20	30022C	int16	ProfileDec	100.00	rps/s
21	60C500	uint32	ProfileAccMax60C5	20000000	PUL/ss
22	60C600	uint32	ProfileDecMax60C6	20000000	PUL/ss
23	607F00	uint32	MaxProfileVel	600000	PUL/s

402001 KppBW[0]: position loop bandwidth

The larger the position loop bandwidth, the faster the position loop response.

However, due to the limitation of the speed loop bandwidth, if the speed loop response cannot keep up, simply increasing the position loop bandwidth will lead to control loop instability.

40102D Pos_Cmd_FT_Coef_MAF

Position command sliding filtering (similar to position smoothing filter time) can eliminate disturbances caused by commands, resulting in smoother operation. However, it may affect Motion Path details. The setting values are 1, 2, 4, 8, 16, 32, 64... (integer multiples of 2).

40101F VelFF_Permillage

Speed feedforward (in per mille) can improve speed tracking capability, enhance speed response, and eliminate steady-state following errors.

5.3.3 Operation Monitoring

While adjusting the control loop parameters, observe the operational performance using the scope.



Sampling Period Configuration:

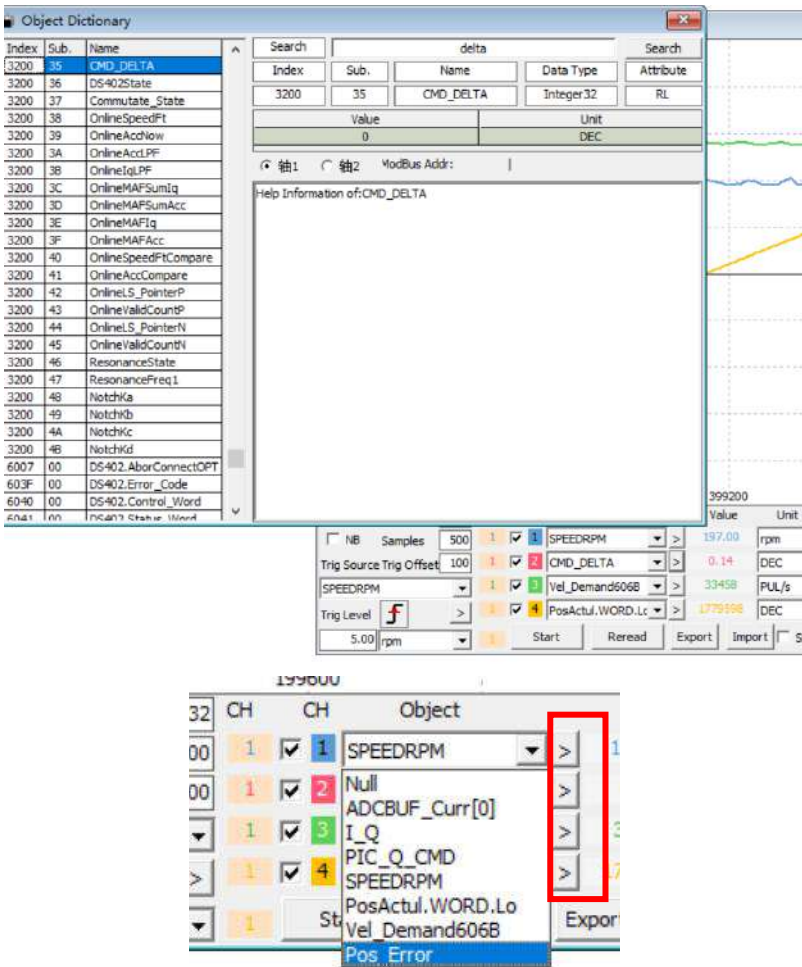
Set the sampling period to 1, 2, 4, 8, 16, 32, 64... (integer multiples of milliseconds) for convenient observation.

Trigger Condition:

Data acquisition can be configured to start once the operating speed reaches a specified value. As shown in the figure above, recording begins 100 sample points before the trigger point when the speed reaches 5 RPM (pre-trigger capture).

Selectable Sampling Channel Object:

You can choose like CMD_DELTA (command delta) as the sampling object. This parameter represents the speed variation value in motion Motion Path planning, allowing you to verify whether the command planning is functioning correctly.

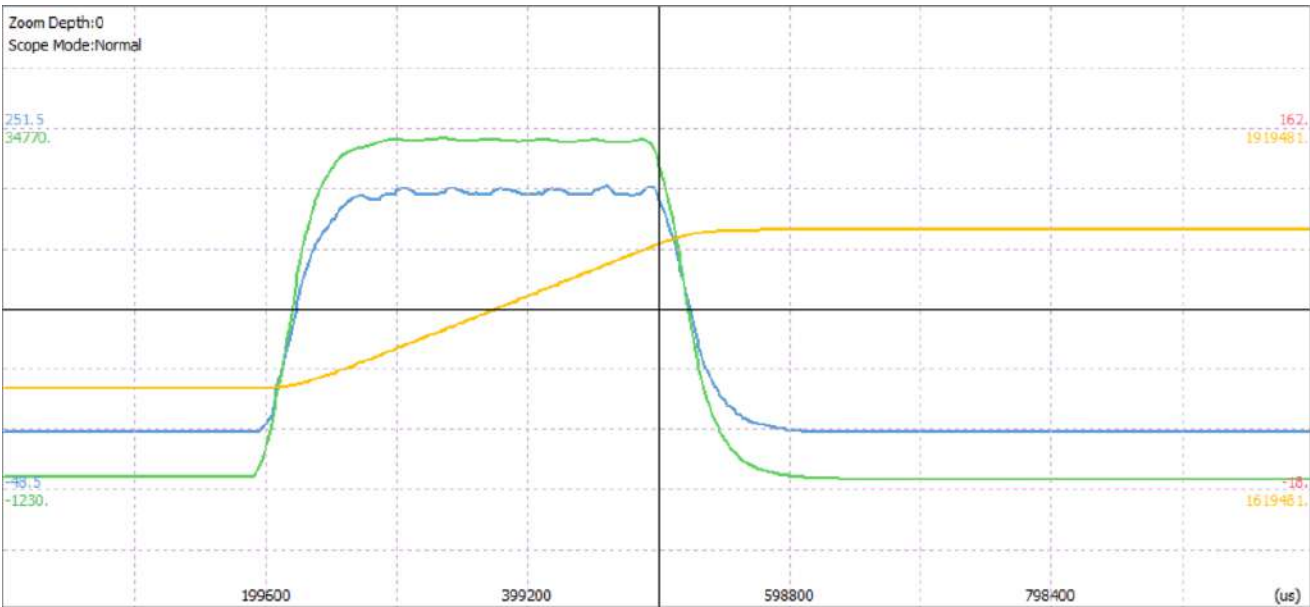


Pos_error is the difference between the commanded position (planned value) and the actual position, reflecting the system's tracking performance. Ideally, this value should remain small and stable (minimal fluctuation).

Unit: Encoder units

Example: 10,000 units = $10,000/8,388,608 \approx 0.0012$ revolutions (assuming a 23-bit encoder resolution).

Note: Additional parameters can be monitored as needed based on specific application requirements.



Chapter 6 Adjustment

6.1. Overview

To adapt to user commands and actual mechanical conditions, it is necessary to adjust the internal gain, filtering, feedforward, and other parameters of the servo drive to achieve ideal operating results.

Below are waveform comparisons for different parameter settings.

Kpp: Position Loop Bandwidth

Vff: Velocity Feedforward

VFBF: Velocity Feedback Filter Bandwidth

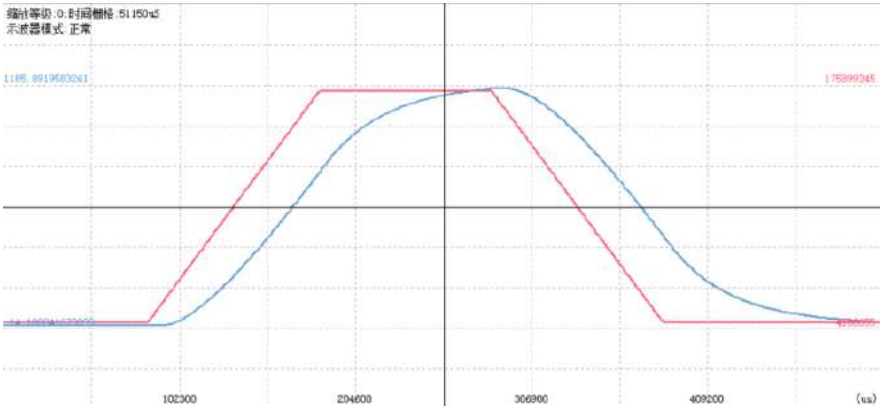
Kvp: Speed Loop Proportional Gain

Tff: Torque Feedforward

TCMDF: Torque Command Filter Bandwidth

Kvi: Speed Loop Integral Gain

VBW: Actual Velocity Loop Bandwidth



Kpp=4Hz

Vff=0

VFBF=500Hz

Kvp=43

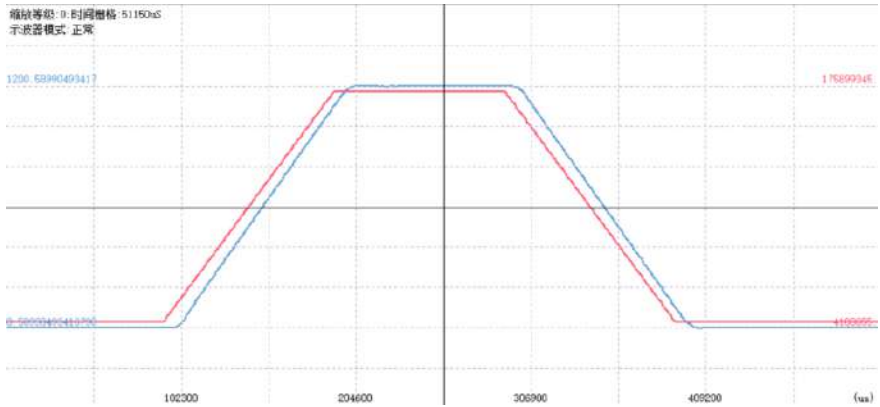
Tff=0

VBW=79Hz

Kvi=1

TCMDF=500Hz

Inertia Ratio=1.03



Kpp=20Hz

Vff=0

VFBF=500Hz

Kvp=43

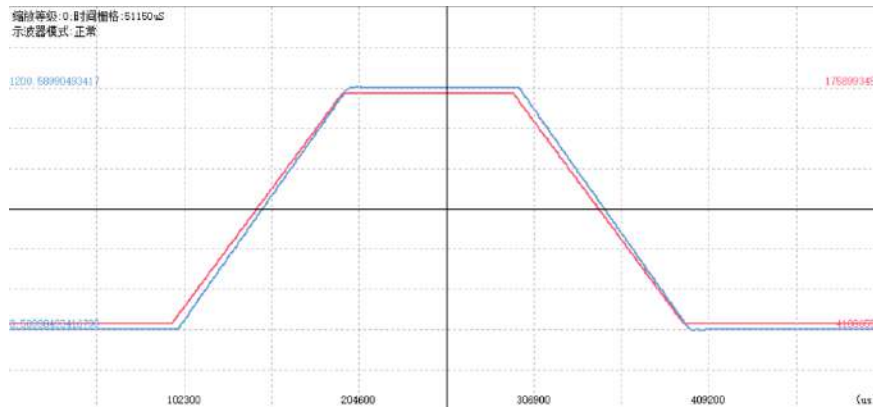
Tff=0

VBW=79Hz

Kvi=1

TCMDF=500Hz

Inertia Ratio=1.03



Kpp=30Hz

Kvp=43

Kvi=1

Vff=0

Tff=0

TCMDF=500Hz

VFBF=500Hz

VBW=79Hz

Inertia Ratio=1.03

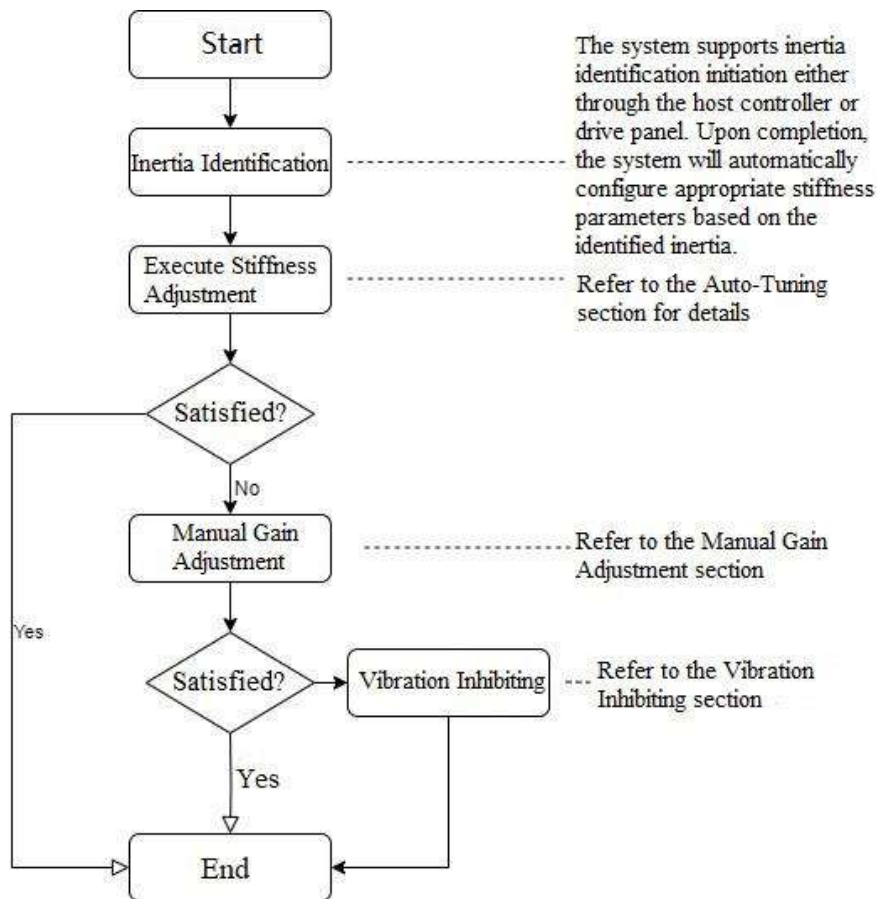
From the above comparison, it can be seen that different motion effects can be achieved through different parameter combinations

※ Note:

- Before adjusting the drive, it is recommended to first run in Jog mode to ensure that the drive and motor can operate normally.

Servo gain is set by combining multiple parameters (speed loop, position loop gain, inertia ratio, filters, etc.), which influence each other. Therefore, setting servo gain must consider the balance between the set values of various parameters.

The gain adjustment process is as follows:



Gain Adjustment Process Description:

Gain Adjustment Process			Function Description	Corresponding Chapter
1	Inertia Identification	Offline	Execute built-in specific actions to obtain inertia	6.2 Inertia Identification
		Online	Real-time monitoring of motion status to obtain inertia	
2	Rigidity Adjustment		Adjust control loop based on inertia and rigidity table	
3	Manual Gain Adjustment (Disabling the rigidity level function is required)	Control Loop Gain	First speed loop, then position loop, finding the balance point between response and stability	6.3 Manual Gain Adjustment
		Loop Filter	Improve vibration, noise, and smoothness of motion	
		Feedforward and Filtering	Optimize tracking capability	
		PDFF	Change speed loop structure, improve positioning capability	
		Motion Compensation	Optimizes start/stop performance	
4	Vibration Inhibiting	Mechanical Resonance	Suppress resonance through notch filters	6.4 Vibration Suppression Function
		End Vibration	Suppress end vibration through shaping filters	

6.2 Inertia Identification

Inertia identification refers to the process where the drive infers the motor load inertia by using information related to current, displacement, and the motor body.

After executing the inertia identification function with a matched motor, the total inertia of the motor and load, the ratio of total inertia to motor rotor inertia, and the torque inertia coefficient can be obtained. The definition of inertia ratio is as follows:

$$\text{Inertia Ratio} = \frac{\text{Motor Inertia} + \text{Load Inertia}}{\text{Motor Inertia}} \quad \text{Inertia Ratio} \geq 1$$

Inertia ratio is a key parameter of the servo system, whose role spans the entire process from mechanical design to servo motor selection to performance tuning.

CD200 drives offer two inertia identification methods:

I. 1-second Inertia Identification

1-second offline inertia identification can be started via the operation panel or host computer. During identification, the motor will move a certain distance, with a default maximum movement range of 0.3 turns, ending after 1 second.

II. Online Inertia Identification

The drive defaults to real-time online inertia identification. As long as the motor performs a movement that meets the conditions, real-time online inertia calculation can be triggered. Therefore, executing 1-second inertia identification will also trigger online inertia calculation. Results can be compared.

✘ Note:

Online inertia identification requires the following conditions to be met for normal triggering:

- Motor speed higher than 150rpm
- Motor acceleration/deceleration higher than 1200rpm/s
- Load inertia and load torque do not change drastically
- If the drive's speed loop gain and load inertia ratio do not match, the motor's acceleration/deceleration may not meet requirements, thus preventing online inertia calculation from triggering. In this case, the speed loop proportional gain needs to be increased.
- If position closed-loop control is used during inertia identification, oscillation may occur due to loop mismatch. Immediate disablement, modification of gain parameters, or switching to speed closed-loop should be performed.
- Identification may be invalid in situations with excessive backlash, friction, or gravity.
- Inertia identification results can be used with rigidity levels for quick adjustment.

6.2.1 1-second Offline Inertia Identification

Inertia Identification Parameter Setting:

Start 1-second inertia identification by setting Pr6.05 in panel F006 group to 1, or by setting "Inertia Measurement (3200h-15h)" to 1 in the host computer [Drive]-[Control Mode]-[Auto-tuning]. The motor will perform about 1 second of oscillation.

Inertia Ratio Setting and Viewing:

The execution result can be viewed via Pr6.06 or host computer 3200h-0Dh. If the result is 1, it means inertia measurement was successful, and the inertia ratio can be further viewed via Pr6.03 or host computer 3100h-27h.

After offline inertia is identified successfully, the drive will automatically select an appropriate rigidity level (Pr6.02 or 4010h-56h) based on the inertia ratio value. The automatically set rigidity range is 3-17, corresponding to the inertia ratio range of 200-20. A larger inertia ratio results in lower rigidity.

※ Note:

The unit of inertia ratio is 0.1

When using the panel, you can directly access Pr6.05 from the main menu using the "▼+SET" shortcut key

If the 1-second offline inertia identification result is negative, and the result is -1 or -2, you can try increasing the oscillation distance (Pr6.07 or 3200h-16h) and re-execute the measurement. If it is -3 or -4, it indicates that the load inertia ratio is too large and requires manual confirmation. This function is associated with parameters

No.	Index	Parameter Name	Setting/Display Value	Default Value	Unit	Access	Validation
Pr6.05	3200h-15h	Inertia Measurement	0→1 Start	0		RW	Immediate
Pr6.06	3200h-0Dh	Inertia Measurement Result	0: Not started 1: Successful -1: Speed amplitude too small -2: Current amplitude too small -3: Inertia ratio exceeding 25 times -4: Inertia ratio exceeding 40 times	0		RO	
Pr6.03	4010h-53h	Inertia Ratio	10-300	30	0.1	RW	Immediate
Pr6.01	4010h-55h	Rigidity Grade	0-31	13		RW	Immediate
Pr6.07	3200h-16h	Inertia Measurement Distance	10-50	30	0.01 Turn	RW	Immediate
Pr6.04	3100h-26h	Current Load Inertia		0	0.01Kg·cm ²	RO	
Pr6.08	3100h-27h	Current Inertia Ratio	10-250	0	0.1	RO	
Pr6.09	3100h-28h	Current KLoad		0		RO	
Pr6.02	4010h-56h	Use Rigidity Level	0: Do not use 1: Use	0		RW	Immediate
Pr6.10	3002h-27h	Speed Loop Proportion 4	0-65535			RO	
Pr6.11	3002h-28h	Speed Loop Integral 4	0-65535			RO	
Pr6.12	3002h-29h	Speed Loop KFR4	0-256			RO	

6.2.2 Online Inertia Identification

Online inertia identification is enabled by default. As long as the load meets the calculation trigger conditions, the online inertia identification function will update the results. Enable online inertia identification by setting Bit0 of Real-time Function Settings (Pr6.17 or 4010h-61h) to 1 (default is 1). Enable online gain adjustment by setting Bit3 to 1, which adjusts the control loop gain based on the user-set rigidity and the currently detected inertia ratio parameter.

Online inertia identification and online gain adjustment are not suitable for situations with drastic inertia changes or excessive backlash. This function is associated with parameters

No.	Index	Parameter Name	Setting/Display Value	Default Value	Unit	Mark	Validation
Pr6.17	4010h-61h	Real-time Function Settings	Bit0: Enable inertia identification Bit1: Enable friction compensation Bit2: Friction compensation source 0: User set 1: Online detection result Bit3: Enable online gain adjustment	1		RW	Immediate
Pr6.18	3200h-1Fh	Forward Load Current		0		RO	
Pr6.19	3200h-25h	Reverse Load Current		0		RO	
Pr6.20	3200h-26h	Online Inertia		0	0.01kg·cm ²	RO	
Pr6.21	3200h-27h	Online Inertia Ratio		0	0.1	RO	
Pr6.10	3002h-27h	Speed Loop Proportion 4	0-65535			RO	
Pr6.11	3002h-28h	Speed Loop Integral 4	0-65535			RO	
Pr6.12	3002h-29h	Speed Loop KFR4	0-256			RO	

6.2.3 Rigidity Adjustment

Servo rigidity refers to the servo drive's ability to eliminate error disturbances. Higher rigidity means faster drive response and stronger tracking capability.

Rigidity essentially corresponds to the gain of the control loop. Here, rigidity adjustment is based on the built-in empirical loop gain parameter table. It involves selecting a suitable combination of loop parameters by setting the index of the parameter table, and then combining it with the user-set or identified inertia ratio to adjust the final internal gain parameters, for the purpose of jointly adjusting position and speed loop parameters by adjusting a single parameter (rigidity index).

Executing 1-second offline inertia identification will automatically set the rigidity index. The rigidity table is as follows:

Rigidity Grade	Position Loop Gain 0.1/s	Speed Loop Proportion 0.1Hz	Speed Loop Integral 0.1ms	Torque Filtering Time 0.01ms
0	20	15	3700	1500
1	25	20	2800	1100
2	30	25	2200	900
3	40	30	1900	800
4	45	35	1600	600
5	55	45	1200	500
6	75	60	900	400
7	95	75	700	300
8	115	90	600	300
9	140	110	500	200
10	175	140	400	200
11	320	180	310	126
12	390	220	250	103
13	480	270	210	84
14	630	350	160	65
15	720	400	140	57
16	900	500	120	45
17	1080	600	110	38
18	1350	750	90	30
19	1620	900	80	25
20	2060	1150	70	20
21	2510	1400	60	16
22	3050	1700	50	13
23	3770	2100	40	11
24	4490	2500	40	9
25	5000	2800	35	8
26	5600	3100	30	7
27	6100	3400	30	7
28	6600	3700	25	6
29	7200	4000	25	6
30	8100	4500	20	5
31	9000	5000	20	5

Associated parameters are

No.	Index	Parameter Name	Setting/Display Value	Default Value	Unit	Access	Validation
Pr6.01	4010h-55h	Rigidity Grade	0-31	13		RW	Immediate
Pr6.02	4010h-56h	Use Rigidity Level	0: Do not use 1: Use	1		RW	Immediate

6.3 Manual Gain Adjustment

※ Note:

Before performing manual gain adjustment, the "Use Rigidity Level" function must be disabled (Pr6.02 set to 0 or 4010h-56h=0)

6.3.1 Control Loop Gain Adjustment

(1) Speed Loop Gain Adjustment

I. Speed Loop Bandwidth

Speed loop bandwidth refers to the drive's maximum response capability to speed commands, based on the current speed loop gain parameter settings and load characteristics. It is defined as the frequency value where the actual response amplitude / command amplitude = -3dB.

For rigid body loads, the calculation relationship between speed loop bandwidth and gain is:

$$k_{vpx} := \frac{8192 J_{driver} \pi^2 F_{bw} ADCRES \sqrt{2}}{I_{Max} k_{t_{driver}} Encoder}, \text{ where } k_{vpx} \text{ is the internal speed loop gain}$$

From this relationship, it can be seen that for the same speed loop bandwidth, the speed loop gain is directly proportional to the load inertia and inversely proportional to the motor torque constant and encoder resolution. That is, when the speed loop bandwidth is constant:

Larger load inertia, larger gain

Larger motor torque constant, smaller gain

Higher motor encoder resolution, smaller gain

However, the speed loop gain cannot be set arbitrarily; the following three aspects need to be considered:

1. Larger gain will also amplify noise, so a balance between noise and gain needs to be achieved during actual setting;
2. The above formula is based on rigid body loads. Actual load frequency characteristics are much more complex, possibly having multiple resonance points. Therefore, suppression of resonance also needs to be considered during setting;
3. When setting, consider the host computer command requirements or actual motion effects, and leave sufficient margin.

Method 1: Using Speed Command Step Response

Step 1: Set Speed Loop Integral Gain 0 (4025h-01h) to 0. Set the drive's operating mode (6060h-00h) to -3. Set "Speed Command rpm (3002h-1Fh)" to a small speed, such as 100 rpm - depending on encoder resolution and whether the load will cause current saturation. If the encoder with 23-bit resolution is set to 200 rpm, the encoder with 17-bit resolution can be set to around 400 rpm.

Step 2: Set the oscilloscope trigger condition to "Speed Loop Command (3002h-0Dh)". Select "Speed Loop Command (3002h-0Dh)" for acquisition channel 1, and "Speed Feedback 1st Order Low-Pass Output (3002h-17h)" for acquisition channel 2.

Step 3: Click the start button of the oscilloscope. Enable the drive via the host computer (it is recommended to use the simulation function in the digital I/O settings), then disable it after about 1 second.

Step 4: Observe the oscilloscope waveform. If overshoot or oscillation is observed in the step response waveform, decrease Speed Loop Proportional Gain 0 (4024h-01h) and repeat Step 3. When Speed Loop Proportional Gain 0 just causes slight overshoot, this setting value is considered acceptable after being slightly reduced.

Step 5: Gradually increase Speed Loop Integral Gain 0 (4025h-01h). Using a method similar to Steps 3 and 4, observe the step response. When slight overshoot appears, the integral gain is considered acceptable.

Method 2: Empirical Judgment Method

Set the servo drive to speed mode, with target speed and "Speed Loop Integral Gain 0" both set to 0, and then enable the drive. Modify "Speed Loop Proportional Gain 0", and touch with your hand or listen with your ear to select a value that results in less vibration or noise. Then, gradually increase "Speed Loop Integral Gain 0" starting from 1, execute the "motion-pause-motion"

cyclic curve for the drive, and observe whether there is obvious jitter when the motor pauses. If there is, decrease "Speed Loop Integral Gain 0".

- "Speed Loop Integral Gain 0" should be set to a minimum of 1

II. Speed Loop Integral Limit

The speed loop integral limit restricts the output current of the speed loop's integral term to a reasonable range (e.g., just above friction torque). Its purpose is to prevent excessive speed overshoot or oscillation caused by continuous integration due to unreasonable acceleration/deceleration commands or loads leading to sustained inability of speed to follow.

Allow the motor to operate in constant velocity mode, then observe the "Current Loop Command q (3000h-05h)" at this time, and set the integral limit to approximately twice the current. Usually, keeping the default value is sufficient. If speed overshoot or oscillation occurs, combine with the software oscilloscope to observe "Speed Loop Integral Output 3002h-13h" and the speed waveform to determine if integral limiting is needed.

- For gravity loads, it is recommended to keep the default value or increase it.

III. Speed Loop Related Filtering

Speed Feedback Filtering (Pr1.09, 4010h-1Dh):

Filters out noise from the speed feedback signal. The default bandwidth is 300Hz. For general applications, the default value is usually sufficient. If the application requires a speed loop bandwidth exceeding 150Hz, this parameter should be increased as needed.

A smaller speed feedback filter setting results in stronger suppression of speed signal noise, but also introduces greater phase lag, leading to instability in the speed control loop.

A larger speed feedback filter setting results in weaker suppression of speed signal noise, but introduces less phase lag, allowing for a higher control loop bandwidth.

The maximum setting for Speed Feedback Filtering (4010h-1Dh) is 2000Hz. By setting "Speed Feedback Channel (Pr1.11, 3002h-26h)" to 0, speed feedback filtering is canceled. Setting it to 1 selects 1st order low-pass data, and setting it to 2 selects 2nd order low-pass data.

Torque Filtering (Pr1.10, 4010h-1Ch):

Filter out noise from the output signal of the speed PI controller. Its function is to eliminate control noise and suppress resonance frequencies.

From a loop stability perspective, torque filtering should be set at more than 2 times the speed loop bandwidth. During actual debugging, adjust from a larger to smaller value based on actual operating noise and vibration.

The maximum value for torque filtering (4010h-1Ch) is 1000Hz. If it is actually higher than 994Hz, the filtering function becomes invalid.

The order can be selected via "Torque Filter Selection (Pr1.12, 4010h-24h)". A first-order filter attenuates noise less than a second-order filter but has less impact on control loop stability.

Notch Filter:

The drive has two built-in notch filters to suppress resonance, thereby extending the speed loop bandwidth. For more details, please refer to the Vibration Suppression section.

(2) Position Loop Gain Adjustment

Based on reasonably set speed loop parameters, proceed with position loop gain adjustment.

For general applications, if the load inertia ratio is known, the theoretical speed loop bandwidth can be calculated from the speed loop gain parameters. Then, 1/4 to 1/3 of this bandwidth can be set as the position loop bandwidth. When setting the position loop bandwidth, the drive automatically calculates the internal position loop gain. If the load inertia ratio is unknown, the position loop bandwidth needs to be set from small to large based on specific repetitive position loop commands, until satisfactory.

Position Loop Bandwidth 0 (Pr1.01, 4020h-01h)

6.3.2 Input Shaping

The input position command can be shaped using "Position Command Low-Pass Filter Coefficient (Pr1.07, 4010h-2Ch)" or "Position Command Smoothing Filter Coefficient (Pr1.08, 4010h-2Dh)" to make it smoother and acceleration/deceleration more reasonable. If the position command sent to the servo is already sufficiently fine, both parameters can be set to 1 to disable the related smoothing functions.

6.3.3 Enhance Position Tracking Capability

When the speed loop bandwidth cannot be further increased, considering the stability of the position loop, its host limit for position loop bandwidth is fixed. It is no longer possible to improve position tracking capability by increasing the position loop proportional gain. Velocity feedforward and torque feedforward functions can be used to further enhance dynamic tracking capability.

(1) Velocity Feedforward Function

This function separates velocity information from the position command and directly uses a portion (Velocity Feedforward Per mille Pr1.17, 4010h-1Fh) as the speed loop's reference command. Since the velocity feedforward command consumes some of the speed loop's response capability, the position loop bandwidth needs to be appropriately reduced after using feedforward to avoid oscillation or overshoot.

(2) Torque Feedforward Function

This function separates acceleration information from the position command. The drive combines this with the set inertia ratio to convert it into torque information, then directly uses a portion (Torque Feedforward Per mille) as the current loop's reference command. Torque feedforward further enhances tracking capability, but since it consumes some current loop response capability, the speed loop gain may need to be appropriately reduced when used.

※ Note:

- The torque feedforward function is limited by the stability of load characteristics and should be used with caution.

(3) Startup Compensation Function

The startup compensation function can accelerate the motor's transition from stop to start. Its principle is to output a gradually changing current over a certain period to speed up the motor's action. This function does not affect the overall tracking performance.

No.	Index	Parameter Name	Setting/Display Value	Default Value	Unit	Mark	Validation
Pr1.44	4010h-76h	Startup Compensation Current	0-1500	0	Internal Unit	RW	Immediate
Pr1.45	4010h-77h	Startup Compensation Time		15	ms	RW	Immediate

Startup compensation current is an internal unit, and its conversion to amperes is:

$$\text{Amperes Current} = \text{Startup Compensation Current} * [3100\text{h-1Ch}] / 8192$$

(4) Friction Compensation Function

The friction compensation function can improve the speed loop's low-frequency tracking capability during startup. Independent settings are required for forward and reverse compensation circuits.

No.	Index	Parameter Name	Setting/Display Value	Default Value	Unit	Mark	Validation
Pr1.46	4010h-5Fh	Forward Friction Current	0-100	0	0.1A	RW	Restart
Pr1.47	4010h-60h	Reverse Friction Current	0-100	0	0.1A	RW	Restart
Pr1.48	4010h-61h	Real-time Function Settings	Bit1: Enable friction compensation Bit2: Friction compensation source 0: User set 1: Online detection result	1		RW	Immediate
Pr1.49	4010h-62h	Forward Maximum Friction Limit		10	0.1A	RW	Immediate
Pr1.50	4010h-63h	Reverse Maximum Friction Limit		10	0.1A	RW	Immediate

Forward/Reverse Friction Current are user-set friction compensation values, corresponding to bit2=0 of the real-time function settings
Forward/Reverse Maximum Friction Limit restricts the online detected friction, corresponding to bit2=1 of the real-time function settings

(5) Speed-up Positioning

To achieve better positioning performance, it is necessary to increase the bandwidth of all control loops as much as possible. The methods previously used to improve tracking capability cannot Validationonly speed-up positioning; reducing the position loop bandwidth will weaken positioning capability. By using the gain switching function, higher position loop and speed loop proportional parameters can be used during positioning, and a stronger integral gain can accelerate the positioning process.

Using the PDFF function to strengthen the speed loop integral capability can also enhance positioning capability

No.	Index	Parameter Name	Setting/Display Value	Default Value	Unit	Mark	Validation
Pr1.19	4010h-4Ch	Gain Switching Mode	0: No switching 1: Switch when speed is 0 2: Switch when position command is constant and position error is less than object 4010h-75h	0		RW	Immediate
Pr1.20	4010h-4Dh	Gain Switching Target	0-3	0		RW	Immediate
Pr1.51	4010h-75h	Gain Switching Positioning Error Comparison Value		2000	Encoder Unit	RW	Immediate

6.4 Vibration Suppression Function

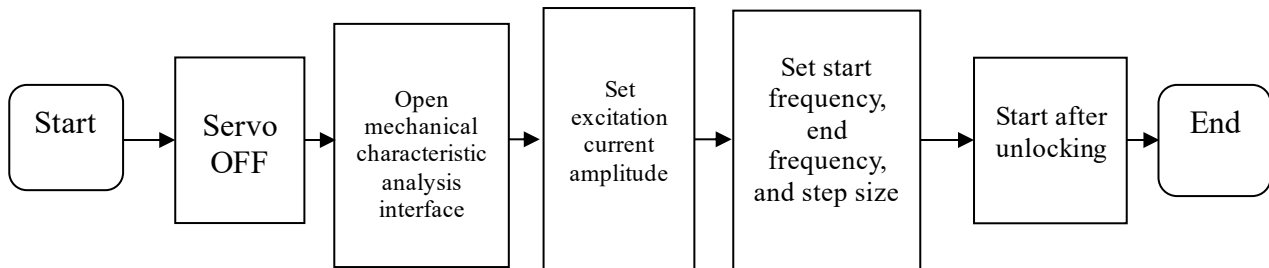
6.4.1 Mechanical Characteristic Analysis

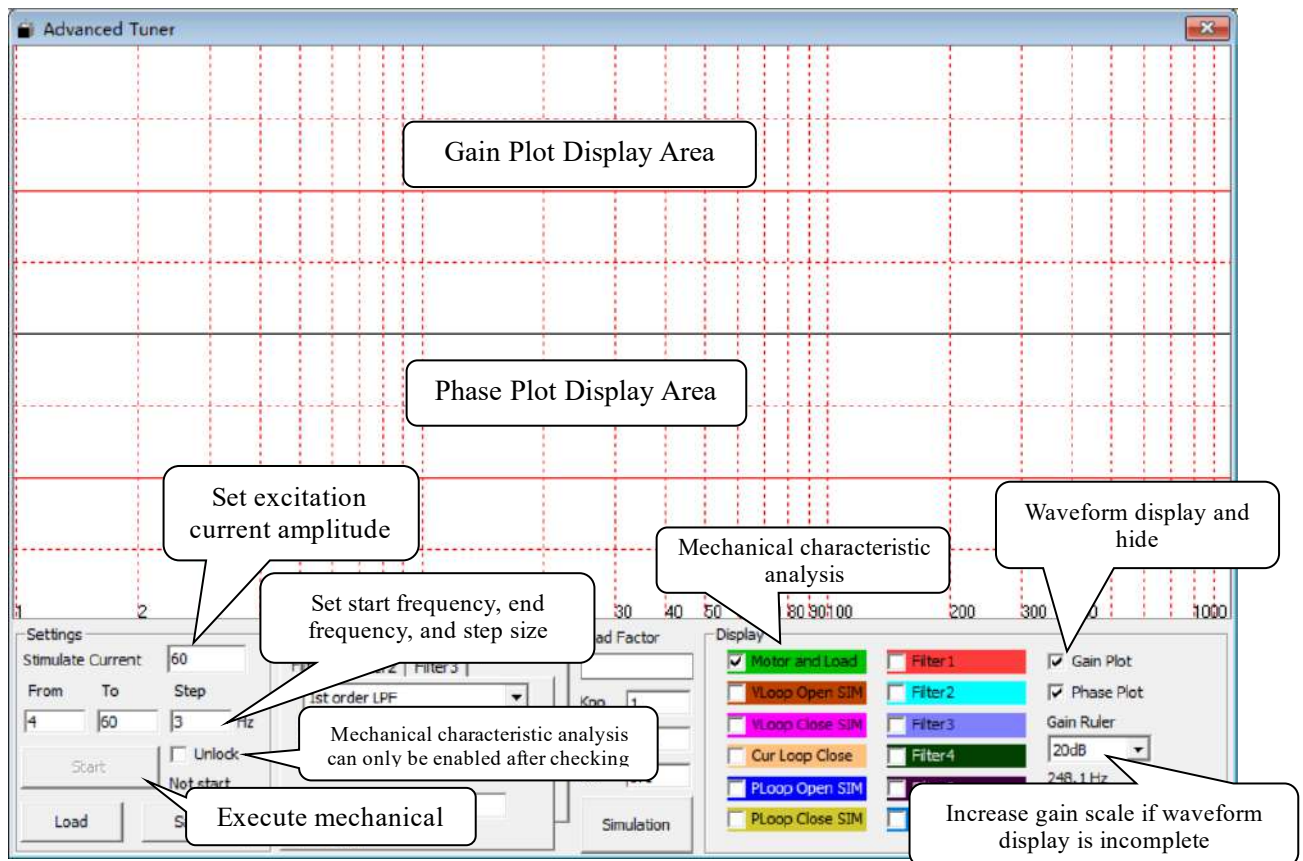
Mechanical characteristic analysis is used to determine mechanical resonance points and system bandwidth, supporting up to 1kHz response characteristic analysis.

Reasonably setting notch filters based on the resonance points identified by mechanical characteristic analysis can Validationonly improve mechanical resonance, making the equipment run more smoothly.

The mechanical characteristic analysis interface is accessed via 'Drive D' → 'Tuner/Expert Mode' in the debugging software menu bar. The waveform display in the mechanical characteristic analysis interface is divided into a gain plot and a phase plot, with the gain plot displayed above and the phase plot below.

The general operation procedure for mechanical characteristic analysis is as follows:





※ Note:

- To avoid excessive vibration during testing, the excitation current amplitude should not be set too high during the first implementation, generally within the range of 300-600;
- If the current excitation is too small, the analyzed waveform will have some distortion;

Safety Instructions:

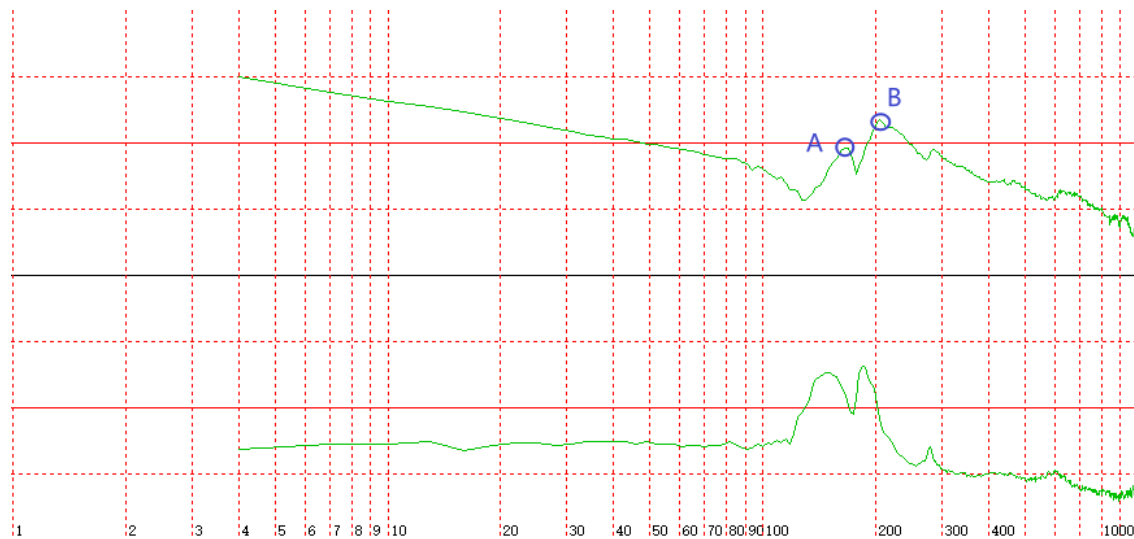
- During mechanical characteristic analysis, the motor automatically runs for a certain time and distance without external signal control. Please perform this operation only under safe conditions to avoid casualties and equipment damage!

6.4.2 Mechanical Resonance Suppression

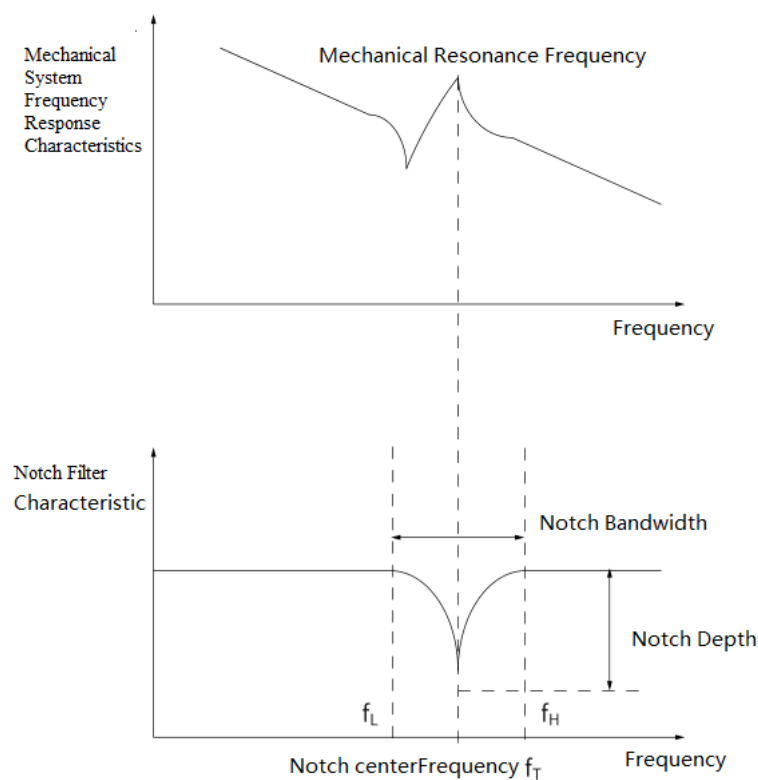
Mechanical resonance between the motor shaft and load typically occurs in elastic transmission mechanisms.

Near resonance frequencies, the speed closed-loop gain can become extremely high, leading to positive feedback and system instability, thus requiring resonance characteristics to be suppressed.

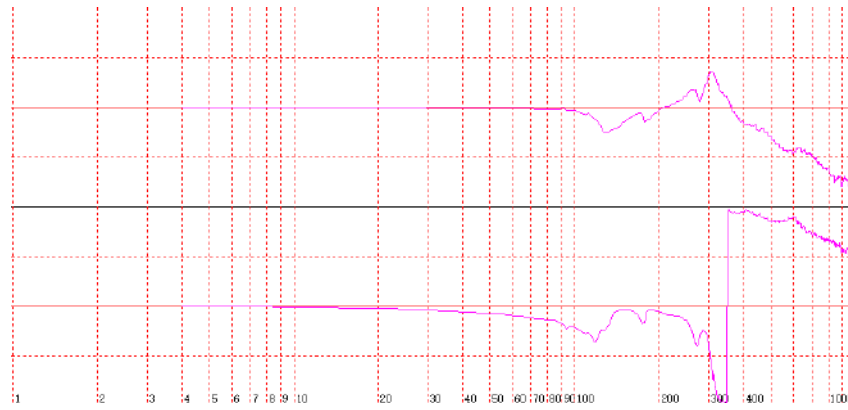
Load resonance frequency can be measured using the mechanical frequency characteristic analysis tool in the host computer. As shown in the figure below



In the figure, two resonance frequency points (A and B) can be found. Resonance can be suppressed by using torque filters or notch filters. The working principle of a notch filter is shown below:

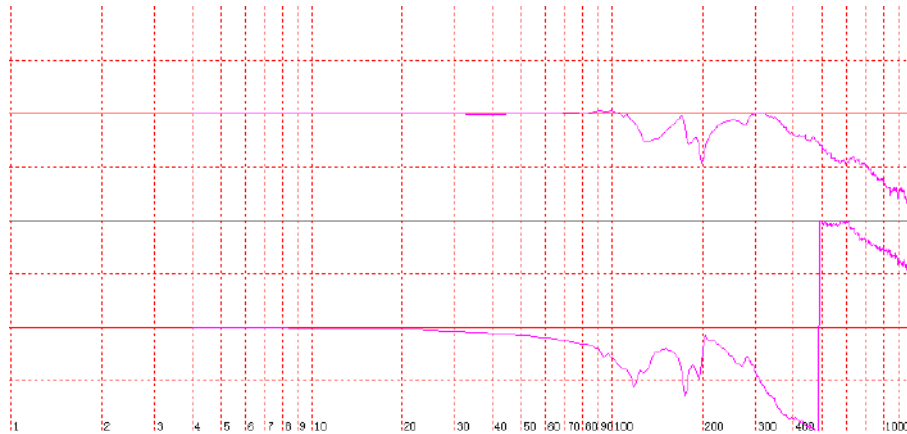


When resonance suppression is not used, the speed closed-loop frequency characteristic is as follows:



As seen from the figure, the system is close to positive feedback and is uncontrollable.

If a notch filter is used:



As seen from the figure, the frequency peak and positive feedback characteristics have been suppressed.

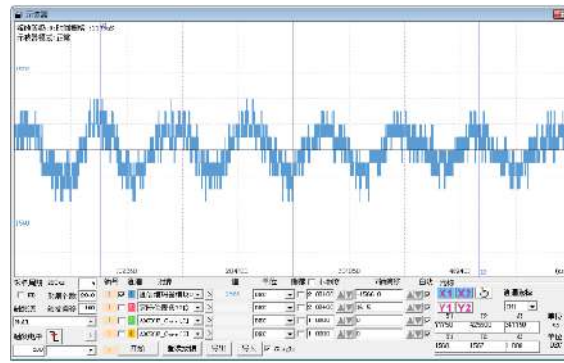
CD200 drives have two built-in notch filters, A and B.

No.	Index	Parameter Name	Setting/Display Value	Default Value	Unit	MARK	Validation
Pr6.23	4010h-6fh	Notch Filter A Width	0-60	45	Hz	RW	Immediate
Pr6.24	4010h-70h	Notch Filter A Depth	0-100	50	dB	RW	Immediate
Pr6.25	4010h-71h	Notch Filter A Frequency	100-8000	8000	Hz	RW	Immediate
Pr6.26	4010h-72h	Notch Filter B Width	0-60	45	Hz	RW	Immediate
Pr6.27	4010h-73h	Notch Filter B Depth	0-100	50	dB	RW	Immediate
Pr6.28	4010h-74h	Notch Filter B Frequency	100-8000	8000	Hz	RW	Immediate

6.4.3 End Vibration Suppression

End vibration typically occurs in transmission mechanisms where the end has a certain weight and elasticity, but not enough to cause mechanical resonance.

In some scenarios, you can use a software oscilloscope to give the end a sudden disturbance with the motor shaft loose, then observe the frequency of position or speed fluctuations. Set the servo to the OFF state, open the oscilloscope function in the debugging software, set the acquisition channel data to "Actual Position (6063h)", manually push the load end in the direction of vibration, and then observe the frequency of the vibration waveform via the oscilloscope function. As shown in the figure on the lower right, the waveform frequency is approximately 14.4Hz.



$$\text{End Vibration Frequency} = \frac{1}{\text{Waveform Period Average}}, \text{Waveform Period Average} = \frac{\text{Time}}{\text{Period Count}}$$

Waveform Period Average = $347750 \div 5 \div 1000000 = 0.06955$ seconds

(※ **Note:** microseconds need to be converted to seconds, so divide by 1000000)

End Vibration Frequency = $1 \div 0.06955 \approx 14.4\text{Hz}$

In scenarios where it cannot be observed with the software oscilloscope, users need to use other means such as cameras or accelerometers to measure the vibration frequency. The trial and error method can also be used to gradually test the actual suppression effect of various frequencies.

Input Shaping to Suppress End Vibration

By setting the input shaping related filters, end vibration can be suppressed. Input shaping filters can suppress relevant frequencies in the command signal, but their Validationness varies greatly on different equipment. Please refer to Section 6.3.2 Input Shaping for details

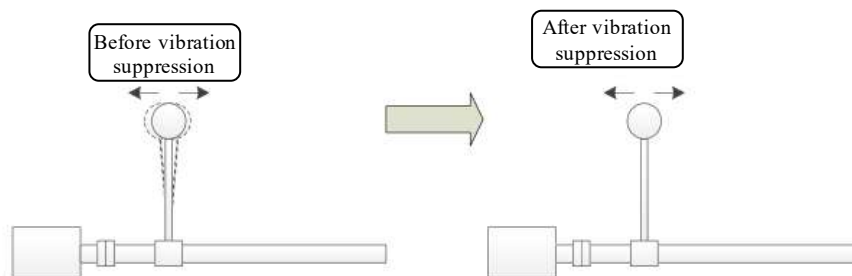
End Vibration Suppression Filter

The drive has two built-in end vibration suppression filters. When using them, the suppression frequency and suppression intensity need to be entered and fine-tuned based on the execution effect.

No.	Index	Parameter Name	Setting/Display Value	Default Value	Unit	Mark	Validation
Pr1.38	4010h-64h	Vibration Suppression Function Setting	0: Not enabled 1: Enable Filter A 2: Enable Filter B 3: Enable Filter AB	0		RW	Immediate
Pr1.39	4010h-65h	Vibration Suppression Frequency A	≤100	100	Hz	RW	Immediate
Pr1.40	4010h-66h	Vibration Suppression Intensity A	1-30	10		RW	Immediate
Pr1.41	4010h-67h	Vibration Suppression Frequency B	≤100	100	Hz	RW	Immediate
Pr1.42	4010h-68h	Vibration Suppression Intensity B	1-30	10		RW	Immediate

※ Caution

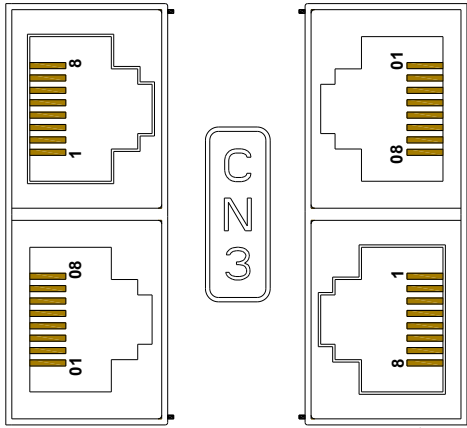
- End vibration suppression function parameters can be set via the panel or debugging software.
- "Vibration Suppression Frequency" is the vibration frequency at the load end when vibration occurs, and the measured end vibration frequency (in 0.1Hz units) is directly written to this parameter
- "Vibration Suppression Intensity" determines the attenuation capability of the "Vibration Suppression Frequency" component in the command, and both must be used together. Excessive settings can cause command distortion.



Chapter 7 Communication

7.1 Interface and Transmission Method

CN3 is the connection terminal for communication functions, consisting of two identical RJ45 interfaces connected in parallel internally

Terminal Position Number CN3		Pin	Signal	Description
	1/01	NC	-	
	2/02	NC	-	
	3/03	NC	-	
	4/04	RS+	RS485 Differential Input Signal Port	
	5/05	RS-		
	6/06	GND	Signal Ground	
	7/07	CANH	CAN Differential Input Signal Port (CD200C series supported)	
	8/08	CANL		
		PE	Casing	-

The recommended RS485 bus topology for CD200 is a linear daisy-chain structure, with all devices connected in series on the main cable (branch lines ≤ 0.5 meters).

Note that **120 Ω termination resistors must be connected at both ends of the bus** to match characteristic impedance and reduce signal reflection; **do not use termination resistors in the middle of the network.**

Shielded twisted-pair cables (single-point grounding) must be used. Baud rate is inversely proportional to length (e.g., 9600bps can reach 1200 meters). The number of nodes should not exceed 32 standard loads. Avoid star topologies (cause reflection); if necessary, extend distance via repeaters or convert topology via hubs.

CD200 supports Modbus RTU transmission mode and its unique TLink protocol (used for digital pulses), and users can configure serial communication parameters (baud rate, parity, etc.).

Baud rate setting range: 4800 bps, 9600 bps, 19200 bps, 38400 bps, 57600 bps, 115200 bps.

CD200C supports the standard CANopen communication protocol, and all parameters of the driver can be accessed via the object dictionary.

The CD200C supports the following message types (communication objects): SDO, PDO, NMT, SYNC, and EMCY.

The CANopen communication baud rate can be set within the range of 50 Kbps, 100 Kbps, 125 Kbps, 250 Kbps, 500 Kbps (default), 800 Kbps, and 1 Mbps.

The transmission distance of the CAN bus is directly related to the baud rate and the communication cable. The recommended relationship between transmission distance and baud rate is shown in the table below.

Baud Rate(bps)	Recommended transmission distance (m)
1M	25
800K	50
500K	100
250K	250
125K	500

7.2 Modbus RTU Communication

Modbus is a serial communication protocol widely used in industrial automation, introduced by Modicon in 1979. It employs a master-slave architecture and supports three transmission modes: RTU (binary), ASCII (text), and TCP/IP (Ethernet). It enables data reading and writing between devices (e.g., servo drives, PLCs) through a simple data frame structure (e.g., function code, register address), characterized by its openness, lightness, and strong compatibility, making it one of the common standards for industrial control systems.

7.2.1 Modbus Communication Data Frame Structure

A Modbus message is a data frame sent by a master to a slave, containing slave address, master command, checksum, and other content. The message format for Modbus protocol over a serial link is as follows:

Interframe Gap	Slave Address	Function Code	Data 1	Data 2		Data n	CRCL	CRCH	Interframe Gap
≥3.5 bytes Transmission Duration	1 byte	1 byte	n bytes, Max 252bytes				2 bytes		≥3.5 bytes Transmission Duration

Modbus RTU frame meanings are shown in the table below:

Name	Description	
Slave Address	Modbus addressing space has 128 different addresses. Address 0 is reserved as the broadcast address; no response is made to broadcast commands. Address 127 can be used for communication when the address is unknown. Broadcast commands support 0x06, 0x10 function codes. It is recommended that broadcast commands only be used for servo enable control Modbus master nodes do not have an address; only slave nodes must have one. This address must be unique on the Modbus serial bus. Slave address setting range is 1-127	
Function Code	Normal send/receive: 03H: Read 16/32-bit parameters 06H: Write 16-bit parameters 10H: Write 32-bit parameters	Abnormal return: 83H: Read parameter value abnormal 86H: Write 16-bit parameter abnormal 90H: Write 32-bit parameter abnormal
Data	Variable length, varying with function code, includes parameters or data for request/response (such as register addresses, quantities), with the high bytes preceding the low bytes	
CRC Check	Cyclic Redundancy Check (CRC). CD200 servo drives use 16-bit CRC check, with low byte first and high byte last	
Silent Interval	A silent interval of at least 3.5-character times (calculated based on baud rate) must be maintained between frames to distinguish continuous frames, but the minimum interval must not be less than 0.5 ms	

7.2.2 Drive Parameter Read/Write

Parameters of CD200 series servo drives are divided into 8-bit, 16-bit, and 32-bit by data length, supporting function codes 03H (read one or more consecutive registers), 06H (write single register), and 10H (write multiple registers). Some parameters cannot be rewritten while the drive is running; an error will be returned if a communication rewrite command is input at this time. 32-bit addresses must be read/written as high and low 16 bits at once; reading/writing only high or low 16 bits is not allowed, meaning only 03H command can be used to read 32 bits, and 10H command to write 32 bits. When writing parameters via communication, generally only the value in memory is modified, not written to EEPROM. If the modified parameters need to be saved after power-off, a parameter save operation must be performed. See 5.1.5 Parameter Saving for details.

(1) 03H: Communication Read Drive Parameters

To read CD200 drive parameters via communication, the send message frame structure is as follows:

Slave Address	Modbus Command	Parameter Address High 8 Bits	Parameter Address Low 8 Bits	Parameter Type High 8 Bits	Parameter Type Low 8 Bits	CRC Check	
1-127	03	1*	-	-	2*	CRCL	CRCH

Note:

1*: Parameter Address Bit14, Bit15 constitute the axis number for a multi-axis system. For a single-axis drive, it defaults to 0. Other bits are the parameter address; when accessing, it should be concatenated according to the parameter address list and axis number.

Bit15	Bit14	Axis No.
0	0	0
0	1	1
1	0	2
1	1	3

2*: Parameter Type:

0x1: Upper computer reads 8-bit or 16-bit parameters from the drive using this type

0x2: Upper computer reads 32-bit parameters from the drive using this type

0x4: Upper computer reads 64-bit parameters from the drive using this type

The parameter address is uniquely bound to the accessible parameter object provided by the drive, while the parameter type serves only as data to verify whether the parameter accessed by the upper computer matches the parameter type corresponding to that address in the drive.

Response frame message is as follows:

Slave Address	Modbus Command	Number of Data Bytes	N Data	CRC Check Low 8 Bits	CRC Check High 8 Bits
1-247	03	1*	2*	CRCL	CRCH

Including:

1*: Number of data bytes

Depending on the parameter type corresponding to the address accessed by the upper computer, the number of data bytes can be 2, 4, or 8. Where 2 corresponds to single-byte and double-byte parameters.

2*: N Data:

Data type corresponding to parameter address	Source data (high byte on left)	Data transmission order (left first)
8-bit parameter	byte0	00 byte0
16-bit parameter	byte1 byte0	byte1 byte0
32-bit parameter	byte3 byte2 byte1 byte0	byte1 byte0 byte3 byte2
64-bit parameter	byte7 byte6 byte5 byte4 byte3 byte2 byte1 byte0	byte1 byte0 byte3 byte2 byte5 byte4 byte7 byte6

Drive abnormal return:

Slave Address	Modbus Command	Error Code	CRC Check Low 8 Bits	CRC Check High 8 Bits
1-247	83	1*	CRCL	CRCH

1* Error Code:

Error Code	Description
01	Data protected by user password was read
02	Upper computer's parameter address error Upper computer's parameter type does not match the drive parameter type bound to the parameter address Length of the command frame sent by the upper computer is incorrect

a. Read the current operating mode of servo drive with station ID 01H (address 0x0498, type Int 8). Commands and response information are as follows:

Host send request frame:

Slave Address	Modbus Command	Parameter Address		Parameter Type High 8 Bits	Parameter Type Low 8 Bits	CRC Check	
		High 8 Bits	Low 8 Bits				
01	03	04	98	00	01	CRCL	CRCH

Slave response frame:

Slave Address	Modbus Command	Number of Data Bytes	Return Data High 8 Bits	Return Data Low 8 Bits	CRC Check Low 8 Bits	CRC Check High 8 Bits
01	03	02	00	01	CRCL	CRCH

This response frame indicates: the slave returns 2 bytes of data, the data content is 0x0001 (current operating mode is 1)

- b. Read the status word of servo drive with station ID 01H (address 0x0488, type U 16). Commands and response information are as follows:

Host send request frame:

Slave Address	Modbus Command	Parameter Address		Parameter Type High 8 Bits	Parameter Type Low 8 Bits	CRC Check	
		High 8 Bits	Low 8 Bits				
01	03	04	88	00	01	CRCL	CRCH

Slave response frame:

Slave Address	Modbus Command	Number of Data Bytes	Return Data High 8 Bits	Return Data Low 8 Bits	CRC Check Low 8 Bits	CRC Check High 8 Bits
01	03	02	12	21	CRCL	CRCH

This response frame indicates: the slave returns 2 bytes of data, the data content is 0x1221 (current status word 0x1221)

- c. Read the feed amount of servo drive with station ID 01H (address 0x0558, type Int 32). Commands and response information are as follows:

Host send request frame:

Slave Address	Modbus Command	Parameter Address		Parameter Type High 8 Bits	Parameter Type Low 8 Bits	CRC Check	
		High 8 Bits	Low 8 Bits				
01	03	05	58	00	02	CRCL	CRCH

Slave response frame:

Slave Address	Modbus Command	Number of Bytes Returned	Return Data Byte1	Return Data Byte0	Return Data Byte3	Return Data Byte2	CRC Check	
01	03	04	27	10	00	00	CRCL	CRCH

This response frame indicates: the slave returns 4 bytes of data, the data content is 0x00002710 (feed amount set value 10000)

(2) 06H: Communication Write 8/16-bit Parameters

This function code is used to modify 8-bit or 16-bit parameters inside the drive. The send message frame structure is as follows:

Slave Address	Modbus Command	Parameter Address High 8 Bits	Parameter Address Low 8 Bits	Write Parameter Value		CRC Check	
				High 8 Bits	Low 8 Bits		
1-247	06	Same as "1 * Parameter Address" above		1 *		CRCL	CRCH

Drive normal return:

The drive returns a response exactly identical to the received command

1*: Write parameter value:

If the drive parameter corresponding to the parameter address is 8-bit data, only the lower 8 bits of the parameter value will be written to the drive; the parameter value H is ignored;

If the drive parameter corresponding to the parameter address is 16-bit data, these two bytes will be reorganized as high and low bytes, then written to the drive.

Drive abnormal return:

Slave Address	Modbus Command	Error Code	CRC Check Low 8 Bits	CRC Check High 8 Bits
1-127	86	1*	CRCL	CRCH

1* Error Code:

Error Code	Description
01	Parameters protected by factory password were written Parameters protected by user password were written
02	Read-only parameters or invalid address were written Attempted to write drive parameters other than 8-bit or 16-bit using 06H command Length of the command frame sent by the upper computer is incorrect

- a. Change the operating mode of servo drive with station ID 01H (address 0x0490, type Int 8) to 3. Commands and response information are as follows:

Host send request frame:

Slave Address	Modbus Command	Parameter Address High 8 Bits	Parameter Address Low 8 Bits	Write Data High 8 Bits	Write Data Low 8 Bits	CRC Check	
01	06	04	90	00	03	CRCL	CRCH

Slave response frame:

Slave Address	Modbus Command	Parameter Address High 8 Bits	Parameter Address Low 8 Bits	Write Data High 8 Bits	Write Data Low 8 Bits	CRC Check	
01	06	04	90	00	03	CRCL	CRCH

This response frame indicates: write the slave drive parameter operating mode setting 0x0490 into data 0x0003 (operating mode changed to 3)

- b. Change the target torque of servo drive with station ID 01H (address 0x04F0, type U 16) to 1000. Commands and response information are as follows:

Host send request frame:

Slave Address	Modbus Command	Parameter Address		Parameter Type High 8 Bits	Parameter Type Low 8 Bits	CRC Check	
		High 8 Bits	Low 8 Bits				
01	06	04	F0	03	E8	CRCL	CRCH

Slave response frame:

Slave Address	Modbus Command	Parameter Address High 8 Bits	Parameter Address Low 8 Bits	Parameter Type High 8 Bits	Parameter Type Low 8 Bits	CRC Check Low 8 Bits	CRC Check High 8 Bits
01	06	04	F0	03	E8	CRCL	CRCH

This response frame indicates: write the slave drive parameter target torque 0x04F0 into data 0x03F0 (target torque 1000%)

(3) 10H: Communication Write 32-bit Parameters

This function code is used to modify 32-bit parameters inside the drive. The send message frame structure is as follows:

Slave Address	Modbus Command	Parameter Address		Number of Registers		Number of Bytes	4-byte Data	CRC Check	
		High 8 Bits	Low 8 Bits	High 8 Bits	Low 8 Bits				
1-127	10	Same as "1* Parameter Address" above		00	02	04	1*	CRCL	CRCH

Drive normal return:

Slave Address	Modbus Command	Parameter Address		Number of Registers		CRC Check	
		High 8 Bits	Low 8 Bits	High 8 Bits	Low 8 Bits		
1-127	10	Same as "1* Parameter Address" above		00	02	CRCL	CRCH

1*: 4-byte Data:

Data type corresponding to parameter address	Source data (high byte on left)	Data transmission order (left first)
32-bit parameter	byte3 byte2 byte1 byte0	byte1 byte0 byte3 byte2

Drive abnormal return:

Slave Address	Modbus Command	Error Code	CRC Check Low 8 Bits	CRC Check High 8 Bits
1-127	90	1*	CRCL	CRCH

1* Error Code:

Error Code	Description
01	Parameters protected by factory password were written Parameters protected by user password were written
02	Read-only parameters or invalid address were written Attempted to write drive parameters other than 32-bit using 10H command Length of the command frame sent by the upper computer is incorrect

a. Change the feed amount of servo drive with station ID 01H (address 0x0558, type U 32) to 131072. Commands and response information are as follows:

Host send request frame:

Slave Address	Modbus Command	Parameter Address		Number of Registers		Number of Bytes	4-byte Data				CRC Check	
		High	Low	High	Low						CRCL	CRCH
01	10	05	58	00	02	04	00	00	00	02	CRCL	CRCH

Slave response frame:

Slave Address	Modbus Command	Parameter Address High 8 Bits	Parameter Address Low 8 Bits	Number of Registers High 8 Bits	Number of Registers Low 8 Bits	CRC Check Low 8 Bits	CRC Check High 8 Bits
01	06	05	58	00	02	CRCL	CRCH

This response frame indicates: write the slave drive parameter feed amount 0x0558 into data 0x00020000 (feed amount 131072)

7.3 TLink Protocol (for digital pulse use)

Traditional pulse control (e.g., differential pulse CW/CCW, PUL/DIR) is limited by hardware receiving circuits, having an upper frequency limit ceiling (usually within 1MHz) and being susceptible to interference, thus limiting performance in high-speed, high-precision control scenarios.

The **TLink bus** uses **all-digital command transmission**, breaking through the hardware receiving limitations of traditional pulse interfaces, achieving **higher frequencies, stronger anti-interference, and simpler wiring**, providing an ideal solution for high-speed, high-precision motion control.

Simphoenix TLink connection uses a high-speed RS485 interface to achieve all-digital pulse command transmission at extremely high baud rates (above 2M), unidirectionally transmitting control signals and pulse command data from the controller, and supports a master-multiple-slave control architecture, as shown below.

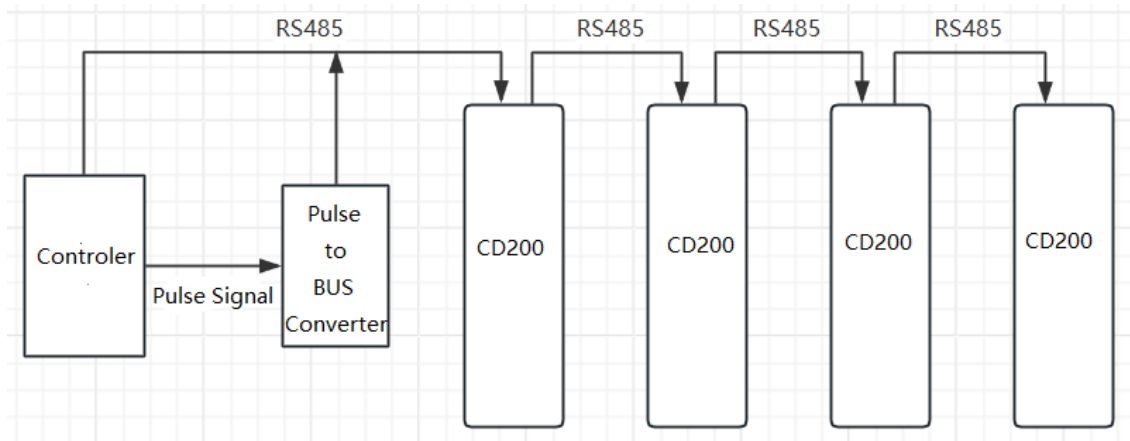


Figure 7-1 TLink Communication Architecture

TLink protocol control has two implementation methods:

- 1、 The motion controller directly outputs control data, which reaches the servo drive via high-speed RS485 to control each servo drive.
- 2、 The pulse signals output by the motion controller are transmitted to a digital pulse bus converter, which converts them into digital signals for transmission.

It should be noted that before enabling the drive, communication must first be established with the motion controller or pulse bus converter, and the communication cycle must be 62.5us

7.3.1 TLink Communication Data Frame Structure

A TLink message is a data frame sent by the master to the slave, containing control commands, DI control commands, pulse commands, checksum, and other content

The message format on the serial link is as follows:

Dual-Axis Control

CMD(0x66)	DIN A	PUL 1	PUL 2	CRC
1 byte	1 byte	2 bytes	2 bytes	1 byte

4-Axis Control

CMD(0x88)	DIN A	PUL 1	PUL 2	DIN B	PUL 3	PUL 4	CRC
1 byte	1 byte	2 bytes	2 bytes	1 byte	2 bytes	2 bytes	1 byte

The meaning of a TLink frame is shown in the table below:

Name	Description
CMD (0x66)	The communication data frame is used for 2-axis pulse control (each channel can be extended to control multiple axes with the same motion trajectory), and the data frame length is 7 bytes
CMD (0x88)	The communication data frame is used for 4-axis pulse control (each channel can be extended to control multiple axes with the same motion trajectory), and the data frame length is 11 bytes
DIN A	The lower 4 bits correspond to the digital input status of Axis 1, and the higher 4 bits correspond to the digital input status of Axis 2
DIN B	The lower 4 bits correspond to the digital input status of Axis 3, and the higher 4 bits correspond to the digital input status of Axis 4
PUL x x=1,2,3,4	Corresponding to 4 digital pulse signals respectively

7.4 CANopen Communication

CANopen is a high-level communication protocol and device specification architecture based on CAN (Controller Area Network), positioned at the application layer of the OSI reference model. It provides standardized communication mechanisms, device function descriptions, and system integration frameworks for real-time industrial control systems.

The CD200C servo drive is equipped with a built-in CANopen communication interface, and its communication system strictly adheres to the following international standards:

Based on the ISO 11898 CAN bus network.

CiA 301 V4.2.0 (CANopen application layer protocol): Defines the basic communication mechanisms shared among devices, including the object dictionary, service data objects (SDO), process data objects (PDO), and network management (NMT), etc. This serves as the critical foundation for device interoperability.

CiA 402 V2.0 (Drive and motion control device profile): A device profile specifically developed for servo drives and motion controllers. The CD200C supports the standard operating modes defined in this specification, such as profile position, profile velocity, profile torque, and homing mode, and incorporates a complete state control mechanism, ensuring compatibility with master devices from different manufacturers.

The CD200C supports the standard CAN 2.0A protocol, with communication baud rates flexibly configurable up to 1 Mbps, fully meeting the requirements for high real-time performance and reliability in industrial field applications.

7.4.1 CANopen Communication Data Frame

CAN transmits data between the host (controller) and bus nodes via data frames. The CD200C drive adopts the standard CAN 2.0A data frame format, supporting 11-bit standard identifiers. The figure below illustrates the structure of the data frame.

SOF	Identifier (ID)	RTR	IDE	R0	DLC	Data	CRC sequence	CRC Delimiter	ACK Slot	ACK Delimiter	EOF
1 bit	11 bit	1 bit	1 bit	1 bit	4 bit	0~8bytes	15 bit	1 bit	1 bit	1 bit	1 bit
Start of Frame	Arbitration Field		Control Field			Data Field	CRC Field		ACK Field		End of Frame

Communication Object Identifier (COB-ID)

In the CANopen protocol, the CAN-ID does not exist in isolation; rather, it serves as the core component of the "Communication Object Identifier" and is assigned clear protocol-layer meanings and strict functional divisions. Unlike the general CAN bus, where the identifier is used only for identification and arbitration, the CANopen protocol precisely defines the value space of the 11-bit standard CAN-ID, making it a direct indicator of communication functions. The core relationship is as follows:

COB-ID (Communication Object Identifier) = Function-defined CAN-ID + Control bits (e.g., RTR bit)

CANopen uses an 11-bit identifier, consisting of a 4-bit function code and a 7-bit node ID. These COB-IDs take effect after the NMT initialization state transitions to the configuration state. In broadcast mode, the Node-ID is zero. The composition of the COB-ID is shown below:

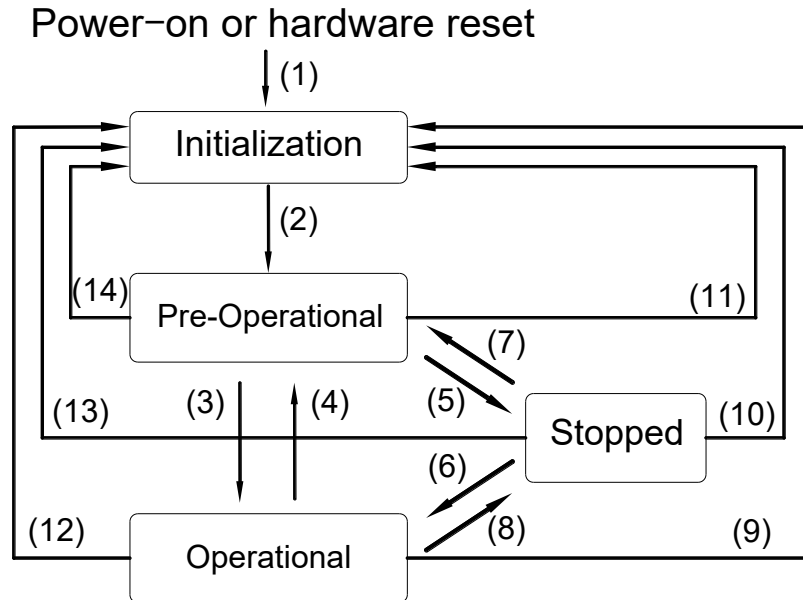
Function code				Node ID						
10	9	8	7	6	5	4	3	2	1	0
4 bit				7 bit						

The function code assignment is shown in the table below:

Function Code	Priority	Communication Object	Typical COB-ID
0000b	Highest	NMT	0x000
0001b	High	SYNC	0x080
0001b	High	EMCY	0x080+NodeID
0011b	High	TPDO1	0x180+NodeID
0100b	High	RPDO1	0x200+NodeID
0101b	Medium	TPDO2	0x280+NodeID
0110b	Medium	RPDO2	0x300+NodeID
0111b	Medium	TPDO3	0x380+NodeID
1000b	Medium	RPDO3	0x400+NodeID
1001b	Medium	TPDO4	0x480+NodeID
1010b	Medium	RPDO4	0x500+NodeID
1011b	Low	SDO response	0x580+NodeID
1100b	Low	SDO request	0x600+NodeID
1110b	Lowest	Heartbeat	0x700+NodeID

7.4.2 Network Management System (NMT)

The Network Management System is responsible for initializing, starting, and stopping the network and the devices within the network, following a master/slave architecture. In a CANopen network, there is one and only one NMT master, which can manage all network nodes, including itself. CANopen devices execute corresponding operations according to the state machine defined by the protocol. Some state transitions occur automatically internally, while others must be triggered by the NMT master sending NMT messages, as shown in the figure below.



Conditions for triggering state transitions are shown in the table below:

(1)	Automatically enters NMT initialization state upon power-up
(2)	NMT initialization complete – automatically enters NMT pre-operational state
(3)	NMT service start remote node command or local control
(4)、(7)	NMT service enter pre-operational state command
(5)、(8)	NMT service stop remote node command
(6)	NMT service start remote node command
(9)、(10)、(11)	NMT service reset node command
(12)、(13)、(14)	NMT service reset communication command

NMT State Description:

Initialization:

- Automatically entered upon CANopen power-up or hardware reset. In this state, the device:
- Loads parameters of each module
- Resets node: restores the manufacturer-defined area and sub-protocol area of the object dictionary to their last saved values
- Resets communication: restores communication parameters in the object dictionary to their last saved values
- After basic CANopen device initialization, it automatically enters the reset application sub-state.

Pre-operational:

- Automatically entered after the device sends the Boot-up message. In this state, the node configuration is primarily set; SDO communication is permitted, while PDO communication is disabled. A CANopen device can switch from this state to the operational state via the NMT start remote node service or through local control.

Operational:

- This is the normal operating state of CANopen, where all modules can function properly.

NMT Stopped:

- This state is used to stop all communication services (except node guarding and heartbeat, if already activated). Additionally, this state can be used to implement specific application behaviors.

The communication objects available in each state are shown in the table below:

Communication Object	Pre-operational	Operational	Stopped
NMT	Yes	Yes	Yes
Error Control	Yes	Yes	Yes
SYNC	Yes	Yes	No
EMCY	Yes	Yes	No
PDO	No	Yes	No
SDO	Yes	Yes	No

7.4.3 NMT Error Control

NMT error control is primarily used to detect whether devices on the network are online and their current status. It includes three mechanisms:

1. Node Guarding

- **Mechanism:** The NMT master periodically queries the status of NMT slaves via remote frames. The master sends a node guarding remote frame every $(0x100C \times 0x100D)$ ms, and the slave must respond; otherwise, the slave is considered offline.
- **Characteristics:** Master-slave model, where each remote frame must be acknowledged.
- **Related objects:**
 - 0x100C: Node guarding time (unit: ms)
 - 0x100D: Node guarding life-time factor

2. Heartbeat Protocol

- **Mechanism:** Uses a producer-consumer model, where nodes periodically send heartbeat messages.
- **Related objects:**
 - 0x1016: Heartbeat consumer setting
 - 0x1017: Heartbeat producer period

Notes:

- **Mutual exclusivity:** Node guarding and heartbeat protocols must not be used simultaneously.
- **Time settings:** The guarding time and heartbeat period should not be set too short to avoid increasing network load.
- **Recommended values:**
 - Guarding time (100Ch) ≥ 100 ms
 - Life-time factor (100Dh) ≥ 2
 - Heartbeat period (1017h) ≥ 500 ms

7.4.4 Service Data Object (SDO)

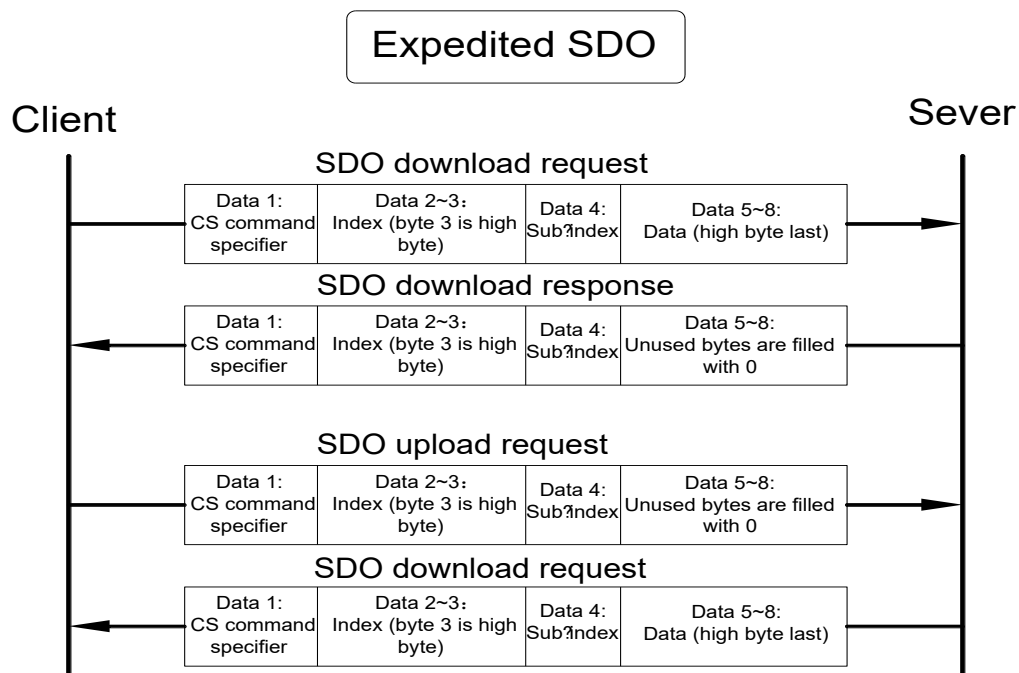
The Service Data Object (SDO) is primarily used for parameter configuration of slave nodes by the CANopen master. Service confirmation is the most distinctive feature of the SDO, as it generates an acknowledgment for each message, ensuring the accuracy of data transmission. If no acknowledgment is received within the timeout period, the sending node will retransmit the original message.

In a CANopen system, typically:

- CANopen slave nodes act as SDO servers.
- CANopen master nodes act as SDO clients.

Through indexes and sub-indexes, the SDO client can access the object dictionary on the SDO server. This allows the CANopen master node to access any parameter in the object dictionary of the slave node, and it is also often used for critical data transmission. For object data not exceeding 4 bytes, expedited SDO transfer can be used; for data exceeding 4 bytes, block or segmented transfer can be employed. The CD200C supports expedited SDO transfer.

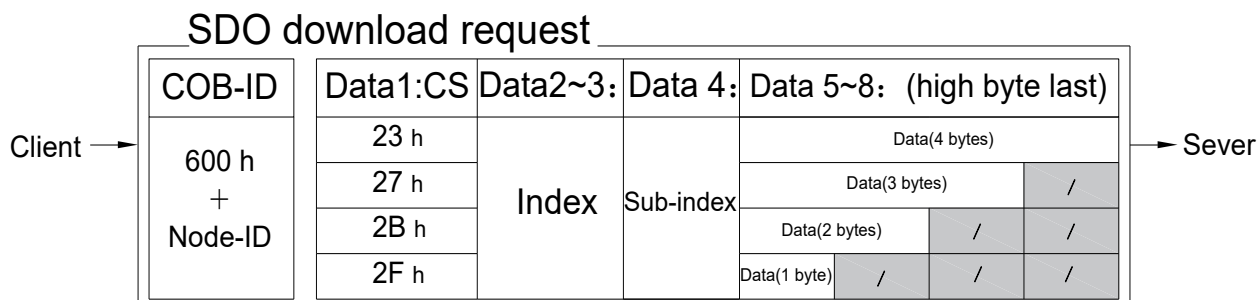
The expedited SDO transfer model is shown in the figure below.



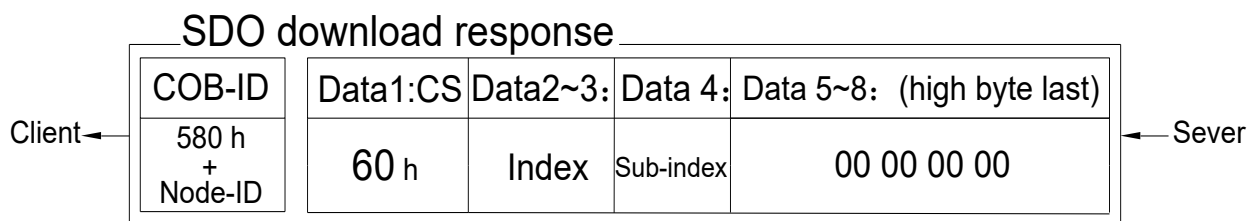
Command Specifier(CS)	
2Fh=write 1 byte	40h=read (request)
2Bh=write 2 bytes	4Fh=read 1 byte
27h=write 3 bytes	4Bh=read 2 bytes
23h=write 4 bytes	47h=read 3 bytes
60h=write successful response	43h=read 4 bytes
80h=abort/error response	

(1) SDO write drive parameters

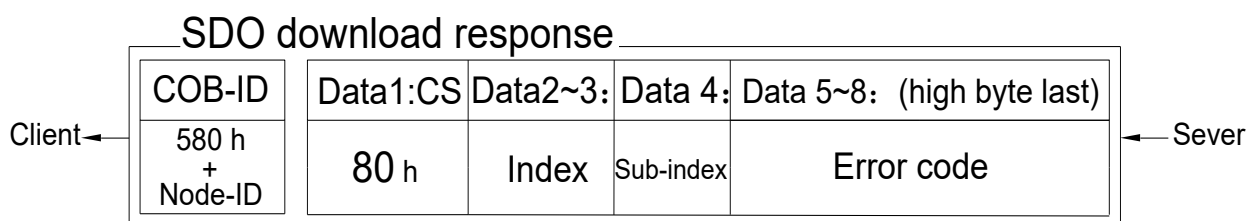
SDO write servo drive parameters. The request message format is as follows:



If the write operation is successful, the servo drive returns the following response message:

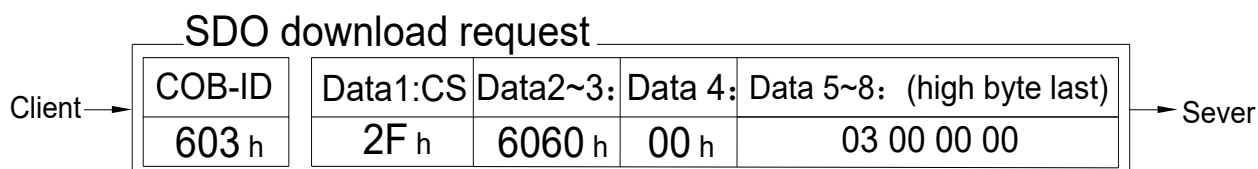


If the SDO write request fails, the command code is 80h, and bytes 5–8 contain the corresponding error code. The response message is as follows:

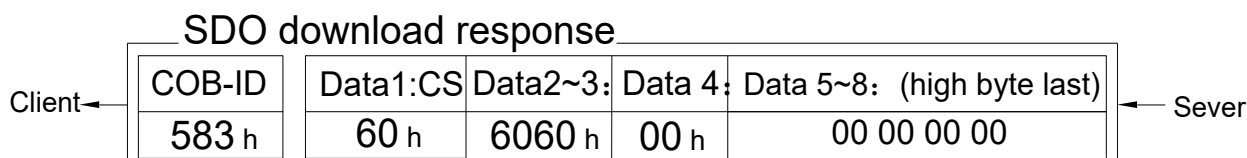


The following example illustrates SDO write drive parameters. In this example, the Node-ID is 3, and it demonstrates the request and response message formats for an SDO write of 8-bit data. All data in the example are in hexadecimal.

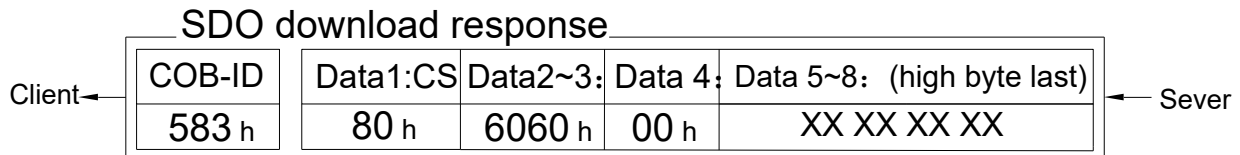
SDO sets the control mode (index 606000h in the object dictionary) to profile velocity mode (setting value 3). The data type of object 606000h is INT 8; therefore, among bytes 5 to 8, only byte 5 contains valid data. The request message is as follows:



If the write is successful, the response message is as follows:



If the write fails, bytes 5–8 contain the error code, and the response message is as follows:

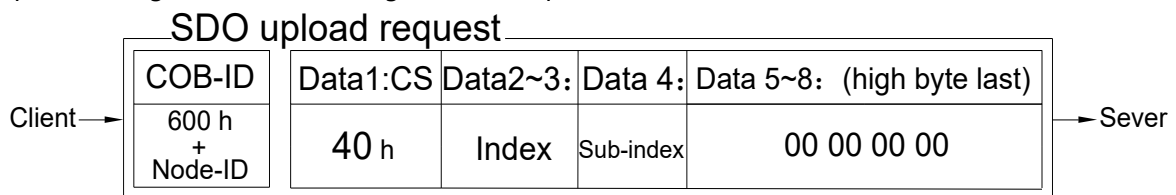


Among them, data bytes 5 to 8 contain the error code indicating the cause of the error. Some error codes are listed below:

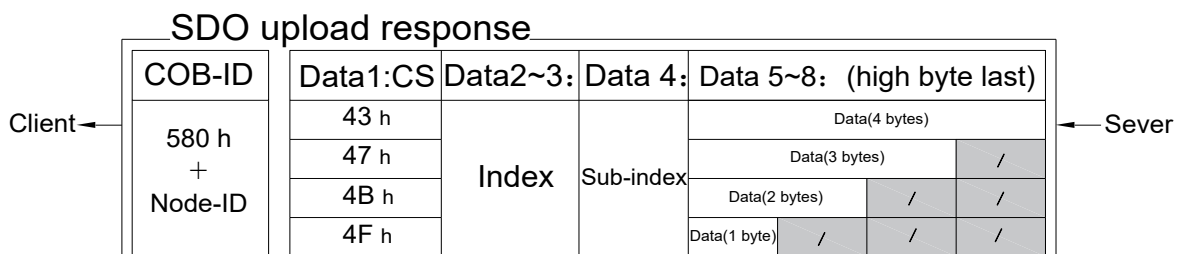
Error Code	Description
06010002h	Read-only entry: attempted to write to a constant or read-only object.
06020000h	Object does not exist: attempted to access an object that does not exist (incorrect index).
06040041h	Cannot map object to PDO: attempted to map an object that is not allowed in the PDO.
06040042h	Mapped PDO exceeds PDO: if the required object is appended to the PDO mapping, it will exceed the 8 bytes of the PDO mapping.
06070012h	Parameter length too long: attempted to write an object with too much data; for example, for an object of type Unsigned8, <CMD>= 23h (4 bytes) would be incorrect, while <CMD>= 2Fh would be correct.
06070013h	Parameter length too short: attempted to write an object with too little data; for example, for an object of type Unsigned32, <CMD>= 2Fh (1 byte) would be incorrect, while <CMD>= 23h would be correct.
06090011h	Sub-index does not exist: attempted to access an invalid sub-index of an object; the index itself, however, exists.
06090031h	Value too large: the value size of certain objects is restricted; in this case, an attempt was made to write a value that is too large to the object. For example, the object 1003h:00 "Pre-defined error field: number of errors" can only be set to value "0"; any other value will result in this error.
06090032h	Value too small: the value size of certain objects is restricted. In this case, an attempt was made to write a value that is too small for the object.
08000000h	General error: a general error that does not belong to any other category.
08000022h	Data cannot be read or stored in this state: PDO parameters can only be changed in the "Stopped" or "Pre-operational" state. Write access is not allowed in the "Operational" state for objects 1400h to 1407h, 1600h to 1607h, 1800h to 1807h, and 1A00h to 1A07h.

(2) SDO Read Drive Parameters

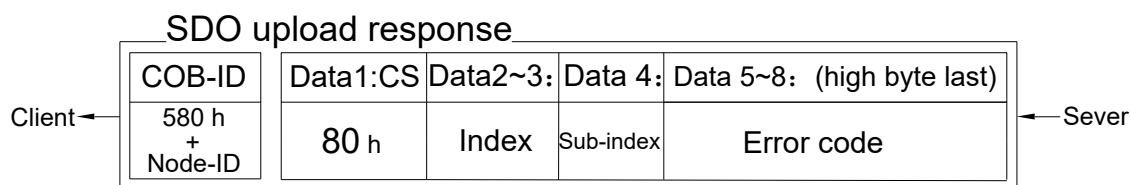
The request message format for reading servo drive parameters via SDO is shown below.



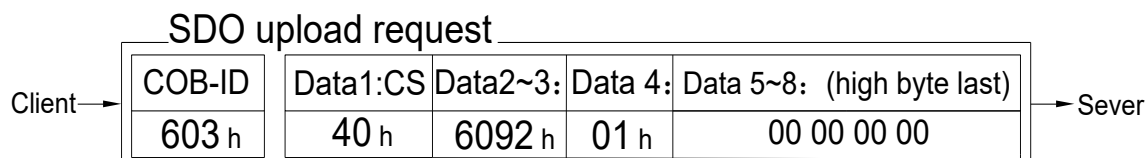
If the read operation is successful, the servo drive returns the following response message:



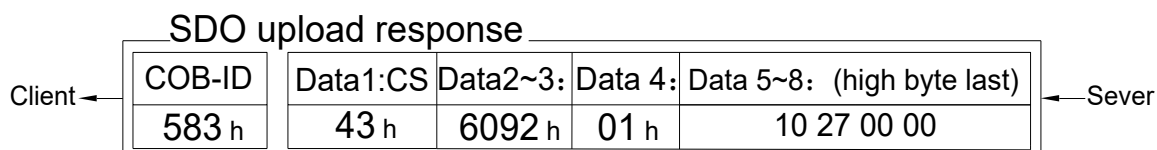
If the SDO read request fails, the returned command code is 80h, and bytes 5 to 8 contain the corresponding error code. The response message is as follows:



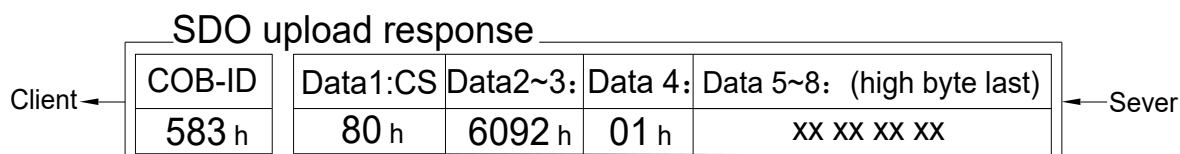
The following example illustrates SDO reading a value from a drive with Node-ID 3. The feed amount (pulses per revolution, with object dictionary index 609201h and data type U32) is read. The example shows the request and response message formats for an SDO read of 32-bit data. The request message is as follows:



If the read is successful, the response message is as follows: the returned value is 10000 (hexadecimal representation is 2710h).



If the read fails, bytes 5 to 8 of the data field contain the error code, and the response message is as follows:



7.4.5 Process Data Object (PDO)

The Process Data Object (PDO) is used for transmitting real-time data, providing direct access to the object dictionary of the servo drive. It is used to transmit short real-time data frames and has a high priority. PDO data must be less than or equal to 8 bytes, and every byte in the CAN message data field is used for data transmission. Therefore, no transmission control information is included at the application layer, resulting in extremely high message utilization. It is the primary data transmission method in CANopen.

Each PDO is described in the object dictionary by two objects: **the communication parameter object and the mapping parameter object**. The PDO communication parameters specify the COB-ID, transmission type, inhibit time, and event timer to be used. The PDO mapping parameters define the mapping relationship of the data in the PDO message, determining the positioning of the data to be transmitted within the CAN message data field. These parameters allow both the producer and consumer of the PDO to know what information is being transmitted, without adding extra protocol control information to the CAN message, thereby maximizing transmission efficiency.

Based on the distinction between reception and transmission, PDOs are classified as RPDO and TPDO. PDOs are ultimately determined in terms of transmission method and content by both communication parameters and mapping parameters. The CD200C series servo drives support PDO transmission using 4

RPDOs and 4 TPDOs. The related object list is as follows:

Name		COB-ID	Communication parameters	Mapping parameters
RPDO	1	200 h + Node-ID	1400 h	1600 h
	2	300 h + Node-ID	1401 h	1601 h
	3	400 h + Node-ID	1402 h	1602 h
	4	500 h + Node-ID	1403 h	1603 h
TPDO	1	180 h + Node-ID	1800 h	1A00 h
	2	280 h + Node-ID	1801 h	1A01 h
	3	380 h + Node-ID	1802 h	1A02 h
	4	480 h + Node-ID	1803 h	1A03 h

PDO Message Content

The content of PDO messages is predefined (or configured at network startup). The CD200C drive supports dynamic PDO mapping.

The PDO mapping procedure is as follows:

1. Disable PDO transmission – write 1 to the most significant bit (MSB) of the PDO COB-ID.
2. Clear the existing mapping entries – set the number of mapped objects to 0.
3. Configure the PDO mapping entries – modify the content of mapping parameter sub-indices 1 to 8 according to the mapping definitions to map the data.
4. Update the number of mapped objects to the actual count.
5. Configure the PDO communication parameters.
6. Enable the PDO – write 0 to the MSB of the PDO COB-ID.

(1) PDO Communication Parameter Object

Sub-index 01h: PDO CAN identifier (COB-ID)

The PDO COB-ID is located at sub-index 01h of the communication parameters (RPDO: 1400h–1403h, TPDO: 1800h–1803h). The MSB determines whether the PDO is valid. The CD200C series servo drives support only point-to-point PDO transmission; therefore, the lower 7 bits of the COB-ID must be the node ID (station address) of the device. For example, for a node with station address 3, the COB-ID of TPDO1 in the disabled state should be 80000184h, and writing 184h to this COB-ID activates the PDO.

Sub-index 02h: PDO transmission type

The PDO transmission type is located at sub-index 02h of the communication parameters (RPDO: 1400h–1403h, TPDO: 1800h–1803h). This parameter determines which transmission mode the PDO follows. The specific correspondence is as shown below:

Transmission type value	Synchronous		Asynchronous
	Cyclic	Acyclic	
0	-	√	-
1~240	√	-	-
241~253	-		-
254/255			√

0: Synchronous cyclic transmission

Sent after receiving a SYNC message and when the inhibit time has elapsed. It is sent upon every SYNC or every n SYNCs (where n is defined by sub-index 06h SYNC Start Value, but is typically used for SYNC counter matching). More precisely, when 06h is 0, the PDO is sent after each SYNC; for cyclic synchronous, a value of 0 generally indicates transmission after every SYNC.

1–240: Synchronous cyclic transmission / periodic synchronous

Sent after receiving a SYNC message. The number n indicates transmission after every n SYNC messages. For example, a value of 1 means after each SYNC, a value of 2 means after every other SYNC.

254: Asynchronous, event-driven, manufacturer-specific

The PDO is sent when the value of one or more application objects mapped to this PDO changes and the inhibit time has elapsed.

255: Asynchronous, event-driven, device profile-specific

The PDO is sent when the specified event timer elapses. The sending behavior is affected by inhibit time and event timer, and the transmission is periodic.

Sub-index 03h: Inhibit time

Sets the minimum time interval between two consecutive PDO transmissions. The unit is 62.5 microseconds (μ s). This is used to prevent PDO messages from being sent too frequently due to rapidly changing data, thereby occupying excessive bus bandwidth. If set to 0, the inhibit time is disabled.

Sub-index 05h: Event timer

For event-driven transmission types, the event timer provides an additional periodic sending mechanism. Even if the mapped data has not changed, the PDO will be sent if the time set by the event timer has elapsed since the last transmission. The unit is 1 millisecond (ms).

(2) PDO Mapping Parameter Object

PDO mapping parameters are used to define the specific content of the data field in the transmitted Process Data Object message. It includes sub-index 00h to record the number of mapped objects, and multiple sub-indices for storing the actual mapping entries. Up to 8 objects can be mapped, forming data of up to 64 bits.

When configuring or modifying PDO mapping, the standard procedure recommended by CiA301 is as follows:

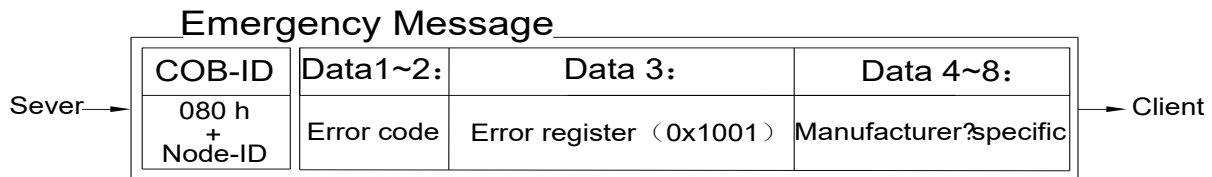
1. First, set the value of this sub-index to 0. This invalidates the PDO and allows mapping entries to be modified.
2. Then, configure or modify the specific mapping entries in sub-indices 01h to 08h.
3. Finally, set the value of this sub-index to the actual number of mapped objects to activate the new mapping.

Each sub-index stores a 32-bit value that defines which object, which sub-index, and the bit length of the data in the corresponding position of the PDO data field, coming from the drive's object dictionary. The structure of this 32-bit value follows the standard CiA format:

- **Bit 31 – Bit 16 (high 16 bits):**
Object Index. The index of the object to be mapped (e.g., 6064h for current position feedback value, 6069h for current speed feedback value).
- **Bit 15 – Bit 08 (middle 8 bits):**
Sub-index. The sub-index of the object to be mapped (e.g., 00h). If the object itself has no sub-index (i.e., it is a simple variable), this value is 00h.
- **Bit 07 – Bit 00 (low 8 bits):**
Data size (Length of object in bits). The number of bits that this object data occupies in the PDO message (e.g., 132 is 32 bits, corresponding to 20h; 116 is 16 bits, corresponding to 10h; BOOLEAN or UNSIGNED8 is 8 bits, corresponding to 08h).

7.4.6 Emergency Message (EMCY)

EMCY is a high-priority, asynchronous, event-triggered PDO message. When the device detects an error or fault condition internally, it immediately (subject to the inhibit time) broadcasts an EMCY message to the network to inform the master and other nodes that an exception has occurred. An emergency message contains 8 bytes, and the message format is as follows:



The Error Register is located in the device's object dictionary (index 0x1001). The following table describes the bit definitions of the error register.

bit0:generic error	Generic error. This bit becomes 1 whenever any error occurs
bit1:Current	Current-related fault
bit2:Voltage	Voltage-related fault
bit3:Temperature	Temperature fault
bit4	Communication timeout or communication error
bit5	Device profile-specific (CiA 402)
bit6	Reserved
bit7	Manufacturer-specific

7.5 Communication Related Parameters

No.	Address	Parameter Name	Setting/Help Information	Default Value	Type	Mark	Validation
Pr1.58	40109E	Disable CAN	0:Enable CAN Others:Disable CAN	255	U8	RWS	Restart
Pr3.06	0x0138	Communication ID	Drive communication ID. Used for 485, CAN, or 232 cascaded buses. The upper computer can set the ID to 127 to connect to drives with unknown communication IDs from 1 to 126	1	U8	RWS	Immediate
Pr3.10	0x0130	Serial Port 2 Baud Rate	Serial Port 2 is an RS485 bus interface	57600	U32	RWS	Restart
Pr3.11	-	Serial Port 2 Protocol	0: TLink Protocol 1: Modbus RTU	1	U8	RWS	Restart
Pr3.12	25000D	CAN baud rate	CAN bus baud rate setting 0: 1 Mbps 1: 800 Kbps 2: 500 Kbps 3: 250 Kbps 4: 125 Kbps 5: 100 Kbps 6: 50 Kbps 7: 20 Kbps 8: 10 Kbps 9: 5 Kbps	2	U8	RWS	Restart
Pr3.41	40100B	Pulse Mode	0: Pulse + Direction 1: CW + CCW 2: AB Phase Quadrature Pulse 3: Digital Pulse	0	U8	RWS	Immediate
Pr3.42	401085	High-speed Pulse Axis Number	Single-drive Multi-axis Control Axis Number Setting 0: Correspond to the 1st pulse input of the converter card	0	U8	RWS	Immediate

			1: Correspond to the 2nd pulse input of the converter card 2: Correspond to the 3rd pulse input of the converter card 3: Correspond to the 4th pulse input of the converter card				
Pr3.45	-	Serial Port 2 Parity Check and Stop Bit Settings	0: 1 Stop Bit, No Parity Check 1: 1 Stop Bit, Odd Parity Check 2: 1 Stop Bit, Even Parity Check 3: 2 Stop Bits, No Parity Check 4: 2 Stop Bits, Odd Parity Check 5: 2 Stop Bits, Even Parity Check	0	U8	RWS	Restart
-	0x0030	Input Pin Simulation	BIT0-BIT7 correspond to DI1-DI8, respectively 0: Invalid 1: Active	0	U16	RWS	Immediate
-	0x0038	Output Pin Simulation	BIT0-BIT3 correspond to DO1-DO4, respectively 0: Invalid 1: Active	0	U16	RWS	Immediate
-	0x0068	DC Bus Voltage	DC Bus Voltage Value	-	U16	RL	-
-	0x0080	System Temperature 1	Internal Temperature Value of Drive	-	I16	RL	-

Chapter 8 Servo Basic Functions

8.1 Electronic Gear

8.1.1 Electronic Gear Ratio

Electronic gear is a function that converts the displacement set by the "command unit" input from the upper computer into the number of pulses required for actual movement.

The "command unit" refers to the minimum displacement of the load. The command unit does not convert displacement into pulses but into units of physical quantities that are easy to understand, such as distance (e.g., μm or $^\circ$).

Through the electronic gear function, each pulse input to the servo unit corresponds to one command unit of workpiece movement. With the use of this function, the motor rotation and movement amount per command unit can be arbitrarily set.

The electronic gear ratio is calculated based on CiA402 defined objects 608Fh (Position encoder resolution), 6091h (Gear ratio), and 6092h (Feed amount).

The relationship between user-defined command units and internal units (pul) is calculated according to the following equation:

$$\text{Electronic Gear Ratio} = \frac{\text{Position encoder resolution} \times \text{Gear ratio}}{\text{Feed constant}}$$

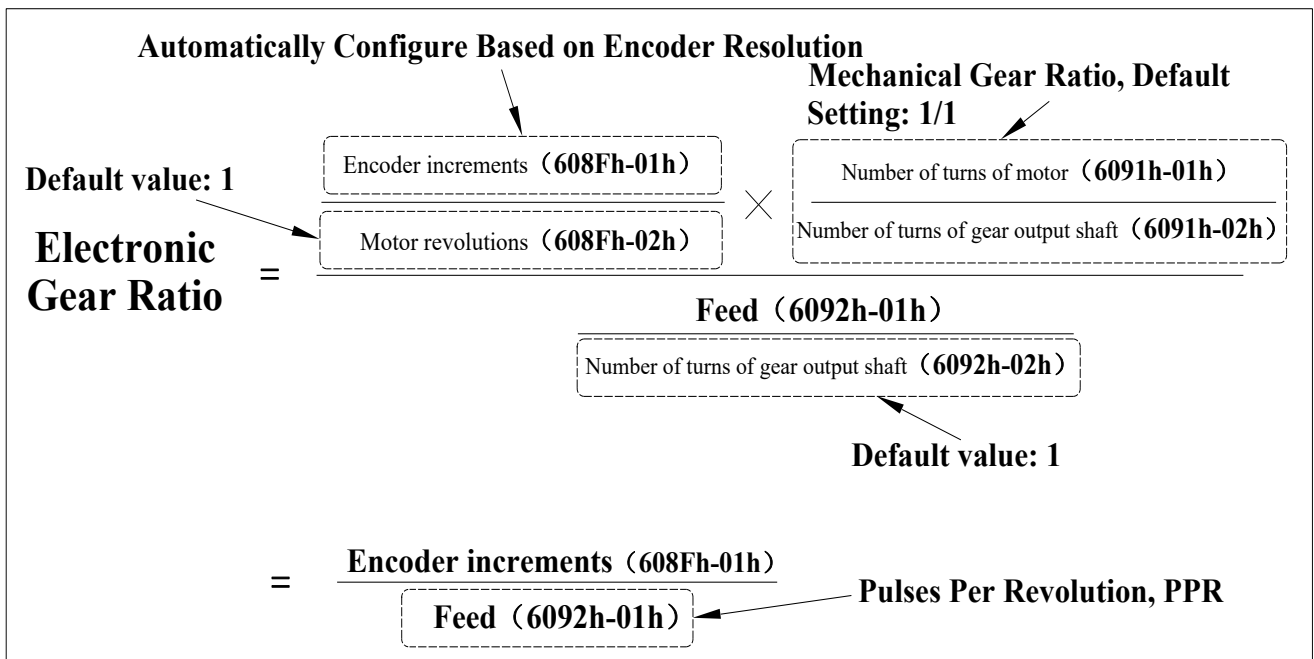
$$\text{Position demand value} \times \text{Electronic Gear Ratio} = \text{Position demand internal value}$$

Note:

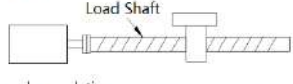
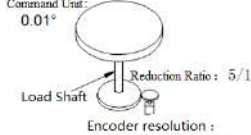
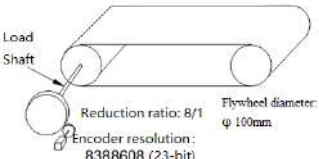
- The electronic gear ratio is Validation in the range of 8000 times to 1/1000 times.
If the set electronic gear ratio is outside this range and enabling is attempted, Er.FF09 (Abnormal Action Setting) will be triggered.
- 608Fh-01h (Encoder increments) is automatically set according to the encoder's resolution.
- 608Fh-02h (Motor revolutions) is fixed at 1.
- When modifying the electronic gear ratio, please disable first.

8.1.2 Common Electronic Gear Setting Methods

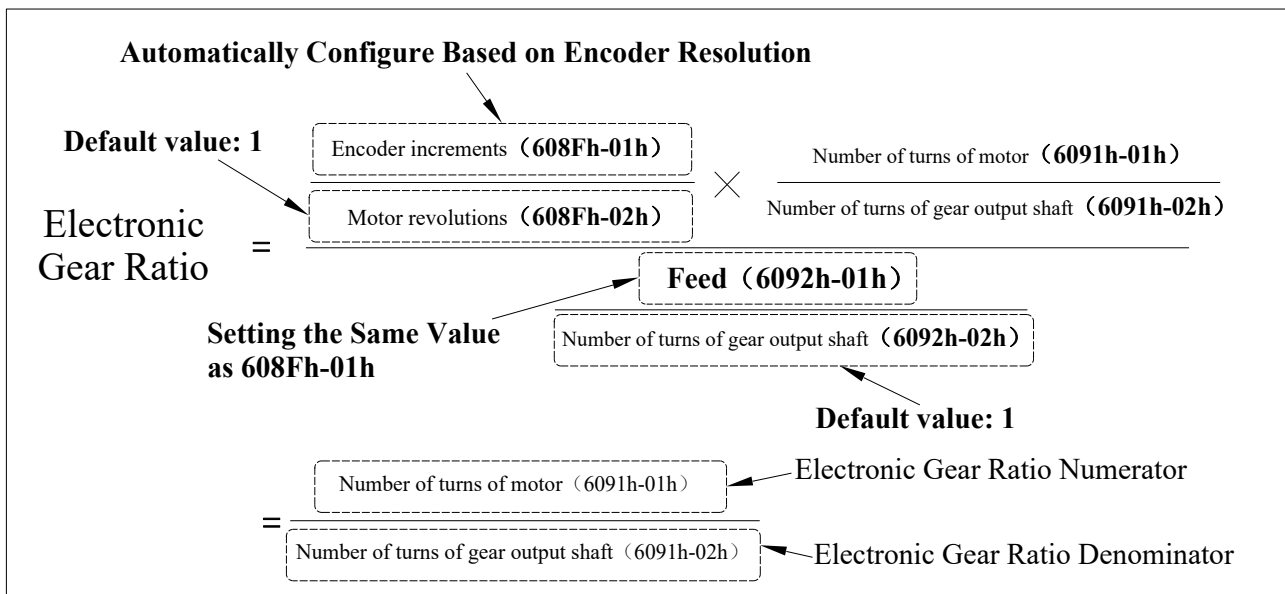
(1) Set the number of command pulses for one turn. The drive's default is this setting.



Examples of settings for three common mechanical systems are as follows:

Content	Mechanical System Configuration		
	Ball Screw	Round Table	Belt + Flywheel
	Command Unit: 0.001mm  Encoder resolution: 8388608 (23-bit) Ball Screw Lead: 6mm	Command Unit: 0.01°  Encoder resolution: 8388608 (23-bit) Reduction Ratio: 5/1	Command Unit: 0.005mm  Encoder resolution: 8388608 (23-bit) Reduction ratio: 8/1 Flywheel diameter: φ100mm
Mechanical Specification	Ball screw lead: 6mm Reduction Ratio: 1/1	1 turn angle: 360° Reduction Ratio: 5/1	Flywheel diameter: 100mm (circumference 314mm) Reduction ratio: 8/1
Encoder Resolution	8388608 (23-bit)	8388608 (23-bit)	8388608 (23-bit)
Command Unit	0.001mm (1μm)	0.01°	0.005mm (5μm)
Load Shaft Rotation 1 Turn Displacement	6mm/0.001mm=6000	360°/0.01°=36000	314mm/0.005mm=62800
Electronic Gear Ratio	$\frac{8388608}{6000} \times \frac{1}{1}$	$\frac{8388608}{36000} \times \frac{5}{1}$	$\frac{8388608}{62800} \times \frac{8}{1}$

(2) Set the electronic gear ratio numerator and denominator.



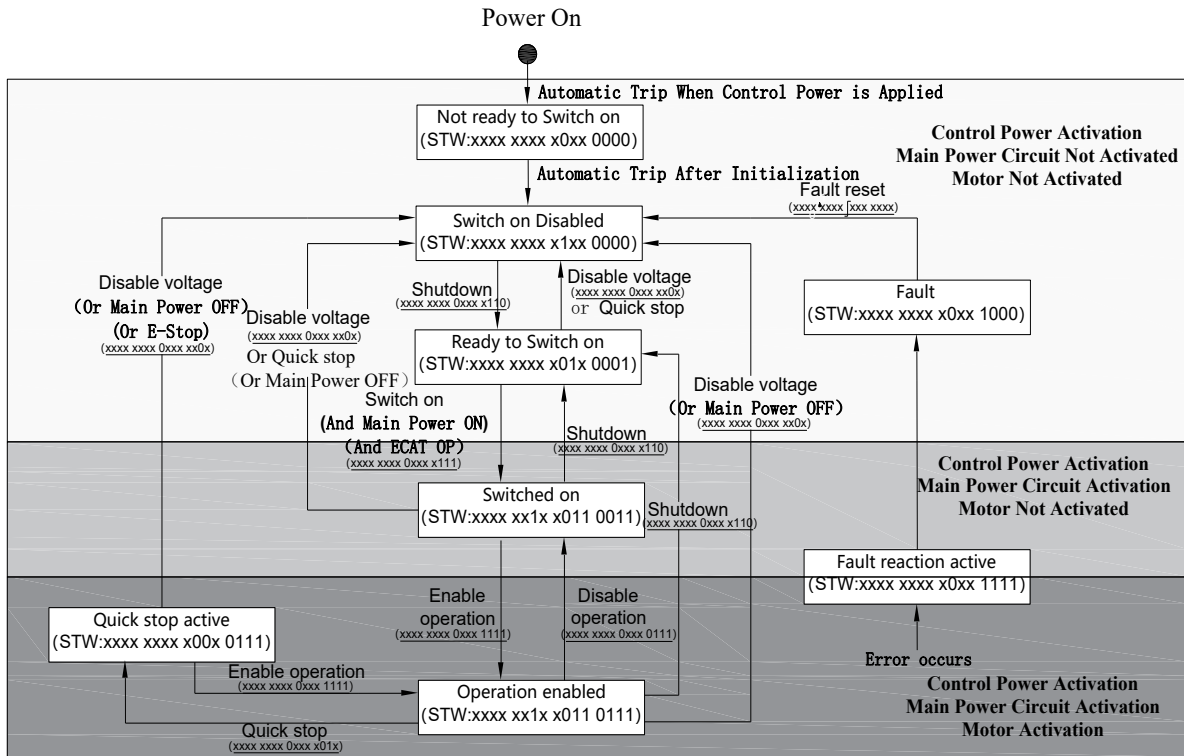
Related Parameters

No.	Index H	Address	Parameter Name	Setting/Help Information	Default Value	Unit	Type	Mark	Validation
Pr3.00	101001	0x0020	Storage	Refer to Pr1.00					
Pr3.01	609101	0x0548	Gear Input Turns	Motor shaft (gear input shaft) rotation turns Refer to "Motion Equivalent Setting"	1	r	U32	RWS	Immediate
Pr3.02	609102	0x0550	Gear Output Turns	Gear output shaft rotation turns Refer to "Motion Equivalent Setting"	1	r	U32	RWS	Immediate
Pr3.03	609201	0x0558	Feed Amount	Number of commands issued by the controller Refer to "Motion Equivalent Setting"	10000	PUL	U32	RWS	Immediate

8.2 Device Control

8.2.1 Power Drive Systems (PDS)

The power drive system (PDS) state transition for CD300 series EtherCAT bus servo drives, as specified by the CiA402 protocol, is shown below. The state transitions are controlled via Controlword (6040h), and the servo drive's device state is monitored using Statusword (6041h).



(Note) • □ indicates a state.

• STW refers to Statusword (6041h).

• ____ (underscore) is the control command for Controlword (6040h).

Each state is described in the table below:

PDS Status		Description
Not ready to Switch on	Initialization (incomplete state)	The device power is on, the drive is in initialization process and performing internal self-check, and the brake is activated. Drive parameters cannot be set, and drive functions cannot be executed.
Switch on Disabled	Initialization (complete state)	The servo drive is fault-free, or the error has been cleared. Drive parameters can be set using SDO communication service.
Ready to Switch on	Main circuit power OFF state	The servo drive is ready, and the motor is not energized. Drive parameters can be set.
Switched on	Servo Enable Off/ Servo Ready	The drive motor is in ready state, the main circuit power will be finally connected in this state, but drive functions cannot be executed. Drive parameters can be set.
Operation enabled	Servo Enable On	The drive is operating normally, a servo operating mode has been enabled, and the motor is energized. When the speed command is not 0, the motor rotates. Drive parameters with "Runtime Change" attribute can be set, others cannot.
Quick stop active	Quick Stop	The quick stop function is activated, and the drive is executing quick stop function. Drive parameters with "Runtime Change" attribute can be set, others cannot.
Fault reaction active	Fault (Alarm) Active	Drive has faulted, stopping according to set method. Motor is still enabled. Drive parameters with "Runtime Change" attribute can be set, others cannot.
Fault	Fault status	Fault Stop complete, all drive functions are inhibited, and drive parameters can be changed to troubleshoot the fault.

Commands controlling the servo drive PDS state transition are set via 6040h (Control Word)

Index	Sub-index	Name/Description	Access	Data Type	Ranges												
6040h	00h	Controlword	RW	U16	0-65535												
		Details of each bit of Controlword are as follows:															
		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
		Manufacture specific							h	fr	oms			eo	qs	ev	so
		Where: so = switch on fr = fault reset															
		ev= enable voltage h = halt															
		qs= quick stop oms = operation mode specific															
		eo= enable operation															
		PDS state transition is triggered by control commands formed by combining bits 7, 3-0 of Controlword															
		Command		Controlword (6040h) bits													
bit7	bit3			bit2	bit1	bit0											
Shutdown		0	X	1	1	0											
Switch on		0	0	1	1	1											
Switch on + Enable operation		0	1	1	1	1											
Disable voltage		0	X	X	0	X											
Quick stop		0	X	0	1	X											
Disable operation		0	0	1	1	1											
Enable operation		0	1	1	1	1											
Fault reset		0→1	X	X	X	X											
Note: X in the table means the bit can be ignored																	

Servo drive status confirmation is done via Statusword (6041h)

Index	Sub-index	Name/Description	Access	Data Type	Ranges												
6041h	00h	Statusword	RO	U16	0-65535												
		Details of each bit of Statusword are as follows:															
		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
		r		oms		ila	oms	rm	r	w	sod	qs	ve	f	oe	so	rtso
		Including:															
		r = reserved								w = warning							
oms = operation mode specific								qs = quick stop									
ila = internal limit active								ve = voltage enabled									
oe = operation enabled								rm = remote									
sod = switch on disabled								so = switched on									
rtso = ready to switch on								f = fault									

8.2.2 Stop Methods

The servo system's stop method can be set via the following objects:

- 605Ah (Quick Stop Option Code)
- 605Bh (Shutdown Option Code)
- 605Ch (Disable Operation Option Code)

Index	Sub-index	Name/Description	Access	Data Type	Ranges	Default Value
605Ah	00h	Quick stop Option Code	RW	INT16	0-15	0
		When the servo state machine executes the Quick stop command from the "Operation enabled" state, the drive will perform the stop operation according to the stop method defined by 605Ah.				
		Value	Description			
		0	Turn off motor excitation signal, motor stops by inertia			
		1	Stop with deceleration according to object 6084h, then remain in "Switch on Disabled" state			
		2	Stop with deceleration according to object 6085h, then remain in "Switch on Disabled" state			
		3/4	The motor stops with maximum current limit and remains in Switch on Disabled state			
		5/9	Stop with deceleration according to object 6084h, then remain in "Quick stop active" state			
		6/10	Stop with deceleration according to object 6085h, then remain in "Quick stop active" state			
		7/8	The motor stops immediately and remains in Quick stop active state			
		11/13	The motor stops with maximum current limit and remains in Quick stop active state			
		12	The motor stops with maximum current limit, reset integrator, and remains in Quick stop active state			
		14	Stop with deceleration according to object 6084h, then remain in "Switched on" state			
		15	Stop with deceleration according to object 6085h, then remain in "Ready to Switch on" state			
605Bh	00h	Shutdown Option Code	RW	INT16	0-1	0
		When the servo state machine executes the Shutdown command from the "Operation enabled" state, the servo will perform the stop operation according to the stop method defined by 605Bh.				
		Value	Description			
		0	Turn off motor excitation signal, motor stops by inertia			
		1	Stop with deceleration according to object 6084h, then remain in "Ready to Switched on" state			
605Ch	00h	Disable Operation Option Code	RW	INT16	0-1	0
		When the servo executes the Disable operation command, the servo will perform the stop operation according to the stop method defined by 605Ch.				
		Value	Description			
		0	Turn off motor excitation signal, motor stops by inertia			
		1	Stop with deceleration according to object 6084h, then remain in "Switched on" state			

8.3 Control Mode Setting

CD200 can determine supported control modes via object 6502h. Servo pre-operational control modes can be set via object 6060h, and the current servo operating mode can be viewed via object 6061h.

Index	Sub-index	Name/Description	Access	Data Type	Ranges	Default Value
6502h	00h	Supported Drive Modes	RO	U32	0-4294967295	0x000003AD
		Each bit definition is as follows:				
		bit	Modes of Operation			Supported or not
		0	Profile Position Mode (PP)			Yes
		1	Velocity Mode (VL)			No
		2	Profile Velocity Mode (PV)			Yes
		3	Profile Torque Mode (PT)			Yes
		5	Homing Mode (HM)			Yes
		6	Interpolated Position Mode (IP)			No
		7	Cyclic Synchronous Position Mode (CSP)			Only CD200C supported
		8	Cyclic Synchronous Velocity Mode (CSV)			Only CD200C supported
		9	Cyclic Synchronous Torque Mode (CST)			Only CD200C supported
		10-31	Reserved			

6060h	00h	Modes of Operation	RW	INT8	-128-127	1
		Set the servo's control mode; unsupported control modes are prohibited from setting.				
		Value	Modes of Operation			
		-4	Pulse Mode			
		0	No mode change/no mode assigned			
		1	Profile Position Mode (PP)			
		3	Profile Velocity Mode (PV)			
		4	Profile Torque Mode (PT)			
		6	Homing Mode (HM)			
		8	Cyclic Synchronous Position Mode (CSP)			
		9	Cyclic Synchronous Velocity Mode (CSV)			
		10	Cyclic Synchronous Torque Mode (CST)			
6061h	00h	Modes of Operation Display	RO	INT8	-128-127	0
		Current Operating Mode Display				
		Value	Modes of Operation Display			
		-4	Pulse Mode			
		0	No mode change/no mode assigned			
		1	Profile Position Mode (PP)			
		3	Profile Velocity Mode (PV)			
		4	Profile Torque Mode (PT)			
		6	Homing Mode (HM)			
		8	Cyclic Synchronous Position Mode (CSP)			
		9	Cyclic Synchronous Velocity Mode (CSV)			
		10	Cyclic Synchronous Torque Mode (CST)			

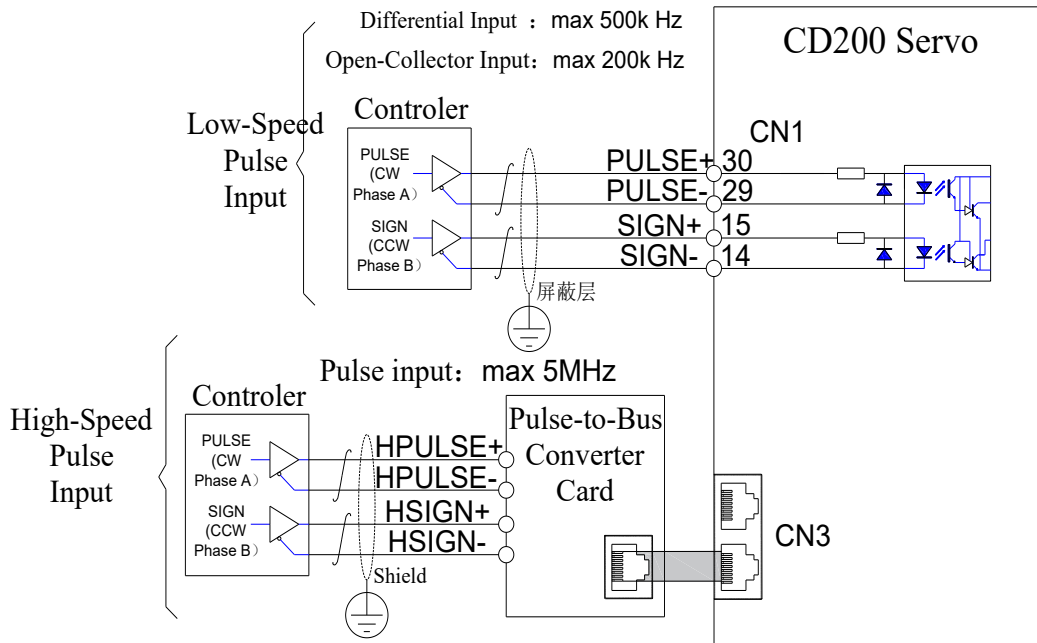
The operating mode can be switched via Modes of Operation (6060h). To prevent abrupt motor actions, change the operating mode when the motor speed is 0. When changing modes while the motor is running, the master must simultaneously update the selected operating mode and all operation mode specific process data objects. When the master selects a new operating mode, the servo unit will immediately switch to the new mode. The action when changing operating mode is shown below.

Operation Mode before Change	Action when Changing Operation Mode
Pulse Mode	The new operating mode will take effect immediately.
Profile Position Mode (Profile Position mode)	Controlword bit 4 = 0: Stop the motor in position control mode. Controlword bit 4 = 1: Immediately start new positioning.
Homing mode (Homing mode)	Controlword bit 4 = 0: Stop the motor in position control mode. Controlword bit 4 = 1: Immediately start origin return.
Profile Torque Mode	The new operating mode will take effect immediately.
Profile Velocity Mode	The new operating mode will take effect immediately.
Cyclic Synchronous Position Mode	The new operating mode will take effect immediately.
Cyclic Synchronous Velocity Mode	The new operating mode will take effect immediately.
Cyclic Synchronous Torque Mode	The new operating mode will take effect immediately.

8.3.1 Pulse Mode (PM)

Pulse mode is a position control mode where the command source is pulse input. Motor speed is determined by the pulse command frequency, and the motor's target position is determined by the total number of pulse commands.

Pulse commands can be given directly via external input, or via a digital pulse converter card which converts high-speed pulse signals into digital pulse signals before transmission, extending the maximum pulse reception frequency up to 5 MHz. To enable pulse mode, set object 6060h to **-4**.



(1) Pulse Command Source Selection

In pulse mode control, the servo unit's pulse input form is selected via parameter Pr3.41 Pulse Mode (object 40100B), as shown below:

Pr3.41	Pulse Command Type	Pulse Waveform
0	Pulse + Direction	
1	CW+CCW	
2	Quadrature Pulse (4x frequency)	
3	Two-channel unidirectional pulse bus	After conversion to digital pulse signals via a two-channel unidirectional pulse conversion card, the command is provided via communication
4	Four-channel bidirectional pulse bus	After conversion to digital pulse signals via a four-channel bidirectional pulse conversion card, the command is provided via communication
5	Two-channel bidirectional pulse bus	After conversion to digital pulse signals via a two-channel bidirectional pulse conversion card, the command is provided via communication

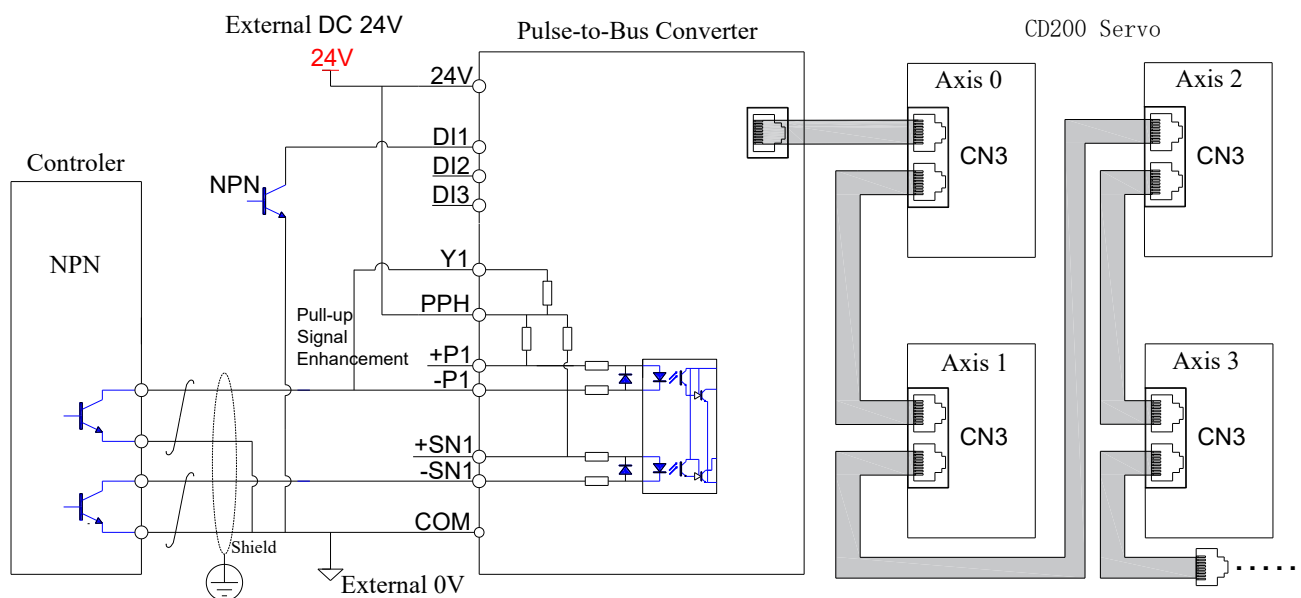
The command pulse input interface circuit is used to receive command pulse signals sent from the upper device to the drive. The output of the upper device can be differential line driver output or open collector output. Its maximum input frequency and minimum pulse width are shown in the table below:

Pulse Input Method	Maximum Single-channel Pulse Frequency (PPS)	Minimum Pulse Width (μ s)				Voltage Specification (V)
		T1	T2	T3	T4	
Differential Input	500k	1	1	2	0.5	5
Open Collector	200k	2.5	2.5	5	1.25	24

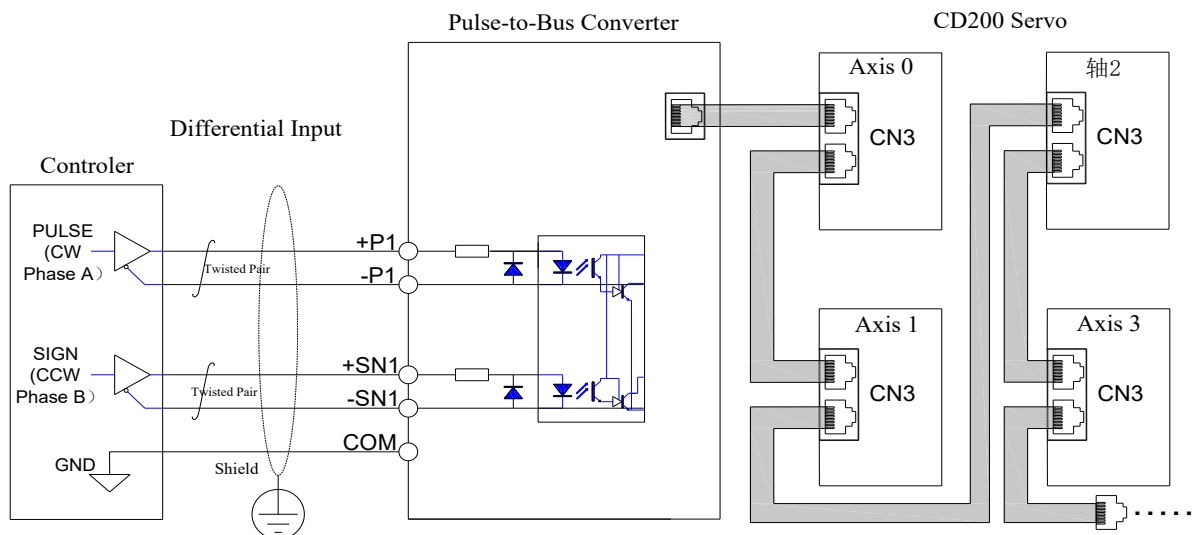
(2) Digital Pulse Input Extension Card

When selecting to configure a digital pulse extension card, set parameter Pr3.41 Pulse Mode to 3. Choose the corresponding model of digital pulse bus converter card based on the upper computer's input pulse format. The converter card and servo drive can be networked via Ethernet cable for communication; refer to Section 7.1 for networking methods. The extension card supports two pulse inputs (quadrature or open collector). Networked drives can select any input pulse signal based on their address. One input pulse signal can be extended to control multiple axes with the same motion trajectory. Each pulse input supports 3 DI digital signal inputs, which are converted by the extension card and communicated to the drive's DI1-DI3 terminals. For detailed usage, please refer to the digital pulse input extension card manual.

- a. Pulse Open Collector Input Converter Card: For other wiring methods, refer to Section 4.3.1



- b. Pulse Differential Input Extension Card:



(3) Related Objects

No.	Index H	Parameter Name	Setting/Help Information	Default Value	Unit	Type	Access	Validation
Pr3.10	250005	Serial Port 2 Baud Rate	Serial Port 2 is an RS485 bus interface	57600		U32	RWS	Restart
Pr3.11	250008	Serial Port 2 Protocol	0: Proprietary protocol 1: Modbus RTU	1		U8	RWS	Restart
Pr3.41	40100B	Pulse Mode	0: Pulse + Direction 1: CW + CCW 2: AB Phase Quadrature Pulse 3: Bus digital pulse	0	-	U8	RWS	Immediate
Pr3.42	401085	High-speed Pulse Axis Number	Single-drive Multi-axis Control Axis Number Setting 0: Correspond to the 1st pulse input of the converter card 1: Correspond to the 2nd pulse input of the converter card 2: Correspond to the 3rd pulse input of the converter card 3: Correspond to the 4th pulse input of the converter card	0	-	U8	RWS	Immediate
Pr3.43	25000B	Serial Port 2 Data Transmission Format	0: 1 Stop bit, No parity check 1: 1 Stop bit, Odd parity check 2: 1 Stop bit, Even parity check 3: 2 Stop bits, No parity check 4: 2 Stop bits, Odd parity check 5: 2 Stop bits, Even parity check	0	-	U8	RWS	Restart
Pr3.44	401088	Serial Port 2 DIN Settings	0: Not enabled 1: Enabled	0		U8	RWS	Restart

8.3.2 Profile Position Mode (PP)

Profile position mode is mainly used for point-to-point positioning applications. In this mode, the host computer sets the target position, operating speed, and acceleration/deceleration. The servo's internal position Motion Path generator will generate position curve commands based on the settings, and the drive will complete position control, speed control, and torque control internally. To enable Profile Position Mode, set object 6060h to 1.

(1) Control Description

Controlword 6040h and Statusword 6041h in PP mode are described as follows:

Index	Sub-index	Name/Description	Access	PDO Mapping	Data Type	Ranges
6040h	00h	Controlword	RW	RxPDO	U16	0-65535
		Bit	Name	Description		
		0	Switch on	Must be set to 1 when enabling the servo		
		1	Enable voltage	Must be set to 1 when enabling the servo		
		2	Quick stop	Must be set to 1 to enable the servo. Setting to 0 results in quick stop		
		3	Operation enable	Must be set to 1 when enabling the servo		
		4	New target position	0→1: Trigger a new target position. 1→0: Clear bit12 of the status word		
		5	Immediate update	0: Wait for the current position command to complete before executing the new command 1: Abort the currently executing command and execute the latest position command		
		6	Position command type	0: Absolute position command, 1: Relative position command		
		7	Fault Reset	Execute a fault reset when changing from 0→1. Other commands are invalid when set to 1		

		Note: Bits 4, 5, 9:						
		Bit9		Bit5	Bit4	Description		
		0		0	0→1	After the current positioning is completed, start the next positioning		
		X		1	0→1	Immediately start the next positioning		
		1		0	0→1	Position to the current target position with the current profile velocity, then start the next positioning		
6041h	00h	Statusword			RO	TxPDO	U16	0-65535
		Bit	Name			Description		
		0	Ready to switch on			0: Invalid, 1: Valid. When valid, servo can be enabled		
		1	Switched on			0: Invalid, 1: Valid. When valid, servo can be enabled		
		2	Operation enabled			0: Invalid, 1: Valid. When valid, indicates servo is enabled		
		3	Servo Fault			0: No fault, 1: Fault		
		4	Voltage enabled			0: Invalid, 1: Valid. When valid, servo can be enabled		
		5	Quick stop			0: Quick stop enabled, 1: Quick stop disabled		
		6	Switch on disabled			0: Invalid, 1: Valid. When valid, servo can be enabled		
		7	Warning			0: No warning, 1: With warning		
		9	Remote Control			0: Invalid, 1: Valid. When active, it indicates the Controlword has taken effect		
		10	Target Reached			0: Position not reached, 1: Position reached		
		11	Internal soft limit status			0: Soft limit not triggered, 1: Soft limit triggered		
		12	New position command reception status			0: Previous setpoint operation completed, the target position can be updated 1: Previous setpoint operation is in progress, the target position cannot be updated		
		13	Position following error			0: No position deviation alarm, 1: Position deviation alarm		
		15	Homing completed			0: Homing not completed, 1: Homing completed		

(2) Related Objects

Index	Sub-index	Name	Access Type	Validation	Data Type
603Fh	00h	Error Code	RO	Not changeable	UINT16
6040h	00h	Control Word	RW	Real-time change	UINT16
6041h	00h	State Word	RO	Not changeable	UINT16
6060h	00h	Control Mode	RW	Real-time change	INT 8
6061h	00h	Current operating mode	RO	Not changeable	INT 8
6064h	00h	Actual Position	RO	Not changeable	INT 32
6065h	00h	Position following error window	RW	Real-time change	UINT 32
6066h	00h	Position following error time window	RW	Real-time change	UINT 32
6067h	00h	In-position window	RW	Real-time change	UINT 32
6068h	00h	In-position time window	RW	Real-time change	UINT 32
606Ch	00h	Velocity loop feedback	RO	Not changeable	INT 32
6077h	00h	Actual torque	RO	Not changeable	INT 16
607Ah	00h	Target Position	RW	Real-time change	INT 32
607Dh	01h	Soft limit position 1	RW	Real-time change	INT 32
	02h	Soft limit position 2	RW	Real-time change	INT 32
607Eh	00h	Command polarity	RW	Real-time change	UINT 8
6081h	00h	Profile velocity	RW	Real-time change	UINT 32
6083h	00h	Profile acceleration	RW	Real-time change	UINT 32
6084h	00h	Profile deceleration	RW	Real-time change	UINT 32
60F4h	00h	Position following error	RO	Not changeable	INT 32

8.3.3 Profile Velocity Mode (PV)

In Profile Velocity mode, the host controller sends the target velocity, acceleration, and deceleration to the drive. Velocity and torque regulation are executed internally by the drive. When enabling Profile Velocity mode, set object 6060h to 3.

(1) Control Description

Controlword 6040h and Statusword 6041h in PV mode are described as follows:

Index	Sub-index	Name/Description	Access	PDO Mapping	Data Type	Ranges
6040h	00h	Controlword	RW	RxPDO	U16	0-65535
		Bit	Name	Description		
		0	Switch on	Must be set to 1 when enabling the servo		
		1	Enable voltage	Must be set to 1 when enabling the servo		
		2	Quick stop	Must be set to 1 to enable the servo. Setting to 0 results in quick stop		
		3	Operation enable	Must be set to 1 when enabling the servo		
		7	Fault Reset	Execute a fault reset when changing from 0→1. Other commands are invalid when set to 1		
6041h	00h	Statusword	RO	TxPDO	U16	0-65535
		Bit	Name	Description		
		0	Ready to switch on	0: Invalid, 1: Valid. When valid, servo can be enabled		
		1	Switched on	0: Invalid, 1: Valid. When valid, servo can be enabled		
		2	Operation enabled	0: Invalid, 1: Valid. When valid, indicates servo is enabled		
		3	Servo Fault	0: No fault, 1: Fault		
		4	Voltage enabled	0: Invalid, 1: Valid. When valid, servo can be enabled		
		5	Quick stop	0: Quick stop enabled, 1: Quick stop disabled		
		6	Switch on disabled	0: Invalid, 1: Valid. When valid, servo can be enabled		
		7	Warning	0: No warning, 1: With warning		
		9	Remote Control	0: Invalid, 1: Valid. When active, it indicates the Controlword has taken effect		
		10	Target Reached	0: Velocity not reached, 1: Velocity reached		
		11	Internal soft limit status	0: Soft limit not triggered, 1: Soft limit triggered		
		12	Velocity information	0: Velocity is not 0, 1: Velocity is 0		

(2) Related Objects

Index	Sub-index	Name	Access Type	Validation	Data Type
603Fh	00h	Error Code	RO	Not changeable	UINT16
6040h	00h	Control Word	RW	Real-time change	UINT16
6041h	00h	State Word	RO	Not changeable	UINT16
6060h	00h	Control Mode	RW	Real-time change	INT 8
6061h	00h	Current operating mode	RO	Not changeable	INT 8
6064h	00h	Actual Position	RO	Not changeable	INT 32
606Ch	00h	Velocity loop feedback	RO	Not changeable	INT 32
606Dh	00h	Velocity window	RW	Real-time change	INT 16
606Eh	00h	Velocity stable time window	RW	Real-time change	INT 16
607Dh	01h	Soft limit position 1	RW	Real-time change	INT 32
	02h	Soft limit position 2	RW	Real-time change	INT 32
607Eh	00h	Command polarity	RW	Real-time change	UINT 8
607Fh	00h	Maximum profile velocity	RW	Real-time change	UINT 32
6081h	00h	Profile velocity	RW	Real-time change	UINT 32
6083h	00h	Profile acceleration	RW	Real-time change	UINT 32
6084h	00h	Profile deceleration	RW	Real-time change	UINT 32
60FFh	00h	Target Speed	RW	Real-time change	INT 32

8.3.4 Profile Torque Mode (PT)

In Profile Torque mode, the host controller sends the target torque 6071h to the servo drive. The servo drive itself plans the torque command curve, and torque regulation is executed internally by the servo. When enabling Profile Torque mode, set object 6060h to 4.

(1) Control Description

Controlword 6040h and Statusword 6041h in PT mode are described as follows:

Index	Sub-index	Name/Description	Access	PDO Mapping	Data Type	Ranges
6040h	00h	Controlword	RW	RxPDO	U16	0-65535
		Bit	Name	Description		
		0	Switch on	Must be set to 1 when enabling the servo		
		1	Enable voltage	Must be set to 1 when enabling the servo		
		2	Quick stop	Must be set to 1 to enable the servo. Setting to 0 results in quick stop		
		3	Operation enable	Must be set to 1 when enabling the servo		
		7	Fault Reset	Execute a fault reset when changing from 0→1. Other commands are invalid when set to 1		
6041h	00h	Statusword	RO	TxPDO	U16	0-65535
		Bit	Name	Description		
		0	Ready to switch on	0: Invalid, 1: Valid. When valid, servo can be enabled		
		1	Switched on	0: Invalid, 1: Valid. When valid, servo can be enabled		
		2	Operation enabled	0: Invalid, 1: Valid. When valid, indicates servo is enabled		
		3	Servo Fault	0: No fault, 1: Fault		
		4	Voltage enabled	0: Invalid, 1: Valid. When valid, servo can be enabled		
		5	Quick stop	0: Quick stop enabled, 1: Quick stop disabled		
		6	Switch on disabled	0: Invalid, 1: Valid. When valid, servo can be enabled		
		7	Warning	0: No warning, 1: With warning		
		9	Remote Control	0: Invalid, 1: Valid. When active, it indicates the Controlword has taken effect		

(2) Related Objects

Index	Sub-index	Name	Access Type	Validation	Data Type
4010h	6A	Torque mode speed limit	RW	Real-time change	UINT 32
603Fh	00h	Error Code	RO	Not changeable	UINT16
6040h	00h	Control Word	RW	Real-time change	UINT16
6041h	00h	State Word	RO	Not changeable	UINT16
6060h	00h	Control Mode	RW	Real-time change	INT 8
6061h	00h	Current operating mode	RO	Not changeable	INT 8
606Ch	00h	Velocity loop feedback	RO	Not changeable	INT 32
6071h	00h	Target Torque	RW	Real-time change	INT 16
6072h	00h	Maximum Torque	RW	Real-time change	INT 16
6074h	00h	Torque command	RO	Not changeable	INT 16
6077h	00h	Actual torque	RO	Not changeable	INT 16
607Eh	00h	Command polarity	RW	Real-time change	UINT 8
6083h	00h	Profile acceleration	RW	Real-time change	UINT 32
6084h	00h	Profile deceleration	RW	Real-time change	UINT 32
60E0h	00h	Positive torque limit	RW	Real-time change	UINT 16
60E1h	00h	Negative torque limit	RW	Real-time change	UINT 16

8.3.5 Homing Mode (HM)

Homing mode is used to find the mechanical origin and establish the positional relationship between the mechanical origin and the mechanical zero point.

- **Mechanical Origin:** A fixed position on the machine, which can correspond to a specific home switch, or the motor Z-pulse signal.
- **Mechanical Zero Point:** The absolute 0 position on the machine.

After successful homing, the motor stop position is the mechanical origin. By setting 607Ch (Home Offset), the relationship between the mechanical origin and the mechanical zero point can be defined:

Mechanical Origin = Mechanical Zero Point + 607Ch (Home Offset)

When 607Ch = 0, it means the mechanical origin and the mechanical zero point coincide.

When enabling this mode, set object 6060h to 6.

(1) Control Description

Controlword 6040h and Statusword 6041h in HM mode are described as follows:

Index	Sub-index	Name/Description		Access	PDO Mapping	Data Type	Ranges
6040h	00h	Controlword		RW	RxPDO	U16	0-65535
			Bit	Name	Description		
			0	Switch on	Must be set to 1 when enabling the servo		
			1	Enable voltage	Must be set to 1 when enabling the servo		
			2	Quick stop	Must be set to 1 to enable the servo. Setting to 0 results in quick stop		
			3	Operation enable	Must be set to 1 when enabling the servo		
			4	Homing enable	0: Inactive, 1: Active. When active, start the homing process; it must remain active throughout the homing process; switching to inactive stops the homing process.		
			7	Fault Reset	Execute a fault reset when changing from 0→1. Other commands are invalid when set to 1		
6041h	00h	Statusword		RO	TxPDO	U16	0-65535
			Bit	Name	Description		
			0	Ready to switch on	0: Invalid, 1: Valid. When valid, servo can be enabled		
			1	Switched on	0: Invalid, 1: Valid. When valid, servo can be enabled		
			2	Operation enabled	0: Invalid, 1: Valid. When valid, indicates servo is enabled		
			3	Servo Fault	0: No fault, 1: Fault		
			4	Voltage enabled	0: Invalid, 1: Valid. When valid, servo can be enabled		
			5	Quick stop	0: Quick stop enabled, 1: Quick stop disabled		
			6	Switch on disabled	0: Invalid, 1: Valid. When valid, servo can be enabled		
			7	Warning	0: No warning, 1: With warning		
			9	Remote Control	0: Invalid, 1: Valid. When active, it indicates the Controlword has taken effect		
			10	Target Reached	0: Homing in progress, 1: Positioned at origin or homing interrupted		
			12	Home found	0: Home signal not found, 1:Home signal found		
			13	Homing error	0: No error occurred during homing, 1: An error occurred during homing		
			15	Homing Complete	0: Homing not completed, 1: Homing completed		
			Note: Bits 10, 12, 13:				
		Bit13	Bit12	Bit10	Description		
		0	0	0	Home return action in progress		
		0	0	1	Home return action interrupted or not yet started		
		0	1	0	Home setting completed, action continues		
		0	1	1	Home return action completed normally		
		1	0	0	Home return error occurred, non-zero speed		
		1	0	1	Home return error occurred, zero speed		

(2) Related Objects

Index	Sub-index	Name	Access Type	Validation	Data Type
603Fh	00h	Error Code	RO	Not changeable	UINT16
6040h	00h	Control Word	RW	Real-time change	UINT16
6041h	00h	State Word	RO	Not changeable	UINT16
605Ah	00h	Quick stop option	RW	Real-time change	UINT 16
6060h	00h	Control Mode	RW	Real-time change	INT 8
6061h	00h	Current operating mode	RO	Not changeable	INT 8
6065h	00h	Position following error window	RW	Real-time change	UINT 32
6066h	00h	Position following error time window	RW	Real-time change	UINT 32
6067h	00h	In-position window	RW	Real-time change	UINT 32
6068h	00h	In-position time window	RW	Real-time change	UINT 32
606Ch	00h	Velocity loop feedback	RO	Not changeable	INT 32
607Ch	00h	Home Offset	RW	Real-time change	INT 32
607Dh	01h	Soft limit position 1	RW	Real-time change	INT 32
	02h	Soft limit position 2	RW	Real-time change	INT 32
6098h	00h	Homing mode	RW	Real-time change	INT 32
6099h	01h	Home Locate speed	RW	Real-time change	UINT 32
	02h	Home Creep Speed	RW	Real-time change	UINT 32
609Ah	00h	Home Acceleration	RW	Real-time change	UINT 32

(3) Homing Mode Introduction (6098h)

Based on encoder index signals, home switch signals, and limit switch signals, CD200 supports the following homing methods:

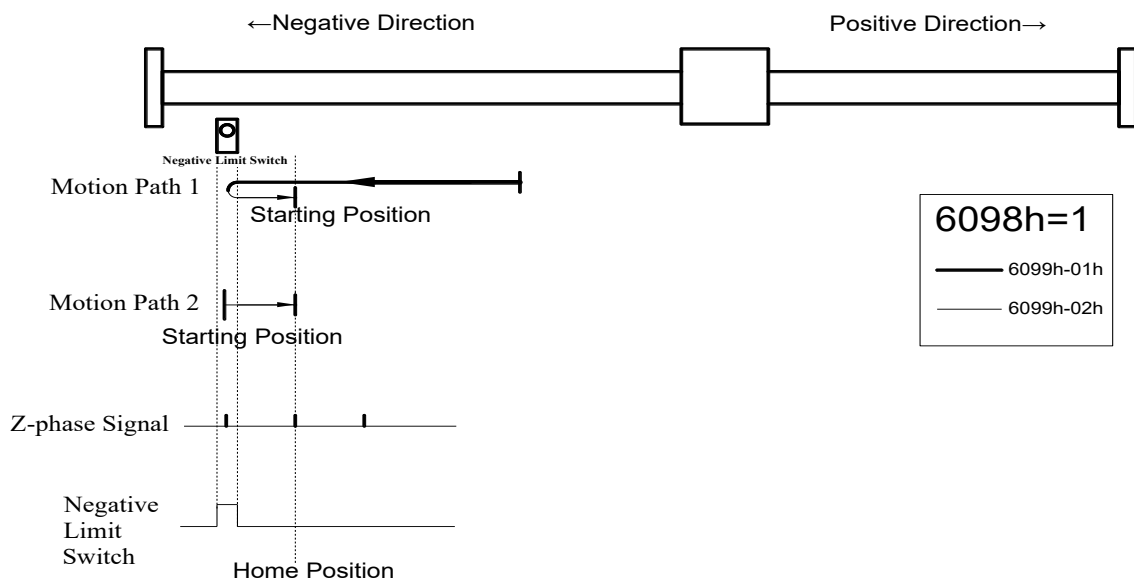
6098h=1

Mechanical origin: Motor Z signal

Deceleration point: Negative limit switch

If the deceleration point signal is inactive at startup, move at home locate speed in the negative direction. After encountering the OFF→ON state of the negative limit switch, decelerate and stop, then change to home creep speed and move in the positive direction. After encountering the ON→OFF state of the negative limit switch while running at home creep speed in the positive direction, continue in the positive direction to find the nearest Z-phase signal as the home position, as shown in Motion Path 1.

If the deceleration point signal is active at startup, move at home creep speed in the positive direction. After encountering the ON→OFF state of the negative limit switch while running in the positive direction, continue in the positive direction to find the nearest Z-phase signal as the home position, as shown in Motion Path 2.



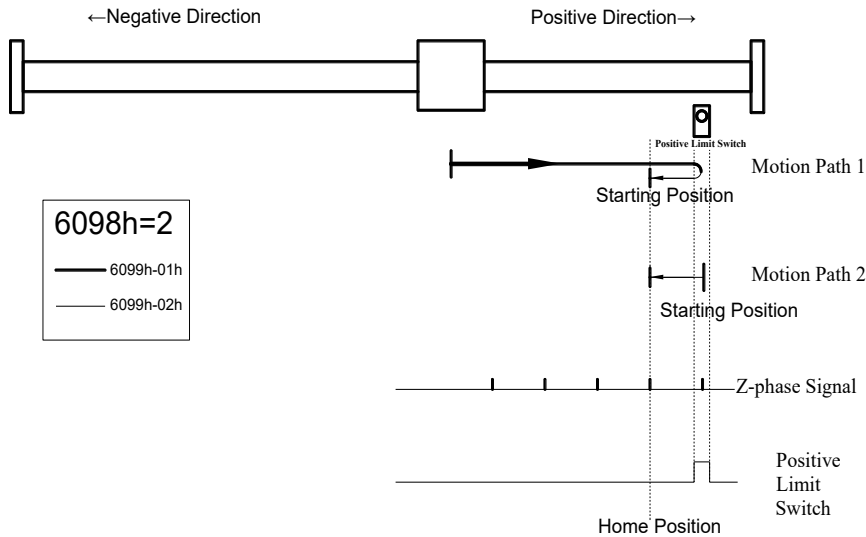
6098h=2

Mechanical : Motor Z signal

Deceleration point: Positive limit switch

If the deceleration point signal is inactive at startup, move at home locate speed in the positive direction. After encountering the OFF→ON state of the positive limit switch, decelerate and stop, then change to home creep speed and move in the negative direction. After encountering the ON→OFF state of the positive limit switch while running at home creep speed in the negative direction, continue in the negative direction to find the nearest Z-phase signal as the home position, as shown in Motion Path 1.

If the deceleration point signal is active at startup, move at home creep speed in the negative direction. After encountering the ON→OFF state of the positive limit switch while running in the negative direction, continue in the negative direction to find the nearest Z-phase signal as the home position, as shown in Motion Path 2.

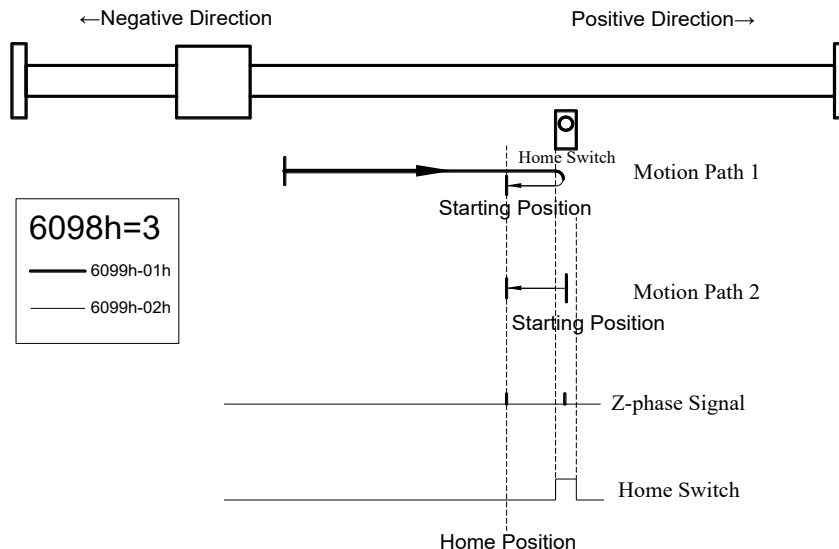
**6098h=3**

Mechanical origin: Motor Z signal

Deceleration point: Home switch

If the home switch is inactive at startup, move at home locate speed in the positive direction. After encountering the OFF→ON state of the home switch while running in the positive direction, decelerate and stop, then change to home creep speed and move in the negative direction. After encountering the ON→OFF state of the home switch while running at home creep speed in the negative direction, continue in the negative direction to find the nearest Z-phase signal as the home position, as shown in Motion Path 1.

If the home switch is active at startup, run at home creep speed in the negative direction. After encountering the ON→OFF state of the home switch while running in the negative direction, it continues searching for the nearest Z-phase signal in the negative direction as the home position, as shown in Motion Path 2.



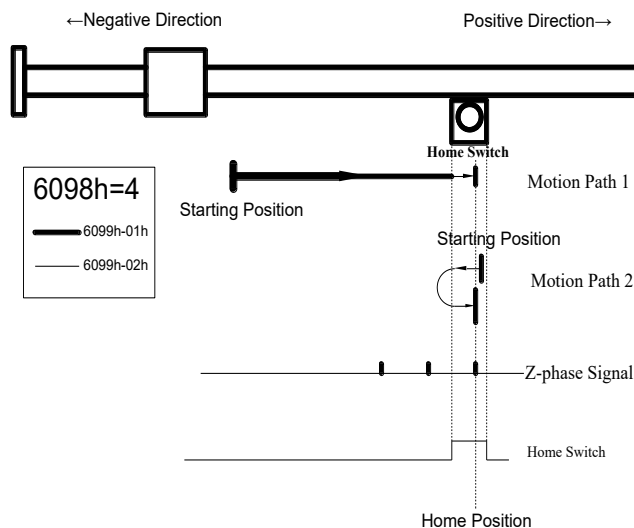
6098h=4

Mechanical origin: Motor Z signal

Deceleration point: Home switch

If the home switch is inactive at startup, run at home locate speed in the positive direction. After encountering the OFF→ON state of the home switch while running in the positive direction, it continues searching at home creep speed in the positive direction for the nearest Z-phase signal as the home position, as shown in Motion Path 1.

If the home switch is active at startup, run at home creep speed in the negative direction. After encountering the ON→OFF state of the home switch while running in the negative direction, decelerate and stop, then run at home creep speed in the positive direction. After encountering the OFF→ON state of the home switch while running at home creep speed in the positive direction, it continues searching for the nearest Z-phase signal in the positive direction as the home position, as shown in Motion Path 2.

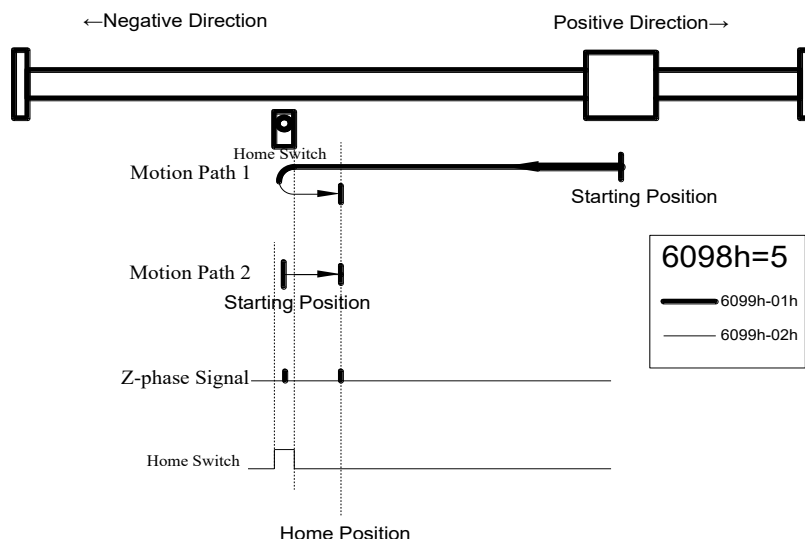
**6098h=5**

Mechanical origin: Motor Z signal

Deceleration point: Home switch

If the home switch is inactive at startup, run at home locate speed in the negative direction. After encountering the OFF→ON state of the home switch while running in the negative direction, decelerate and stop, then switch to home creep speed and run in the positive direction. After encountering the ON→OFF state of the home switch while running at home creep speed in the positive direction, it continues searching for the nearest Z-phase signal in the positive direction as the home position, as shown in Motion Path 1.

If the home switch is active at startup, run at home creep speed in the positive direction. After encountering the ON→OFF state of the home switch while running in the positive direction, it continues searching for the nearest Z-phase signal in the positive direction as the home position, as shown in Motion Path 2.



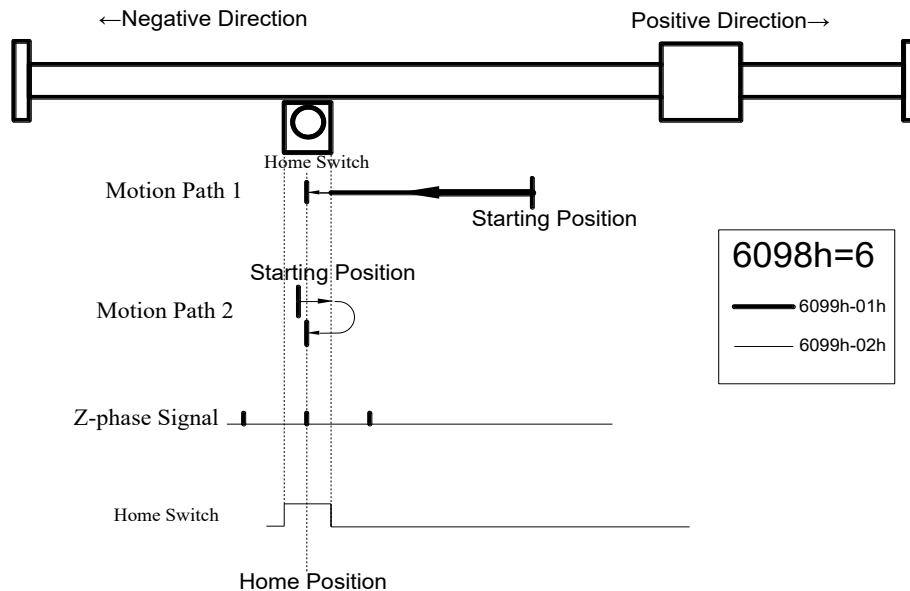
6098h=6

Mechanical origin: Motor Z signal

Deceleration point: Home switch

If the home switch is inactive at startup, run at home locate speed in the negative direction. After encountering the OFF→ON state of the home switch while running in the negative direction, switch to home creep speed and it continues searching for the nearest Z-phase signal in the negative direction as the home position, as shown in Motion Path 1.

If the home switch is active at startup, then run at home creep speed in the positive direction. After encountering the ON→OFF state of the home switch while running in the positive direction, decelerate and stop, then run at home creep speed in the negative direction. After encountering the OFF→ON state of the home switch while running at home creep speed in the negative direction, it continues searching for the nearest Z-phase signal in the negative direction as the home position, as shown in Motion Path 2.

**6098h=7**

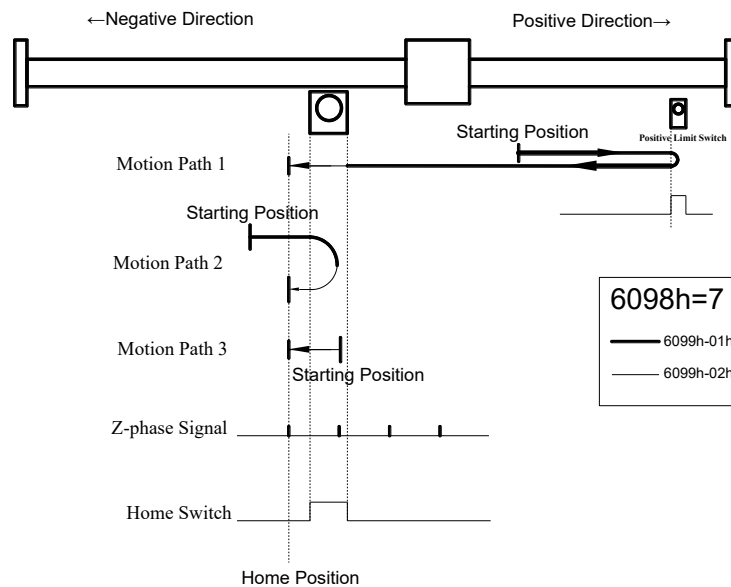
Mechanical origin: Motor Z signal

Deceleration point: Home switch

If the home switch is inactive at startup and the positive limit switch is inactive, then run at home locate speed in the positive direction. If encountering the positive limit switch, decelerate and stop, then reverse direction and continue running at home locate speed in the negative direction. After encountering the OFF→ON state of the home switch while running in the negative direction, decelerate, then run at home creep speed in the negative direction. After encountering the ON→OFF state of the home switch while running at home creep speed in the negative direction, continue in the negative direction to find the nearest Z-phase signal as the home position, as shown in Motion Path 1.

If the home switch is inactive at startup and the positive limit switch is inactive, then run at home locate speed in the positive direction. After encountering the OFF→ON state of the home switch while running in the positive direction, decelerate and stop, then switch to home creep speed and run in the negative direction. After encountering the ON→OFF state of the home switch while running at home creep speed in the negative direction, continue in the negative direction to find the nearest Z-phase signal as the home position, as shown in Motion Path 2.

If the home switch is active at startup, then run at home creep speed in the negative direction. After encountering the ON→OFF state of the home switch while running in the negative direction, it continues searching for the nearest Z-phase signal in the negative direction as the home position, as shown in Motion Path 3.



6098h=8

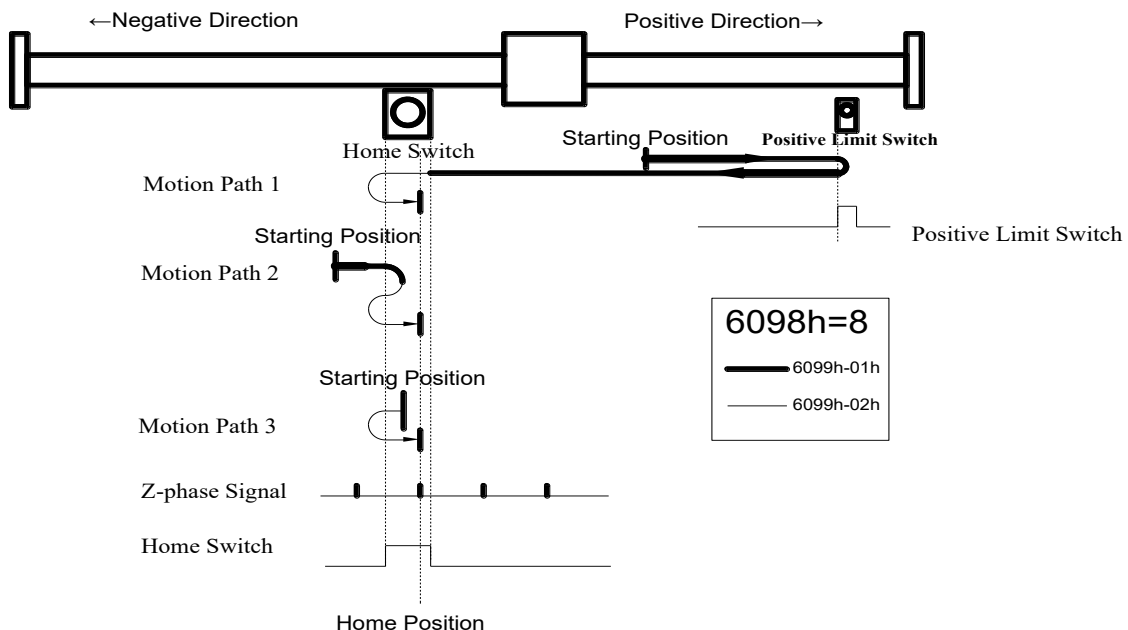
Mechanical origin: Motor Z signal

Deceleration point: Home switch

If the home switch is inactive at startup and the positive limit switch is inactive, then run at home locate speed in the positive direction. After encountering the positive limit switch, decelerate and stop, then run at home locate speed in the negative direction. After encountering the OFF→ON state of the home switch while running in the negative direction, decelerate, then run at home creep speed in the negative direction. After encountering the ON→OFF state of the home switch while running at home creep speed in the negative direction, decelerate and stop, then switch to home creep speed and run in the positive direction. After encountering the OFF→ON state of the home switch while running in the positive direction, it continues searching for the nearest Z-phase signal in the positive direction as the home position, as shown in Motion Path 1.

If the home switch is inactive at startup and the positive limit switch is inactive, then run at home locate speed in the positive direction. After encountering the OFF→ON state of the home switch while running in the positive direction, decelerate and stop, then switch to home creep speed and run in the negative direction. After encountering the ON→OFF state of the home switch while running at home creep speed in the negative direction, decelerate and stop, then switch to home creep speed and run in the positive direction. After encountering the OFF→ON state of the home switch while running in the positive direction, it continues searching for the nearest Z-phase signal in the positive direction as the home position, as shown in Motion Path 2.

If the home switch is active at startup, then run at home creep speed in the negative direction. After encountering the ON→OFF state of the home switch while running in the negative direction, decelerate and stop, then run at home creep speed in the positive direction. After encountering the OFF→ON state of the home switch while running in the positive direction, it continues searching for the nearest Z-phase signal in the positive direction as the home position, as shown in Motion Path 3.



6098h=9

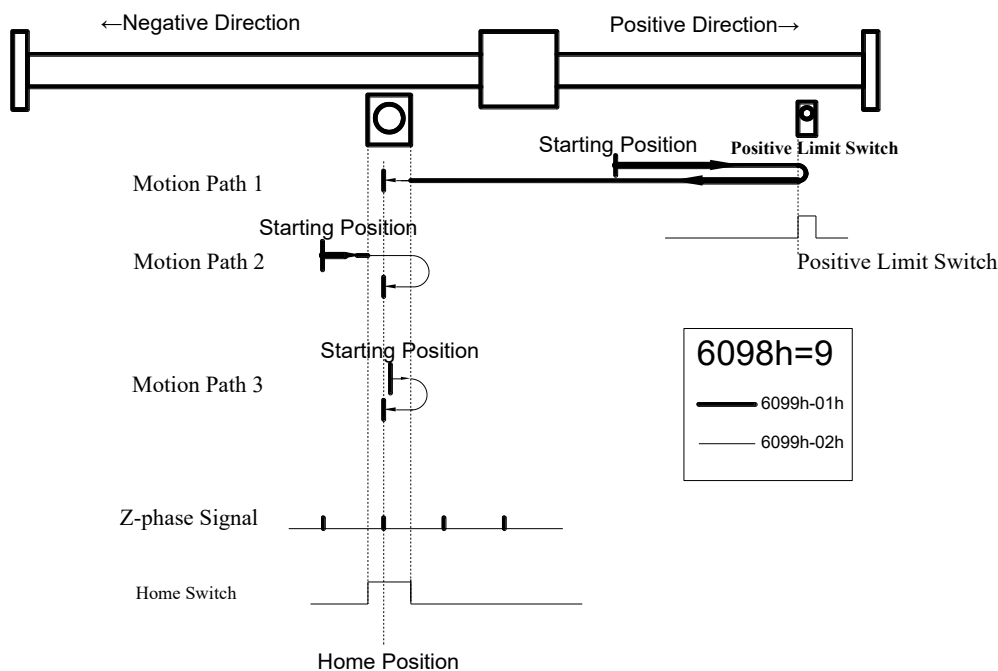
Mechanical origin: Motor Z signal

Deceleration point: Home switch

If the home switch is inactive at startup and the positive limit switch is inactive, then run at home locate speed in the positive direction. After encountering the positive limit switch, decelerate and stop, then run at home locate speed in the negative direction. After encountering the OFF→ON state of the home switch while running in the negative direction, decelerate, and search for the nearest Z-phase signal at home creep speed as the home position, as shown in Motion Path 1.

If the home switch is inactive at startup and the positive limit switch is inactive, then run at home locate speed in the positive direction. After encountering the OFF→ON state of the home switch while running in the positive direction, it switches to home creep speed and runs in the positive direction. When running at home creep speed in the positive direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, then runs at home creep speed in the negative direction. After encountering the OFF→ON state of the home switch while running in the negative direction, it continues searching for the nearest Z-phase signal in the positive direction as the home position, as shown in Motion Path 2.

If the home switch is active at startup, it runs at home creep speed in the positive direction. When running in the positive direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, then switches to home creep speed and runs in the negative direction. After encountering the OFF→ON state of the home switch while running at home creep speed in the negative direction, it continues searching for the nearest Z-phase signal in the negative direction as the home position, as shown in Motion Path 3.

**6098h=10**

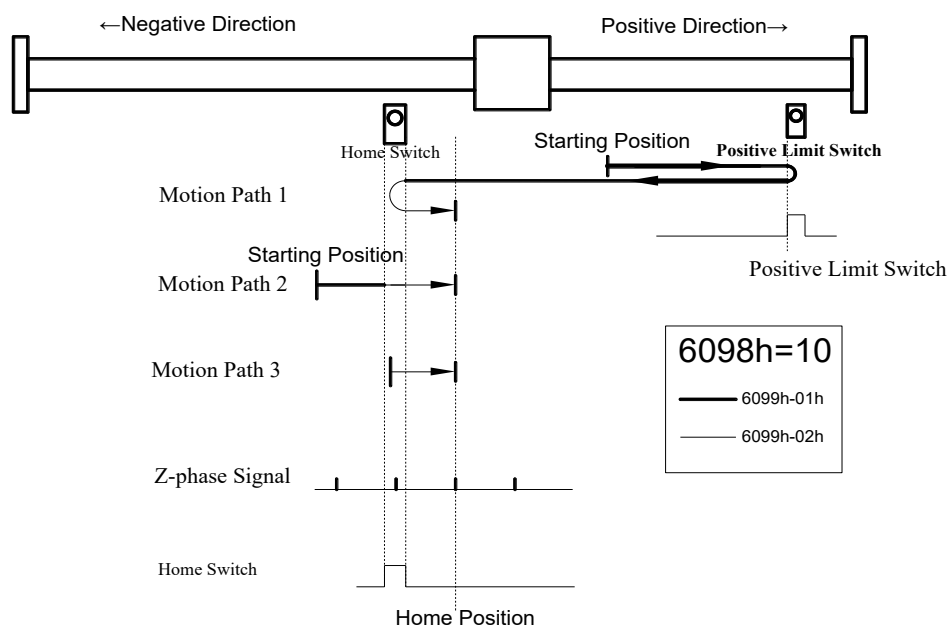
Mechanical origin: Motor Z signal

Deceleration point: Home switch

If the home switch is inactive at startup and the positive limit switch is inactive, then run at home locate speed in the positive direction. After encountering the positive limit switch while running in the positive direction, it decelerates and stops, then runs at home locate speed in the negative direction. While running in the negative direction, after encountering the OFF→ON state of the home switch, it decelerates and stops, then switches to home creep speed and runs in the positive direction. After encountering the ON→OFF state of the home switch while running at home creep speed in the positive direction, it continues searching for the nearest Z-phase signal in the positive direction as the home position, as shown in Motion Path 1.

If the home switch is inactive at startup and the positive limit switch is inactive, then run at home locate speed in the positive direction. After encountering the OFF→ON state of the home switch while running in the positive direction, it switches to home creep speed and runs in the positive direction. After encountering the ON→OFF state of the home switch while running at home creep speed in the positive direction, it continues searching for the nearest Z-phase signal in the positive direction as the home position, as shown in Motion Path 2.

If the home switch is active at startup, it runs at home creep speed in the positive direction. After encountering the ON→OFF state of the home switch while running in the positive direction, it continues searching for the nearest Z-phase signal in the positive direction as the home position, as shown in Motion Path 3.

**6098h=11**

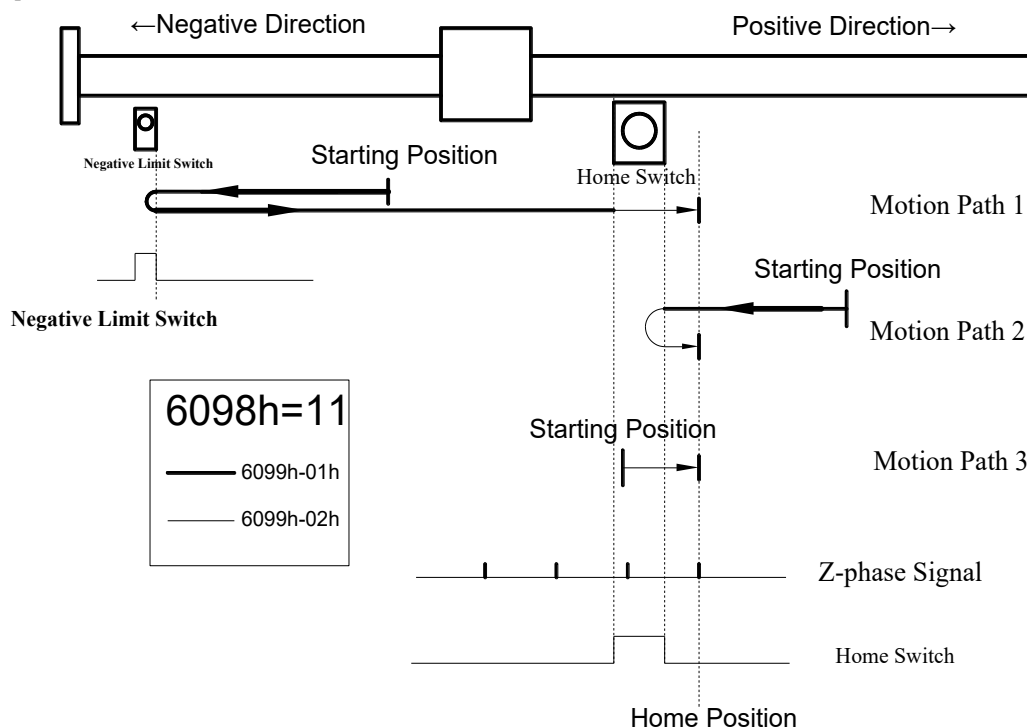
Mechanical origin: Motor Z signal

Deceleration point: Home switch

If the home switch is inactive at startup and the negative limit switch is inactive, it runs at home locate speed in the negative direction. After encountering the negative limit switch while running in the negative direction, it decelerates and stops, then runs at home locate speed in the positive direction. After encountering the OFF→ON state of the home switch, it switches to home creep speed and runs in the positive direction. After encountering the ON→OFF state of the home switch while running at home creep speed in the positive direction, it continues searching for the nearest Z-phase signal in the positive direction as the home position, as shown in Motion Path 1.

If the home switch is inactive at startup and the negative limit switch is inactive, it runs at home locate speed in the negative direction. After encountering the OFF→ON state of the home switch while running in the negative direction, it decelerates and stops, then switches to home creep speed and runs in the positive direction. After encountering the ON→OFF state of the home switch while running at home creep speed in the positive direction, it continues searching for the nearest Z-phase signal in the positive direction as the home position, as shown in Motion Path 2.

If the home switch is active at startup, it runs at home creep speed in the positive direction. After encountering the ON→OFF state of the home switch while running in the positive direction, it continues searching for the nearest Z-phase signal in the positive direction as the home position, as shown in Motion Path 3.



6098h=12

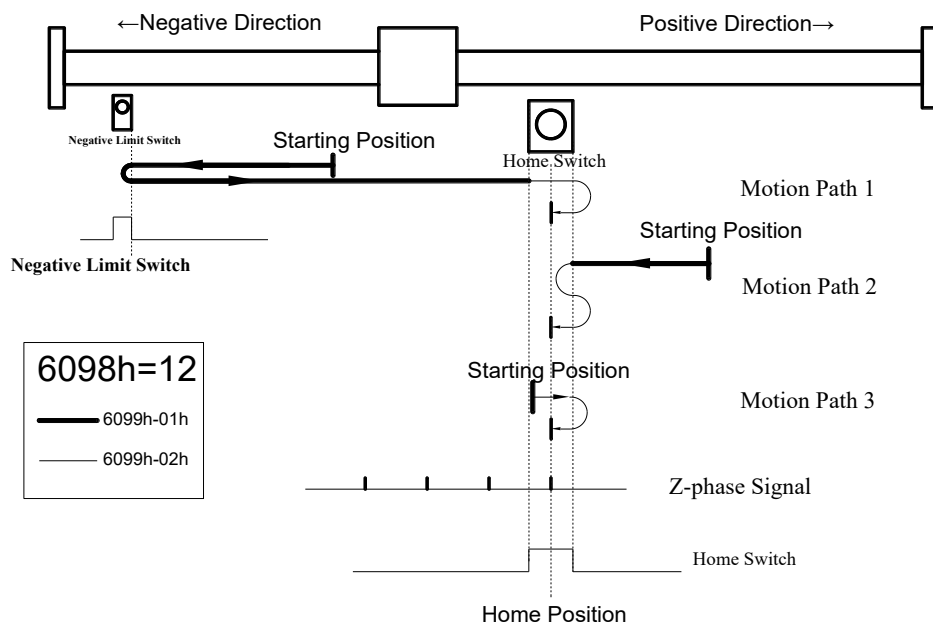
Mechanical origin: Motor Z signal

Deceleration point: Home switch

If the home switch is inactive at startup and the negative limit switch is inactive, it runs at home locate speed in the negative direction. After encountering the negative limit switch while running in the negative direction, it decelerates and stops, then runs at home locate speed in the positive direction. When running in the positive direction, after encountering the OFF→ON state of the home switch, it switches to home creep speed and continues running in the positive direction. When running at home creep speed in the positive direction, after encountering the ON→OFF state of the home switch, it stops, then switches to home creep speed and runs in the negative direction. After encountering the OFF→ON state of the home switch while running at home creep speed in the negative direction, it continues searching for the nearest Z-phase signal in the negative direction as the home position, as shown in Motion Path 1.

If the home switch is inactive at startup and the negative limit switch is inactive, it runs at home locate speed in the negative direction. After encountering the OFF→ON state of the home switch while running at home creep speed in the negative direction, it continues searching for the nearest Z-phase signal in the negative direction as the home position, as shown in Motion Path 2.

If the home switch is active at startup, it runs at home creep speed in the positive direction. After encountering the ON→OFF state of the home switch while running in the positive direction, it stops, then runs at home creep speed in the negative direction. When running at home creep speed in the negative direction, after encountering the OFF→ON state of the home switch, it continues searching for the nearest Z-phase signal in the negative direction as the home position, as shown in Motion Path 3.

**6098h=13**

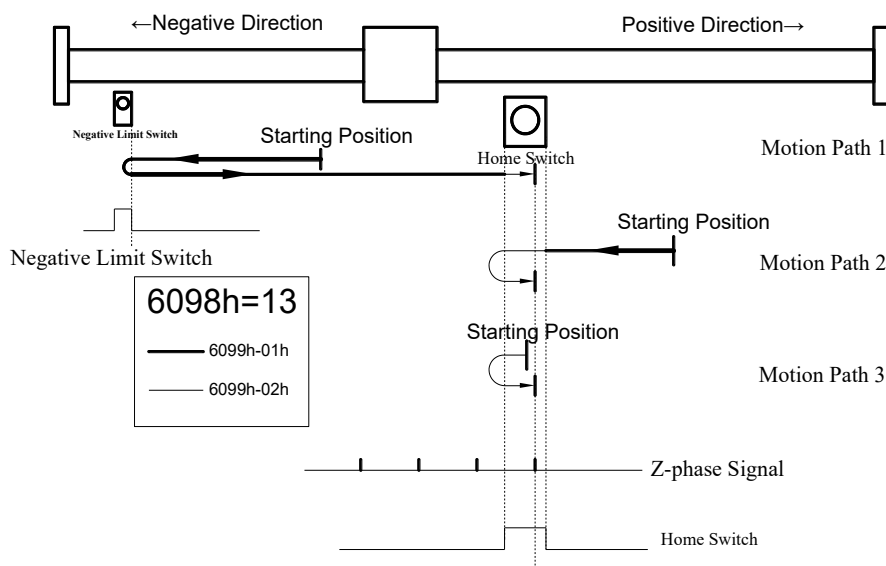
Mechanical origin: Motor Z signal

Deceleration point: Home switch

If the home switch is inactive at startup and the negative limit switch is inactive, it runs at home locate speed in the negative direction. After encountering the negative limit switch while running in the negative direction, it decelerates and stops, then runs at home locate speed in the positive direction. After encountering the OFF→ON state of the home switch while running in the positive direction, it decelerates, then switches to home creep speed and runs in the positive direction to find the nearest Z-phase signal as the home position, as shown in Motion Path 1.

If the home switch is inactive at startup and the negative limit switch is inactive, it runs at home locate speed in the negative direction. After encountering the OFF→ON state of the home switch while running in the negative direction, it switches to home creep speed and continues running in the negative direction. When running at home creep speed in the negative direction, after encountering the ON→OFF state of the home switch, it stops, then switches to home creep speed and runs in the positive direction. After encountering the OFF→ON state of the home switch while running in the positive direction, it continues searching for the nearest Z-phase signal in the positive direction as the home position, as shown in Motion Path 2.

If the home switch is active at startup, then run at home creep speed in the negative direction. After encountering the ON→OFF state of the home switch while running in the negative direction, it stops, then switches to home creep speed and runs in the positive direction. After encountering the OFF→ON state of the home switch while running at home creep speed in the positive direction, it continues searching for the nearest Z-phase signal in the positive direction as the home position, as shown in Motion Path 3.

**6098h=14**

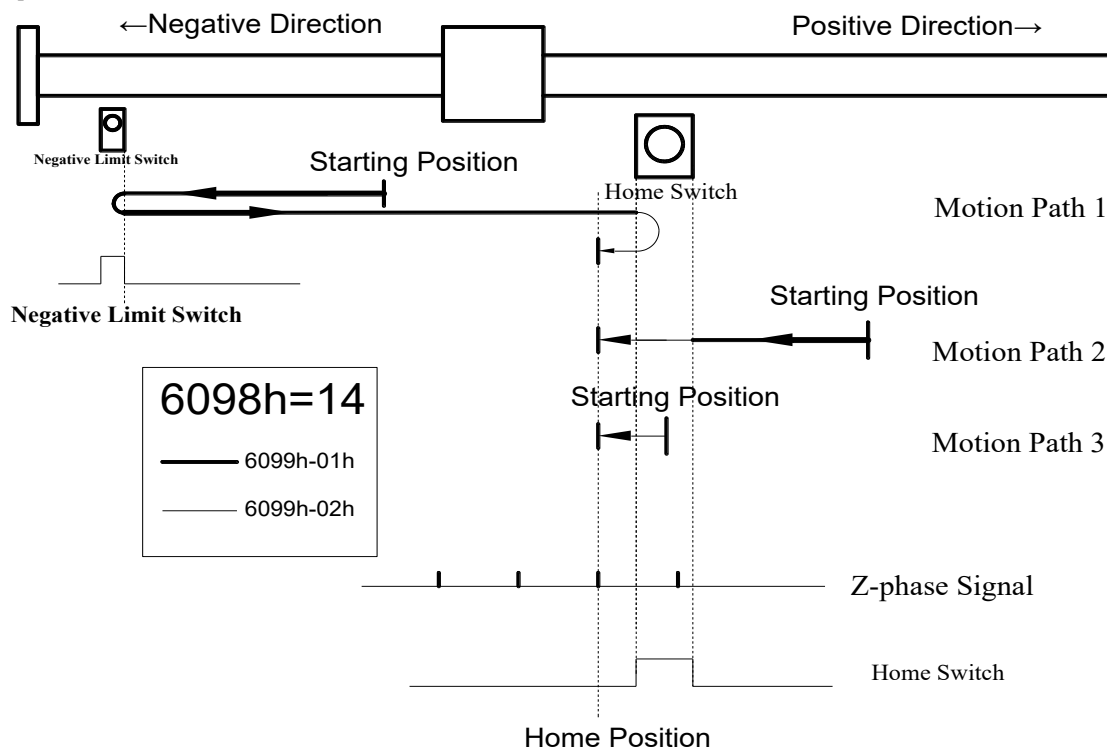
Mechanical origin: Motor Z signal

Deceleration point: Home switch

If the home switch is inactive at startup and the negative limit switch is inactive, it runs at home locate speed in the negative direction. After encountering the negative limit switch while running in the negative direction, it decelerates and stops, then runs at home locate speed in the positive direction. After encountering the OFF→ON state of the home switch while running in the positive direction, it stops, then switches to home creep speed and runs in the negative direction. When running at home creep speed in the negative direction, after encountering the ON→OFF state of the home switch, it continues searching for the nearest Z-phase signal in the negative direction as the home position, as shown in Motion Path 1.

If the home switch is inactive at startup and the negative limit switch is inactive, it runs at home locate speed in the negative direction. After encountering the OFF→ON state of the home switch while running in the negative direction, it switches to home creep speed and runs in the negative direction. When running at home creep speed in the negative direction, after encountering the ON→OFF state of the home switch, it continues searching for the nearest Z-phase signal in the negative direction as the home position, as shown in Motion Path 2.

If the home switch is active at startup, then run at home creep speed in the negative direction. After encountering the ON→OFF state of the home switch while running in the negative direction, it continues searching for the nearest Z-phase signal in the negative direction as the home position, as shown in Motion Path 3.



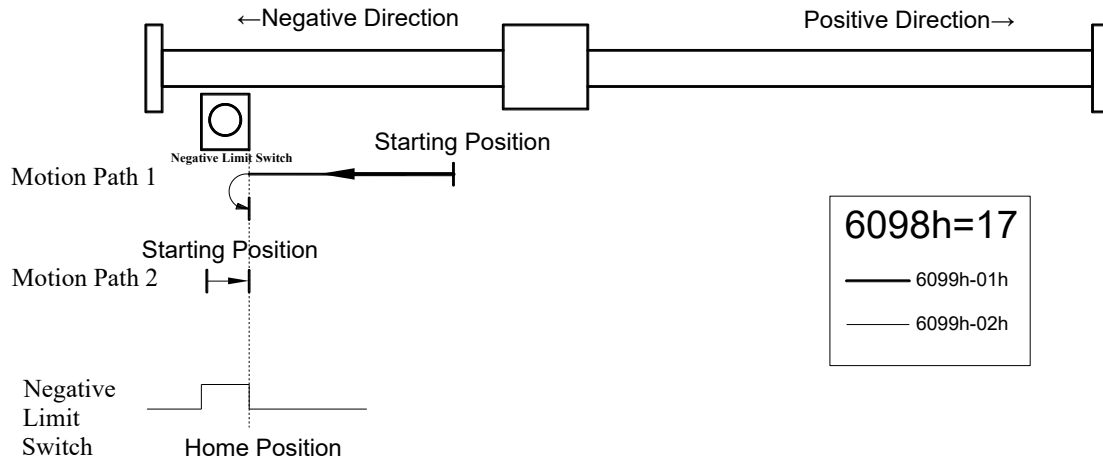
6098h=17

Mechanical origin: Negative limit switch

Deceleration point: Negative limit switch

If the negative limit switch is inactive at startup, it runs at home locate speed in the negative direction. After encountering the OFF→ON state of the negative limit switch, it decelerates and stops, then switches to home creep speed and runs in the positive direction. When running at home creep speed in the positive direction, upon encountering the ON→OFF state of the negative limit switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 1.

If the negative limit switch is active at startup, it runs at home creep speed in the positive direction. When running in the positive direction, upon encountering the ON→OFF state of the negative limit switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 2.

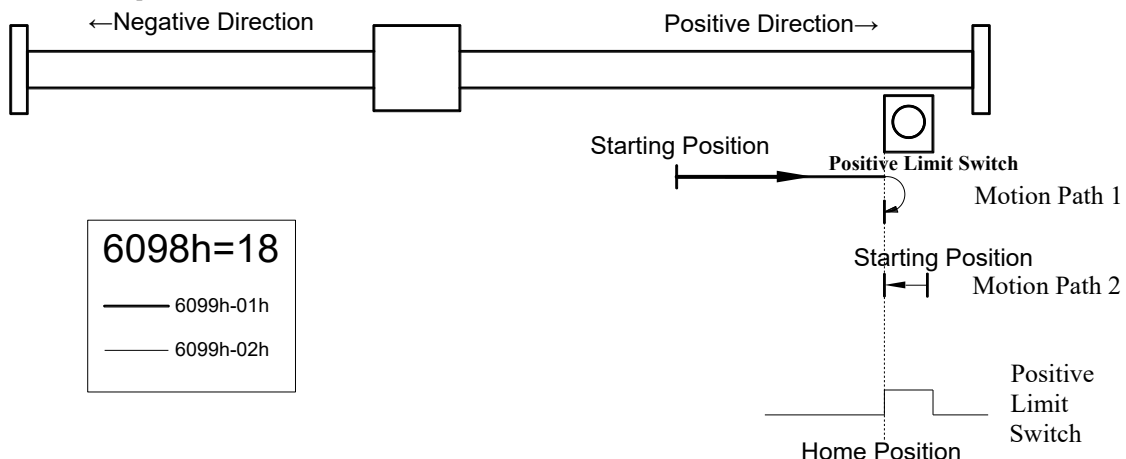
**6098h=18**

Mechanical origin: Positive limit switch

Deceleration point: Positive limit switch

If the positive limit switch is inactive at startup, it runs at home locate speed in the positive direction. After encountering the OFF→ON state of the positive limit switch, it decelerates and stops, then switches to home creep speed and runs in the negative direction. When running at home creep speed in the negative direction, upon encountering the ON→OFF state of the positive limit switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 1.

If the positive limit switch is active at startup, it runs at home creep speed in the negative direction. When running at home creep speed in the negative direction, upon encountering the ON→OFF state of the positive limit switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 2.

**6098h=19**

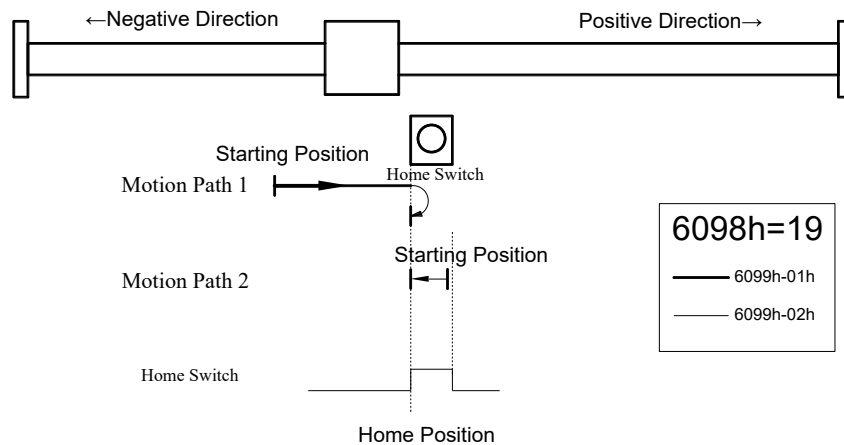
Mechanical origin: Home switch

Deceleration point: Home switch

If the home switch is inactive at startup, move at home locate speed in the positive direction. After encountering the OFF→ON state of the home switch while running in the positive direction, decelerate and stop, then change to home creep speed and move in the negative direction. When running at home creep speed in the negative direction, upon encountering the ON→OFF state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 1.

If the home switch is active at startup, it runs at home creep speed in the negative direction. When running in the negative direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, using the stopping position as the home position,

as shown in Motion Path 2.



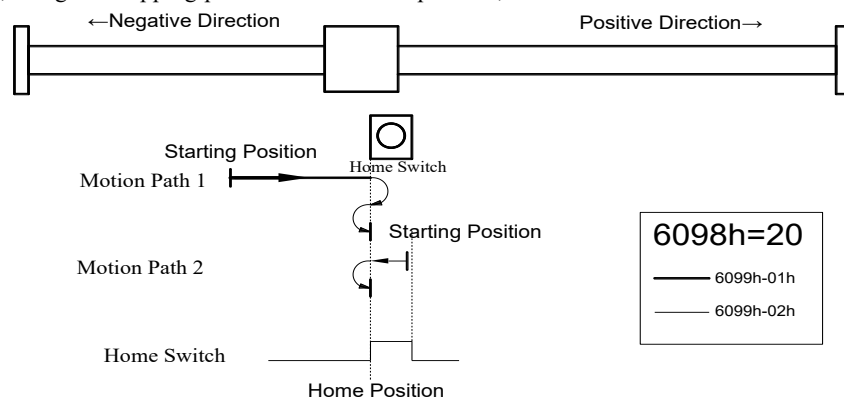
6098h=20

Mechanical origin: Home switch

Deceleration point: Home switch

If the home switch is inactive at startup, move at home locate speed in the positive direction. After encountering the OFF→ON state of the home switch while running in the positive direction, decelerate and stop, then change to home creep speed and move in the negative direction. When running at home creep speed in the negative direction, upon encountering the ON→OFF state of the home switch, it decelerates and stops, then runs at home creep speed in the positive direction. When running at home creep speed in the positive direction, upon encountering the ON→OFF state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 1.

If the home switch is active at startup, it runs at home creep speed in the negative direction. When running in the negative direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, then runs at home creep speed in the positive direction. When running at home creep speed in the positive direction, upon encountering the OFF→ON state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 2.



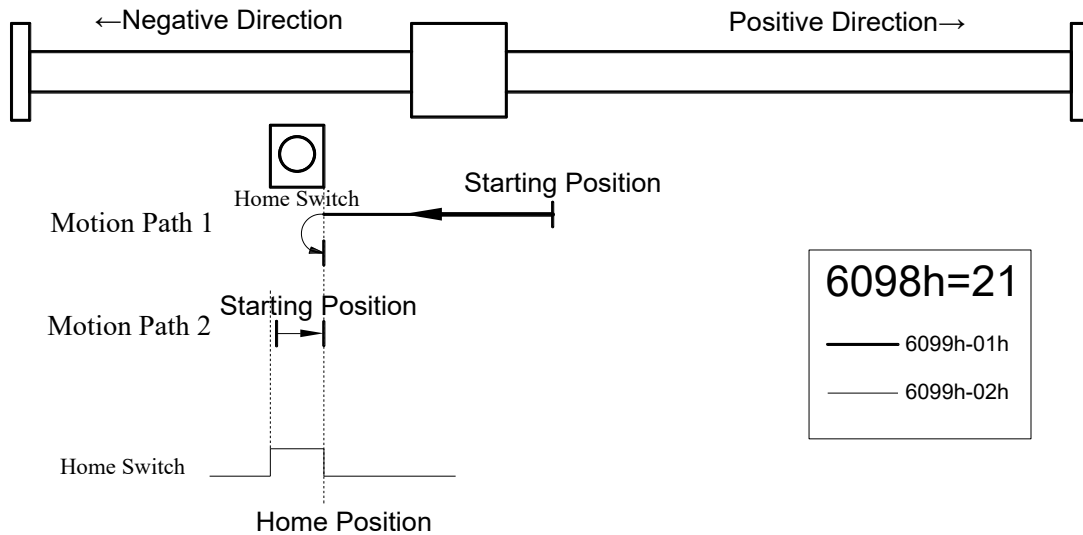
6098h=21

Mechanical origin: Home switch

Deceleration point: Home switch

If the home switch is inactive at startup, run at home locate speed in the negative direction. After encountering the OFF→ON state of the home switch while running in the negative direction, decelerate and stop, then switch to home creep speed and run in the positive direction. When running at home creep speed in the positive direction, upon encountering the ON→OFF state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 1.

If the home switch is active at startup, it runs at home creep speed in the positive direction. When running in the positive direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 2.

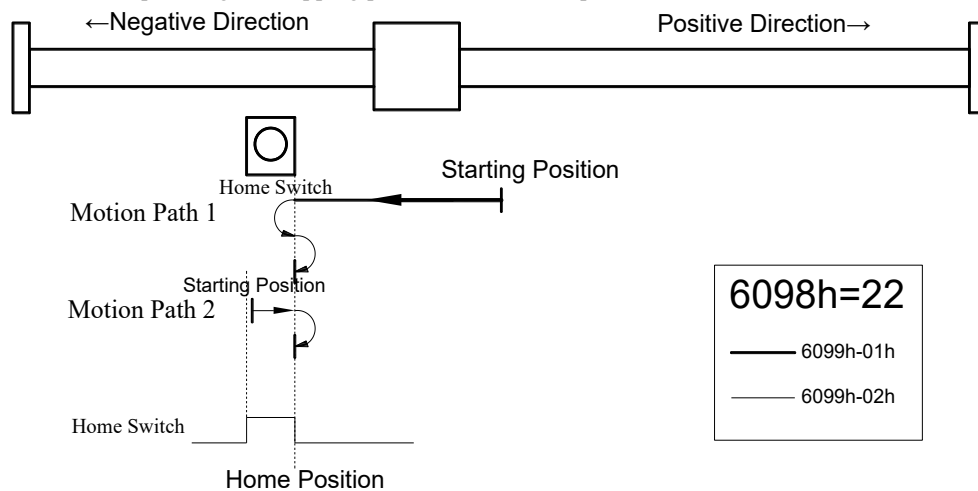
**6098h=22**

Mechanical origin: Home switch

Deceleration point: Home switch

If the home switch is inactive at startup, run at home locate speed in the negative direction. After encountering the OFF→ON state of the home switch while running in the negative direction, decelerate and stop, then switch to home creep speed and run in the positive direction. When running at home creep speed in the positive direction, upon encountering the ON→OFF state of the home switch, it decelerates and stops, then switches to home creep speed and runs in the negative direction. When running at home creep speed in the negative direction, upon encountering the OFF→ON state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 1.

If the home switch is active at startup, it runs at home creep speed in the positive direction. When running in the positive direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, then switches to home creep speed and runs in the negative direction. When running at home creep speed in the negative direction, upon encountering the OFF→ON state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 2.

**6098h=23**

Mechanical origin: Home switch

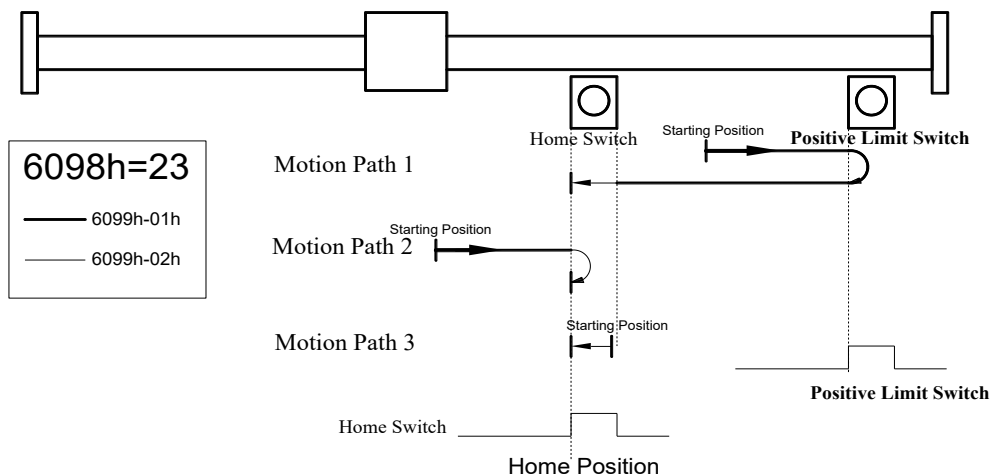
Deceleration point: Home switch

If the home switch is inactive and the positive limit switch is inactive at startup, it runs at home locate speed in the positive direction. After encountering the positive limit switch while running in the positive direction, it decelerates and stops, then runs at home locate speed in the negative direction. After encountering the OFF→ON state of the home switch while running in the negative direction, decelerate, then run at home creep speed in the negative direction. When running at home creep speed in the negative direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 1.

If the home switch is inactive and the positive limit switch is inactive at startup, it runs at home locate speed in the positive direction. After encountering the OFF→ON state of the home switch while running in the positive direction, decelerate and stop, then switch to home creep speed and run in the negative direction. When running at home creep speed in the negative direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion

Path 2.

If the home switch is active at startup, then run at home creep speed in the negative direction. When running in the negative direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 3.



6098h=24

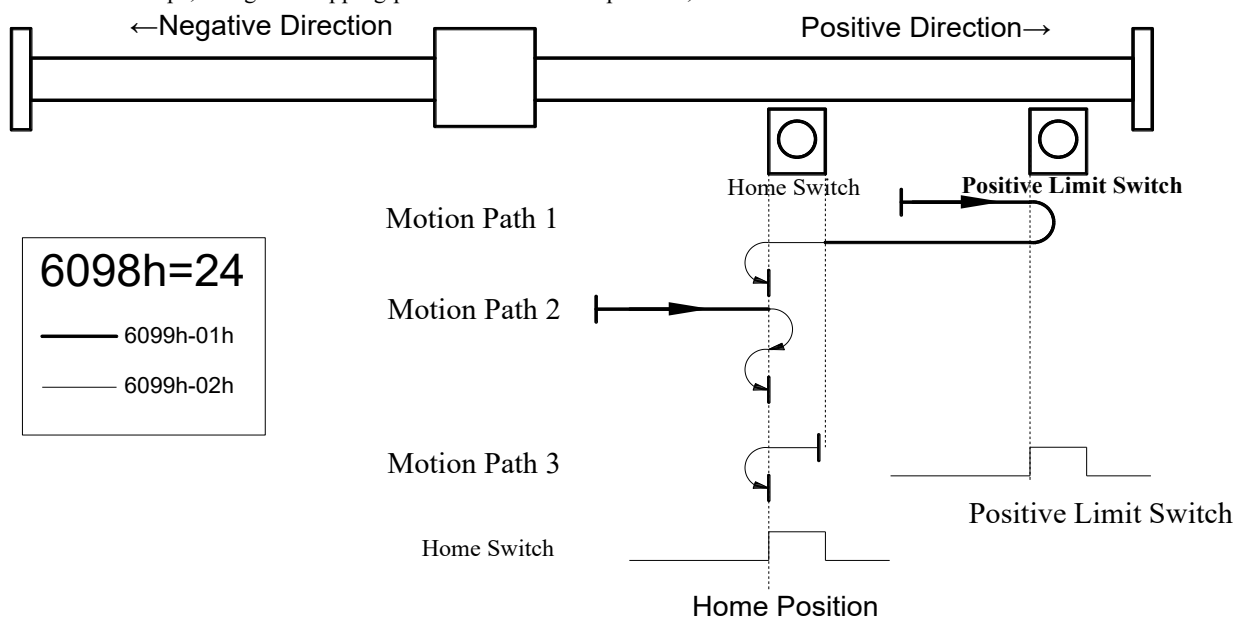
Mechanical origin: Home switch

Deceleration point: Home switch

If the home switch is inactive and the positive limit switch is inactive at startup, it runs at home locate speed in the positive direction. After encountering the positive limit switch while running in the positive direction, it decelerates and stops, then runs at home locate speed in the negative direction. After encountering the OFF→ON state of the home switch while running in the negative direction, decelerate, then run at home creep speed in the negative direction. When running at home creep speed in the negative direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, then runs at home creep speed in the positive direction. When running at home creep speed in the positive direction, upon encountering the OFF→ON state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 1.

If the home switch is inactive and the positive limit switch is inactive at startup, it runs at home locate speed in the positive direction. After encountering the OFF→ON state of the home switch while running in the positive direction, decelerate and stop, then switch to home creep speed and run in the negative direction. When running at home creep speed in the negative direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, then runs at home creep speed in the positive direction. When running at home creep speed in the positive direction, upon encountering the OFF→ON state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 2.

If the home switch is active at startup, then run at home creep speed in the negative direction. When running in the negative direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, then runs at home creep speed in the positive direction. When running at home creep speed in the positive direction, upon encountering the OFF→ON state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 3.



6098h=25

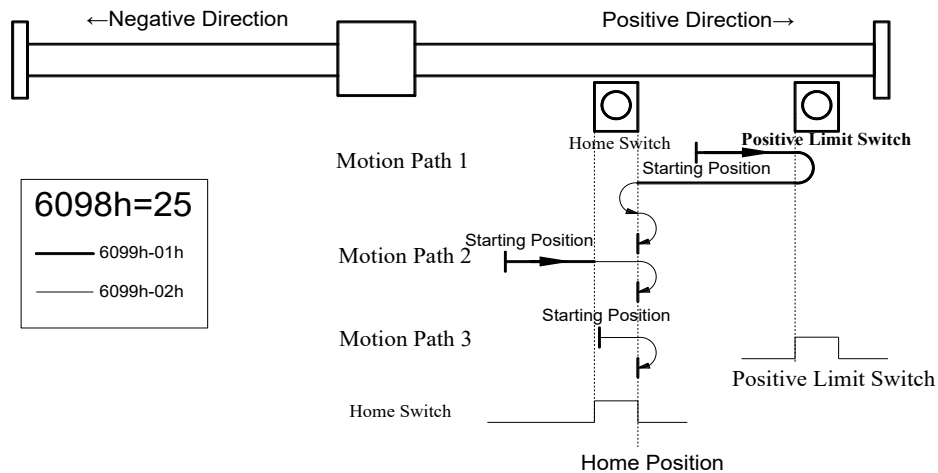
Mechanical origin: Home switch

Deceleration point: Home switch

If the home switch is inactive at startup and the positive limit switch is inactive, then run at home locate speed in the positive direction. After encountering the positive limit switch while running in the positive direction, it decelerates and stops, then runs at home locate speed in the negative direction. When running in the negative direction, after encountering the OFF→ON state of the home switch, it decelerates and stops, then runs at home creep speed in the positive direction. When running at home creep speed in the positive direction, upon encountering the ON→OFF state of the home switch, it decelerates and stops, then switches to home creep speed and runs in the negative direction. When running in the negative direction, upon encountering the OFF→ON state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 1.

If the home switch is inactive at startup and the positive limit switch is inactive, then run at home locate speed in the positive direction. When running in the positive direction, after encountering the OFF→ON state of the home switch, it decelerates, then switches to home creep speed and continues running in the positive direction. When running at home creep speed in the positive direction, upon encountering the ON→OFF state of the home switch, it decelerates and stops, then switches to home creep speed and runs in the negative direction. When running in the negative direction, upon encountering the OFF→ON state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 2.

If the home switch is active at startup, it runs at home creep speed in the positive direction. When running in the positive direction, upon encountering the ON→OFF state of the home switch, it decelerates and stops, then switches to home creep speed and runs in the negative direction. When running in the negative direction, upon encountering the OFF→ON state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 3.

**6098h=26**

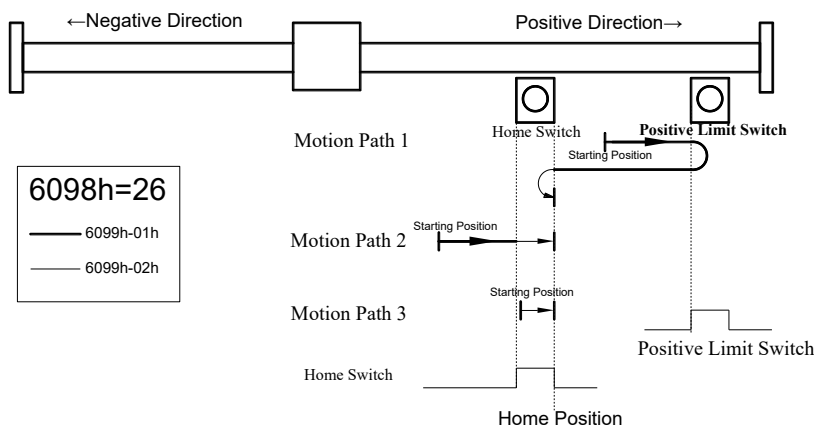
Mechanical origin: Home switch

Deceleration point: Home switch

If the home switch is inactive at startup and the positive limit switch is inactive, then run at home locate speed in the positive direction. After encountering the positive limit switch while running in the positive direction, it decelerates and stops, then runs at home locate speed in the negative direction. When running in the negative direction, after encountering the OFF→ON state of the home switch, it decelerates and stops, then runs at home creep speed in the positive direction. When running at home creep speed in the positive direction, upon encountering the ON→OFF state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 1.

If the home switch is inactive at startup and the positive limit switch is inactive, then run at home locate speed in the positive direction. When running in the positive direction, after encountering the OFF→ON state of the home switch, it decelerates, then switches to home creep speed and continues running in the positive direction. When running at home creep speed in the positive direction, upon encountering the ON→OFF state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 2.

If the home switch is active at startup, it runs at home creep speed in the positive direction. When running in the positive direction, upon encountering the ON→OFF state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 3.

**6098h=27**

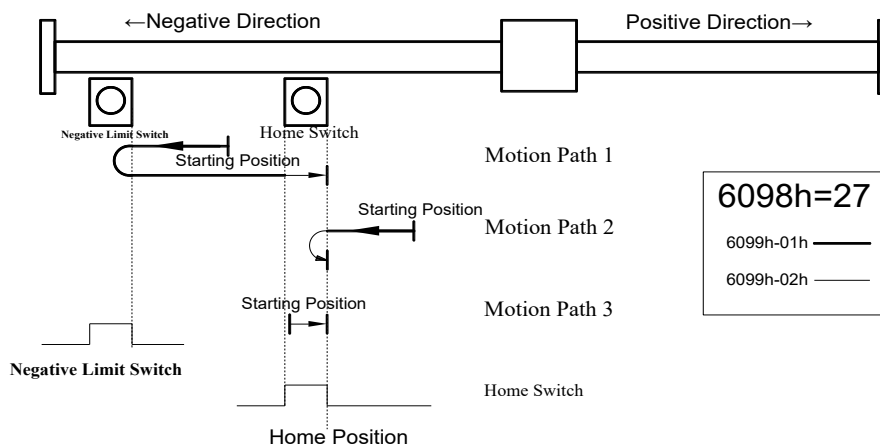
Mechanical origin: Home switch

Deceleration point: Home switch

If the home switch is inactive at startup and the negative limit switch is inactive, it runs at home locate speed in the negative direction. After encountering the negative limit switch while running in the negative direction, it decelerates and stops, then runs at home locate speed in the positive direction. When running in the positive direction, after encountering the OFF→ON state of the home switch, it decelerates, then runs at home creep speed and continues running in the positive direction. When running at home creep speed in the positive direction, upon encountering the ON→OFF state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 1.

If the home switch is inactive at startup and the negative limit switch is inactive, it runs at home locate speed in the negative direction. When running in the negative direction, after encountering the OFF→ON state of the home switch, it decelerates and stops, then switches to home creep speed and runs in the positive direction. When running at home creep speed in the positive direction, upon encountering the ON→OFF state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 2.

If the home switch is active at startup, it runs at home creep speed in the positive direction. When running in the positive direction, upon encountering the ON→OFF state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 3.

**6098h=28**

Mechanical origin: Home switch

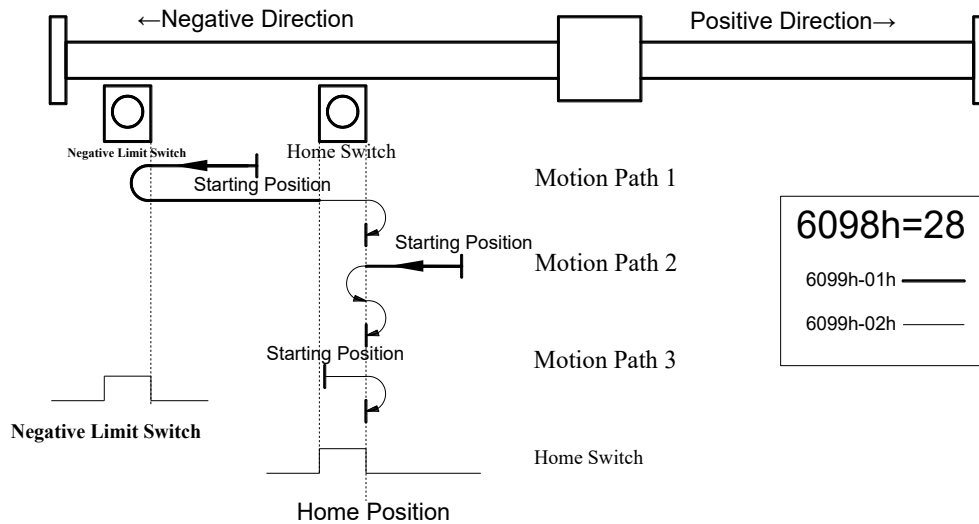
Deceleration point: Home switch

If the home switch is inactive at startup and the negative limit switch is inactive, it runs at home locate speed in the negative direction. After encountering the negative limit switch while running in the negative direction, it decelerates and stops, then runs at home locate speed in the positive direction. When running in the positive direction, after encountering the OFF→ON state of the home switch, it decelerates, then runs at home creep speed and continues running in the positive direction. When running at home creep speed in the positive direction, upon encountering the ON→OFF state of the home switch, it decelerates and stops, then runs at home creep speed in the negative direction. When running at home creep speed in the negative direction, upon encountering the OFF→ON state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 1.

If the home switch is inactive at startup and the negative limit switch is inactive, it runs at home locate speed in the negative direction. When running in the negative direction, after encountering the OFF→ON state of the home switch, it decelerates and stops, then switches to home creep speed and runs in the positive direction. When running at home creep speed in the positive direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, then runs at home creep speed in the negative direction.

When running at home creep speed in the negative direction, after encountering the OFF→ON state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 2.

If the home switch is active at startup, it runs at home creep speed in the positive direction. When running in the positive direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, then runs at home creep speed in the negative direction. When running at home creep speed in the negative direction, after encountering the OFF→ON state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 3.



6098h=29

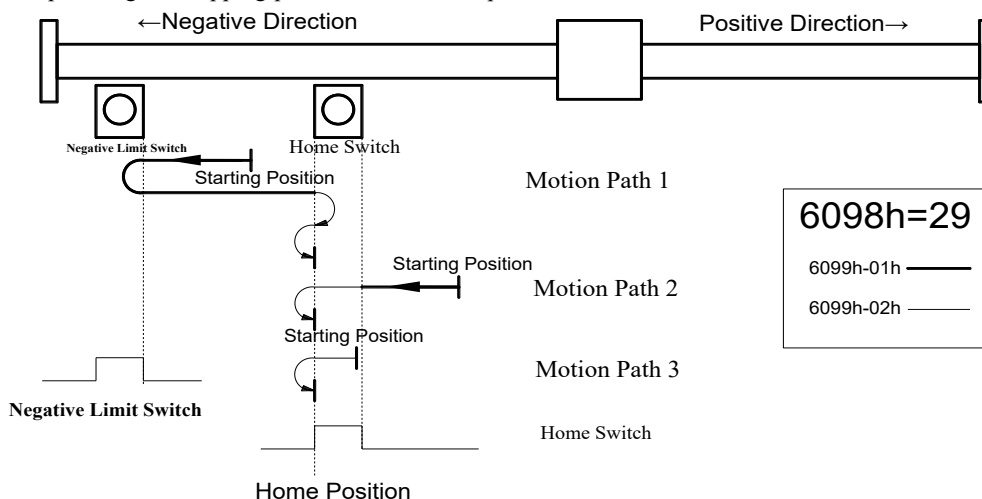
Mechanical origin: Home switch

Deceleration point: Home switch

If the home switch is inactive at startup and the negative limit switch is inactive, it runs at home locate speed in the negative direction. After encountering the negative limit switch while running in the negative direction, it decelerates and stops, then runs at home locate speed in the positive direction. When running in the positive direction, after encountering the OFF→ON state of the home switch, it decelerates and stops, then switches to home creep speed and runs in the negative direction. When running at home creep speed in the negative direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, then runs at home creep speed in the positive direction. When running at home creep speed in the positive direction, after encountering the OFF→ON state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 1.

If the home switch is inactive at startup and the negative limit switch is inactive, it runs at home locate speed in the negative direction. When running in the negative direction, after encountering the OFF→ON state of the home switch, it decelerates, then switches to home creep speed and continues running in the negative direction. When running at home creep speed in the negative direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, then runs at home creep speed in the positive direction. When running at home creep speed in the positive direction, after encountering the OFF→ON state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 2.

If the home switch is active at startup, then run at home creep speed in the negative direction. When running in the negative direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, then runs at home creep speed in the positive direction. When running at home creep speed in the positive direction, after encountering the OFF→ON state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 3.



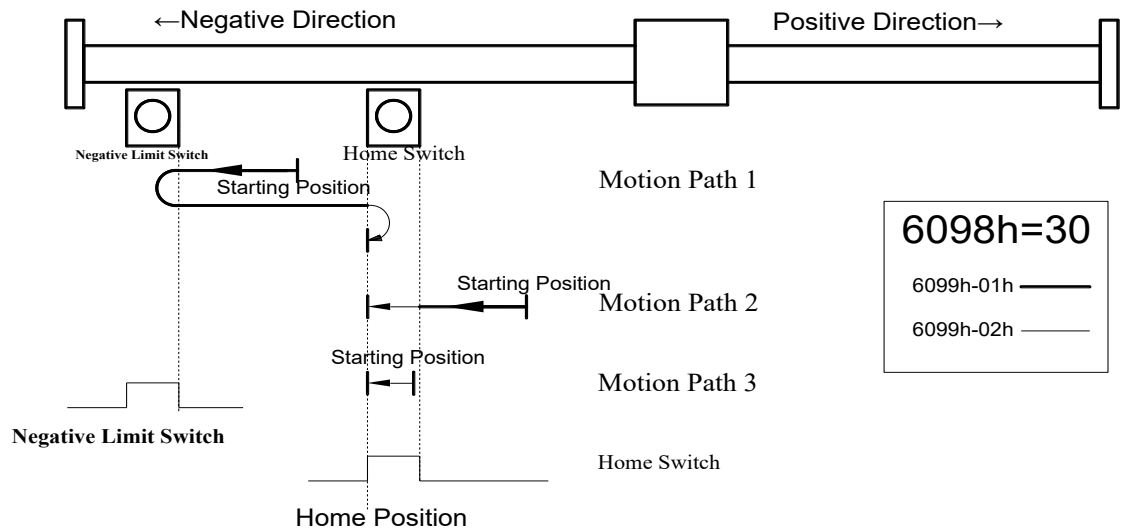
6098h=30

Mechanical origin: Home switch
Deceleration point: Home switch

If the home switch is inactive at startup and the negative limit switch is inactive, it runs at home locate speed in the negative direction. After encountering the negative limit switch while running in the negative direction, it decelerates and stops, then runs at home locate speed in the positive direction. When running in the positive direction, after encountering the OFF→ON state of the home switch, it decelerates and stops, then switches to home creep speed and runs in the negative direction. When running at home creep speed in the negative direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 1.

If the home switch is inactive at startup and the negative limit switch is inactive, it runs at home locate speed in the negative direction. When running in the negative direction, after encountering the OFF→ON state of the home switch, it decelerates, then switches to home creep speed and continues running in the negative direction. When running at home creep speed in the negative direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 2.

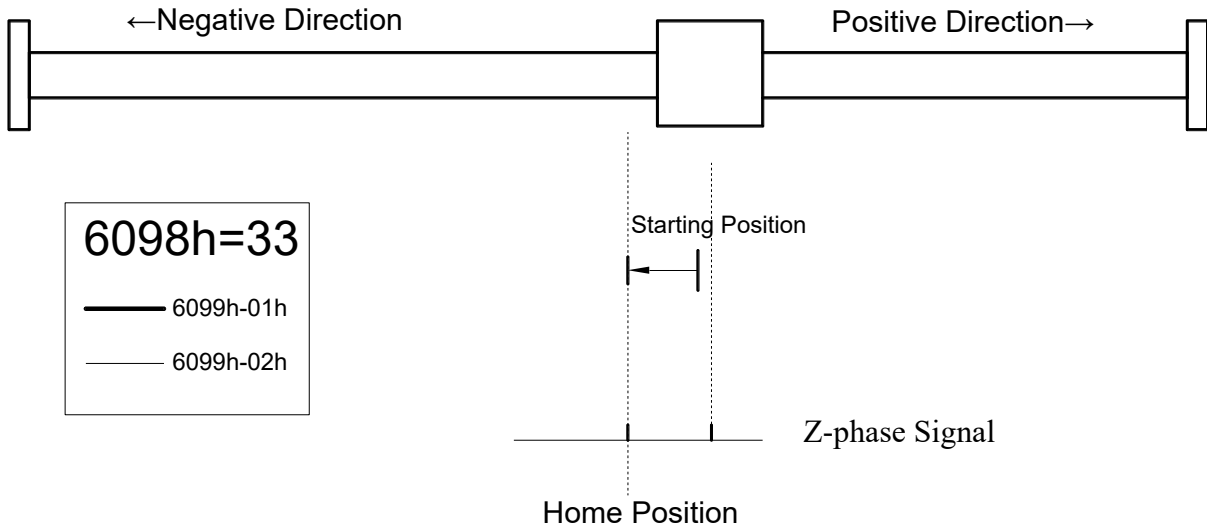
If the home switch is active at startup, then run at home creep speed in the negative direction. When running in the negative direction, after encountering the ON→OFF state of the home switch, it decelerates and stops, using the stopping position as the home position, as shown in Motion Path 3.



6098h=33

Mechanical origin: Motor Z signal
Deceleration point: None

At startup, it searches for the nearest Z-phase signal as the home position, running at home creep speed (6099h-02h) in the negative direction.

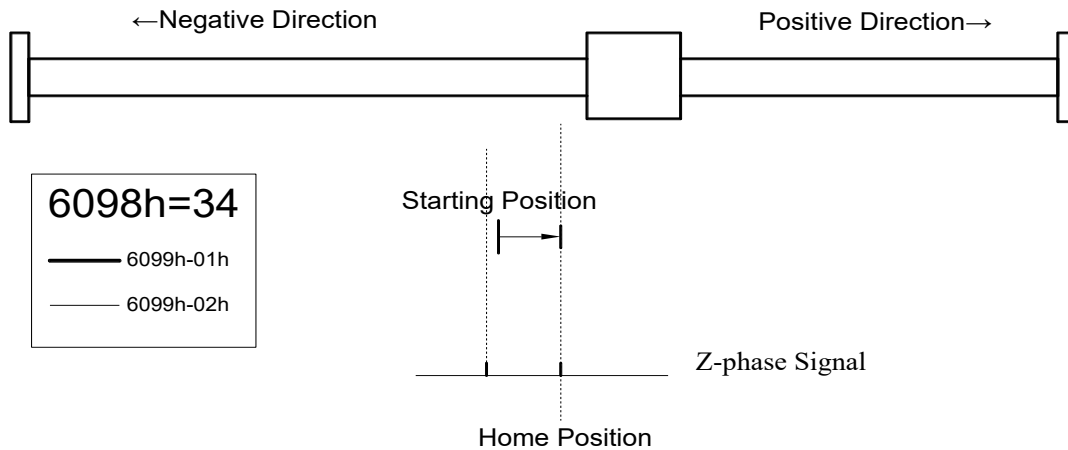


6098h=34

Mechanical origin: Motor Z signal

Deceleration point: None

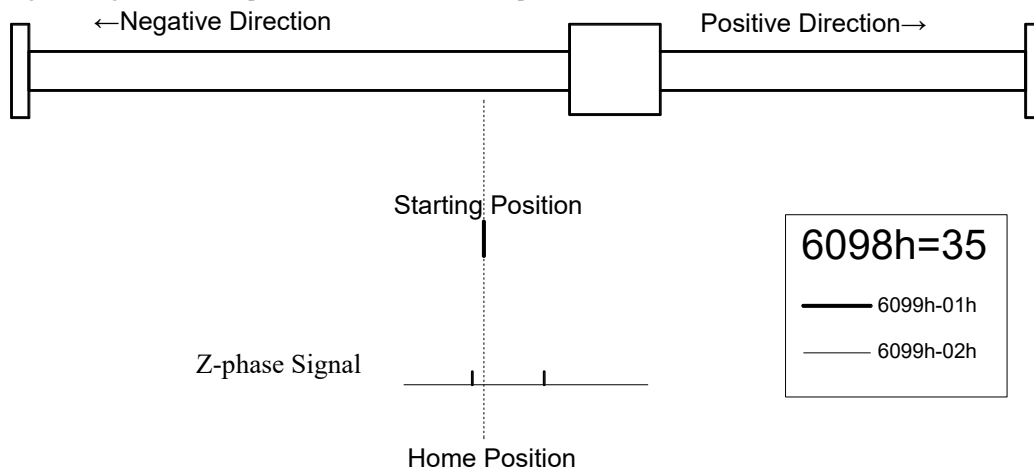
At startup, it searches for the nearest Z-phase signal as the home position, running at home creep speed (6099h-02h) in the positive direction.

**6098h=35**

Mechanical origin: Current position

Deceleration point: None

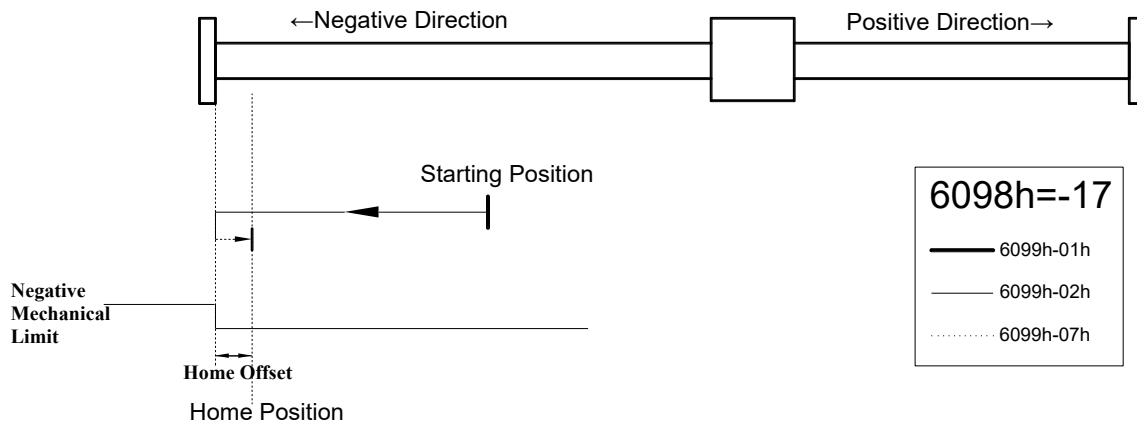
After triggering homing, the current position is set as the home position.

**6098h= -17**

Mechanical origin: Negative mechanical limit

Deceleration point: None

After triggering homing, it runs at home creep speed (6099h-02h) in the negative direction. After hitting the negative mechanical limit, when the motor speed is 0 and the current is greater than "Origin Finding Current (6099h-04h)", it moves in the positive direction by the set origin offset distance at the offset positioning speed (6099h-07h), using this position as the home position reference point.

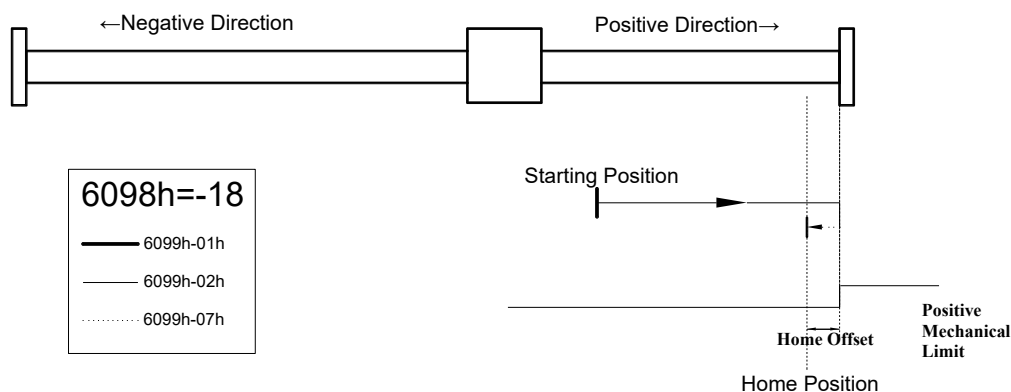


6098h= -18

Mechanical origin: Positive mechanical limit

Deceleration point: None

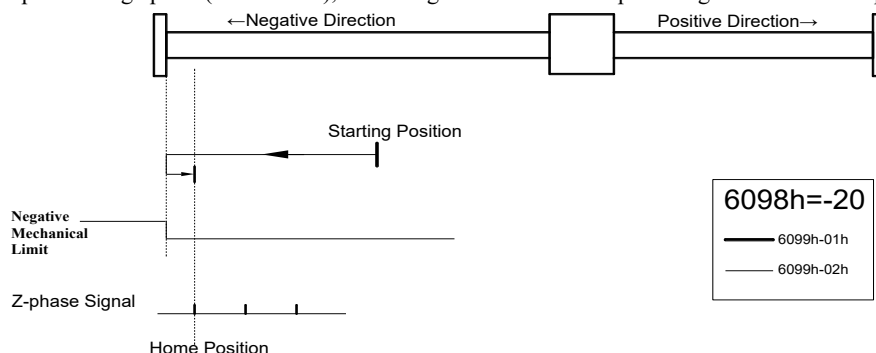
After triggering homing, it runs at home creep speed (6099h-02h) in the positive direction. After hitting the positive mechanical limit, when the motor speed is 0 and the current is greater than "Origin Finding Current (6099h-04h)", it moves in the negative direction by the set origin offset distance at the offset positioning speed (6099h-07h), using this position as the home position reference point.

**6098h= -20**

Mechanical origin: Motor Z signal

Deceleration point: None

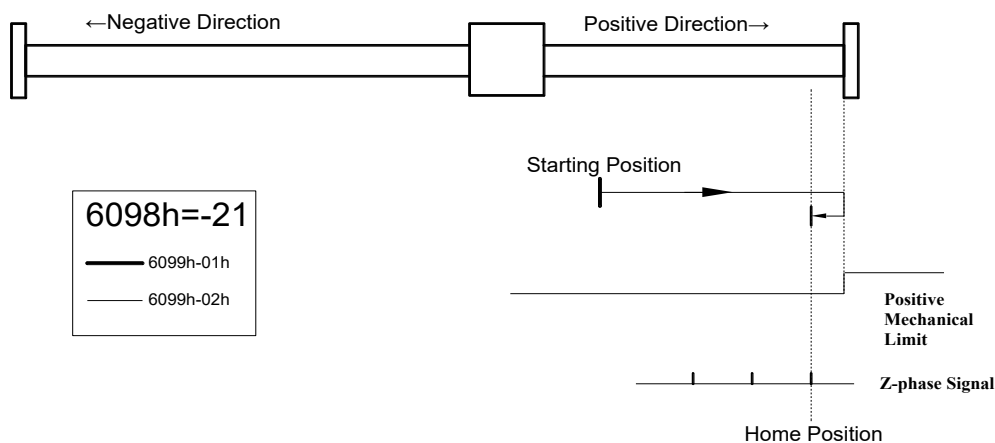
After triggering homing, it runs at home creep speed (6099h-02h) in the negative direction. After hitting the negative mechanical limit, when the motor speed is 0 and the current is greater than "Origin Finding Current (6099h-04h)", it switches to the positive direction and runs at the offset positioning speed (6099h-07h), searching for the nearest Z-phase signal as the home position.

**6098h= -21**

Mechanical origin: Motor Z signal

Deceleration point: None

After triggering homing, it runs at home creep speed (6099h-02h) in the positive direction. After hitting the positive mechanical limit, when the motor speed is 0 and the current is greater than "Origin Finding Current (6099h-04h)", it switches to the negative direction and runs at the offset positioning speed (6099h-07h), searching for the nearest Z-phase signal as the home position.



8.3.6 Cyclic Synchronous Position Mode (CSP)

In Cyclic Synchronous Position mode, the host computer controller completes position command planning and then periodically sends the planned target position 607Ah to the servo drive. Position, velocity, and torque control are completed internally by the servo drive. To enable Cyclic Synchronous Position mode, set object 6060h to 8.

(1) Control Description

Controlword 6040h and Statusword 6041h in CSP mode are described as follows:

Index	Sub-index	Name/Description	Access	PDO Mapping	Data Type	Ranges
6040h	00h	Controlword	RW	RxPDO	U16	0-65535
		Bit	Name	Description		
		0	Switch on	Must be set to 1 when enabling the servo		
		1	Enable voltage	Must be set to 1 when enabling the servo		
		2	Quick stop	Must be set to 1 to enable the servo. Setting to 0 results in quick stop		
		3	Operation enable	Must be set to 1 when enabling the servo		
		7	Fault Reset	Execute a fault reset when changing from 0→1. Other commands are invalid when set to 1		
6041h	00h	Statusword	RO	TxPDO	U16	0-65535
		Bit	Name	Description		
		0	Ready to switch on	0: Invalid, 1: Valid. When valid, servo can be enabled		
		1	Switched on	0: Invalid, 1: Valid. When valid, servo can be enabled		
		2	Operation enabled	0: Invalid, 1: Valid. When valid, indicates servo is enabled		
		3	Servo Fault	0: No fault, 1: Fault		
		4	Voltage enabled	0: Invalid, 1: Valid. When valid, servo can be enabled		
		5	Quick stop	0: Quick stop enabled, 1: Quick stop disabled		
		6	Switch on disabled	0: Invalid, 1: Valid. When valid, servo can be enabled		
		7	Warning	0: No warning, 1: With warning		
		9	Remote Control	0: Invalid, 1: Valid. When active, it indicates the Controlword has taken effect		
		10	Target Reached	0: Position not reached, 1: Position reached		
		11	Internal soft limit status	0: Soft limit not triggered, 1: Soft limit triggered		
		12	Slave Following Command	0: Target position not followed, 1: Target position followed already		
		13	Following Position Error Alarm	0: No position deviation alarm, 1: Position deviation alarm		
		15	Homing completed	0: Homing not completed, 1: Homing completed		

(2) Related Objects

Index	Sub-index	Name	Access Type	Validation	Data Type
603Fh	00h	Error Code	RO	Not changeable	UINT16
6040h	00h	Control Word	RW	Real-time change	UINT16
6041h	00h	State Word	RO	Not changeable	UINT16
605Ah	00h	Quick stop option	RW	Real-time change	UINT 16
6060h	00h	Control Mode	RW	Real-time change	INT 8
6061h	00h	Current operating mode	RO	Not changeable	INT 8
6064h	00h	Actual Position	RO	Not changeable	INT 32
6065h	00h	Position following error window	RW	Real-time change	UINT 32
606Ch	00h	Velocity loop feedback	RO	Not changeable	INT 32
6077h	00h	Actual torque	RO	Not changeable	INT 16
607Ah	00h	Target Position	RW	Real-time change	INT 32
607Dh	01h	Soft limit position 1	RW	Real-time change	INT 32
	02h	Soft limit position 2	RW	Real-time change	INT 32
607Eh	00h	Command polarity	RW	Real-time change	UINT 8

6080h	00h	Maximum motor velocity	RW	Real-time change	UINT 32
6085h	00h	Emergency Stop Deceleration	RW	Real-time change	UINT 32
60B0h	00h	Position Offset	RW	Real-time change	INT 32
60B1h	00h	Velocity Offset	RW	Real-time change	INT 32
60B2h	00h	Torque Offset	RW	Real-time change	INT 16
60F4h	00h	Position following error	RO	Not changeable	INT 32

(3) Recommended Configuration

RPDO	TPDO	Description
6040h: Control word	6041h: Status word	Required
607Ah: Target position	6064h: Position actual value	Required
6060h: Modes of operation	6061h: Modes of operation display	Optional

8.3.7 Cyclic Synchronous Velocity Mode (CSV)

In Cyclic Synchronous Velocity mode, the host computer controller periodically and synchronously sends the calculated target velocity 60FFh to the servo drive. Velocity and torque regulation are performed internally by the servo. To enable Cyclic Synchronous Velocity mode, set object 6060h to 9.

(1) Control Description

Controlword 6040h and Statusword 6041h in CSV mode are described as follows:

Index	Sub-index	Name/Description	Access	PDO Mapping	Data Type	Ranges	
6040h	00h	Controlword		RW	RxPDO	U16	0-65535
		Bit	Name	Description			
		0	Switch on	Must be set to 1 when enabling the servo			
		1	Enable voltage	Must be set to 1 when enabling the servo			
		2	Quick stop	Must be set to 1 to enable the servo. Setting to 0 results in quick stop			
		3	Operation enable	Must be set to 1 when enabling the servo			
		7	Fault Reset	Execute a fault reset when changing from 0→1. Other commands are invalid when set to 1			
6041h	00h	Statusword		RO	TxPDO	U16	0-65535
		Bit	Name	Description			
		0	Ready to switch on	0: Invalid, 1: Valid. When valid, servo can be enabled			
		1	Switched on	0: Invalid, 1: Valid. When valid, servo can be enabled			
		2	Operation enabled	0: Invalid, 1: Valid. When valid, indicates servo is enabled			
		3	Servo Fault	0: No fault, 1: Fault			
		4	Voltage enabled	0: Invalid, 1: Valid. When valid, servo can be enabled			
		5	Quick stop	0: Quick stop enabled, 1: Quick stop disabled			
		6	Switch on disabled	0: Invalid, 1: Valid. When valid, servo can be enabled			
		7	Warning	0: No warning, 1: With warning			
		9	Remote Control	0: Invalid, 1: Valid. When active, it indicates the Controlword has taken effect			
		11	Internal soft limit status	0: Soft limit not triggered, 1: Soft limit triggered			
		12	Slave Following Command	0: Target velocity not followed, 1: Target velocity followed already			

(2) Related Objects

Index	Sub-index	Name	Access Type	Validation	Data Type
603Fh	00h	Error Code	RO	Not changeable	UINT16
6040h	00h	Control Word	RW	Real-time change	UINT16
6041h	00h	State Word	RO	Not changeable	UINT16
605Ah	00h	Quick stop option	RW	Real-time change	UINT 16
6060h	00h	Control Mode	RW	Real-time change	INT 8
6061h	00h	Current operating mode	RO	Not changeable	INT 8
6064h	00h	Actual Position	RO	Not changeable	INT 32
606Bh	00h	Speed Loop Command	RO	Not changeable	INT 32
606Ch	00h	Velocity loop feedback	RO	Not changeable	INT 32
606Dh	00h	Velocity Reached Threshold	RW	Real-time change	INT 16
606Eh	00h	Velocity Reached Time	RW	Real-time change	INT 16
6077h	00h	Actual torque	RO	Not changeable	INT 16
607Eh	00h	Command polarity	RW	Real-time change	UINT 8
6080h	00h	Maximum motor velocity	RW	Real-time change	UINT 32
6085h	00h	Emergency Stop Deceleration	RW	Real-time change	UINT 32
60B1h	00h	Velocity Offset	RW	Real-time change	INT 32
60B2h	00h	Torque Offset	RW	Real-time change	INT 16
60E0h	00h	Positive torque limit	RW	Real-time change	UINT 16
60E1h	00h	Negative torque limit	RW	Real-time change	UINT 16
60FFh	00h	Target Speed	RW	Real-time change	INT 32

(3) Recommended Configuration

RPDO	TPDO	Description
6040h : Control word	6041h : Status word	Required
60FFh : Target velocity		Required
6060h : Modes of operation	6061h : Modes of operation display	Optional
	6064h : Position actual value	Optional
	606Ch : Velocity actual value	Optional

8.3.3 Cyclic Synchronous Torque Mode (CST)

In Cyclic Synchronous Torque mode, the host computer controller periodically and synchronously sends the calculated target torque 6071h to the servo drive. Torque regulation is performed internally by the servo. To enable Cyclic Synchronous Torquemode, set object 6060h to 10.

(1) Control Description

Controlword 6040h and Statusword 6041h in CST mode are described as follows:

Index	Sub-index	Name/Description	Access	PDO Mapping	Data Type	Ranges	
6040h	00h	Controlword	RW	RxPDO	U16	0-65535	
		Bit	Name	Description			
		0	Switch on	Must be set to 1 when enabling the servo			
		1	Enable voltage	Must be set to 1 when enabling the servo			
		2	Quick stop	Must be set to 1 to enable the servo. Setting to 0 results in quick stop			
		3	Operation enable	Must be set to 1 when enabling the servo			
		7	Fault Reset	Execute a fault reset when changing from 0→1. Other commands are invalid when set to 1			

6041h	00h	Statusword	RO	TxPDO	U16	0-65535
		Bit	Name	Description		
		0	Ready to switch on	0: Invalid, 1: Valid. When valid, servo can be enabled		
		1	Switched on	0: Invalid, 1: Valid. When valid, servo can be enabled		
		2	Operation enabled	0: Invalid, 1: Valid. When valid, indicates servo is enabled		
		3	Servo Fault	0: No fault, 1: Fault		
		4	Voltage enabled	0: Invalid, 1: Valid. When valid, servo can be enabled		
		5	Quick stop	0: Quick stop enabled, 1: Quick stop disabled		
		6	Switch on disabled	0: Invalid, 1: Valid. When valid, servo can be enabled		
		7	Warning	0: No warning, 1: With warning		
		9	Remote Control	0: Invalid, 1: Valid. When active, it indicates the Controlword has taken effect		
		12	Slave Following Command	0: Target torque not followed, 1: Target torque followed already		

(2) Related Objects

Index	Sub-index	Name	Access Type	Validation	Data Type
4010h	6A	Torque mode speed limit	RW	Real-time change	UINT 32
603Fh	00h	Error Code	RO	Not changeable	UINT16
6040h	00h	Control Word	RW	Real-time change	UINT16
6041h	00h	State Word	RO	Not changeable	UINT16
605Ah	00h	Quick stop option	RW	Real-time change	UINT 16
6060h	00h	Control Mode	RW	Real-time change	INT 8
6061h	00h	Current operating mode	RO	Not changeable	INT 8
606Ch	00h	Velocity actual value	RO	Not changeable	INT 32
6071h	00h	Target Torque	RW	Real-time change	INT 16
6072h	00h	Maximum Torque	RW	Real-time change	INT 16
6074h	00h	Torque command	RO	Not changeable	INT 16
6077h	00h	Actual torque	RO	Not changeable	INT 16
607Eh	00h	Command polarity	RW	Real-time change	UINT 8
6085h	00h	Emergency Stop Deceleration	RW	Real-time change	UINT 32
60E0h	00h	Positive torque limit	RW	Real-time change	UINT 16
60E1h	00h	Negative torque limit	RW	Real-time change	UINT 16

(3) Recommended Configuration

RPDO	TPDO	Description
6040h: Control word	6041h: Status word	Required
6071h: Target torque		Required
6060h: Modes of operation	6061h: Modes of operation display	Optional
	6064h: Position actual value	Optional
	606Ch: Velocity actual value	Optional
	6077h: Torque actual value	Optional

8.4 Application Functions

8.4.1 Probe Function

The probe function, also known as position latching, is a feature where the servo drive records the position information at the moment an externally specified DI signal or motor Z signal changes, and stores it in designated registers. The CD200 supports 2 probe functions.

(1) Control Description

Related settings for the probe function are done via object (60B8h). The meaning of each bit is shown in the table below:

Bit	Description	ReMarks
0	Probe 1 Enable: 0: Probe 1 not enabled 1: Probe 1 enabled	bit0-bit5: Probe 1 related settings - When using DI as the probe trigger signal, the DI source cannot be changed after the probe is enabled. - For absolute encoders, the Z signal refers to the zero point of the motor's single-turn position feedback. - When bit0/bit8 transitions from "0→1", position information is latched according to the setting of (bit1-bit5)/(bit9-bit13) and stored in objects 60BAh (Probe 1 rising edge position), 60BBh (Probe 1 falling edge position), 60BCh (Probe 2 rising edge position), and 60BDh (Probe 2 falling edge position) - If the probe function setting is changed, please return bit0/bit8 to 0, then change it from "0→1" again for the change to take effect
1	Probe 1 Trigger Mode: 0: Single trigger, only when the trigger signal is active for the first time 1: Continuous trigger	
2	Probe 1 Trigger Signal Selection: 0: DI input signal 1: Motor encoder Z signal	
3	Reserved	
4	Probe 1 Rising Edge Enable: 0: Do not use Probe 1 rising edge latch 1: Use Probe 1 rising edge latch	
5	Probe 1 Falling Edge Enable: 0: Do not use Probe 1 falling edge latch 1: Use Probe 1 falling edge latch	
6-7	Reserved	Bit8-bit13: Probe 2 related settings
8	Probe 2 Enable: 0: Probe 2 not enabled 1: Probe 2 enabled	
9	Probe 2 Trigger Mode: 0: Single trigger, only when the trigger signal is active for the first time 1: Continuous trigger	
10	Probe 2 Trigger Signal Selection: 0: DI input signal 1: Motor encoder Z signal	
11	Reserved	
12	Probe 2 Rising Edge Enable: 0: Do not use Probe 1 rising edge latch 1: Use Probe 1 rising edge latch	
13	Probe 2 Falling Edge Enable: 0: Do not use Probe 1 falling edge latch 1: Use Probe 1 falling edge latch	
14-15	Reserved	

Object (60B9h) allows viewing probe running status. The meaning of each bit is shown in the table below:

Bit	Description	ReMarks
0	Probe 1 Enable: 0: Probe 1 not enabled 1: Probe 1 enabled.	bit0-bit2: Probe 1 Running Status
1	Probe 1 Rising Edge Trigger Data Valid: 0: Rising edge trigger data invalid 1: Rising edge trigger data valid	
2	Probe 1 Falling Edge Trigger Data Valid: 0: Falling edge trigger data invalid 1: Falling edge trigger data valid	
3-7	Reserved	
8	Probe 2 Enable: 0: Probe 2 not enabled 1: Probe 2 enabled	Bit8-bit10: Probe 2 Running Status
9	Probe 2 Rising Edge Trigger Data Valid: 0: Rising edge trigger data invalid 1: Rising edge trigger data valid	
10	Probe 2 Falling Edge Trigger Data Valid: 0: Falling edge trigger data invalid 1: Falling edge trigger data valid	
11-15	Reserved	

(2) Related Objects

Index	Sub-index	Name	Access Type	Validation	Data Type
60B8h	00h	Probe Settings	RW	Real-time change	UINT16
60B9h	00h	Probe Status	RO	Not changeable	UINT16
60BAh	00h	Probe 1 Rising Edge Position	RO	Not changeable	INT 32
60BBh	00h	Probe 1 Falling Edge Position	RO	Not changeable	INT 32
60BCh	00h	Probe 2 Rising Edge Position	RO	Not changeable	INT 32
60BDh	00h	Probe 2 Falling Edge Position	RO	Not changeable	INT 32
60D1h	00h	Probe 1 Rising Edge Time	RO	Not changeable	UINT 32
60D2h	00h	Probe 1 Falling Edge Time	RO	Not changeable	UINT 32
60D3h	00h	Probe 2 Rising Edge Time	RO	Not changeable	UINT 32
60D4h	00h	Probe 2 Falling Edge Time	RO	Not changeable	UINT 32
60D5h	00h	Probe 1 Rising Edge Count	RO	Not changeable	UINT 16
60D6h	00h	Probe 1 Falling Edge Count	RO	Not changeable	UINT 16
60D7h	01h	Probe 2 Rising Edge Count	RO	Not changeable	UINT 16
60D8h	00h	Probe 2 Falling Edge Count	RO	Not changeable	UINT 16

8.4.2 Position Limit Function

CD200 supports traditional hardware limit functions: the extreme position is given by an external DI signal, connecting the external limit switch signal to the servo drive's CN1 interface (default DI3 for positive limit, DI4 for negative limit). It also supports the use of software limit functions, defining soft limits relative to the mechanical origin using object 607Dh as an absolute position. However, it must be noted that hardware limits and software limits cannot be used simultaneously; software limits are in Validation when hardware limits are used.

(1) Control Description

For software limit function to work properly, the following conditions must be met:

- In control modes (PP, PV, CSP, CSV)
- 607Dh-01h < 607Dh-02h

(2) Related Objects

Set the software limit function using parameters Pr.3.34(4010h-4A), Pr.3.35(607Dh-01h), Pr.3.36(607Dh-02h):

No.	Index	Sub-index	Name/Description	Description
Pr3.34	4010h	4Ah	Software Limit Enable	0: Disabled, 1: Enabled
Pr3.35	607Dh	01h	Soft limit position 1	Set the minimum position value for the soft limit function, restricting the motor's reverse movement range
Pr3.36	607Dh	02h	Soft limit position 2	Set the maximum position value for the soft limit function, restricting the motor's forward movement range

8.4.3 Forced DO Output

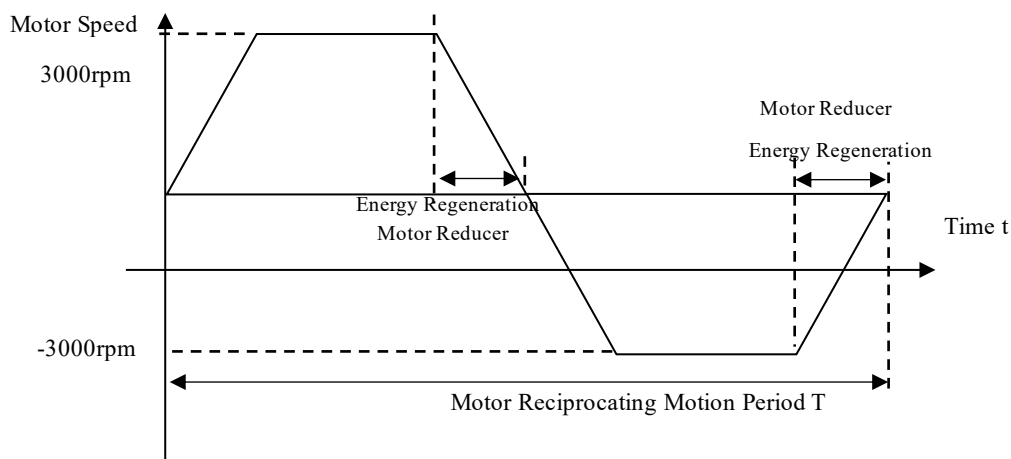
(1) Control Description

When ESM switches to OP, forced DO output is determined jointly by 60FEh-01h/60FEh-02h. DOs can be selected bitwise for forced DO output. CD200 can set 1 brake output and 3 forced DO outputs.

Index	Sub-index	Name/Description	Access	PDO Mapping	Data Type
60FEh	01h	Physical Outputs	RW	RxPDO	U32
		Bit	Name	Description	
		0	Brake Enable	0: Brake not active 1: Forced control of brake enable	
		16	DO1	0: DO no output 1: Forced DO output	
		17	DO2		
		18	DO3		
60FEh	02h	Bit Mask	RW	RxPDO	U32
		When the corresponding position in 60FEh-01h is set to 1, the setting in 60FEh-01h becomes Validation			

8.4.4 Virtual Braking Function

When the motor's torque and speed directions are opposite, energy is fed back from the motor to the drive, causing the bus voltage to rise. When it rises to the braking point, energy needs to be dissipated through a braking resistor or the virtual braking function to prevent further bus voltage increase, otherwise it may trigger an overvoltage alarm or even damage the servo drive. Taking the motor running from 3000rpm to standstill at no load as an example, the motor speed curve is as follows:



For drives experiencing overvoltage alarms, if the load is not heavy, virtual braking function can be considered first. The braking current should be gradually increased until the overvoltage alarm disappears. Using the virtual braking function eliminates the need for an external braking resistor, saving costs.

※ Note:

- A larger value for virtual braking current (2421h-07h) results in stronger braking capability (more regenerative energy can be consumed), but it also means the motor's operating temperature may be higher. Therefore, users should test and select an appropriate virtual braking current value based on actual working conditions, ensuring that the motor's operating temperature does not reach its maximum allowed value under corresponding conditions and has a certain temperature rise margin.
- Motor noise may increase when virtual braking is active.

Related Object

Index	Parameter Name	Set Value	Default Value	Unit	Access	Validation
2421h-06h	Virtual Braking Function	0: Not enabled Not 0: Enabled	0		RW	Immediate
2421h-07h	Virtual Braking Current	≤100	10	0.1A	RW	Immediate

8.4.5 Frequency Division Output

Encoder frequency division output is when the servo drive internally processes the feedback encoder position pulses and outputs them in the form of A/B phase quadrature pulses, to be used as position feedback in the upper device.

For details on frequency division output wiring, please refer to Section 4.3.5 Encoder Frequency Division Output.

Frequency Division Output Parameter Settings

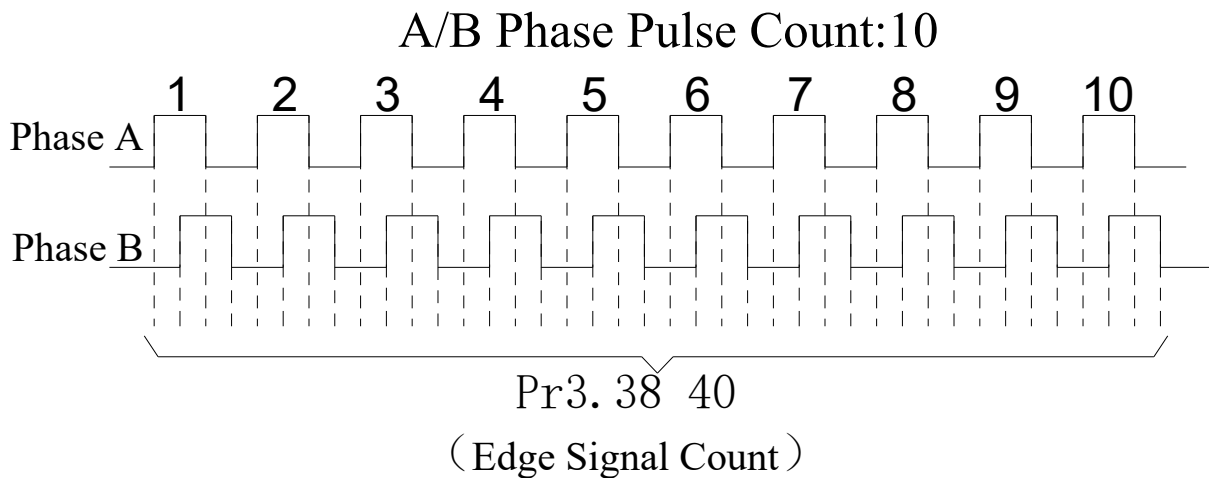
No.	Index	Parameter	Setting/Help Information	Default Value	Unit	Type	Validation
Pr3.38	40105B	Encoder Frequency Division Resolution	Number of A/B phase single-turn pulses for frequency division output, usually should not be set above 10000	10000	edge	U32	Restart
Pr3.39	401079	Frequency Division Output Z Pulse Width	Width of the frequency division output Z pulse signal (number of edge signals)	50	edge	U16	Restart

a. Encoder Frequency Division Resolution

Internally, the servo drive processes the number of pulses per turn from the encoder, and after frequency division, outputs the value of the number of pulse edges to the A/B phases.

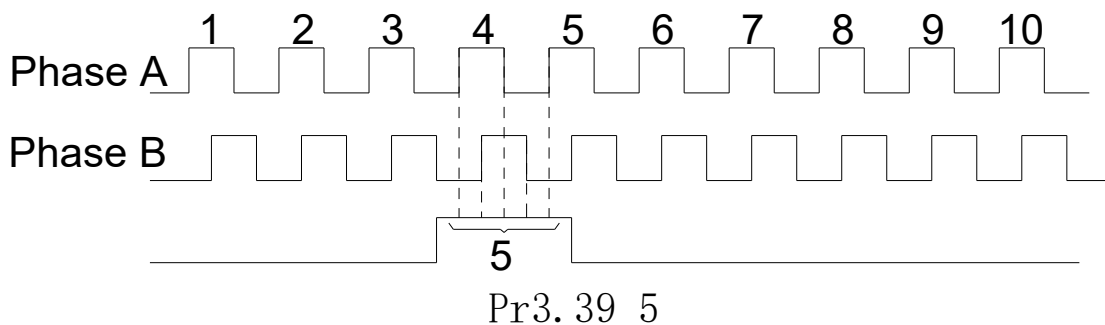
Output Example:

When Pr3.38 is set to 40, for one turn of the motor, phase A and phase B each output 10 pulse signals, and the number of pulse edge signals is 40, as shown below:



b. Frequency Division Output Z Pulse Width

The **frequency division output Z-phase pulse** is a benchmark signal generated once per motor turn, used to determine the zero position or identify a position. CD200 provides an adjustable Z-pulse output width function, used to broaden the encoder Z signal to meet different upper computer application requirements. The Z-pulse width setting value is the number of included edge signals starting from the rising edge of phase A. When set to 5, it is as follows:



Chapter 9 Parameters

9.1 Panel Parameters

ReMarks:

- ✧ The first parameter Prx.00 of all panel parameter groups corresponds to the storage function
- ✧ Indices are all represented in hexadecimal
- ✧ The "Mark" column RW indicates Read/Write, RO indicates Read-Only, and RWS indicates Read/Write and Save
- ✧ The second row of the "Mark" column indicates the parameter type. The letter 'U' indicates unsigned, the letter 'I' indicates signed, and the number indicates the bit width
- ✧ Objects that are operated in hexadecimal will be specially Marked in the "Type" attribute

9.1.1 F001 Group (Control Loop Parameter Group) Parameter List

No.	Index H	Parameter Name	Setting/Help Information	Default Value	Unit	Type	Access	Validation
Pr1.00	101001	Storage	Return value 2: indicates storage task is being executed Write 1: Store changed modules Write 0x786D68: Store checksum data Write 0x786D6872: Initialize and store checksum data Write 22: Restore entire machine to factory settings Write 3: Restore I/O and communication modules to factory settings Write 4: Reset axis 1 control and motor parameters Write 8: Reset axis 1 motor parameters Write 12: Clear fault records	0		U32 HEX	RW	Immediate
Pr1.01	402001	Position Loop Bandwidth 0	Correspond to position loop response capability Please refer to the "Adjustment" section for setting method	764	0.01Hz	U16	RW	Immediate
Pr1.02	402401	Speed Loop Proportional Gain 0	Correspond to speed loop response capability Please refer to the "Adjustment" section for setting method	Determined by motor parameters		U32	RWS	Immediate
Pr1.03	402501	Speed Loop Integral Gain 0	Correspond to speed loop steady-state following capability Please refer to the "Adjustment" section for setting method	Determined by motor parameters		U16	RWS	Immediate
Pr1.04	401028	Current Loop Proportional Gain	Correspond to current loop response capability This parameter is automatically calculated based on motor parameters by default It can be manually adjusted when using custom motors For manual adjustment, Pr1.06 needs to be enabled	Determined by motor parameters		U16	RWS	Immediate
Pr1.05	401027	Current Loop Integral Gain	Correspond to current loop steady-state following capability This parameter is automatically calculated based on motor parameters by default It can be manually adjusted when using custom motors For manual adjustment, Pr1.06 needs to be enabled	Determined by motor parameters		U16	RWS	Immediate
Pr1.06	401026	Current Loop PI Auto Calculation	0: Automatically calculate current loop PI parameters after each power-up 1: Use manually set current loop PI parameters	0		U8	RWS	Restart
Pr1.07	40102C	Position Command Low-Pass Filter	Correspond to position command low-pass filter coefficient Larger setting value means stronger filtering, greater lag Setting value -1 is the actual filtering coefficient 1-20000	1	62.5us	U16	RWS	Immediate
Pr1.08	40102D	Position Command Smoothing Filter	Correspond to position command moving average filter coefficient Larger setting value means stronger filtering, greater lag Setting value -1 is the actual filtering coefficient 1-2048	1	62.5us	U16	RWS	Immediate
Pr1.09	40101D	Speed Feedback Filter Bandwidth	Set the bandwidth of the speed feedback filter Larger setting value means weaker filtering, smaller lag Reasonable setting can reduce noise Too small a setting may affect loop stability 10-2000	300	Hz	U16	RWS	Immediate
Pr1.10	40101C	Torque Filter Bandwidth	Set the bandwidth of the current command filter Larger setting value means weaker filtering, smaller lag Reasonable setting can eliminate noise and resonance Too small a setting may affect loop stability	418	Hz	U16	RWS	Immediate
Pr1.11	300226	Speed Feedback Channel	0: Original speed information 1: Information after 1st order low-pass filtering 2: Information after 2nd order low-pass filtering Note: If encoder resolution exceeds 65536, setting to 2 has the same effect as setting to 1	2		U8	RWS	Immediate

Pr1.12	401024	Torque Filter Order	0: First-order low-pass 1: Second-order low-pass	1		U8	RWS	Immediate
Pr1.13	606500	Position following error window	One of the detection conditions for position following alarm, used with 606600 Trigger position following alarm detection when position error exceeds this setting	50000	PUL	U32	RWS	Immediate
Pr1.14	606600	PositionFollowing Error Time Window	One of the detection conditions for position following alarm Used with 606500 Trigger alarm when position error exceeds the setting value of 6065 and persists for this time window	10	ms	U16	RWS	Immediate
Pr1.15	606700	In-position Window	One of the detection conditions for in-position detection Used with 606800	10	PUL	U32	RWS	Immediate
Pr1.16	606800	In-position time window	One of the detection conditions for in-position detection Used with 606700	15	ms	U16	RWS	Immediate
Pr1.17	40101F	Velocity Feedforward Permille	Setting velocity feedforward can improve dynamic tracking performance Note: Setting value / 1024 corresponds to true permille	0	‰	U16	RWS	Immediate
Pr1.18	401020	Velocity Feedforward Filter Bandwidth	Filters velocity information in the position command 6-2000	100	Hz	U16	RWS	Immediate
Pr1.19	40104C	Gain Switching Mode	0: Do not use automatic gain switching If "Gain Select" is configured for the input function, the gain group corresponding to the "Gain Select" function is used. Otherwise, if rigidity level is used, the rigidity gain group is selected; otherwise, gain group 0 is selected 1: Switch gain to the group corresponding to 40104D when speed is 0 2: In position mode, when position error is less than the value corresponding to 401075, and the position command is unchanged, switch gain to the group corresponding to 40104D. For non-position modes, use gain group 0 or the rigidity gain group	0		U8	RWS	Immediate
Pr1.20	40104D	Gain Switching Target	Set the gain group number selected when automatic gain switching is enabled 0-3			U8	RWS	Immediate
Pr1.21	310005	Current Gain Selection	Display the number of the currently used gain group	0		U8	RO	
Pr1.22	402002	Position Loop Bandwidth 1	Reference Position Loop Bandwidth 0					
Pr1.23	402003	Position Loop Bandwidth 2	Reference Position Loop Bandwidth 0					
Pr1.24	402004	Position Loop Bandwidth 3	Reference Position Loop Bandwidth 0					
Pr1.25	402402	Speed Loop Proportional Gain 1	Reference Speed Loop Proportional Gain 0					
Pr1.26	402403	Speed Loop Proportional Gain 2	Reference Speed Loop Proportional Gain 0					
Pr1.27	402404	Speed Loop Proportional Gain 3	Reference Speed Loop Proportional Gain 0					
Pr1.28	402502	Speed Loop Integral Gain 1	Reference Speed Loop Integral Gain 0					
Pr1.29	402503	Speed Loop Integral Gain 2	Reference Speed Loop Integral Gain 0					
Pr1.30	402504	Speed Loop Integral Gain 3	Reference Speed Loop Integral Gain 0					
Pr1.31	402702	Integral Limit 1	Set the speed loop integral output current limit Should not exceed the current command limit when set	Drive limited	0.1A	U16	RWS	Immediate
Pr1.32	402703	Integral Limit 2	Reference "Integral Limit 1"					
Pr1.33	402704	Integral Limit 3	Reference "Integral Limit 1"					
Pr1.34	401025	Current Command Limit	Limit the drive's maximum output current	Drive limited	0.1A	U16	RWS	Immediate
Pr1.35	401021	Torque Feedforward Permille	Setting torque feedforward can improve dynamic tracking performance Torque feedforward setting assumes correct inertia ratio Note: Setting value / 1024 corresponds to true permille	0	‰	U16	RWS	Immediate
Pr1.36	401022	Torque Feedforward Filter Bandwidth	Filter acceleration in the position command 6-2000	100	Hz	U16	RWS	Immediate
Pr1.37	401016	Speed Mode Command Source	0: Speed command originates from the profile generator (operating mode 3) or immediate speed command source (operating mode -3) 1: Speed command originates from analog input 2: Speed command originates from pulse frequency 4: Speed command originates from Pr1.55(40109B)	0		U8	RWS	Immediate

Pr1.38	401064	Vibration Suppression Setting	Bit0-1: 0 does not enable vibration suppression function 1 Enable vibration filter A 2 Enable vibration filter B 3 Enable vibration filter A+B	0		U16 HEX	RWS	Immediate
Pr1.39	401065	Vibration Suppression Frequency A	Set frequency of vibration filter A 10-1000	100	0.1Hz	U16	RWS	Immediate
Pr1.40	401066	Vibration Suppression Intensity A	Set intensity of vibration filter A 1-30	10		U16	RWS	Immediate
Pr1.41	401067	Vibration Suppression Frequency B	Reference Group A	100	0.1Hz	U16	RWS	Immediate
Pr1.42	401068	Vibration Suppression Intensity B	Reference Group A	10		U16	RWS	Immediate
Pr1.43	40101E	Integral Limit 0	Reference Integral Limit 1					
Pr1.44	401076	Startup Compensation Current	Accelerate motor from 0 speed Validation in position mode	0	Internal Unit	U16	RWS	Immediate
Pr1.45	401077	Startup Compensation Time	Constant time for startup compensation current, which will gradually decay to 0 later	15	ms	U8	RWS	Immediate
Pr1.46	40105F	Forward Friction Current	This compensation current can be set to accelerate motor startup	0	0.1A	U8	RWS	Restart
Pr1.47	401060	Reverse Friction Current	This compensation current can be set to accelerate motor startup	0	0.1A	U8	RWS	Restart
Pr1.48	401061	Real-time Function Settings	Bit0: Enable online load detection Bit1: Enable friction compensation Bit2: Friction compensation source, 0 from 40105F and 401060, 1 from online detected value Bit3: Enable online gain adjustment, and adjust gain based on online measured inertia and user-selected rigidity level	1		U8 HEX	RWS	Immediate
Pr1.49	401062	Forward Maximum Friction Current	Limit the maximum forward friction current detected online	10	0.1A	U8	RWS	Restart
Pr1.50	401063	Reverse Maximum Friction Current	Limit the maximum reverse friction current detected online	10	0.1A	U8	RWS	Restart
Pr1.51	401075	Gain Switching Positioning Error Comparison Value	Used when gain switching mode is 2 Need to be adjusted according to actual conditions	2000	inc	U16	RWS	Immediate
Pr1.52	242213	Pulse command hardware filtering	Setting format is hexadecimal 0xAB. B represents the filter value, settable from 0 to F. A represents the division value, settable from 0 to 3. With the same filter value, a larger division value results in stronger filtering	00		U8 HEX	RWS	Immediate
Pr1.53	401099	Multi-turn cycle function	Set the usage method of the absolute encoder. 0: Used in absolute system (absolute mode). 2: Used in absolute system (absolute mode), but allows arbitrary setting of the multi-turn count upper limit and ignores multi-turn count overflow (infinite rotation absolute mode).	0		U8	RWS	Restart
Pr1.54	40109A	Multi-turn cycle maximum value	Set the upper limit of absolute multi-turn data in infinite rotation absolute mode. If the multi-turn data exceeds this set value, it will reset to zero; conversely, when reversing at 0, it will become the set value of this parameter.	50	round	U16	RWS	Restart
Pr1.55	40109B	Speed command in velocity mode	When the operating mode is 3 (speed mode) and the Pr1.37=4, this is the speed command set value.	0	rpm	U16	RWS	Immediate
Pr1.57	401044	Pulse overclocking	Input pulse frequency comparison point; if exceeded, an alarm will be triggered.	1000	KHz	U16	RWS	Immediate
Pr1.58	40109E	CAN disable	0: Enable CAN; other values: disable CAN	255		U8	RWS	Immediate
Pr1.59	401097	Pulse speed command filter coefficient	The larger the value, the lower the filter bandwidth	10		U16	RWS	Immediate
Pr1.60	401069	Enable torque speed limiting		1		U8	RWS	Immediate
Pr1.61	40106A	Speed limit in torque mode		3000	rpm	U16	RWS	Immediate
Pr1.62	40106B	Torque mode compensation current limit				U16	RWS	Immediate
Pr1.63	40106C	Torque speed proportional gain		100		U16	RWS	Immediate
Pr1.64	40106D	Torque speed integral gain		1		U16	RWS	Immediate

Pr1.65	401098	Gravity integral compensation current	Used to counteract vertical axis gravity	0	Ap	I8	RWS	Immediate
Pr1.66	4010AF	Gear expansion coefficient numerator	Used for gear ratio switching.	1		U16	RWS	Immediate
Pr1.67	4010B0	Gear expansion coefficient denominator	Used for gear ratio switching.	1		U16	RWS	Immediate

9.1.2 F002 Group (Motor Parameter Group) Parameter List

No.	Index H	Parameter Name	Setting/Help Information	Default Value	Unit	Type	Access	Validation
Pr2.00	101001	Storage	Refer to Pr1.00					
Pr2.01	310019	Current Motor Model	Display the currently connected motor model			U16	RO	
Pr2.02	47FE01	Motor Model Setting	Display or sets the motor model 0 means custom motor			U16	RWS	Restart
Pr2.03	47FE02	Motor Body ID	Correspond to motor body code			U16	RWS	Restart
Pr2.04	47FE03	Motor Encoder ID	Correspond to encoder code			U16	RWS	Restart
Pr2.05	47FE04	Motor Bandwidth	It can be fixedly set to 2000Hz	2000	Hz	U16	RWS	Restart
Pr2.06	47FE05	Rotor Inertia	Motor rotor inertia		kg·mm ²	U16	RWS	Restart
Pr2.07	47FE06	Back EMF Coefficient	Describe the relationship between motor speed and back EMF		0.1V/krpm	U16	RWS	Restart
Pr2.08	47FE07	Torque Coefficient	Describe the relationship between current (RMS) and torque		0.01Nm/A	U16	RWS	Restart
Pr2.09	47FE08	Motor Inductance	Line-to-line inductance		0.1mH	U16	RWS	Restart
Pr2.10	47FE09	Motor Overload Time	Motor's continuous operating time slightly above rated current		s	U16	RWS	Restart
Pr2.11	47FE0A	Motor Pole Pair Number	Number of magnetic pole pairs in the motor rotor			U16	RWS	Restart
Pr2.12	47FE0B	Rated Current of Motor	Current during motor continuous operation		0.1Arms	U16	RWS	Restart
Pr2.13	47FE0C	Rated Power of Motor	Rated Power of Motor		W	U16	RWS	Restart
Pr2.14	47FE0D	Motor Resistance	Line-to-line resistance		0.1Ω	U16	RWS	Restart
Pr2.15	47FE0E	Baud Rate Index	Describe the baud rate used by the encoder communication protocol	0		U8	RWS	Restart
Pr2.16	47FE0F	Motor Maximum Speed	Describe the maximum allowable motor speed		rpm	U16	RWS	Restart
Pr2.17	47FE10	Multi-turn Resolution	Use when the encoder is a multi-turn absolute encoder Otherwise, set to 0		round	U32	RWS	Restart
Pr2.18	47FE11	Single-turn Resolution	Encoder position resolution within one revolution		inc	U32	RWS	Restart
Pr2.19	47FE12	Encoder Type	Currently only support type 1	1		U8	RWS	Restart
Pr2.20	47FE13	Excitation Mode	0: Dither excitation 1: Use default excitation information	1		U16	RWS	Restart
Pr2.21	47FE14	Brake ID	Motor brake code Set to 0 if no brake	0		U16	RWS	Restart
Pr2.22	47FE15	Temperature Sensor ID	Motor built-in temperature sensor code Set to 0 if no temperature sensor	0		U16	RWS	Restart
Pr2.23	300102	Brake Delay	Set the brake release delay When Brake ID is not 0, the drive will attempt to control the brake drive signal to release the brake when enabled	100	ms	U16	RWS	Immediate
Pr2.24	40102A	Motor Model Auto-Recognition	0: Do not auto-recognize motor 1: Auto-recognize motor. The drive identifies the current encoder, then automatically reads data from inside the encoder.	1		U8	RWS	Immediate

9.1.3 F003 Group (Drive Configuration Parameter Group) Parameter List

No.	Index H	Parameter Name	Setting/Help Information	Default Value	Unit	Type	Access	Validation
Pr3.00	101001	Storage	Refer to Pr1.00					
Pr3.01	609101	Gear Input Turns	Motor shaft (gear input shaft) rotation turns Refer to "Motion Equivalent Setting"	1	r	U32	RWS	Immediate
Pr3.02	609102	Gear Output Turns	Gear output shaft rotation turns Refer to "Motion Equivalent Setting"	1	r	U32	RWS	Immediate
Pr3.03	609201	Feed	Number of commands issued by the controller Refer to "Motion Equivalent Setting"	10000	PUL	U32	RWS	Immediate
Pr3.04	609202	Rotations of Shaft	"Fee" corresponds to the number of rotations of the gear output shaft Refer to "Motion Equivalent Setting"	1	r	U32	RWS	Immediate

Pr3.05	607E00	Rotation Direction	0: Command positive polarity 1: Command negative polarity	0		U8	RWS	Restart
Pr3.06	25000C	Communication ID	Drive communication ID. Used for 485, CAN, or 232 cascaded buses. The host computer can set the ID to 127 to connect to drives with unknown communication IDs from 1 to 126	1		U8	RWS	Immediate
Pr3.07	401031	Operating Mode at Power-up	Set the drive's default operating mode at power-up	1		S8	RWS	Restart
Pr3.08	250004	Serial Port 1 Baud Rate	Serial port 1 is the drive's debug port, use a 485 interface. Baud rate cannot be changed arbitrarily Support 128000, 256000, 512000	256000		U32	RWS	Restart
Pr3.09	25000E	Temporary Baud Rate	When decide to use a non-default baud rate, first set the temporary baud rate code, if communication is normal then execute Serial Port 1 baud rate modification	1		U8	RW	Immediate
Pr3.10	250005	Serial Port 2 Baud Rate	Serial Port 2 is an RS485 bus interface	57600		U32	RWS	Restart
Pr3.11	250008	Serial Port 2 Protocol	0: Proprietary protocol 1: Modbus RTU	1		U8	RWS	Restart
Pr3.12	25000D	CAN Baud Rate	CAN bus baud rate setting			U8	RWS	Restart
Pr3.13	250010	Manufacturer Password	For drive factory configuration use			U32	WO	Immediate
Pr3.14	242105	User Password	Protect application data. When user password protection is not released, application data cannot read or write, user password returns 9527, otherwise returns 0	0		U16	RWS	Immediate
Pr3.15	300103	Relay Control Delay	Manufacturer Parameters	120	ms	U16	RWS	Immediate
Pr3.16	401012	Encoder Multi-turn Disable	Validation when the encoder is a multi-turn absolute encoder 0: Position feedback is multi-turn absolute 1: Disable multi-turn data, position feedback is single-turn	0		U8	RWS	Restart
Pr3.17	401013	Auto Enable	0: Do not auto-enable 1: Auto-enable at power-up 2: Automatically execute enable after each fault reset	0		U8	RWS	Restart
Pr3.18	401014	Auto Fault Reset	0: Do not auto-reset fault 1: Automatically clear after a certain time after fault occurs	0		U8	RWS	Immediate
Pr3.19	401015	Fault Reset Time	Validation when auto fault reset is enabled	5000	ms	U16	RWS	Immediate
Pr3.20	40102B	Auto Run Position Table	0: Do not auto-run position table ≥1: Auto-run position table at power-up 2: Auto-run position table after fault reset 3: Automatically execute position table after re-homing	0		U8	RWS	Immediate
Pr3.21	609903	Auto Origin Finding	0: Do not auto-homing ≥1: Auto-homing at power-up 2: Auto-homing after being enabled when origin is lost 3: Always auto-homing when fault is cleared	0		U8	RWS	Immediate
Pr3.22	242101	Use internal braking resistor	0: Not used 1: Use Used by default if an internal braking resistor is present	0/1		U8	RWS	Immediate
Pr3.23	242102	Braking Resistor Resistance	External braking resistor resistance	20	Ω	U8	RWS	Immediate
Pr3.24	242103	Braking Resistor Overload Time	External braking resistor overload time 1-255	15	s	U8	RWS	Immediate
Pr3.25	242104	Braking Resistor Rated Power	External braking resistor power	100	w	U16	RWS	Immediate
Pr3.26	210007	Braking Voltage	Trigger voltage to open the braking circuit Manufacturer Parameters		0.1V	U16	RWS	Immediate
Pr3.27	210008	Braking Release Voltage	Trigger voltage to close the braking circuit Manufacturer Parameters		0.1V	U16	RWS	Immediate
Pr3.28	401057	Current Comparison Value A	When this object is 0, the current comparison function is inactive	0	0.1A	S16	RWS	Restart
Pr3.29	401058	Current Comparison Rule	0: Compare actual current value 1: Compare absolute current value	0		U8	RWS	Immediate
Pr3.30	401059	Current Comparison Time	Time window required for current comparison output result	50	ms	U16	RWS	Immediate
Pr3.31	401034	Stop in step	0: Not enabled 1: When the drive stops moving, automatically switch to current lock state to prevent closed-loop jitter	0		U8	RWS	Restart
Pr3.32	40103A	Step Stop Start Current	Current when entering step stop mode	30	0.1A	U16	RWS	Immediate
Pr3.33	401045	Step Stop End Current	Holding current in step stop mode	10	0.1A	U16	RWS	Immediate
Pr3.34	40104A	Software Limit Enable	0: Do not enable 1: Enable	0		U8	RWS	Immediate
Pr3.35	607D01	Soft limit position 1	Correspond to the soft limit negative position	0	PUL	S32	RWS	Immediate
Pr3.36	607D02	Soft limit position 2	Correspond to the soft limit positive position	0	PUL	S32	RWS	Immediate
Pr3.37	401078	Locked Rotor Current	Stall Detection Threshold	Hardware limited	0.1A	U16	RWS	Restart

Pr3.38	40105B	Pulse Regeneration Quantity	Quantity of quadrature pulse regeneration per revolution	10000	edge	U32	RWS	Restart
Pr3.39	401079	Regeneration Index Width	Width of the pulse regeneration index signal	50	edge	U16	RWS	Restart
Pr3.40	242108	Dynamic Braking Mode	0: Continuously enable relay after enabling 1: Relay controlled by enable state 2: Automatically enable relay after power-up	0		U8	RWS	Immediate
Pr3.41	40100B	Pulse mode	0: Pulse + direction 1: CW + CCW 2: AB phase quadrature pulse 3: Two-channel unidirectional pulse bus 4: Four-channel bidirectional pulse bus 5: Two-channel bidirectional pulse bus with analog	0	-	U8	RWS	Immediate
Pr3.42	401085	High-speed pulse axis number	One-Drive-Multi-Axis Control Axis Number Setting 0: Corresponds to the 1st pulse input of the conversion card 1: Corresponds to the 2nd pulse input of the conversion card 2: Corresponds to the 3rd pulse input of the conversion card 3: Corresponds to the 4th pulse input of the conversion card	0	-	U8	RWS	Immediate
Pr3.43	25000B	Serial port 2 data Transmission format	0: Stop bit 1, no parity 1: Stop bit 1, odd parity 2: Stop bit 1, even parity 3: Stop bit 2, no parity 4: Stop bit 2, odd parity 5: Stop bit 2, even parity	0	-	U8	RWS	Restart
Pr3.44	401088	Enable pulse bus DIN	0: Disable 1: Enable	0	-	U8	RWS	Restart
Pr3.45	250006	port 1 communication protocol	Serial Port 1 Communication Protocol Manufacturer protocol	0		U8	RWS	Restart
Pr3.46	401004	Speed pulse frequency coefficient	Speed Pulse Frequency Coefficient rpm/kHz	100	rpm/kHz	I16	RWS	Immediate
Pr3.47	242106	Virtual braking function	0: Virtual braking function disabled Non-zero: Virtual braking function enabled Note: This function can be enabled when a slight braking capability is required.	0		U8	RWS	Restart
Pr3.48	242107	Virtual braking current		0	0.1A	U8	RWS	Restart
Pr3.49	605A00	Emergency stop option	Emergency Stop Options 0: Drive disable (free stop) 1: Decelerate using RAMP0 (6084), then disable 2: Decelerate using RAMP1 (6085), then disable 3, 4: Decelerate with maximum current limit, then disable 5, 9: Decelerate using RAMP0 (6084) and maintain emergency stop 6, 10: Decelerate using RAMP1 (6085) and maintain emergency stop 7, 8: Decelerate with maximum current limit and maintain emergency stop 11, 13: Decelerate with maximum current limit and maintain emergency stop 12: Decelerate with maximum current limit, clear integral, and maintain emergency stop 14: Decelerate using RAMP0 (6084) and remain in SWITCHED_ON state 15: Decelerate using RAMP0 (6084) and remain in READY_TO_SWITCH_ON state	0		I16	RWS	Immediate
Pr3.50	605B00	Shutdown option	0: Free stop 1: Decelerate to 0 and remain in READY_TO_SWITCH_ON (ready2on)	0		I16	RWS	Immediate
Pr3.51	605C00	Disable option	0: Free stop 1: Decelerate to 0 and remain in SWITCHED_ON	0		I16	RWS	Immediate
Pr3.52	608300	Profile acceleration		1000000	PUL/ss	U32	RWS	Immediate
Pr3.53	608400	Profile deceleration		1000000	PUL/ss	U32	RWS	Immediate
Pr3.54	608500	Emergency stop deceleration		20000000	PUL/ss	U32	RWS	Immediate
Pr3.55	607F00	Maximum profile velocity		600000	PUL/s	U32	RWS	Immediate

Pr3.56	608000	Maximum speed command (rpm)		3500	rpm	U32	RWS	Immediate
Pr3.57	608100	Profile velocity		50000	PUL/s	U32	RWS	Immediate
Pr3.58	607A00	Target position		0	PUL	I32	RWS	Immediate
Pr3.59	607100	Target torque		0	%	I16	RWS	Immediate
Pr3.60	607200	Maximum torque		3000	%	I16	RWS	Immediate
Pr3.61	40109D	Home switch position	Setting is referenced to position 6064h.	0	PUL	I32	RWS	Immediate
Pr3.62	40107A	PWM frequency Switch enablecode	0: Cannot switch (PWM frequency control mode disabled) 1: Can switch PWM frequency control mode (enabled)	1		U8	RWS	Restart

9.1.4 F004 Group (Input Pin Configuration) Parameter List

No.	Index H	Parameter Name	Setting/Help Information		Default Value	Unit	Type	Access	Validation
Pr4.00	101001	Storage	Refer to Pr1.00						
Pr4.01	240001	DIN1 Action Mode	0: Low level	1: High level	1		U8	RWS	Immediate
			2: Rising edge	3: Falling edge					
Pr4.02	240002	DIN2 Action Mode	Same as above		2		U8	RWS	Immediate
Pr4.03	240003	DIN3 Action Mode	Same as above		1				
Pr4.04	240004	DIN4 Action Mode	Same as above		1		U8	RWS	Immediate
Pr4.05	240005	DIN5 Action Mode	Same as above		1				
Pr4.06	240006	DIN6 Action Mode	Same as above		1		U8	RWS	Immediate
Pr4.07	240007	DIN7 Action Mode	Same as above		1				
Pr4.08	240008	DIN8 Action Mode	Same as above		1		U8	RWS	Immediate
Pr4.09	240101	DIN1 FilterCoefficient	Apply a low-pass filter to the DIN pins		100	ms	U16	RWS	Immediate
Pr4.10	240102	DIN2 FilterCoefficient	Apply a low-pass filter to the DIN pins		100	ms	U16	RWS	Immediate
Pr4.11	240103	DIN3 FilterCoefficient	Apply a low-pass filter to the DIN pins		2	ms	U16	RWS	Immediate
Pr4.12	240104	DIN4 FilterCoefficient	Apply a low-pass filter to the DIN pins		2	ms	U16	RWS	Immediate
Pr4.13	240105	DIN5 FilterCoefficient	Apply a low-pass filter to the DIN pins		2	ms	U16	RWS	Immediate
Pr4.14	240106	DIN6 FilterCoefficient	Apply a low-pass filter to the DIN pins		10	ms	U16	RWS	Immediate
Pr4.15	240107	DIN7 FilterCoefficient	Apply a low-pass filter to the DIN pins		2	ms	U16	RWS	Immediate
Pr4.16	240108	DIN8 FilterCoefficient	Apply a low-pass filter to the DIN pins		2	ms	U16	RWS	Immediate
Pr4.17	240201	DIN1 Function Index	0: No function	1: Enable	1		U8	RWS	Immediate
			2: Reset	3: Index 0					
			4: Index 1	5: Index 2					
			6: Index 3	7: Index 4					
			8: Mode Index 0	9: Mode Index 1					
			10: Gain Index 0	11: Gain Index 1					
			12: Integral Clear	13: Positive Limit Switch					
			14: Negative Limit Switch	15: Origin Signal					
			16: Start Origin Finding	17: Command Reversal					
			18: Emergency Stop	19: Trigger 0					
			20: Trigger 1	21: Condition 0					
			22: Condition 1	23: Pause					
			24: Pulse gear	25: Activate Position Table					
			26: Interrupt Position Table	27: Pause Position Table					
			28: Probe 1	29: Probe 2					
			30: Full/half closed-loop switching	31: Pulse input inhibit					
Pr4.18	240202	DIN2 Function Index	Same as above		2		U8	RWS	Immediate
Pr4.19	240203	DIN3 Function Index	Same as above		3		U8	RWS	Immediate
Pr4.20	240204	DIN4 Function Index	Same as above		4		U8	RWS	Immediate
Pr4.21	240205	DIN5 Function Index	Same as above		5		U8	RWS	Immediate
Pr4.22	240206	DIN6 Function Index	Same as above		16		U8	RWS	Immediate
Pr4.23	240207	DIN7 Function Index	Same as above		15		U8	RWS	Immediate
Pr4.24	240208	DIN8 Function Index	Same as above		18		U8	RWS	Immediate
Pr4.25	240301	DIN1 Polarity	0: Negative polarity	1: Positive polarity	1		U8	RWS	Immediate
Pr4.26	240302	DIN2 Polarity	0: Negative polarity	1: Positive polarity	1		U8	RWS	Immediate
Pr4.27	240303	DIN3 Polarity	0: Negative polarity	1: Positive polarity	1		U8	RWS	Immediate
Pr4.28	240304	DIN4 Polarity	0: Negative polarity	1: Positive polarity	1		U8	RWS	Immediate
Pr4.29	240305	DIN5 Polarity	0: Negative polarity	1: Positive polarity	1		U8	RWS	Immediate
Pr4.30	240306	DIN6 Polarity	0: Negative polarity	1: Positive polarity	1		U8	RWS	Immediate
Pr4.31	240307	DIN7 Polarity	0: Negative polarity	1: Positive polarity	1		U8	RWS	Immediate
Pr4.32	240308	DIN8 Polarity	0: Negative polarity	1: Positive polarity	1		U8	RWS	Immediate

Pr4.33	40107D	Hybrid deviation clear setting	Set internal encoder multi-turn count	0		U16	RWS	Immediate
Pr4.34	401018	Full closed-loop encoder baud rate	Full closed-loop encoder baud rate	0		U8	RWS	Immediate
Pr4.35	401081	External division numerator	Default is 0, which equals the encoder resolution automatically identified by the current drive (17-bit = 131072, 23-bit = 8388608)	0		U32	RWS	Immediate
Pr4.36	401082	External division denominator	Number of external encoder pulses per revolution of the servo motor; takes effect after saving and restarting.	10000		U32	RWS	Immediate
Pr4.37	401080	Maximum hybrid deviation		16000	PUL	U32	RWS	Immediate
Pr4.38	40107F	Hybrid filter coefficient		1		U16	RWS	Immediate
Pr4.39	40107E	Hybrid gain offset	Correction for hybrid gain	0		I16	RWS	Immediate
Pr4.40	40107B	Full closed-loop index signal mode	0: Do not use Z signal 1: Use periodic Z signal, mostly low level 2: Use periodic Z signal, mostly high level	1		U8	RWS	Immediate
Pr4.41	40107C	Full closed-loop counting direction	0: Forward; 1: Reverse	0		U8	RWS	Immediate
Pr4.42	401019	Full closed-loop encoder resolution	Full closed-loop encoder resolution	10000		U32	RWS	Immediate
Pr4.43	20003B	Full closed-loop Z forward rising edge position low				I32	RO	
Pr4.44	20003D	Full closed-loop Z reverse rising edge position low				I32	RO	
Pr4.45	20003A	Full closed-loop position low 32 bits				I32	RO	
Pr4.46	401017	Full closed-loop encoder model	0: Disable full closed-loop function; 1: Enable full closed-loop function. Takes effect after saving and restarting.			U8	RWS	Immediate
Pr4.47	200040	Full closed-loop quadrature fault				U8	RO	
Pr4.48	300239	Initiate homing via keypad				U8	RWS	Immediate
Pr4.49	310031	Absolute clear / Absolute zero reset	For multi-turn absolute encoder motors, perform fault and position clear when the drive is not enabled.			U8	RWS	Restart
Pr4.50	4010B2	Gantry correction mode	The correction function must be used with two pulse servo drives and in pulse control scenarios, and the motors must be equipped with multi-turn absolute encoders. Before enabling this function, ensure that the absolute positions of the two axes meet the user's required settings.	0		U8	RWS	Restart
Pr4.51	4010B3	Gantry correction speed				U8	RWS	Restart
Pr4.52	4010B4	Gantry correction maximum deviation				I32	RWS	Restart
Pr4.53	4010B5	Gantry correction slave address				U8	RWS	Restart

9.1.5 F005 Group (Output Pin Configuration) Parameter List

No.	Index H	Parameter Name	Setting/Help Information		Default Value	Unit	Type	Access	Validation
Pr5.00	101001	Storage	Refer to Pr1.00						
Pr5.01	240701	DOUT1 Polarity	0: Negative polarity	1: Positive polarity	1		U8	RWS	Immediate
Pr5.02	240702	DOUT2 Polarity	0: Negative polarity	1: Positive polarity	1		U8	RWS	Immediate
Pr5.03	240703	DOUT3 Polarity	0: Negative polarity	1: Positive polarity	1		U8	RWS	Immediate
Pr5.04	240704	DOUT4 Polarity	0: Negative polarity	1: Positive polarity	1		U8	RWS	Immediate
Pr5.05	240705	DOUT5 Polarity	0: Negative polarity	1: Positive polarity	1		U8	RWS	Immediate
Pr5.06	240706	DOUT6 Polarity	0: Negative polarity	1: Positive polarity	1		U8	RWS	Immediate
Pr5.07	240707	DOUT7 Polarity	0: Negative polarity	1: Positive polarity	1		U8	RWS	Immediate
Pr5.08	240708	DOUT8 Polarity	0: Negative polarity	1: Positive polarity	1		U8	RWS	Immediate
Pr5.09	240801	DOUT1 Function Index	0: No Function	1: Ready	1		U8	RWS	Immediate
			2: Fault	3: Target Reached					
			4: Speed Saturated	5: Current Saturated					
			6: Motor 0 Speed	7: Brake Control					
			8: Regenerative Braking	9: Z Signal Output					
			10: Torque Mode Speed Limit	11: Motor Enable					
			12: In Limit	13: Origin Found					
			14: Homing	15: Output Index 0					
			16: Output Index 1	17: Output Index 2					
Pr5.10	240802	DOUT2 Function Index	Same as above		2		U8	RWS	Immediate

Pr5.11	240803	DOUT3 Function Index	Same as above	3		U8	RWS	Immediate
Pr5.12	240804	DOUT4 Function Index	Same as above	7		U8	RWS	Immediate
Pr5.13	240805	DOUT5 Function Index	Same as above	11		U8	RWS	Immediate
Pr5.14	240806	DOUT6 Function Index	Same as above	13		U8	RWS	Immediate
Pr5.15	240807	DOUT7 Function Index	Same as above	6		U8	RWS	Immediate
Pr5.16	240808	DOUT8 Function Index	Same as above	12		U8	RWS	Immediate

9.1.6 F006 Group (Auto-Tuning and Real-time Function Settings) Parameter List

No.	Index H	Parameter Name	Setting/Help Information	Default Value	Unit	Type	Access	Validation
Pr6.00	101001	Storage	Refer to Pr1.00					
Pr6.01	401055	Rigidity Grade	0-31 Refer to "Rigidity Gain Table"	17		U8	RWS	Immediate
Pr6.02	401056	Use Rigidity Level	0: Not used 1: Use	0		U8	RWS	Immediate
Pr6.03	401053	Inertia Ratio	Define the content about inertia ratio in "Adjustment" Users can manually set inertia ratio	30	0.1	U16	RWS	Immediate
Pr6.04	310026	Current Load Inertia	Total load inertia deduced from inertia ratio	3*motor	Kg*mm ²	U16	RO	
Pr6.05	320015	Start Inertia Measurement	0: Not started. Automatically reset to 0 after execution 1: Start	0		U8	RWS	Immediate
Pr6.06	32000D	Measuring Results	1: Measurement successful 0: Measurement not started or in progress -1: Speed or current response amplitude too small -2: Measured torque inertia coefficient abnormal -3: Inertia ratio too large, exceeding 25 times -4: Inertia ratio exceeding 40 times	0		S8	RO	
Pr6.07	320016	Movement distance during inertia measurement	10-50	30	0.01 Turn	U8	RWS	Immediate
Pr6.08	310027	Current Inertia Ratio	Measurement result of current inertia ratio	30	0.1	U16	RO	
Pr6.09	310028	Current KLoad	Current torque inertia coefficient			U32	RO	
Pr6.10	300227	Speed Loop Proportion 4	Validation when "Use Rigidity Level" is 1			U32	RO	
Pr6.11	300228	Speed Loop Integral 4	Validation when "Use Rigidity Level" is 1			U16	RO	
Pr6.12	300229	Speed Loop KFR4	Validation when "Use Rigidity Level" is 1	256		U16	RO	
Pr6.13	401061	Real-time Function Settings	Refer to Pr1.48	1		U8	RWS	Immediate
Pr6.14	32001F	Positive Damping Current	Online detected positive damping current	0	ADC/4	S16	RO	
Pr6.15	320025	Negative Damping Current	Online detected negative damping current	0	ADC/4	S16	RO	
Pr6.16	320026	Online Inertia	Total load inertia deduced from online inertia ratio	0	Kg*mm ²	U16	RO	
Pr6.17	320027	Online Inertia Ratio	Online real-time detected inertia ratio	0		U16	RO	
Pr6.19	40106F	Notch Filter A Width	Set the frequency range of the notch filter	45	Hz	U8	RWS	Immediate
Pr6.20	401070	Notch Filter A Depth	Set the filter strength of the notch filter	50	dB	U8	RWS	Immediate
Pr6.21	401071	Notch Filter A Frequency	100-8000. 8000 indicates function disabled	8000	Hz	U16	RWS	Immediate
Pr6.22	401072	Notch Filter B Width	Set the frequency range of the notch filter	45	Hz	U8	RWS	Immediate
Pr6.23	401073	Notch Filter B Depth	Set the filter strength of the notch filter	50	dB	U8	RWS	Immediate
Pr6.24	401074	Notch Filter B Frequency	100-8000. 8000 indicates function disabled	8000	Hz	U16	RWS	Immediate
Pr6.25	401093	Stiffness compensation bandwidth	0-15: Stiffness compensation disabled >15: Stiffness compensation enabled	0		U8	RWS	Immediate
Pr6.26	401094	Stiffness compensation gain	Enhance system rigidity, actively increase the motor's resistance stiffness, and improve dynamic response.	4000		U16	RWS	Immediate
Pr6.27	40109F	Model switch	0: Model tracking control disabled 1: First-order rigid body model tracking control enabled 2: Inertia model tracking control enabled	0		U16	RWS	Immediate
Pr6.28	4010A0	Model bandwidth	Determines the system response speed. The higher the bandwidth, the faster the system response, but it may also amplify mechanical vibration. Range: 1-500 Hz	20	Hz	U16	RWS	Immediate
Pr6.29	4010A1	Model damping	Used to suppress overshoot and oscillation of the system in response to commands, enabling smooth positioning to the target position.	1000		U16	RWS	Immediate

Pr6.30	4010A2	Model anti-resonance frequency	The frequency point corresponding to the "resonance frequency" at which the system's vibration response is minimal. Active when inertia model is enabled	50	Hz	U16	RWS	Immediate
Pr6.31	4010A3	Model resonance frequency	The frequency at which the mechanical system resonates, i.e., the point where vibration is greatly amplified due to energy concentration. The servo system needs to suppress it via a filter Active when inertia model is enabled	70	Hz	U16	RWS	Immediate
Pr6.32	4010A4	Model speed compensation	Speed feedforward compensation. It applies a feedforward command directly to the speed loop, enabling the motor to respond faster to speed changes and thus shorten positioning time	1024	‰	U16	RWS	Immediate
Pr6.33	4010A5	Model forward compensation	Acceleration feedforward compensation	1024	‰	U16	RWS	Immediate
Pr6.34	4010A6	Model reverse compensation	Deceleration feedforward compensation	1024	‰	U16	RWS	Immediate
Pr6.35	4010A7	Vibration suppression frequency	Used to set the mechanical vibration frequency to be suppressed. It can be determined by observing waveforms using the servo software.	100	Hz	U16	RWS	Immediate
Pr6.36	4010A8	Vibration suppression gain	Adjusts the intensity of vibration suppression. The larger the gain, the stronger the suppression effect, but excessive gain may cause system instability.	100		U16	RWS	Immediate
Pr6.37	4010A9	Vibration suppression damping	Adjusts the speed of vibration decay, affecting the smoothness of vibration suppression. Range:0-384	0		U16	RWS	Immediate
Pr6.38	4010AA	Vibration suppression offset 1		0		U16	RWS	Immediate
Pr6.39	4010AB	Vibration suppression offset 2		0		U16	RWS	Immediate
Pr6.40	4010AC	Vibration suppression frequency 1	Used to set a second set of vibration suppression frequencies to handle multiple different vibration frequency points	100	Hz	U16	RWS	Immediate
Pr6.41	4010AD	Vibration suppression damping 1		0		U16	RWS	Immediate

9.1.7 F007 Group (Jog and Constant Display Settings) Parameter List

No.	Index H	Parameter Name	Setting/Help Information	Default Value	Unit	Type	Access	Validation
Pr7.00	101001	Storage	Refer to Pr1.00					
Pr7.01	310024	Always Display Object	0: Display RPM speed 1: Display bus voltage 2: Display temperature 3: DIN final input signal 4: DOUT final output status 5: Drive load rate 6: Motor load rate Other monitoring objects can be extended	0		U8	RW	Immediate
Pr7.02	310022	Jog Speed	Set the speed for jog movement	100	rpm	U16	RW	Immediate
Jog		Jog mode entry	Press SET to enter Jog mode Refer to "Jog Mode" for details					
Softn	100A00	Software version	Software version (x.xx format)				RO	
Pr7.03	20003F	Customer code	Customized code				RO	
Pr7.04	606300	Actual position				I32	RO	
Pr7.05	300306	Position loop command				I32	RO	
Pr7.06	20001B	Pulse input 0 counter				I32	RO	
Pr7.07	300039	1 ms pulse count	Pulse input count per ms			I16	RO	
Pr7.08	30003A	Disable control	1: Disable drive	0		U8	RW	Immediate
Pr7.09	300032	I Q LFT			Arms	I32	RO	
Pr7.10	20000D	Firmware CRC checksum				U32	RO	
Pr7.11	101803	XML version				U32	RO	
Pr7.12	606400	Position feedback	Actual position feedback (command units)		PUL	I32	RO	

9.1.8 F008 Group (Position Table Function) Parameter List

No.	Index H	Parameter Name	Setting/Help Information	Default Value	Unit	Type	Access	Validation
Pr8.00	101001	Storage	Refer to Pr1.00					
Pr8.01	31000B	Position Table Run Mode	0: Position Table 1: I/O Position 2: I/O Speed	0		U8	RWS	Immediate
Pr8.02	31000C	Position Table Trigger Mode	0: Trigger Signal 1: Time Trigger	0		U8	RWS	Immediate
Pr8.03	31000D	Position Table Trigger Delay	Delay when triggering by time	100	ms	U16	RWS	Immediate
Pr8.04	40102B	Position Table Auto Run	0: Do not auto-run 1: Execute auto-run once at power-up 2: After fault cleared 3: After auto-origin finding	0		U8	RWS	Immediate
PPPP1		Position Setting Entry	Setting range PP1.01-PP1.30	0	inc	S32	RWS	Immediate
PPPP2		Speed Setting Entry	Setting range PP2.01-PP2.30	0	rpm	S16	RWS	Immediate
PPPP3		Delay Register Entry	Setting range PP3.01-PP3.30	0	ms	U32	RWS	Immediate
PPPP4		Acceleration/Deceleration Index Entry	Setting range PP4.01-PP4.30	0		U8	RWS	Immediate
PPPP5		Control Register Entry	Setting range PP5.01-PP5.30	0		U16	RWS	Immediate
PPPP6		Cycle Count Entry	Setting range PP6.01-PP6.30	0		U16	RWS	Immediate
PPPP7		Acceleration Setting Entry	Setting range PP7.01-PP7.30	0	rps/s	U16	RWS	Immediate
PPPP8		Deceleration Setting Entry	Setting range PP8.01-PP8.30	0	rps/s	U16	RWS	Immediate
PPPP9		Cycle Count Entry	Setting range PP9.01-PP9.30	0		U16	RO	

9.1.9 F009 Group (Origin Function Settings) Parameter List

No.	Index H	Parameter Name	Setting/Help Information	Default Value	Unit	Type	Access	Validation
Pr9.00	101001	Storage	Refer to Pr1.00					
Pr9.01	607C00	Origin Offset	Through this object, the final origin is offset from the physical signal generation point		PUL	S32	RWS	Immediate
Pr9.02	609800	Origin mode	Refer to "Origin Mode Section"	0		S8	RWS	Immediate
Pr9.03	609901	Origin Rough Positioning Speed	A faster speed used for rough origin search	35000	PUL/s	U32	RWS	Immediate
Pr9.04	609902	Origin Fine Positioning Speed	A slower speed used for precise origin positioning	15000	PUL/s	U32	RWS	Immediate
Pr9.05	609903	Auto Origin Finding	0: Disabled 1: When power-up 2: Origin lost	0		U8	RWS	Restart
Pr9.06	609A00	Origin Acceleration	This object is used for acceleration and deceleration during the origin finding process	1000000	rpss	U32	RWS	Immediate
Pr9.07	609904	Origin Finding Current	Validation when using mechanical limits for origin finding	20	0.1A	U16	RWS	Immediate
Pr9.08	609905	Origin Offset Mode	0: Move to the origin offset position, then clear the position 1: Do not move, directly set the current position as negative origin offset	0		U8	RWS	Immediate
Pr9.09	609906		For manufacturer use					
Pr9.10	609907	Origin Offset Positioning Speed	The speed when moving to the origin offset position	200	rpm	U16	RWS	Immediate
Pr9.11	40105A		For manufacturer use					

9.1.A F00A Group (Auxiliary Parameter Settings) Parameter List

No.	Index H	Parameter Name	Setting/Help Information	Default Value	Unit	Type	Access	Validation
PrA.00	101001	Storage	Refer to Pr1.00					
PrA.01	20002B	Encoder Reset	Write non-0 data to execute multi-turn data clear and encoder fault clear actions	0		U8	W	Immediate
PrA.02	20000F	Encoder Multi-turn Data	Display encoder multi-turn data	0		U32	RO	
PrA.03	200004	Encoder Single-turn Data	Display encoder single-turn data	0		U32	RO	
PrA.04	200017	Encoder Internal Fault	Display encoder internal fault word	0		U16	RO	

PrA.05	20001F	SAE Operation	User can only write 3, 4, 7 3: Clear encoder fault 4: Clear multi-turn data 7: Clear both fault and multi-turn data Some faults require repeated power-up	0		U32	RW	Immediate
PrA.06	30001D	Current Command A	N/A					
PrA.07	310001	Execute excitation information search	N/A					
PrA.08	310002	Apply excitation search results	N/A					
PrA.09	310004	Hall Status Access	Not applicable. Complete when it is 3F and changes 12 times					
PrA.10	101004	Restart Processor	Write 1 to perform hot restart	0		U32	RW	
PrA.11	310010	DIN Control Word Source	Corresponding control word when DIN pin is enabled	103F		U16 HEX	RWS	Immediate
PrA.12	606000	Working Mode	Drive Target Operating Mode Refer to Operating Mode section			S8	RWS	Immediate
PrA.13	606100	Current Operating Mode	The operating mode currently in use by the drive			S8	RO	Immediate
PrA.14	210019	Aging Current	This parameter is internally limited when written Manufacturer Parameters	Drive Rated	0.1A	U8	RWS	Immediate
PrA.15	21001A	Aging Command	Write 1, enter aging mode after next restart Manufacturer Parameters	0		U8	RWS	Restart
PrA.16	21001B	Aging State	0: Not aged 1: Aging complete 2: Aging failed	0		U8	RO	

9.1.B F00b Group (Historical Error Display Group)

No.	Example	Help Information
xE.yyyy		x represents the x-th historical error, 0 is the latest E is the abbreviation for Error yyyy is the error code, specifically check error code description For detailed historical error information, please use the debugging software's fault display

9.1.C F00C Group (Tuning Parameters) Parameter List

No.	Index H	Parameter Name	Setting/Help Information	Default Value	Unit	Type	Access	Validation
PrC.00	101001	Storage	Refer to PrI.00					
PrC.01	609201	Feed	Number of commands issued by the controller Refer to "Motion Equivalent Setting"	10000	PUL	U32	RWS	Immediate
PrC.02	402001	Position loop bandwidth 0	Correspond to position loop response capability Please refer to the "Adjustment" section for setting method	764	0.01Hz	U16	RW	Immediate
PrC.03	40101F	Speed feedforward ratio (per mille)	Setting velocity feedforward can improve dynamic tracking performance Note: Setting value/1024 corresponds to true permille	0	‰	U16	RWS	Immediate
PrC.04	401021	Torque feedforward ratio (per mille)	Setting torque feedforward can improve dynamic tracking performance Torque feedforward setting assumes correct inertia ratio Note: Setting value/1024 corresponds to true permille	0	‰	U16	RWS	Immediate
PrC.05	40101D	Speed feedback filter bandwidth	Set the bandwidth of the speed feedback filter Larger setting value means weaker filtering, smaller lag Reasonable setting can reduce noise Too small a setting may affect loop stability 10-2000	300	Hz	U16	RWS	Immediate

PrC.06	40101C	Torque filter bandwidth	Set the bandwidth of the current command filter Larger setting value means weaker filtering, smaller lag Reasonable setting can eliminate noise and resonance Too small a setting may affect loop stability	418	Hz	U16	RWS	Immediate
PrC.07	402401	Speed loop proportional gain 0	Correspond to speed loop response capability Please refer to the "Adjustment" section for setting method	Determined by motor parameters		U32	RWS	Immediate
PrC.08	402501	Speed loop integral gain 0	Correspond to speed loop steady-state following capability Please refer to the "Adjustment" section for setting method	Determined by motor parameters		U16	RWS	Immediate
PrC.09	40105E	PWM frequency control mode	PWM switch frequency setting 0:16K single, 1:8K double, 2:16K double, 3:8K double to 16K single automatic switching, 4:8K double to 16K double automatic switching					Immediate
PrC.10	401055	Stiffness level	0-31 Refer to "Rigidity Gain Table"	17		U8	RWS	Immediate
PrC.11	401056	Use rigidity level	0:Manual gain parameter setting – refer to Section 6.3 "Manual Gain Setting". 1:Use the "Rigidity Gain Table". 2/3/4:Enable the adjustment-free function, which can automatically accommodate complex mechanical systems with different rigidity and inertia. The settings 2, 3, and 4 correspond to relatively high, high, and very high adjustment-free response levels, respectively. It is necessary to comprehensively consider factors such as inertia matching, mechanical rigidity, and external disturbances when selecting the appropriate adjustment-free mode and adjustment-free level.	0		U8	RWS	Immediate
PrC.12	401053	Inertia ratio	Define the content about inertia ratio in "Adjustment" Users can manually set inertia ratio	30	0.1	U16	RWS	Immediate
PrC.13	47FE14	Brake ID	Motor brake code Set to 0 if no brake	0		U16	RWS	Restart
PrC.14	21000E	Low temperature protection temperature		-20		I16	RWS	Restart
PrC.15	4010AE	Adjustment-free rigidity level	The adjustment-free level can be set from 1 to 12, corresponding to position loop response from low to high.	6		U8	RWS	Immediate
PrC.16	401094	Stiffness compensation gain	Enhance system rigidity, actively increase the motor's resistance stiffness, and improve dynamic response.	4000		U16	RWS	Immediate

9.1.D F00d Group (Gantry synchronization parameters) Parameter List

No.	Index H	Parameter Name	Setting/Help Information	Default Value	Unit	Type	Access	Validation
PrD.00	101001	Storage	Refer to Pr1.00					
PrD.01	401091	Enable gantry	0:Disable 1:Set as gantry master axis 2:Set as gantry slave axis 3:Set as gantry slave axis (consistent with master axis command)	0		U8	RWS	Immediate
PrD.02	401089	Gantry parameter synchronization	0:Synchronized 1:Unsynchronized	0		U8	RW	Immediate
PrD.03	40108A	Gantry slave axis synchronization direction	0:Slave not reversed 1:Slave reversed	0		U8	RWS	Immediate
PrD.04	40108B	Gantry synchronization type	0:Rigid gantry 1:Flexible gantry	0		U8	RWS	Immediate

PrD.05	40108C	Gantry synchronization filter coefficient	Range:1–255	1		U8	RWS	Immediate
PrD.06	40108D	Maximum synchronization deviation		100	PUL	U32	RWS	Immediate
PrD.07	40108E	Gantry position bandwidth		15	Hz	U16	RWS	Immediate
PrD.08	40108F	Gantry integral gain		0		U8	RWS	Immediate
PrD.09	401090	Gantry data channel	0:Use RS485 channel (Ethernet port) 1:Use auxiliary communication encoder interface (CN1)	0		U8	RWS	Restart
PrD.10	401092	Gantry compensation gain		1		I8	RWS	Immediate
PrD.11	300323	Gantry synchronization deviation				I32	RO	

9.2 Object Dictionary

Index H	Parameter Name	Setting/Help Information	Default Value	Unit	Type	Access	Validation
300210	Speed Error	Speed loop command - speed feedback		Internal	S32	RO	
300222	Speed Mode Profile Velocity	Speed profile generated by the internal speed profile generator		Internal	S32	RO	
30020D	Speed Loop Command_Internal	Speed loop's command input		Internal	S32	RO	
300217	Speed Feedback 1st Order Low-Pass	Feedback speed after first-order low-pass filtering		Internal	S32	RO	
300305	Position Loop Command	Position Loop's Command Input		inc	S64	RO	
300306	Position Loop Command Low 32 bits	Lower 32 bits of 64-bit position loop command		inc	S32	RO	
30001D	Current Command A	Manufacturer Parameters		0.1A	S16	RWS	Immediate
300321	Actual Position Low 32 bits	Lower 32 bits of 64-bit position data	0	inc	S32	RO	
300322	Actual Position High 32 bits	Higher 32 bits of 64-bit position data	0	inc	S32	RO	
300307	Actual Position	64-bit Position Data	0	inc	S64	RO	
300309	Position Error	Position loop command - actual position	0	inc	S32	RO	
300318	VPG Position Planning Output	Position profile generated by the internal position profile generator	0	inc	S32	RO	
300319	VPG Speed Planning	Speed profile during the VPG planning position curve process	0	inc	S32	RO	
30031A	VPG Target Position	Planning target of the internal profile generator	0	inc	S32	RO	
30031B	Velocity Feedforward Command	Velocity feedforward command generated by velocity feedforward calculation	0	Internal	S32	RO	
30031D	Electronic Gear Numerator	Derived from 402 gear ratio related objects			U32	RO	
30031E	Electronic Gear Denominator	Derived from 402 gear ratio related objects			U32	RO	
603F00	Error Code	Current servo error code	0		U16	RO	
604000	Control Word	402 state machine control word			U16	RW	Immediate
604100	State Word	402 state machine status word			U16	RW	Immediate
605A00	Quick stop option	Emergency stop command action selection	0		U16	RWS	Immediate
605B00	Shutdown Option	Shutdown command action selection	0		U16	RWS	Immediate
605C00	Disable Option	Disable operation command action selection	0		U16	RWS	Immediate
606000	Working Mode	Select drive operating mode	1		I8	RWS	Immediate
606100	Current Operating Mode	Current operating mode display			I8	RO	
606300	Actual Position	Real-time motor absolute position feedback (encoder units)		DEC	I32	RO	
606400	Position Feedback	Actual position feedback (command units)		PUL	I32	RO	
606500	Position following error window	Position following error alarm threshold	50000	PUL	U32	RWS	Immediate
606600	Position following error time window	Position following error alarm window time	10	ms	U16	RWS	Immediate
606700	Position reached window	Position reached detection threshold	10	PUL	U32	RWS	Immediate
606800	Position reached time window	Position reached window time	15	ms	U16	RWS	Immediate
606B00	Speed Loop Command	Generate internal command velocity value with velocity command Motion Path		PUL/s	I32	RO	
606C00	Velocity loop feedback	Encoder actual feedback velocity value		PUL/s	I32	RO	

607100	Target Torque	CST mode command input, limited by 607200.		‰	S16	RWS	Immediate
607200	Maximum Torque	0~3000. Limit the range for 607100.		‰	U16	RWS	Immediate
607400	Torque command	Display target torque value, 1000 corresponds to 1x rated torque		‰	I16	RO	
607700	Actual torque	Servo internal actual torque feedback		‰	I16	RO	
607A00	Target Position	Command interface for position control modes (PPM, CSP).		PUL	S32	RWS	Immediate
607D01	Soft limit position 1	Soft limit minimum position limit		PUL	I32	RWS	Immediate
607D02	Soft limit position 2	Soft limit maximum position limit		PUL	I32	RWS	Immediate
607E00	Control loop command and feedback in reverse direction	Control loop command value and feedback value reversed	0	DEC	U8	RWS	Restart
608000	Maximum Speed Command	Limit the speed of the controller		rpm	U32	RWS	Immediate
608100	Profile velocity	Profile speed used in "Position Profile Mode"		PUL/s	U32	RWS	Immediate
608300	Profile acceleration	Profile acceleration used in "Position Profile Mode"		PUL/s ²	U32	RWS	Immediate
608400	Profile deceleration	Profile deceleration used in "Position Profile Mode"		PUL/s ²	U32	RWS	Immediate
608500	Emergency Stop Deceleration	Deceleration when emergency stop command is set to decelerate to stop		PUL/s ²	U32	RWS	Immediate
609200	Feed Amount	Number of feed commands per revolution issued by the controller	10000	PUL	U32	RWS	Immediate
609800	Origin mode	Set the homing method		DEC	I8	RWS	Immediate
60B800	Probe Settings	Set probe function, see Probe Function for details		HEX	U16	RWS	Immediate
60B900	Probe Status	Probe running status, see Probe Function for details		HEX	U16	RO	
60E000	Positive torque limit	Positive maximum torque limit value		‰	U16	RWS	Immediate
60E100	Negative torque limit	Reverse maximum torque limit value		‰	U16	RWS	Immediate
60F400	Position following error	Position following error real-time value		PUL	I32	RO	
60FD00	External signal input status	Feedback limit, origin, enable, and other signal states		HEX	U32	RO	
60FE01	DO Forced Output	Forced control DO output state		HEX	U32	RWS	Immediate
60FE02	DO Forced Output Mask	Mask 60FEh-01h, control is active when corresponding bit is 1		HEX	U32	RWS	Immediate
60FF00	Target Speed	Command interface for speed control modes (PVM, CSV)		PUL/s	S32	RW	Immediate
650200	Supported Operating Modes	Operating modes supported by the drive		HEX	U32	RO	

Chapter 10 Troubleshooting

10.1 Warning Codes

10.1.1 Warning Code List

When a servo drive issues a warning, display the warning code with an AL. prefix. This indicates that the drive detects an abnormality. Inspect and eliminate the cause of the warning, otherwise it will affect the continuous normal use of the servo system.

Warning Code	Name
AL.0001	Battery Warning
AL.0002	Drive Overload Warning
AL.0003	Motor Overload Warning

10.1.2 Warning Handling

➤ **AL.0001: Battery Warning**

Fault mechanism: Multi-turn absolute encoder battery voltage is below 3.0V

Confirmation method: Measure battery voltage (EVE ER14505 AA rated voltage 3.6V)

Solution: Power on the drive, replace the battery in a non-operating state. The warning will clear automatically after battery replacement

Note:

If replace the battery while powered off, the drive will alarm Er.FF02 (Encoder Internal Fault) upon power-up. In this case, first set PrA.05=3 (2000h-1Fh) to reset the encoder fault, then re-perform the homing function operation

➤ **AL.0002: Drive Overload Warning**

Fault mechanism: Drive average load rate exceeds overload warning threshold

Fault Cause	Inspection Method	Solution
1. The load the drive needs to operate exceeds its capacity	Confirm the overload characteristics of the motor or servo drive Check if the servo drive's average load rate (3000h-25h) is continuously greater than 95.0%	Reduce load weight or reduce load inertia to ensure that the load is within the drive's rated capacity; Replace with a larger capacity servo drive and matched motor
2. The output capability of drive does not match actual load operation	Check mechanical inertia ratio or perform inertia identification, check inertia ratio Pr6.03(4100h-53h) Confirm single operating cycle time during servo motor cyclic operation	Reasonably arrange equipment operating cycles, avoid prolonged continuous operation, and ensure the drive has sufficient cooling time
3. Motor stalls due to mechanical factors, causing excessive load during operation	Use X Servo Configurator to check operating commands and motor speed (3002h-25h) ·Operating command in position mode: 6062h ·Operating command in speed mode: 606Bh ·Operating command in torque mode: 6074h Confirm if, in the corresponding mode, the operating command is not 0 or very large, while the motor speed is 0	Eliminate mechanical factors
4. Servo drive failure	Check if the drive cooling fan is working properly and if the drive's operating environment is normal; Power up again after a period of power-off	Eliminate other fault factors, if fault reports after power up again, please contact Simphoenix Electric technical support to replace the servo drive

➤ **AL.0003: Motor Overload Warning**

Fault mechanism: Motor average load rate exceeds overload warning threshold

Fault Cause	Inspection Method	Solution
1. Motor wiring or encoder wiring errors or poor connections	Compare with the correct wiring diagram, check wiring between motor, drive, and encoder	Connect cables according to the correct wiring diagram; Prioritize using Simphoenix Electric's standard cables; When using self-made cables, please follow the hardware wiring guide to produce and connect
2. Load is too heavy, motor output Validation torque exceeds rated torque, continuous operation for a long time	Confirm the overload characteristics of the motor or servo drive; Check if the servo drive's average load rate (3000h-25h) is continuously greater than 95.0%	Replace with a larger capacity servo drive and matched motor; or reduce the load, increase acceleration and deceleration time.
3. Too frequent acceleration/deceleration or excessive load inertia	Check mechanical inertia ratio or perform inertia identification, check inertia ratio Pr6.03(4100h-53h) Confirm single operating cycle time during servo motor cyclic operation	Extend acceleration/deceleration time
4. Improper gain adjustment or excessive rigidity	Observe if the motor vibrates or makes abnormal sounds during operation	Re-adjust gain
5. Motor stalls due to mechanical factors, causing excessive load during operation	Use X Servo Configurator to check operating commands and motor speed (3002h-25h) ·Operating command in position mode: 6062h ·Operating command in speed mode: 606Bh ·Operating command in torque mode: 6074h Confirm if, in the corresponding mode, the operating command is not 0 or very large, while the motor speed is 0	Eliminate mechanical factors
6. Servo drive failure	Power up again after a period of power-off	If fault reports after power-up again, please contact Simphoenix Electric technical support to replace the servo drive

10.2 Error Codes

When the servo drive issues a fault alarm, a fault code with an Er. prefix is displayed. At this time, the motor will stop operating

10.2.1 Error Code List

Fault Code	Name
<u>Er.2250</u>	Bus current overcurrent
<u>Er.2320</u>	Motor phase current overcurrent
<u>Er.2350</u>	Drive Overload
<u>Er.3130</u>	Input Phase Loss
<u>Er.3210</u>	Main circuit overvoltage
<u>Er.3220</u>	Main circuit undervoltage
<u>Er.4310</u>	Drive Overheating
<u>Er.4320</u>	Drive low temperature
<u>Er.5210</u>	Current detection circuit fault
<u>Er.6310</u>	Storage module fault
<u>Er.7110</u>	Braking resistor protection
<u>Er.7121</u>	Motor Stalling
<u>Er.7122</u>	Excitation error
<u>Er.7305</u>	Pulse encoder anomaly
<u>Er.7306</u>	Input pulse frequency anomaly
<u>Er.7307</u>	Full-closed loop Z signal anomaly
<u>Er.7510</u>	Encoder communication loss
<u>Er.7511</u>	Encoder communication interference
<u>Er.8611</u>	Position tracking error
<u>Er.8700</u>	Bus Communication Fault
<u>Er.FF00</u>	Drive configuration error
<u>Er.FF01</u>	Motor Overload
<u>Er.FF02</u>	Encoder Internal Fault
<u>Er.FF03</u>	Encoder model identification fault
<u>Er.FF04</u>	Encoder data read anomaly
<u>Er.FF05</u>	Encoder data checksum error
<u>Er.FF06</u>	Encoder data version error
<u>Er.FF08</u>	Data import protection
<u>Er.FF09</u>	Configuration parameter setting error
<u>Er.FF0A</u>	Output Phase Loss
<u>Er.FF0B</u>	Motor setting error
<u>Er.FF0C</u>	Full-closed loop setting error
<u>Er.FF0D</u>	Excessive mixed control deviation
<u>Er.FF0E</u>	Full-Closed Loop Quadrature Signal Loss

10.2.2 Troubleshooting

➤ **Er.2250: Bus Current Overcurrent****Fault mechanism:** Bus current detection feedback value exceeds protection threshold

Fault Cause	Inspection Method	Solution
1. Gain setting is not reasonable, and motor is oscillating	Check if there is vibration or sharp noise during motor startup and operation; also use X Servo Configurator to monitor parameters like "Current Feedback"	1. Motor parameter setting error, check motor parameter settings; 2. Current loop parameters anomaly, readjust current loop parameters; 3. Speed loop parameters anomaly, oscillation generated from servo; 4. Servo drive anomaly, replacement of servo drive is required.
2. Encoder wiring anomaly	Check encoder cable for aging, corrosion, or loose connectors; turn off servo enable signal, manually rotate motor shaft, and check if drive display speed changes with motor shaft rotation	Re-solder, firmly plug in, or replace encoder cable
3. Servo drive failure	1. Disconnect the motor cable and power up again, but the fault still reports; 2. Check the external braking resistor configuration for a resistance value that is too low or a short circuit in the wiring (at the main circuit input terminals P+ and C).	Replace the servo drive Re-select braking resistor resistance and model Rewire

➤ **Er.2320: Motor Phase Current Overcurrent****Fault mechanism:** Phase current detection feedback value exceeds protection threshold

Fault Cause	Inspection Method	Solution
1. Motor cable damaged, grounded, or Poor contact	1. Check if the servo drive's UVW output and motor side connections are loose; 2. After ensuring that the drive's power cables and motor cables are securely connected, measure the insulation resistance between the servo drive's UVW terminals and the ground wire (PE) to confirm that it is in the megaohm (MΩ) range	Tighten loose or disconnected wiring Replace problematic motor or motor cable if insulation is poor
2. The motor is burned out	Disconnect the motor cable, measure if the motor UVW phase resistance is balanced, and check if there is a short circuit between motor UVW phases	Correctly connect motor cables If the motor is faulty, please replace the motor
3. Drive power module fault	Disconnect the motor cable and power up again, but the fault still reports;	Please contact Simphoenix Electric technical support to replace the servo drive

➤ **Er.2350: Drive Overload****Fault mechanism:** Drive average load rate exceeds overload warning threshold

Fault Cause	Inspection Method	Solution
1. The load the drive needs to operate exceeds its capacity	Confirm the overload characteristics of the motor or servo drive Check if the servo drive's average load rate (3000h-25h) is continuously greater than 100.0%	Reduce load weight or reduce load inertia to ensure that the load is within the drive's rated capacity; Replace with a larger capacity servo drive and matched motor
2. The output capability of drive does not match actual load operation	Check mechanical inertia ratio or perform inertia identification, check inertia ratio Pr6.03(4100h-53h) Confirm single operating cycle time during servo motor cyclic operation	Reasonably arrange equipment operating cycles, avoid prolonged continuous operation, and ensure the drive has sufficient cooling time
3. Motor stalls due to mechanical factors, causing excessive load during operation	Use X Servo Configurator to check operating commands and motor speed (3002h-25h) ·Operating command in position mode: 6062h ·Operating command in speed mode: 606Bh ·Operating command in torque mode: 6074h Confirm if, in the corresponding mode, the operating command is not 0 or very large, while the motor speed is 0	Eliminate mechanical factors

Fault Cause	Inspection Method	Solution
4. Servo drive failure	Check if the drive cooling fan is working properly and if the drive's operating environment is normal; Power up again after a period of power-off	Eliminate other fault factors, if fault reports after power up again, please contact Simphoenix Electric technical support to replace the servo drive

➤ **Er.3130: Input Phase Loss**

Fault mechanism: Abnormal three-phase input to the drive (applicable to 380V servo drives)

Fault Cause	Inspection Method	Solution
1. Poor connection of the three-phase input wiring	Check if the servo drive's main circuit input terminals (R S T) and the upstream input cables are good and securely connected	Replace cables and correctly connect main circuit power lines
2. 380V servo drive running on single-phase power	Check the servo drive's input power specifications, check actual input voltage specifications, and measure if the main circuit input voltage complies with the following specifications: 380V servo drive: RMS value: 380V-440V AC All three phases need to be measured	According to power specifications, replace or adjust power

➤ **Er.3210: Main Circuit Overvoltage (DC Bus Voltage)**

Fault mechanism: Detected DC bus voltage exceeds alarm threshold:

220V servo drive: Overvoltage point 405V

380V servo drive: Overvoltage point 805V

Fault Cause	Inspection Method	Solution
1. Main circuit input voltage too high	Check servo drive input power specifications, and measure if the main circuit cable servo drive (R S T) input voltage complies with the following specifications: 220V servo drive: RMS value: 220V-240V AC 380V servo drive: RMS value: 380V-440V AC	According to power specifications, replace or adjust power
2. Power supply is unstable or affected by lightning strike	Troubleshoot if the drive input power supply has been affected by lightning, measure if the input power supply is stable, and if it meets the above specifications	Access a surge suppressor, then power on again. If the fault still occurs, replace the servo drive
3. Braking resistor selection inappropriate or failed	If built-in braking resistor is used (Pr3.22=1/2421h-01h=1), confirm if P+ and D are reliably connected with a wire. If yes, then measure the resistance between C and D; If external braking resistor is used (Pr3.22=0/2421h-01h=0), measure the resistance of the external braking resistor between P+ and C, and compare it with the recommended value; For braking resistor specifications, please refer to 2.6 Regenerative Braking Resistor Selection	1. If the resistance is "∞" (infinity), the internal wire of the braking resistor is broken, replace with a new braking resistor; 2. If built-in braking resistor is used, adjust to use external braking resistor, and remove the short-circuit wire between P+ and D. The resistance value can be chosen to be consistent with the built-in braking resistor, and the power needs to be not less than the built-in braking resistor; 3. If external braking resistor is used, replace with an external braking resistor of appropriate specifications and reconnect between P+ and C; 4. Be sure to set Pr3.23 (External Braking Resistor Resistance) and Pr3.25 (External Braking Resistor Power) to match the parameters of the actual external braking resistor used.

Fault Cause	Inspection Method	Solution
4. The motor is running in rapid acceleration/deceleration state, and the maximum braking energy exceeds absorbable value	Confirm the acceleration/deceleration time during operation, measure the DC bus voltage, and confirm if the voltage exceeds fault value during deceleration phase	First, ensure that the main circuit input voltage is within the specification, secondly, and adjust the acceleration/deceleration time if allowed
5. Large deviation in bus voltage sampling value	Measure if the DC bus voltage value is normal, compare with the drive's monitored bus voltage value	Please contact Simphoenix Electric technical support
6. Servo drive failure	Repeatedly power off and reconnect main circuit power, but the fault still reports	Replace the servo drive

➤ Er.3220: Main Circuit Undervoltage (DC Bus Voltage)

Fault mechanism: Detected DC bus voltage is below alarm threshold

220V servo drive: Undervoltage point 180V

380V servo drive: Undervoltage point 380V

Fault Cause	Inspection Method	Solution
1. Main circuit power supply unstable or power loss	Check servo drive input power specifications, and measure if the main circuit cable power side and servo drive side (RST) input voltage comply with the following specifications: 220V servo drive: RMS value: 220V-240V AC 380V servo drive: RMS value: 380V-440V AC	According to power specifications, replace or adjust power
2. Power supply voltage drops during operation	Monitor servo drive input power voltage, and check if the same main circuit power supply has too many other settings enabled, causing insufficient power capacity and voltage drop	Increase power capacity
3. Phase loss, servo drive that should operate on 3-phase power actually operate on single-phase power	Check if the main circuit wiring is correct and reliable	Replace cables and correctly connect main circuit power lines
4. Servo drive failure	Measure if the DC bus voltage value is normal, and compare with the drive's monitored bus voltage value; Repeatedly power off and reconnect main circuit power, but the fault still reports	Please contact Simphoenix Electric technical support to replace the servo drive

➤ Er.4310: Drive Overheat

Fault mechanism: Servo drive power module temperature is higher than overtemperature protection value (88°C)

Fault Cause	Inspection Method	Solution
1. The ambient temperature is too high	Measure ambient temperature	Improve servo drive cooling conditions, and reduce ambient temperature
2. After overload, reset overload fault by turning off power, and repeat multiple times	Check fault records for overload faults or warnings	Change fault reset method, wait 30s after overload before resetting. Increase servo drive and motor capacity, extend acceleration/deceleration time, and reduce load.
3. Fan operation anomaly	Check if the fan is operating during run	Please contact Simphoenix Electric technical support to replace the servo drive
4. Unreasonable installation space for servo drive	Compare with 3.3.2 Installation Space Requirements, confirm if servo drive installation is reasonable	Install according to servo drive installation standards
5. Servo drive failure	Still reports fault after 5 minutes of power off and restart	Please contact Simphoenix Electric technical support to replace the servo drive

➤ **Er.4320: Drive Low Temperature**

Fault mechanism: Servo drive power module temperature is below low temperature protection value (-10°C)

Fault Cause	Inspection Method	Solution
1. Ambient temperature too low	Measure ambient temperature	Improve servo drive operating conditions, increase ambient temperature Power on and preheat for a period of time, start operation after the alarm clears
2. Servo drive failure	Still reports fault after 5 minutes of power off and restart	Please contact Simphoenix Electric technical support to replace the servo drive

➤ **Er.5210: Current Detection Circuit Fault**

Fault mechanism: Phase current sensor sampling value anomaly

Fault Cause	Inspection Method	Solution
1. Short circuit or other problems occurred on encoder or I/O interface power pins	Troubleshoot if there is any anomaly in encoder and I/O interface wiring	Re-process encoder and I/O interface wiring
2. Servo drive failure	Attempt power cycling several times, but the fault still reports	Please contact Simphoenix Electric technical support to replace the servo drive

➤ **Er.6301: Storage Module Fault**

Fault mechanism: Internal logic power supply issue occurred during data storage execution

Fault Cause	Inspection Method	Solution
1. Short circuit or other problems occurred on encoder or I/O interface power pins	Troubleshoot if there is any anomaly in encoder and I/O interface wiring	Re-process encoder and I/O interface wiring
2. Perform firmware upgrade	Alarm occurs after firmware upgrade is completed	Perform the initialization operation
3. Servo drive failure	Attempt power cycling several times, but the fault still reports	Please contact Simphoenix Electric technical support to replace the servo drive

➤ **Er.7110: Braking Resistor Protection**

Fault mechanism: Braking resistor actual power exceeds rated power

Fault Cause	Inspection Method	Solution
1. Regenerative energy exceeds dissipation limit	Whether the motor speed is too fast; whether the load inertia is too large; monitor actual bus voltage value during braking	Reduce load weight or reduce load inertia to ensure that the load is within the drive's rated capacity; Replace with a larger capacity servo drive and matched motor
2. Braking resistor wiring error	Check wiring against 2.4.1 Braking Resistor Wiring	Correctly connect braking resistor
3. Braking resistor selection or parameter setting is unreasonable	Refer to 2.6 Regenerative Braking Resistor Selection to confirm braking resistor specifications, and verify parameter values for Pr3.23 (External Braking Resistor Resistance) and Pr3.25 (External Braking Resistor Power)	Select appropriate braking resistor and correctly set parameters

➤ Er.7121: Motor Stall

Fault mechanism: Actual speed of motor is zero, but the operating current reaches the limit, and persists for a certain time

Fault Cause	Inspection Method	Solution
1. Servo drive U V W output phase sequence wired incorrectly	Verify if the motor phase sequence is correct	Rewire according to correct wiring
2. Motor stall due to mechanical factors	Use X Servo Configurator to monitor if the operating command and motor speed match, and monitor current feedback waveform	Troubleshoot mechanical factors causing jamming, occasional sticking, eccentricity, etc. For motors with brakes, check if the brake is working properly

➤ Er.7122: Motor Excitation Error

Fault mechanism: Execute special excitation function failed

Cause: During motor excitation, motor phase loss or motor parameter error is detected (motor pole pairs, encoder resolution, etc.)

Solution: Troubleshoot encoder and motor information input; check motor wiring

➤ Er.7305: Pulse Encoder Anomaly

Fault mechanism: Pulse encoder internal parameter anomaly

Fault Cause	Inspection Method	Solution
1. Pulse encoder wiring error or loose cable	1. Check encoder wiring 2. Check if field vibration is too large, causing encoder cable to loosen, or even vibrate and damage the encoder. 3. Replace with a working encoder cable. If the fault no longer occurs after replacement, it indicates that the original encoder cable was damaged	1. Rewire according to the correct wiring diagram, and ensure that encoder terminal connections are tight. Prioritize using Simphoenix Electric's standard cables; if non-standard cables are used, check if they meet specifications, e.g., use shielded twisted pair . When routing cables, try to separate strong and weak currents. Do not bundle motor cables and encoder cables. Ensure good ground contact for motor and servo drive.
2. Encoder Z signal interference	Check field wiring: Are there large devices nearby that produce interference, or are there multiple power/frequency conversion devices and other interference sources in the cabinet?	2. Check if the plugs at both ends of the encoder have good contact, if any pins are recessed. 3. Replace with a working encoder cable. If not, the encoder itself has a significant problem, and replace the servo motor.

➤ Er.7306: Input Pulse Frequency Anomaly

Fault mechanism: Input pulse frequency exceeds the limit

Solution: Troubleshoot controller pulse parameter configuration, and system interference

➤ Er.7307: Full-closed loop Z Signal Anomaly

Fault mechanism: When using periodic full-closed loop Z signal, signal detection is abnormal due to disconnection, etc.

Solution: Troubleshoot full-closed loop encoder wiring, and system interference

➤ Er.7510: Encoder Communication Loss

Fault mechanism: No encoder signal detected

Fault Cause	Inspection Method	Solution
1. Encoder wiring error or loose cable	1. Check encoder wiring 2. Check if field vibration is too large, causing encoder cable to loosen, or even vibrate and damage the encoder. 3. Replace with a working encoder cable. If the fault no longer occurs after replacement, it indicates that the original encoder cable was damaged	1. Rewire according to the correct wiring diagram, and ensure that encoder terminal connections are tight. Prioritize using Simphoenix Electric's standard cables; if non-standard cables are used, check if they meet specifications, e.g., use shielded twisted pair. When routing cables, try to separate strong and weak currents. Do not bundle motor cables and encoder cables. Ensure good ground contact for motor and servo drive.
2. Encoder damaged	Power up after 30S power off. If the fault still occurs, the motor encoder may be damaged	2. Check if the plugs at both ends of the encoder have good contact, if any pins are recessed.
3. Drive failure	Power up after 30S power off. If the fault still occurs, the drive may be faulty	3. Replace with a working encoder cable. If not, the encoder itself has a significant problem, and replace the servo motor.

➤ Er.7511: Encoder Communication Interference

Fault mechanism: Encoder communication data anomaly

Fault Cause	Inspection Method	Solution
1. Encoder signal interfered	Check field wiring: Are there large devices nearby that produce interference, or are there multiple power/frequency conversion devices and other interference sources in the cabinet?	Rewire according to the correct wiring diagram, and ensure encoder terminal connections are tight. Prioritize using Simphoenix Electric's standard cables; if non-standard cables are used, check if they meet specifications, e.g., use shielded twisted pair . When routing cables, try to separate strong and weak currents. Do not bundle motor cables and encoder cables. Ensure good ground contact for motor and servo drive.
2. Encoder damaged	If the fault reoccurs, the motor encoder may be damaged	Please contact Simphoenix Electric technical support

➤ Er.8611: Position Tracking Error

Fault mechanism: Position response cannot follow command

Fault Cause	Inspection Method	Solution
1. Motor stall due to mechanical factors	Use X Servo Configurator to monitor if the operating command and motor speed match, and monitor current feedback waveform	Troubleshoot mechanical factors
2. Servo drive parameter settings are unreasonable	Check servo drive ·Position loop gain: ·Speed loop gain: Pr1.02 (4024h-01h) ·Profile speed: 6081h ·Position deviation threshold: Pr1.13(6065h) ·Profile acceleration/deceleration: 6083h/6084h ·Positive/negative torque limit: 60E0h/60E1h Whether the set values are reasonable	1. Re-perform servo gain adjustment 2. Set appropriate position deviation threshold Pr1.13 (6065h) according to operating conditions 3. Adjust parameters such as command acceleration/deceleration time, torque limits

Fault Cause	Inspection Method	Solution
3. Host computer position command inappropriate	Position Control Mode: ·CSP mode: Check gear ratio 6091h-01h/6091h-02h to determine the increment value of position command corresponding to a single synchronous cycle, and convert to velocity information; ·PP mode: Check gear ratio 6091h-01h/6091h-02h to determine 6081h (profile run speed); ·HM mode: Check gear ratio 6091h-01h/6091h-02h to determine 6099.01h and 6099.02h.	·CSP: Reduce the position command increment corresponding to a single synchronous cycle. When planning commands in the host computer, increase the position ramp. ·PP: Reduce 6081h, or increase acceleration/deceleration (6083h, 6084h). ·HM: Reduce 6099h-01h and 6099h-02h, or increase origin acceleration/deceleration (609Ah). ·Adjust gear ratio according to actual conditions.
4. Servo drive/motor fault	Monitor operating waveforms using X Servo Configurator's oscilloscope function: position command, position feedback, speed command, torque command, etc.	If position command is non-zero but position feedback is always zero, please replace servo drive/motor

➤ Er.8700: BUS Communication Fault

Fault mechanism: EtherCAT communication loss

Fault Cause	Inspection Method	Solution
1. Unstable physical connection of data link, or loss of process data due to unplugging network cable	Check if servo drive network cable connection is reliable and firm, if there is severe vibration on site; confirm if unplug or plug network cable; confirm if it is the specified network cable specification	Replace with a more reliable network cable
2. Data loss due to EMC interference, poor network cable quality, or poor connection	Check host computer network status; Detect if servo drive is reliably grounded	Ensure reliable grounding of servo drive, rectify EMC

➤ Er.FF01: Motor Overload

Fault mechanism: The average load rate of motor exceeds overload alarm threshold

Fault Cause	Inspection Method	Solution
1. Motor wiring or encoder wiring errors or poor connections	Compare with the correct wiring diagram, check wiring between motor, drive, and encoder	Connect cables according to the correct wiring diagram; Prioritize using Simphoenix Electric's standard cables; When using self-made cables, please follow the hardware wiring guide to produce and connect
2. Load is too heavy, motor output Validation torque exceeds rated torque, continuous operation for a long time	Confirm the overload characteristics of the motor or servo drive; Check if the servo drive's average load rate (3000h-25h) is continuously greater than 100.0%	Replace with a larger capacity servo drive and matched motor; or reduce the load, increase acceleration and deceleration time.
3. Too frequent acceleration/deceleration or excessive load inertia	Check mechanical inertia ratio or perform inertia identification, check inertia ratio Pr6.03(4100h-53h) Confirm single operating cycle time during servo motor cyclic operation	Extend acceleration/deceleration time
4. Improper gain adjustment or excessive rigidity	Observe if the motor vibrates or makes abnormal sounds during operation	Re-adjust gain
5. Motor stalls due to mechanical factors, causing excessive load during operation	Use X Servo Configurator to check operating commands and motor speed (3002h-25h) ·Operating command in position mode: 6062h ·Operating command in speed mode: 606Bh ·Operating command in torque mode: 6074h Confirm if, in the corresponding mode, the operating command is not 0 or very large, while the motor speed is 0	Eliminate mechanical factors
6. Servo drive failure	If fault reports after powering off for a period and then power up again	Please contact Simphoenix Electric technical support to replace the servo drive

Fault	Fault Cause	Solution
Er.FF00: Drive Configuration Error	The drive did not execute full factory procedure or configuration data lost	Please contact Simphoenix Electric technical support
Er.FF02: Encoder Internal Fault	A fault of the encoder is detected	Please contact Simphoenix Electric technical support to replace the motor
Er.FF03: Encoder Model Identification Fault	Failed to correctly identify its type upon power-up or re-connection of encoder	Please contact Simphoenix Electric technical support to re-perform motor excitation operation
Er.FF04: Encoder Data Read Anomaly	Error reading internal encoder data	Please contact Simphoenix Electric technical support
Er.FF05: Encoder Data Checksum Error	Motor encoder not initialized by manufacturer or information modified	Please contact Simphoenix Electric technical support
Er.FF06: Encoder Data Version Error	The drive does not support encoder internal data version	Update drive firmware, please refer to 5.2.4 Firmware Download
Er.FF08: Data Import Protection	Actively generate this error when performing batch data import	Execute save and restart after data import, and set <i>PrA.10 to 1 or operate 1010h-04h=1</i>
Er.FF09: Configuration Parameter Setting Error	Set unreasonable motion equivalent and enable it	Re-set reasonable motion equivalent data, refer to Section 8.1 Electronic Gear for settings
Er.FF0A: Output Phase Loss	One or more phases of the motor power lines (U/V/W) have abnormal connections	Confirm if power lines have breaks or poor contact, and reconnect; Contact Simphoenix Electric technical support to replace the servo motor
Er.FF0B: Motor Setting Error	Occur when motor encoder type is set to incremental encoder but drive hardware do not support it	Correctly set motor parameters, then restart the drive
Er.FF0C: Full-closed loop Z Signal Anomaly	Apply full-closed loop settings to a drive without a full-closed loop hardware circuit	Correctly set drive parameters, then restart
Er.FF0E: Full-Closed Loop Quadrature Signal Loss	Abnormal detection of quadrature input signals occurs during power-on due to issues such as wire breaks.	Check connections of the full-closed loop encoder, rectify EMC

➤ Er.FF0D: Excessive Mixed Control Deviation

Fault mechanism: Caused by excessive position error during full-closed loop mixed control

Fault Cause	Inspection Method	Solution
1. Motor stall due to mechanical factors	Use X Servo Configurator to monitor if the operating command and motor speed match, and monitor current feedback waveform	Troubleshoot mechanical factors
2. Servo drive parameter settings are unreasonable	Check servo drive ·Position loop gain: 4020h-01h ·Speed loop gain: Pr1.02 (4024h-01h) ·Profile speed: 6081h ·Position deviation threshold: 4010h-80h ·Profile acceleration/deceleration: 6083h/6084h ·Positive/negative torque limit: 60E0h/60E1h Whether the set values are reasonable	1. Re-perform servo gain adjustment 2. Set appropriate maximum mixed control position deviation threshold (4010h-80h) according to operating conditions 3. Adjust parameters such as command acceleration/deceleration time, torque limits
3. Host computer position command inappropriate	Position Control Mode: ·CSP mode: Check gear ratio 6091h-01h/6091h-02h to determine the increment value of position command corresponding to a single synchronous cycle, and convert to velocity information; ·PP mode: Check gear ratio 6091h-01h/6091h-02h to determine 6081h (profile run speed); ·HM mode: Check gear ratio 6091h-01h/6091h-02h to determine 6099.01h and 6099.02h.	·CSP: Reduce the position command increment corresponding to a single synchronous cycle. When planning commands in the host computer, increase the position ramp. ·PP: Reduce 6081h, or increase acceleration/deceleration (6083h, 6084h). ·HM: Reduce 6099h-01h and 6099h-02h, or increase origin acceleration/deceleration (609Ah). ·Adjust gear ratio according to actual conditions.
4. Servo drive/motor fault	Monitor operating waveforms using X Servo Configurator's oscilloscope function: position command, position feedback, speed command, torque command, etc.	If position command is non-zero but position feedback is always zero, please replace servo drive/motor

Maintenance

Due to the influence of various factors such as ambient temperature, humidity, dust, vibration, and aging or wear of internal components of the servo drive, the servo drive may have potential faults. To ensure long-term, stable operation of the servo drive, regular maintenance and upkeep must be performed during storage and use.

If the servo drive has undergone long-distance transportation, check if components are intact and screws are tightened before use. During normal use, regularly clean dust inside the servo drive and check for loose screws, etc.

High voltage is present during servo drive operation, and incorrect operations may lead to serious personal injury. Maintenance operations can only be performed by cutting off the servo drive power supply and waiting until the servo drive panel's digital display extinguishes (after ten minutes).



- Inspections must be performed by professional technical personnel, and the servo drive's power supply must be cut off.
- For servo drives stored for more than half a year, when powering on, the voltage should be slowly increased via a voltage regulator, otherwise there is a risk of electric shock and explosion (of internal electrolytic capacitors).

Daily Maintenance and Upkeep

Through daily inspections and maintenance, various abnormal situations can be discovered promptly, their causes identified in time, and potential faults eliminated early, ensuring normal equipment operation and extending the service life of the servo drive. Please refer to the table below for daily inspections and maintenance.

Inspection and Maintenance Checklist

Inspection Subject	Inspection Cycle		Inspection Items	Judgment Criteria
	Any time	Regular		
Operating Environment	√		1. Temperature, humidity 2. Dust, moisture 3. Gas	1. When temperature > 45°C, the servo drive cover should be opened, humidity < 95%, no frost 2. No abnormal odor, no flammable or explosive gases
Cooling System		√	1. Installation environment 2. Fan of servo drive	1. Good ventilation in the installation environment, no obstruction in the air duct 2. Body fan operates normally, no abnormal noise
Servo Drive	√		1. Vibration, temperature rise 2. Noise 3. Wires, terminals	1. Stable vibration, normal air outlet temperature 2. No abnormal noise, no abnormal odor 3. Tightened screws are not loose
Motor	√		1. Vibration, temperature rise 2. Noise	1. Stable operation, normal temperature 2. No abnormal, uneven noise
Input or Output Parameters	√		1. Input voltage 2. Output current	1. Input voltage within specified range 2. Output current below rated value



- The servo drive has undergone electrical insulation tests before leaving the factory, and users do not need to perform additional withstand voltage tests. Otherwise, internal components may be damaged.
- If insulation testing of the servo drive is necessary, all input and output terminals must be reliably short-circuited. Insulation testing on individual terminals is strictly prohibited. Use a 500V megohmmeter for testing.
- Do not use a megohmmeter to measure the control circuit. The servo drive contains electrostatic-sensitive components—direct contact is prohibited.
- When performing insulation testing on the motor, the connection between the motor and the servo drive must be disconnected.

Inspection and Replacement of Vulnerable Parts

Some components inside the servo drive may wear out or degrade in performance during use. To ensure stable and reliable operation of the servo drive, preventive maintenance should be performed, and components replaced when necessary.

Filter Capacitor

Possible causes of damage: High ambient temperature, large pulsating current, electrolyte aging.

Judgment criteria: Does the servo drive frequently experience faults like overcurrent or overvoltage during loaded operation? Is there any liquid leakage? Is the safety valve protruding? Are the measurements of static capacitance and insulation resistance abnormal?

- The pulsating current in the main circuit will affect the performance of aluminum electrolytic filter capacitors. The extent of the effect is related to ambient temperature and usage conditions. Servo drives used under normal conditions should have their electrolytic capacitors replaced every 3 to 4 years.
- When the electrolytic capacitor's electrolyte leaks, the safety valve protrudes, or the capacitor body expands, it should be replaced immediately.

Cooling Fan

Possible causes of damage: Bearing wear, blade aging, etc.

Judgment criteria: When the servo drive is powered off, check for cracks or other abnormalities in the fan blades and other parts; when the servo drive is powered on, check if the fan operation is normal, and if there is abnormal vibration, noise, etc.

- The service life of all cooling fans inside the servo drive is approximately 15,000 hours (i.e., approximately two years of continuous use of the servo drive). If the fan produces abnormal sound or vibration, it should be replaced immediately.

Storage

If the servo drive is purchased but not used immediately or stored for a long time, the following precautions should be observed:

Storage environment should comply with the table below:

Environmental Characteristics	Requirements	Remarks
Ambient Temperature	-10°C-45°C	Long-term storage temperature should not exceed 45°C to prevent capacitor characteristics from deteriorating. Environments where condensation or freezing may occur due to sudden temperature changes should be avoided.
Relative Humidity	5-95%	Measures such as sealing with plastic film and desiccants can be adopted.
Storage Environment	Do not expose to direct sunlight, free from dust, corrosive or flammable gases, oil, steam, gas, dripping water, vibration, and low in salt.	

If the servo drive is not used for a long time, it should be powered on every six months to restore the characteristics of the filter capacitors, and at the same time, check the other functions of the servo drive. When powering on, the voltage should be gradually increased via an autotransformer, and the power-on time should be more than half an hour.



- The performance of the internal filter capacitors in the servo drive may degrade if it remains unused for an extended period.

Warranty

The manufacturer will provide warranty service for the servo drive body under the following conditions:

Free repair is provided for failures or damages occurring under normal use within the warranty period (18 months from the date of purchase). After 18 months, a reasonable repair fee will be charged.

Even within the warranty period, a certain repair fee will be charged for faults caused by the following reasons:

- * **Faults caused by not following the operation manual or exceeding standard specifications;**
- * **Faults caused by unauthorized self-repair or modification;**
- * **Faults caused by improper storage;**
- * **Faults caused by using the servo drive for abnormal functions;**
- * **Machine damage caused by fire, salt corrosion, gas corrosion, earthquake, storm, flood, lightning, abnormal voltage, or other force majeure.**

Even beyond the warranty period, the manufacturer also provides lifelong paid repair services.

More

For more product information, please follow the product center and our official WeChat account (scan QR code).

Shenzhen Simphoenix Electric Technology Co., Ltd.

Address: Factory Building A, Huichao Industrial Zone, Gushu Second Road, Xixiang, Bao'an District, Shenzhen

Tel: (86) 0755-26919258

Fax: (86) 0755-26919882

Website: www.simphoenix.com.cn



Wanwei Electric (Huizhou) Co., Ltd.

No. 23 Songbailing Avenue, Start-up Area, China-Korea Huizhou Industrial Park, Zhongkai High-tech Zone, Huizhou City, Guangdong Province, China. Tel: (86) 0752-2600100

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